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VISION

"To foster professional level competence in all areas of Electronics and Communication Engineering and to benchmark the Department as a centre for nurturing Women Engineers in the Country"

MISSION

M1: To impart value based Technical education and training.

M2: To impart Theoretical Knowledge, Practical Knowledge and Entrepreneurship Skills.

M3: Fostering culture of innovation and research for development of society.

M4: To sensitize the Students regarding Social, Moral and Professional ethics.

M5: To provide industry standard certifications on skills to enhance students knowledge make them prepared for placements

Program Educational Objective's

PEO 1: To inculcate students to excel in professional career and/or higher education by acquiring knowledge in the field of Electronics and Communication.

PEO 2: To make the students capable of managing their profession based on existing as well as new emerging technologies in the area of Electronics and Communication Engineering.

PEO 3: To Produce Technically competent graduates with Ability to analyse, design, develop, optimise and implement Electronics and Communication systems.

PEO 4: To prepare the students to be able to exhibit professionalism, ethical attitude, communication skills, team work in their profession and to adapt to current trends by engaging in life-long learning.

Electronic Principles and Circuits		Semester	3
Course Code	BEC303	CIE Marks	50
Teaching Hours/Week (L:T:P: S)	3:0:2	SEE Marks	50
Total Hours of Pedagogy	40 hours Theory + 8-10 Lab slots	Total Marks	100
Credits	04	Exam Hours	
Examination nature (SEE)	Theory/Practical		
Course objectives: This course will enable students to • Design and analyse the BJT circuits as an amplifier and voltage regulation. • Design of MOSFET Amplifiers and analyse the basic amplifier configurations using small signal equivalent circuit models • Design of operational amplifiers circuits as Comparators, DAC and filters. • Understand the concept of positive and negative feedback. • Analyze Power amplifier circuits in different modes of operation. • Construct Feedback and Oscillator circuits using FET. • Understand the thyristor operation and the different types of thyristors.			
Teaching-Learning Process (General Instructions) These are sample Strategies, which teacher can use to accelerate the attainment of the various course outcomes. 1. Lecture method (L) does not mean only traditional lecture method, but different type of teaching methods may be adopted to develop the outcomes. 2. Show Video/animation films to explain evolution of communication technologies. 3. Encourage collaborative (Group) Learning in the class 4. Ask at least three HOTS (Higher order Thinking) questions in the class, which promotes critical thinking 5. Adopt Problem Based Learning (PBL), which fosters students' Analytical skills, develop thinking skills such as the ability to evaluate, generalize, and analyze information rather than simply recall it. 6. Show the different ways to solve the same problem and encourage the students to come up with their own creative ways to solve them. 7. Discuss how every concept can be applied to the real world - and when that's possible, it helps improve the students' understanding.			
MODULE-1			
BJT AC models: Base Biased Amplifier, Emitter Biased Amplifier, Small Signal Operation, AC Beta, AC Resistance of the emitter diode, Two transistor models, Analyzing an amplifier. Voltage Amplifiers: Voltage gain, Multistage Amplifiers. CC and CB Amplifiers: CC Amplifier, Output Impedance, Cascading CE and CC, Darlington Connections, Voltage regulation, The Common base Amplifier. [Text1]			
MODULE-2			

MOSFET

Biasing in MOS amplifier circuits: Fixing VGS, Fixing VG, Drain to Gate feedback resistor.

Small signal operation and modelling: The DC bias point, signal current in drain, voltage gain, small signal equivalent circuit models, transconductance, The T equivalent circuit model.

MOSFET Amplifier configuration: Basic configurations, characterizing amplifiers, CS amplifier with and without source resistance, The Common Gate Amplifier, Source follower.

MODULE-3

Linear Opamp Circuits: Summing Amplifier and D/A Converter, Nonlinear Op-amp Circuits: Comparator with zero reference, Comparator with non-zero references. Comparator with Hysteresis.

Oscillator: Theory of Sinusoidal Oscillation, The Wein-Bridge Oscillator, RC Phase Shift Oscillator, The Colpitts Oscillator, Hartley Oscillator, Crystal Oscillator.

The 555 timer: Monostable Operation, Astable Operation.

[Text1]

MODULE-4

Negative Feedback: Four Types of Negative Feedback, VCVS Voltage gain, Other VCVS Equations, ICVS Amplifier, VCIS Amplifier, ICIS Amplifier (No Mathematical Derivation).

Active Filters: Ideal Responses, First Order Stages, VCVS Unity Gain Second Order Low pass Filters, VCVS Equal Component Low Pass Filters, VCVS High Pass Filters, MFB Bandpass Filters, Bandstop Filters.

[Text1]

MODULE-5

Power Amplifiers: Amplifier terms, Two load lines, Class A Operation, Class B operation, Class B push pull emitter follower, Class C Operation.

Thyristors: The four layer Diode, SCR, SCR Phase control, Bidirectional Thyristors, IGBTs, Other Thyristors.

[Text1]

PRACTICAL COMPONENT OF IPCC (Experiments can be conducted either using any circuit simulation software or discrete components)

Sl.NO	Experiments
1	Design and Test (i) Bridge Rectifier with Capacitor Input Filter (ii) Zener voltage regulator
2	Design and Test Biased Clippers – a) Positive, b) Negative, c) Positive-Negative Positive and Negative Clampers with and without Reference.
3	Plot the transfer and drain characteristics of a JFET and calculate its drain resistance, mutual conductance and amplification factor.
4	Plot the transfer and drain characteristics of n-channel MOSFET and calculate its parameters, namely; drain resistance, mutual conductance and amplification factor.
5	Design and test (i) Emitter Follower, (ii) Darlington Connection
6	Design and plot the frequency response of Common Source JFET/MOSFET amplifier
7	Test the Opamp Comparator with zero and non zero reference and obtain the Hysteresis curve.
8	Design and test Full wave Controlled rectifier using RC triggering circuit.
9	Design and test Precision Half wave and full wave rectifiers using Opamp
10	Design and test RC phase shift oscillator

Course outcomes (Course Skill Set):

At the end of the course, the student will be able to:

- Understand the characteristics of BJTs and FETs for switching and amplifier circuits.
- Design and analyze amplifiers and oscillators with different circuit configurations and biasing conditions.
- Understand the feedback topologies and approximations in the design of amplifiers and oscillators.
- Design of circuits using linear ICs for wide range applications such as ADC, DAC, filters and timers.
- Understand the power electronic device components and its functions for basic power electronic circuits.

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). The student is declared as a pass in the course if he/she secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

The IPCC means the practical portion integrated with the theory of the course. CIE marks for the theory component are **25 marks** and that for the practical component is **25 marks**.

CIE for the theory component of the IPCC

- 25 marks for the theory component are split into **15 marks** for two Internal Assessment Tests (Two Tests, each of 15 Marks with 01-hour duration, are to be conducted) and **10 marks** for other assessment methods

mentioned in 22OB4.2. The first test at the end of 40-50% coverage of the syllabus and the second test after covering 85-90% of the syllabus.

- Scaled-down marks of the sum of two tests and other assessment methods will be CIE marks for the theory component of IPCC (that is for **25 marks**).
- The student has to secure 40% of 25 marks to qualify in the CIE of the theory component of IPCC.

CIE for the practical component of the IPCC

- **15 marks** for the conduction of the experiment and preparation of laboratory record, and **10 marks** for the test to be conducted after the completion of all the laboratory sessions.
- On completion of every experiment/program in the laboratory, the students shall be evaluated including viva-voce and marks shall be awarded on the same day.
- The CIE marks awarded in the case of the Practical component shall be based on the continuous evaluation of the laboratory report. Each experiment report can be evaluated for 10 marks. Marks of all experiments' write-ups are added and scaled down to **15 marks**.
- The laboratory test (**duration 02/03 hours**) after completion of all the experiments shall be conducted for 50 marks and scaled down to **10 marks**.
- Scaled-down marks of write-up evaluations and tests added will be CIE marks for the laboratory component of IPCC for **25 marks**.
- The student has to secure 40% of 25 marks to qualify in the CIE of the practical component of the IPCC.

SEE for IPCC

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours**)

5. The question paper will have ten questions. Each question is set for 20 marks.
6. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
7. The students have to answer 5 full questions, selecting one full question from each module.
8. Marks scored by the student shall be proportionally scaled down to 50 Marks

The theory portion of the IPCC shall be for both CIE and SEE, whereas the practical portion will have a CIE component only. Questions mentioned in the SEE paper may include questions from the practical component.

- The minimum marks to be secured in CIE to appear for SEE shall be 10 (40% of maximum marks-25) in the theory component and 10 (40% of maximum marks -25) in the practical component. The laboratory component of the IPCC shall be for CIE only. However, in SEE, the questions from the laboratory component shall be included. The maximum of 04/05 sub-questions are to be set from the practical component of IPCC, the total marks of all questions should not be more than 20 marks.
- SEE will be conducted for 100 marks and students shall secure 35% of the maximum marks to qualify for the SEE. Marks secured will be scaled down to 50.
- The student is declared as a pass in the course if he/she secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Suggested Learning Resources:**Books**

1. Albert Malvino, David J Bates, Electronic Principles, 7th Edition, Mc Graw Hill Education, 2017, ISBN:978-0-07-063424-4.
2. Microelectronic Circuits, Theory and Applications, Adel S Sedra, Kenneth C Smith, 6th Edition, Oxford, 2015. ISBN:978-0-19-808913-1

Web links and Video Lectures (e-Resources):

1. Integrated Electronics: Analog and Digital Circuits and Systems, Jacob Millman, Christos C. Halkias, McGraw-Hill, 2015.
2. Electronic Devices and Circuit, Boylestad & Nashelsky, Eleventh Edition, Pearson, January 2015.

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

Course Outcome

After the completion of the Course, the students will be able to

- CO1** Design BJT Amplifier circuits with different circuit configurations and biasing conditions.
- CO2** Design MOSFET Amplifier circuits with different circuit configurations and biasing conditions.
- CO3** Design Oscillator, Opamp and 555 Timer circuits for various applications.
- CO4** Apply the design of filter circuits and feedback topologies in amplifier and oscillator circuits.
- CO5** Understand the characteristics and functions of power electronic devices and thyristors for basic power electronic circuits.

Module - 1

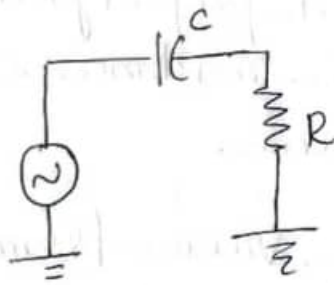
AC Models

Base-Biased Amplifier

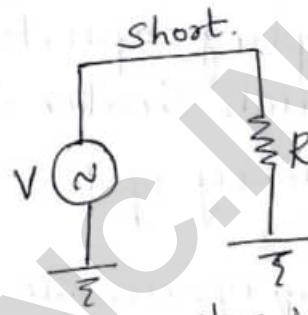
Coupling Capacitor

Coupling capacitor couples or transmits the ac signal ~~through it~~ into an amplifier without disturbing its Q-point.

→ Consider the circuit shown below.



a) coupling capacitor



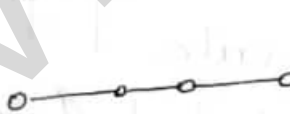
(b) capacitor is an ac short



Capacitor



DC → for DC signal capacitor acts as open ckt.



AC → for AC signal capacitor acts as short ckt.

fig(c).

→ Consider fig(a) in which ac voltage source connected to a capacitor and a resistor.

→ ~~The impedance~~ X_c is,

$$X_c = \frac{1}{2\pi fC} \quad \text{--- (1)}$$

from eqⁿ (1) impedance of capacitor is inversely proportional to f .

to the frequency, hence the capacitor effectively blocks dc voltage and transmits ac voltage.

→ At high frequency, the capacitive reactance is much smaller than the resistance.

And in this case, almost all the ac source voltage appears across the resistor.

→ When capacitor is used in this way, it is referred to as coupling capacitor

→ For a coupling capacitor to work properly, its reactance must be much smaller than the resistance at the lowest frequency of the ac source.

for EX :- If freq of ac ~~src~~ source varies from 20Hz to 20KHz, the worst case occurs at 20Hz.

→ So select a capacitor whose reactance at 20Hz is much smaller than the resistance.

i.e Good coupling: $X_c < 0.1R$

this is called 10:1 rule

→ When the 10:1 rule is satisfied fig(a) can be replaced with fig(b) as The magnitude of impedance of fig(a) is given by

$$Z = \sqrt{R^2 + X_c^2}$$

→ When 10:1 rule (worst-case) condition is satisfied

$$Z = \sqrt{R^2 + (0.1R)^2} = \sqrt{R^2 + 0.01R^2} = \sqrt{1.01R^2}$$

$$Z = 1.005R$$

→ Since the impedance is within half of a percent of R at the lowest freq, the current in fig(a) is only half a percent less than the current in fig(b)

→ As any well-designed circuit satisfies the 10:1 rule all coupling capacitor can be approximated to an ac short. [fig(b)]

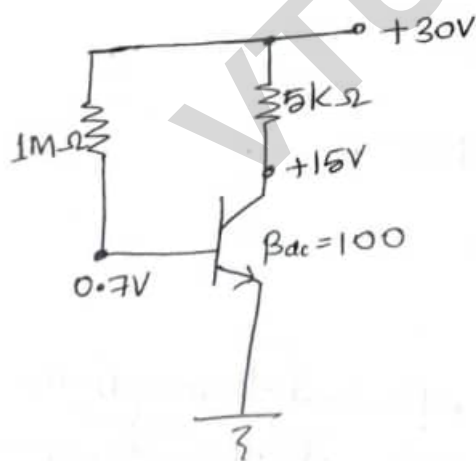
Coupling capacitor :- Wkt DC vlg has a zero freq, the reactance of a coupling capacitor is infinite at zero freq.

∴ For dc analysis → the capacitor is replaced with open ckt.

& for ac analysis → the capacitor is shorted.
as in fig(c)

DC circuit

Consider the Base-biased circuit



→ The Base current is given by

$$I_B = \frac{30V}{1M\Omega} = 30\mu A$$

and for $\beta_{DC} = 100$, the collector current I_C is given as

$$I_C = \beta I_B = (100)(30 \mu A)$$

$$I_C = 3 \text{ mA}$$

and collector voltage given by

$$V_C = V_{CC} - I_C R_C$$

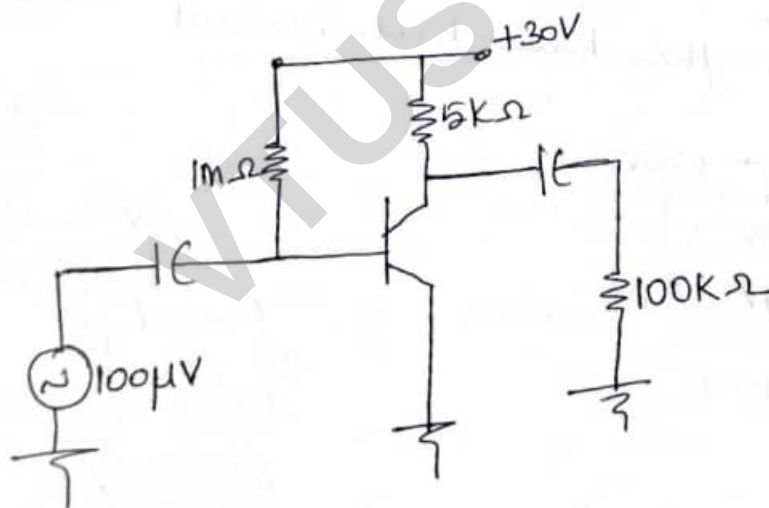
$$= 30 - (3 \text{ mA})(5 \text{ k}\Omega)$$

$$V_C = 15 \text{ V}$$

$\therefore$ Q-point is located at 3 mA & 15 V.

Amplifying circuit

Once the Q-point is located on DC load line the amplifying ckt is built by adding coupling capacitor as shown below.



The ac source voltage is $100 \mu V$, passed through the coupling capacitor which is ac short, all the ac sec V/g appears b/w the base and the ground.

→ This ac voltage produces ac base current that is added with existing dc base current.

and for $\beta_{ac} = 100$, the collector current I_c is given as

$$I_c = \beta I_B = (100)(30 \mu A)$$

$$I_c = 3mA$$

and collector voltage given by

$$V_c = V_{cc} - I_c R_c$$

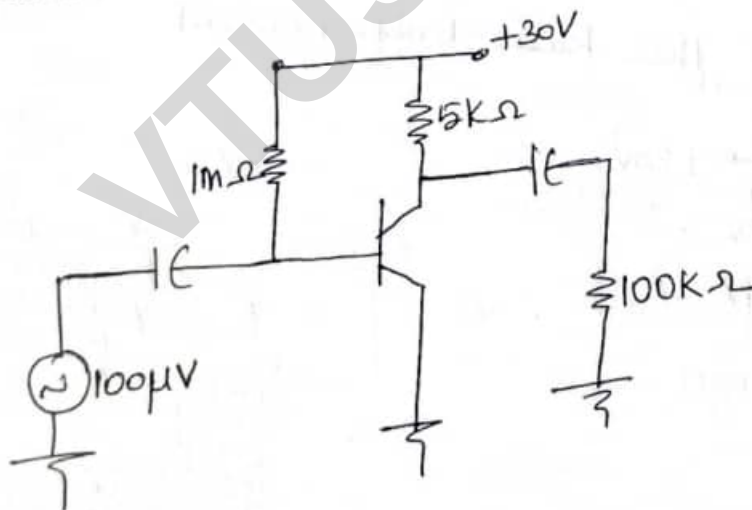
$$= 30 - (3m)(5K\Omega)$$

$$V_c = 15V$$

$\therefore$ Q-point is located at 3mA & 15V.

Amplifying circuit

Once the Q-point is located on DC load line the amplifying ckt is built by adding coupling capacitors as shown below.



The ac source voltage is $100 \mu V$, passed through the coupling capacitor which is ac short, all the ac V_{be} appears b/w the base and the ground.

→ This ac voltage produces ac base current that is added with existing dc base current.

and for $\beta_{DC} = 100$, the collector current I_C is given as

$$I_C = \beta I_B = (100)(30 \mu A)$$

$$I_C = 3mA$$

and collector voltage given by

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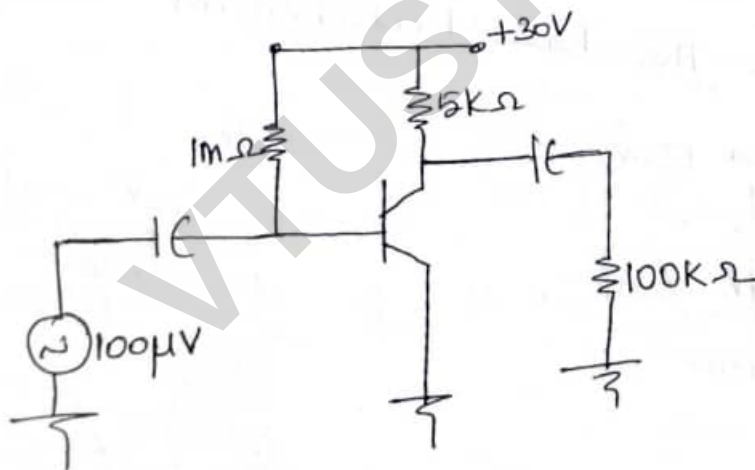
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$$I_c = 3mA$$

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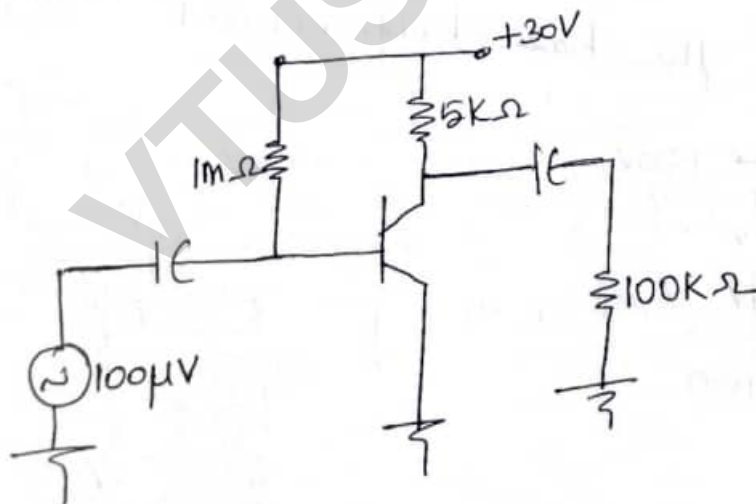
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Amplifying circuit

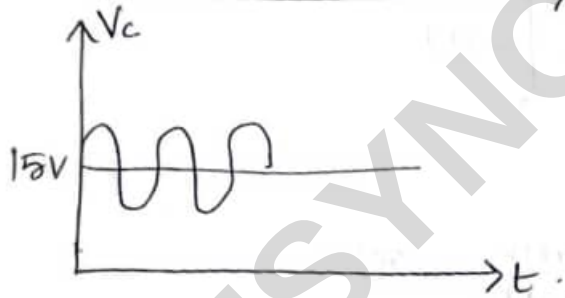
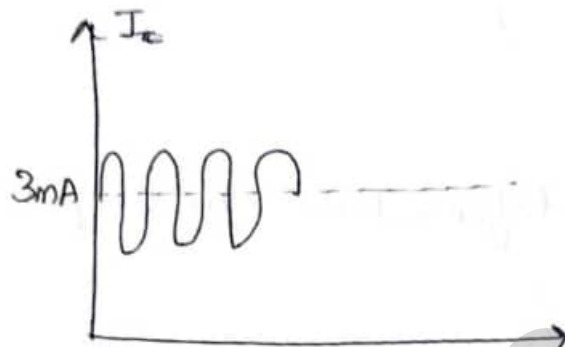
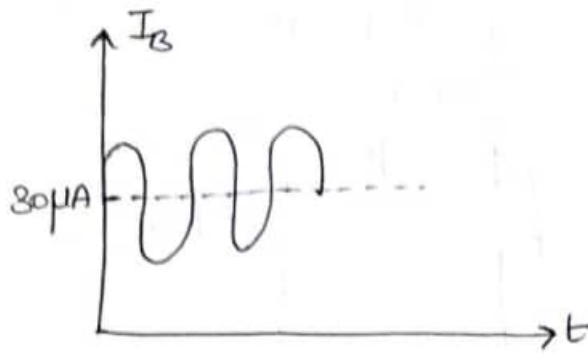
Once the Q-point is located on DC load line the amplifying ckt is built by adding coupling capacitors as shown below.



The ac source voltage is 100μV, passed through the coupling capacitor which is ac short, all the ac V_{be} appears b/w the base and the ground.

→ This ac voltage produces ac base current that is added with existing dc base current.

∴ The total base current is sum of dc base current + ac base current. as illustrated in fig below.

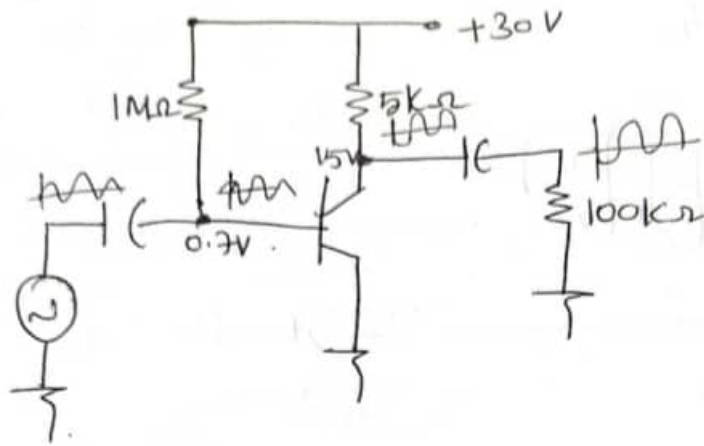


→ On positive half cycle, the ac base current adds to the $30 \mu\text{A}$ of dc base current, & on negative half cycle it subtracts from it.

→ ~~III~~ ^hy collector current & collector v/g is shown in fig.

→ The ^{ac} collector voltage is inverted 180° out of phase as on +ve half cycle of ac base current, the collector current increases, producing more v/g across the collector resistor.

→ ~~P~~ This means there that there is less v/g b/w the collector & gnd. ~~III~~ ^hy on +ve half cycle reverse happens.



Voltage Gain

Voltage Gain is defined as ac output v/g divided by ac input v/g

$$A_v = \frac{V_{out}}{V_{in}} \quad \text{--- (1)}$$

Let $V_{out} = 50mV$

$$\text{then } A_v = \frac{50mV}{100\mu V} = 500$$

∴ ac o/p v/g is 500 times larger than ac i/p v/g.

then V_{out} & V_{in} can be found from eqn (1) as

$$V_{in} = \frac{V_{out}}{A_v} \quad \& \quad V_{out} = A_v V_{in}$$

Emitter - Biased Amplifier

Base - Biased amplifier has an unstable Q-pt & hence not used as an amplifier.

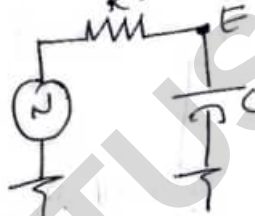
→ And emitter - biased amplifiers such as VDB or TSEB with stable Q-point are preferred.

Bypass capacitor

→ The function of bypass capacitor is to create ac ground.

→ Similar to coupling capacitor, bypass capacitor appears open ckt to DC & short ckt to ac.

→ Consider the circuit shown below.



→ here ac voltage source is connected to a resistor & a capacitor

→ $R \rightarrow$ the Thevenin resistance as seen by the capacitor

→ when the frequency is high from the eqn $X_c = \frac{1}{2\pi fC}$ the

Capacitance X_c is smaller than resistance. & hence ~~all~~ almost all the ac voltage that appears across resistor is shorted to the ground through capacitor C.

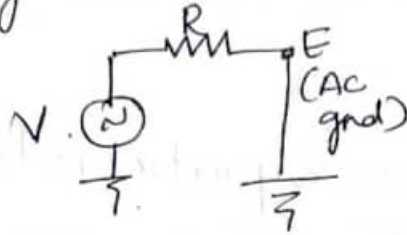
→ The capacitor used in this way is called Bypass capacitor.

Capacitor as it bypasses or shorts the point E to ground.

→ For Good bypassing

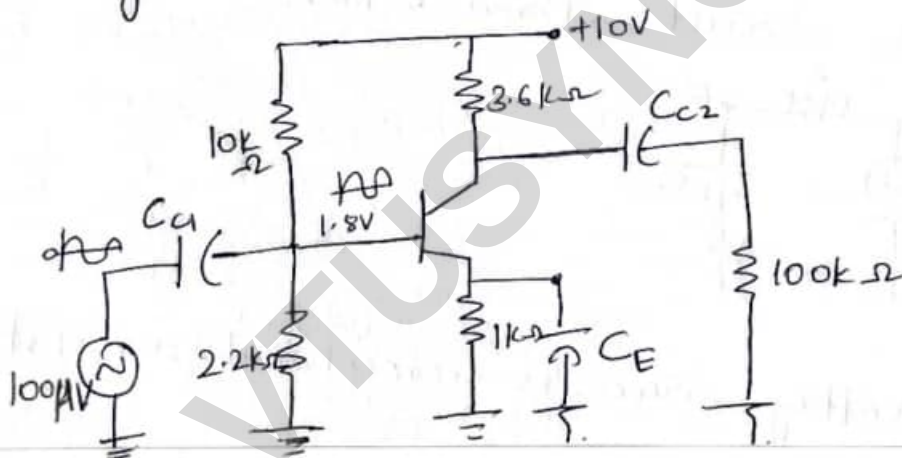
$$X_C < 0.1R \text{ at the lowest freq of the ac source.}$$

→ When this condition is satisfied, we can replace Capacitor by a short ckt as shown below.



VDB Amplifier

Fig below shows Voltage-Divider Bias (VDB) ampk.



→ The DC operation analysis of the above ckt is done by making all capacitors open circuit.

→ The DC voltages & currents are calculated using the equations of VDB as follows :-

$$V_B = \frac{R_2}{R_1 + R_2} V_{CC} = \left[\frac{2.2K}{10K + 2.2K} \right] 10V = 1.8V$$

$$V_E = V_B - V_{BE} = 1.8 - 0.7V = 1.1V$$

$$I_C \approx I_E = \frac{V_E}{R_E} = \frac{1.1V}{1K\Omega} = 1.1mA$$

$$V_C = V_{CC} - I_C R_C = 10V - (1.1mA)(3.6k) = 6.04V.$$

The circuit shows the Voltage-Divider Bias amplifier with two coupling capacitors and one Emitter Bypass capacitor. The ac open of the ckt is:

- ac voltage source of $100\mu V$ is coupled to base through the coupling capacitor C_1 , as shown in ckt
- Because of the bypass capacitor C_E , all of this ac voltage appears across the base-emitter diode
- without bypass capacitor, ac ~~current~~ base current would be much smaller.
- The ac base current then produces an amplified ac collector voltage at collector terminal which is then coupled to load resistor through another coupling capacitor C_2 .
- we notice the waveforms. The ac source voltage is a small sinusoidal voltage with average value $0V$.
- The base voltage is an ac voltage superimposed on dc voltage of $+1.8V$
- The collector v/g is an amplified & inverted ac v/g superimposed on the dc v/g of $+6.04V$.
- The load v/g is same as collector voltage but it has an average value of Zero volts.

TSEB amplifier

Consider the TSEB amplifier circuit shown below

→ The dc analysis of the circuit is done by making all capacitors open circuit

→ The DC voltages & currents are calculated by using eqns of TSEB as follows.

$$V_B \approx 0V$$

$$V_E = -0.7V$$

$$I_E = \frac{V_{RE}}{R_E}$$

$$V_C = V_{CC} - I_C R_C$$

$$V_{RE} = -0.7V - (-2V)$$

$$I_C \approx I_E$$

$$= 1.3V$$

$$V_C = 10V - (1.3m)(3.6k\Omega) \therefore I_E = \frac{1.3V}{3.0k\Omega}$$

$$= 5.32V$$

$$I_E = 1.3mA$$

→ The ac analysis is as follows.

→ There are two coupling capacitors & an emitter bypass capacitor.

TSEB amplifier

Consider the TSEB amplifier circuit shown below

→ The dc analysis of the circuit is done by making all capacitors open circuit

→ The DC voltages & currents are calculated by using eqns of TSEB as follows.

$$V_B \approx 0V$$

$$V_E = -0.7V$$

$$V_C = V_{CC} - I_C R_C$$

$$I_C \approx I_E$$

$$V_C = 10V - (1.3mA)(3.6k\Omega)$$

$$= 5.32V$$

$$I_E = \frac{V_{RE}}{R_E}$$

$$V_{RE} = -0.7V - (-2V)$$
$$= 1.3V$$

$$\therefore I_E = \frac{1.3V}{1.0k\Omega}$$

$$I_E = 1.3mA$$

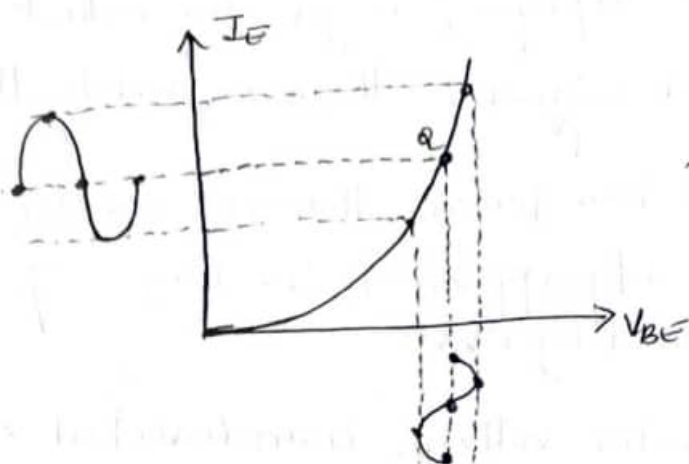
→ The ac analysis is as follows.

→ There are two coupling capacitors & an emitter bypass capacitor.

- The coupling capacitor C_1 couples a signal into the base of transistor.
- The signal is amplified to get the collector v/g.
- This amplified signal is then coupled to the load.
- To understand waveform, the ac source v/g is small sinusoidal v/g applied to the base of transistor which is approximately 0V.
- The total collector voltage is an inverted sine wave riding on the dc collector v/g of +5.32V.
- The o/p load v/g is the same amplified s/w with no dc component.

Small-Signal Operation

Fig (a) shows the graph of current versus V_{BE} for the base-emitter diode.



fig(a)

→ When an ac V_{BE} is coupled into the base of a transistor, an ac voltage appears across the base-emitter diode. This produces the sinusoidal variation in V_{BE} as in fig(a).

→ Instantaneous Operating point.

When the voltage V_{BE} increases to its +ve peak, the instantaneous operating point moves from Q pt to the upper point as in fig(a).

→ ~~When~~ on otherhand, when the voltage V_{BE} decreases to its -ve peak, the Q -point moves to lower point.

→ $\therefore$ the total base-emitter voltage is an ac voltage centred on a dc V_{BE} .

→ The size of the ac voltage determines how far the instantaneous point moves away from Q -point.

Distortion

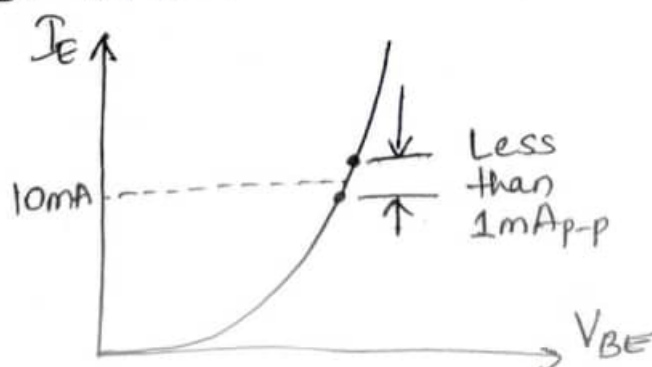
- The ac voltage on the base produces the ac emitter current as in fig(a)
- This ac emitter current has the same freq as the ac base voltage and approximately same shape (sinusoidal)
- But the ac emitter current is not a perfect replica of the ac base voltage because of the curvature of the graph.
- Since the graph is curved upward, the +ve half cycle of ac emitter current is elongated (stretched) & -ve half cycle is compressed.
- This stretching & compressing of alternate half cycles is called distortion
- This is undesirable in high-fidelity amplifiers because it changes the sound of voice & music.

Reducing Distortion

One way to reduce distortion of fig(a) is to keep ac base voltage small.

→ When voltage is kept small, the movement of instantaneous operating point is reduced.

→ If the signal is small enough, then the graph appears to be linear as in fig(b)



fig(b)

10 percent Rule

The total emitter current in amplifier is consist of a dc component & an ac component, which is given as

$$I_E = I_{EQ} + i_e$$

where, $I_E \rightarrow$ the total emitter current
 $I_{EQ} \rightarrow$ the dc emitter current
 $i_e \rightarrow$ the ac emitter current.

$\rightarrow$ For minimizing distortion, the peak-to-peak value of i_e must be small compared to I_{EQ}
 $\therefore$ the small-signal operation is defⁿ is given by

$$\text{small signal: } i_{e(p.p)} < 0.1 I_{EQ}$$

i.e the signal is said to be small ac signal when peak-to-peak ac emitter current is less than 10 percent of dc emitter current.

$\rightarrow$ We refer amplifiers that satisfy the 10 percent rule as small-signal amplifiers.

10 percent Rule

The total emitter current in amplifier is consist of a dc component & an ac component, which is given as

$$I_E = I_{EQ} + i_e$$

where, $I_E \rightarrow$ the total emitter current
 $I_{EQ} \rightarrow$ the dc emitter current
 $i_e \rightarrow$ the ac emitter current.

$\rightarrow$ For minimizing distortion, the peak-to-peak value of i_e must be small compared to I_{EQ}

$\therefore$ the small-signal operation is defⁿ is given by

$$\text{small signal: } i_{e(pp)} < 0.1 I_{EQ}$$

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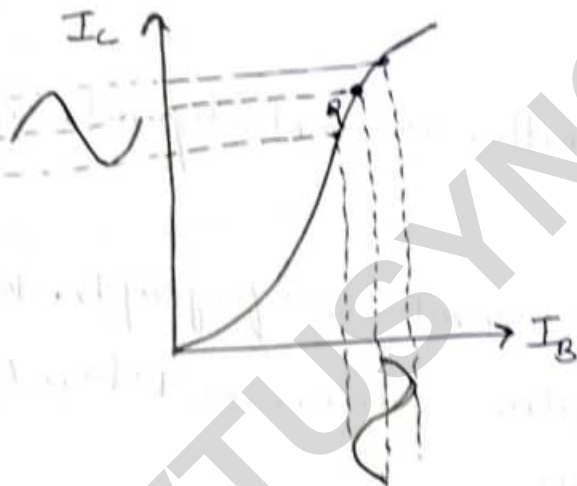
AC Beta

→ The dc current gain is defined as

$$\beta_{dc} = \frac{I_c}{I_B}$$

The currents in this formula are the currents at the Q-point as in fig(a).

→ Because of the curvature in the graph of I_c versus I_B , the dc current gain depends on the location of Q-pt.



→ The ac current gain is defined as

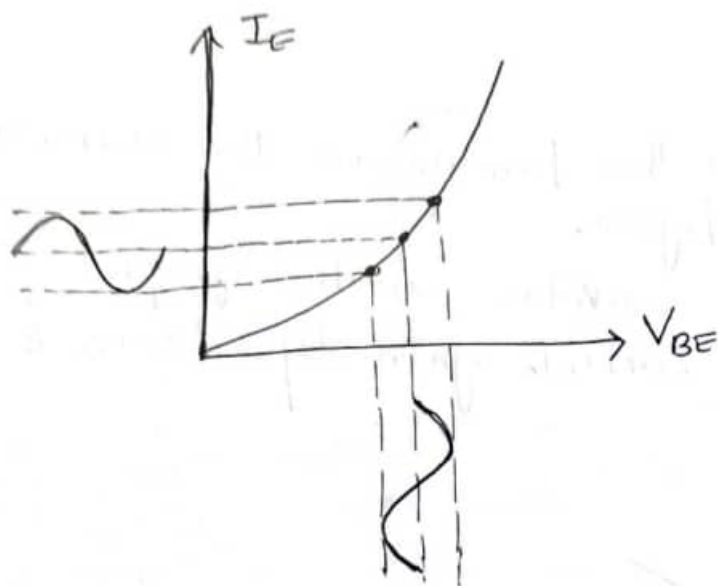
$$\beta = \frac{i_c}{i_b}$$

→ The ac signal uses only a small part of the graph on both sides of the Q-point whereas the dc signal uses almost all of the graph. Hence ac current gain is different from dc current gain.

→ on datasheet β_{dc} is listed as h_{FE} & β is shown as h_{fe} .

AC Resistance of the Emitter Diode.

Fig(a) shows a graph of V_{BE} v/s I_E



→ The size of ac emitter current depends on the location of Q-point.

→ ~~As the~~ ~~Q-point~~ Because of curvature of graph, the ac emitter current value increases when Q-point is higher up on the graph.

→ i.e., the total emitter current = dc component + ac component

$$I_E = I_{EQ} + i_e$$

Similarly, the total base-emitter voltage = dc component + ac component

$$\text{i.e. } V_{BE} = V_{BEQ} + v_{be}$$

→ A fixed value for v_{be} produces more i_e when Q-point is biased higher up the curve also ac resistance of emitter diode decreases when

dc emitter current increases.

∴ The ac emitter resistance of emitter diode is defined as

$$r_e' = \frac{V_{be}}{i_e}$$

→ The prime (') in r_e' is to indicate that the resistance is inside the transistor.

For instance, Let $V_{be} = 5\text{mV}_{p-p}$, the Q-point is set up in such a way to produce $i_e = 100\mu\text{A}_{p-p}$

$$\therefore r_e' = \frac{5\text{mV}}{100\mu\text{A}} = 50\Omega$$

As another example Let $V_{be} = 5\text{mV}_{p-p}$ & Q-point is set up higher on graph to produce $i_e = 200\mu\text{A}_{p-p}$

$$\therefore r_e' = \frac{5\text{mV}}{200\mu\text{A}} = 25\Omega$$

∴ The ac emitter resistance decreases when the dc emitter current increases, because V_{be} is constant

→ In solid-state physics & calculus, the r_e' is given by

$$r_e' = \frac{25\text{mV}}{I_E}$$

it applies to all transistor types.

& in practice r_e' ranges b/w $\frac{25\text{mV}}{I_E}$ & $\frac{50\text{mV}}{I_E}$

→ r_e' is important, as it determines the voltage gain.
 smaller r_e' , the higher the voltage gain.

Two Transistor Models

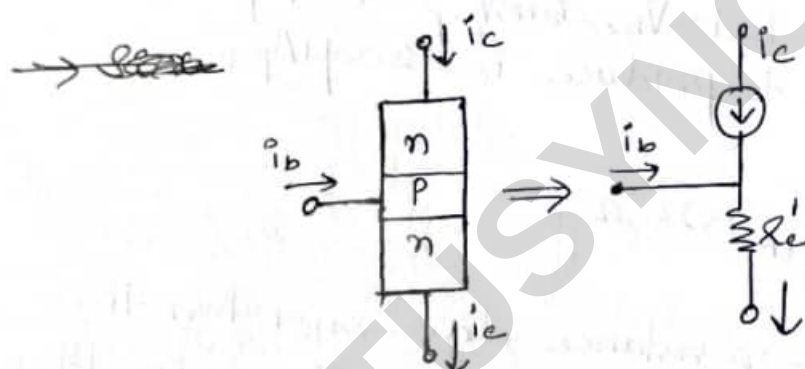
We need model of transistors that simulates.

The model for transistor is needed to simulate its behaviour when ac signal is present.

- 1] T Model
- 2] π Model

T-Model [Ebers-Moll Model]

For small ac signal, the emitter diode of a transistor acts like an ac resistance r_e' , & the collector diode acts like a current source i_c .



fig(a):- T-model

Since the Ebers-Moll model looks like T, it is also called as T-Model.

→ When analyzing a transistor amplifier for small ac signal, we can replace each transistor by a T-Model.

→ When an ac input signal drives a transistor amplifier, an ac base-emitter voltage V_{be} is produced across the emitter diode as in fig(a).

Two Transistor Models

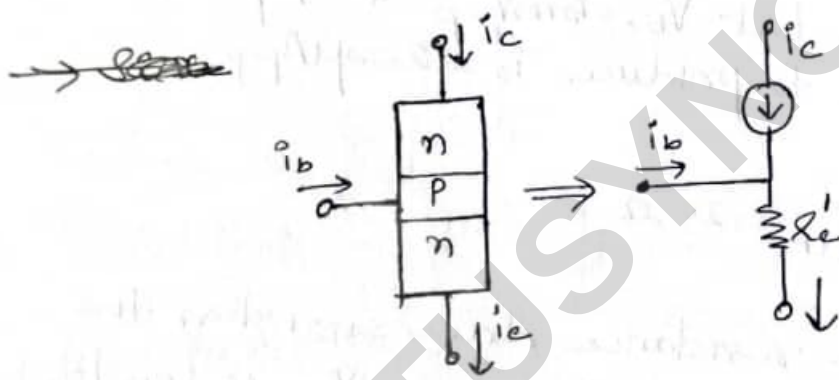
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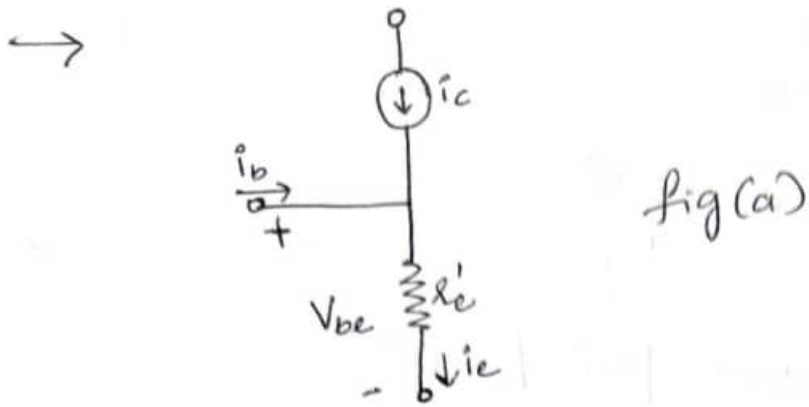


fig(a) :- T-model

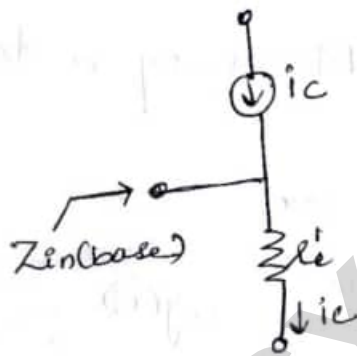
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→ When analyzing a transistor amplifier for small ac signal, we can replace each transistor by a T-Model.

→ When an ac input signal drives a transistor amplifier, an ac base-emitter voltage V_{be} is produced across the emitter diode as in fig(a).



→ The V_{be} voltage produces ac base current i_b .
 $\therefore$ ac Voltage V_{be} is loaded with input impedance of the base. as in fig(b)



→ Looking into the base of the transistor, the ac voltage source sees an input impedance $Z_{in(base)}$.
 At low frequencies, this impedance is purely resistive & defined as

$$Z_{in(base)} = \frac{V_{be}}{i_b} \quad \text{--- (1)}$$

→ Applying Ohm's law to the emitter diode of fig(a), we get

$$V_{be} = i_e r_e' \quad \text{--- (2)}$$

→ substituting eqⁿ (2) in (1), we get

$$Z_{in(base)} = \frac{i_e r_e'}{i_b}$$

as we know

$$i_e \approx i_c$$

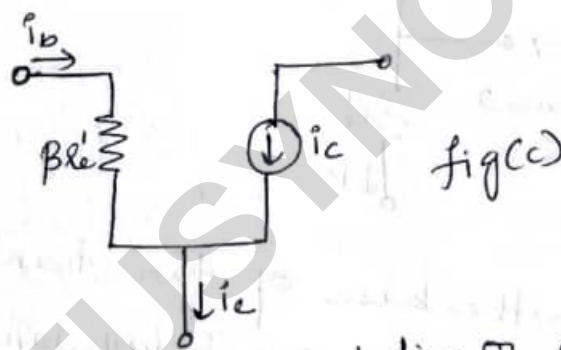
$$Z_{in(base)} = \frac{i_c r_e'}{i_b \beta}$$

$$\boxed{Z_{in(base)} = \beta r_e'} \quad \text{--- (3)}$$

$\therefore$ Input impedance of base is equal to the ac current gain times the ac resistance of the emitter diode.

π Model

Fig (c) shows the π -Model of a transistor.



It is actually visual representation of eqⁿ (3).

- π -Model is easier to use than T-Model because the input impedance of $\beta r_e'$ is not explicitly shown on circuit.
- On otherhand, the π -Model clearly shows that an input impedance of $\beta r_e'$ will load the ac voltage source driving the base.
- Since π & T Models are ac equivalent circuits for a transistor, we can use either one when analyzing an amplifier.

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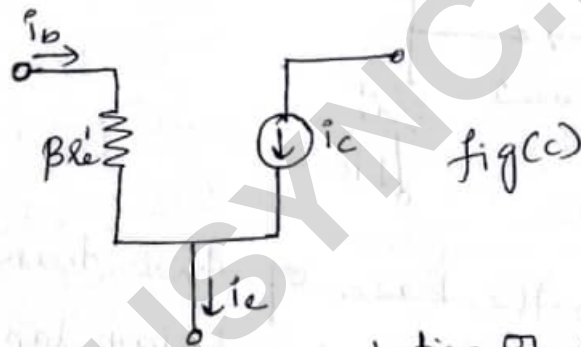
$$Z_{in(base)} = \frac{i_c e'}{i_b} \beta$$

$$\boxed{Z_{in(base)} = \beta e'} \quad \text{--- (3)}$$

$\therefore$ Input impedance of base is equal to the ac current gain times the ac resistance of the emitter diode.

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- Since π & T Models are ac equivalent circuits for a transistor, we can use either one when analyzing an amplifier.

Analyzing an Amplifier

Amplifier analysis is complicated because both dc & ac sources are in same ckt.

→ Hence we simplify this analysis by first calculating the effect of the dc sources and then the effect of the ac sources.

→ Then we apply superposition theorem to get the total effect of all sources acting simultaneously.

Step-1

* DC Equivalent circuit:-

In DC analysis, we calculate the DC Voltages & currents.

→ To calculate this, we have to replace all capacitors by open circuit. The circuit remains after this step is the DC Equivalent circuit.

* Step-2

AC Effect of a DC Voltage source:-

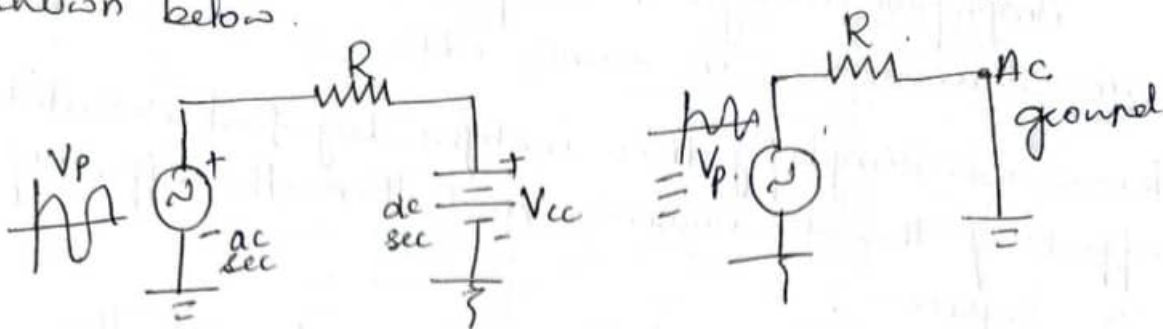
→ For the ac current, the dc voltage source acts like an ac short, this is because a dc voltage source has a constant voltage across it.

→ And any ac current flowing through it cannot produce an ac voltage across it.

→ i.e. if no ac voltage can exist, the dc voltage source is equivalent to an ac short.

∴ When analyzing the ac operation of an amplifier,

Each dc voltage supply point acts like an ac ground as shown below.



Step-3:-

AC Equivalent circuit

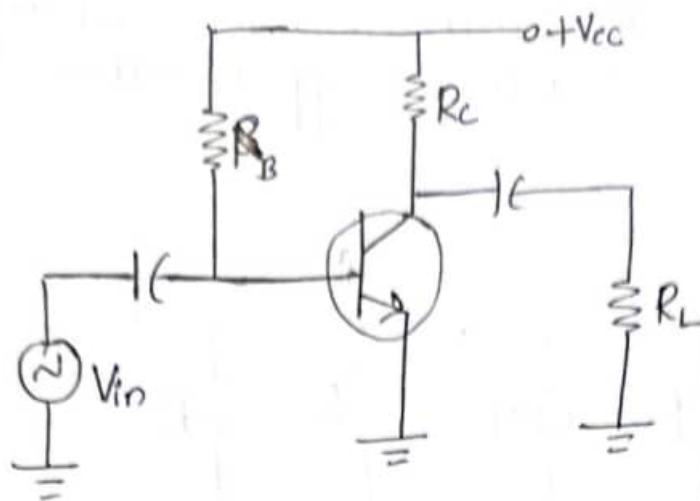
After analysing the dc equivalent circuit completely, the next step is to analyze the ac equivalent ckt.

- ~~The first st~~
- To analyze ac equivalent circuit, we short all Capacitors. first. and then we short all dc voltage sources and short it to ac ground.
- In the circuit that remains after this step, we replace transistor by either the π -model or the T-model.
- Once this is done the resulting circuit is referred as ac equivalent circuit.

* Now let us analyze Base-biased Amplifier, VDB amplifier and TSEB Amplifiers.

i] Base - Biased Amplifier.

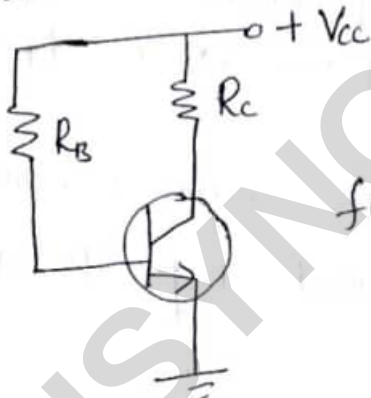
Consider Base-biased amplifier circuit as shown below fig(a).



fig(a) Base-biased amplifier.

Step 1:- DC equivalent circuit.

Replace all capacitors by open circuit and rewrite the resulting circuit.



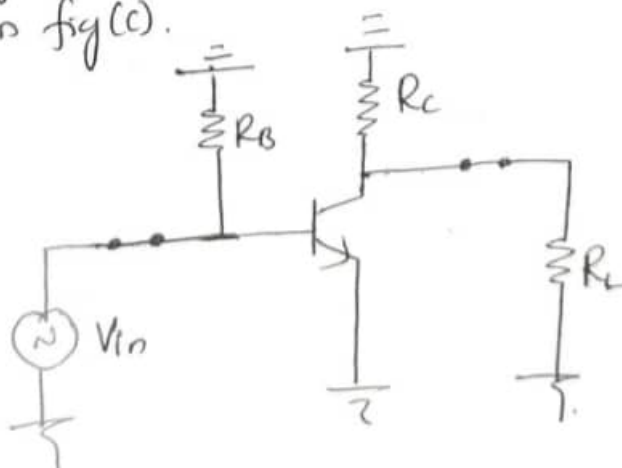
fig(b):- DC equivalent ckt.

→ The above circuit is DC equivalent circuit.

Step-2:- AC Equivalent circuit

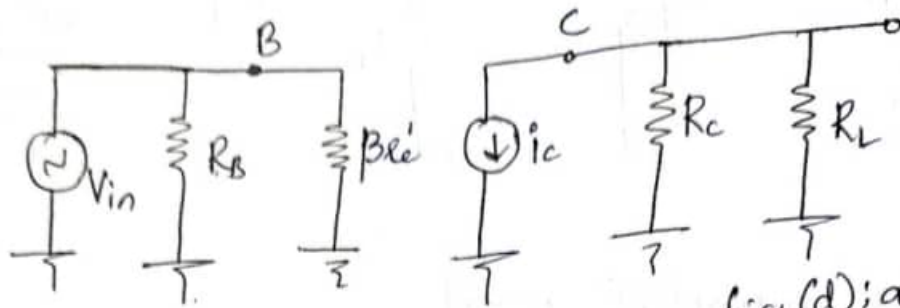
→ Now let us do the ac analysis by considering fig(a) circuit.

→ For ac analysis we short all capacitors and dc voltage source +V_cc is connected to ac ground as below in fig(c).



fig(c)

→ Now replace the transistor by its π -model and leave the circuit component without affecting its actual operation.

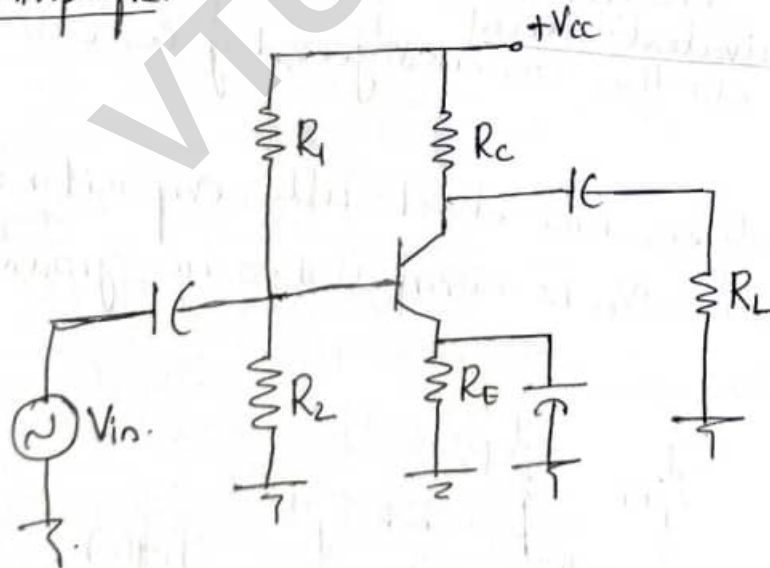


fig(d): ac equivalent ckt.

→ The above is ac equivalent circuit, in which V_{in} the input voltage appears across R_B in parallel with βi_e .

→ In collector circuit, the current source pumps an ac current of i_c through R_C in parallel with R_L .

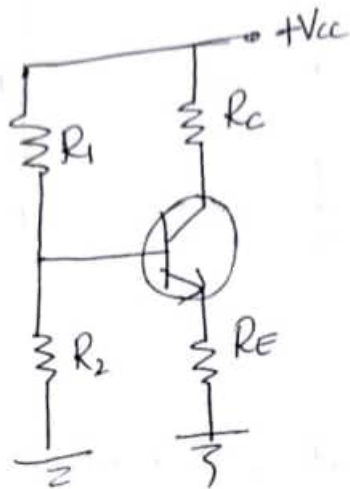
ii) VDB Amplifier



fig(a): VDB Amplifier.

1] DC equivalent circuit

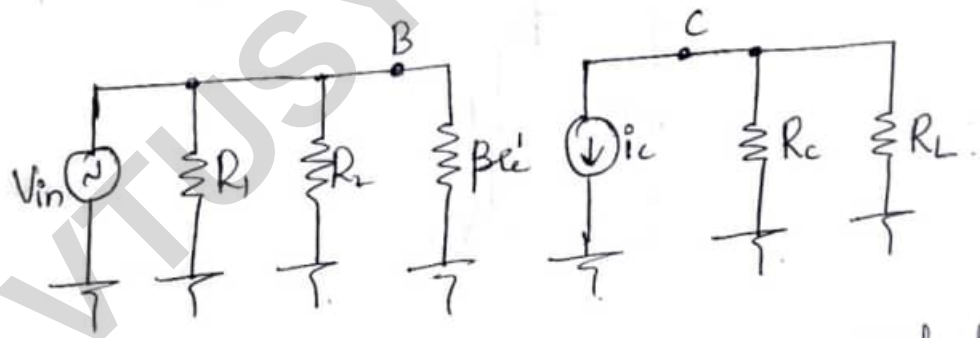
Open circuit all capacitor & redraw the resulting dc equivalent circuit.



fig(b):- DC equivalent ckt

2] ac equivalent circuit

short circuit all capacitors in fig(a) & connect a dc voltage source to ac ground also replace the transistor by its π -model to obtain ac equivalent circuit.



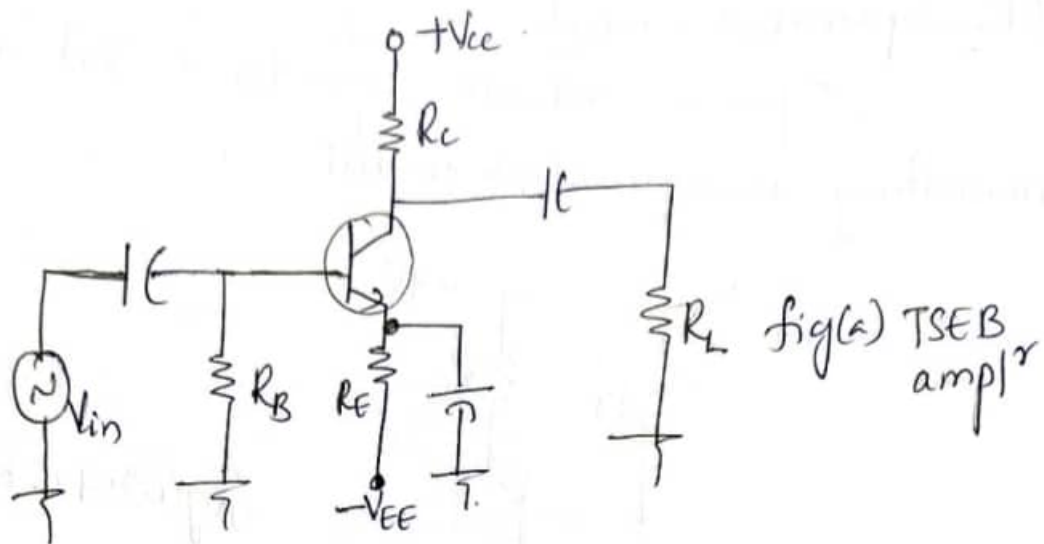
fig(c) :- ac equivalent circuit.

→ In base circuit, ac input voltage appears across $R_B \parallel \beta r_i$

→ In collector circuit, the current source pumps ac current of i_c th¹ $R_C \parallel R_L$.

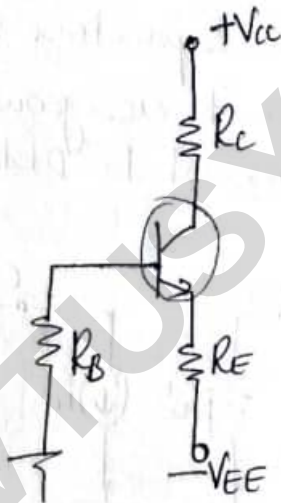
iii] TSEB Amplifier

Consider the TSEB Amplifier circuit as shown in fig(a) below.



1] DC equivalent circuit

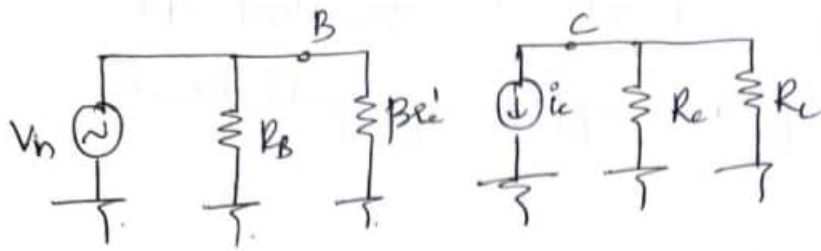
Replace all capacitors by open circuit & redraw the resulting DC equivalent circuit from fig(a)



fig(b) DC equivalent ckt

2] AC equivalent circuit

- i) Replace all capacitor by short circuit
 - ii) Replace all dc voltage sources (\$+V_{CC}\$ & \$-V_{EE}\$) with ac ground
 - iii) Replace transistor by its π -model in fig(c)
- & then redraw the resulting ac equivalent circuit as in fig(c)



→ In base circuit ac vlg appears across R_B in $\parallel$ with $\beta R_E'$

→ In collector circuit, the collector current ~~source~~ pumps an ac current of i_c th^r $R_C \parallel R_L$.

CE Amplifiers

The three different amplifier namely Base-biased, VDB and TSEB circuits are examples of a Common-emitter (CE) amplifier.

→ In CE amplifier, its emitter is at ac ground, and ac signal is coupled into the base and the amplified output signal appears at the collector.

→ Two other basic transistor amplifiers are Common-Base (CB) and Common-Collector (CC) amplifiers.

→ CB^{ampl^r} has its base at ac ground & CC^{ampl^r} has its collector at ac ground.

→ They are useful in some applications, but are not as popular as the CE amplifiers.

General Steps to write ac equivalent ckt

→ Below steps are applied for all CE, CC & CB amplifiers.

- 1] Short all coupling & bypass capacitors
- 2] Visualize all dc supply voltages as ac grounds
- 3] Replace the transistor by its π or T Models
- 4] Draw the ac equivalent ckt.

AC Quantities on the Datasheet

H-parameters

When the transistor was first invented, an approach known as the h-parameters (hybrid parameters) were used to analyze & design transistor circuits.

→ The 4 useful h-parameters are:-

h_{fe} → forward transfer current ratio

h_{ie} → input resistance

h_{re} → reverse transfer voltage ratio

h_{oe} → Output conductance.

→ There are some other parameter methods also exists such as α which uses quantities like β & β'

Relations b/w R & H parameters.

→ β in terms of h-parameter is given as
 $\beta = h_{fe}$

→ The datasheet lists a minimum h_{fe} of 100 & a max^m of 400, $\therefore \beta$ ranges from 100 to 400.

→ These values are for a collector current of 1mA & collector-emitter voltage of 10V.

→ r_e' in terms of h-parameter is given as

$$r_e' = \frac{h_{ie}}{h_{fe}}$$

Let max^m value of $h_{ie} = 10\text{K}\Omega$ & $h_{fe} = 400$

$$\text{then } r_e' = \frac{10\text{K}\Omega}{400} = 25\Omega$$

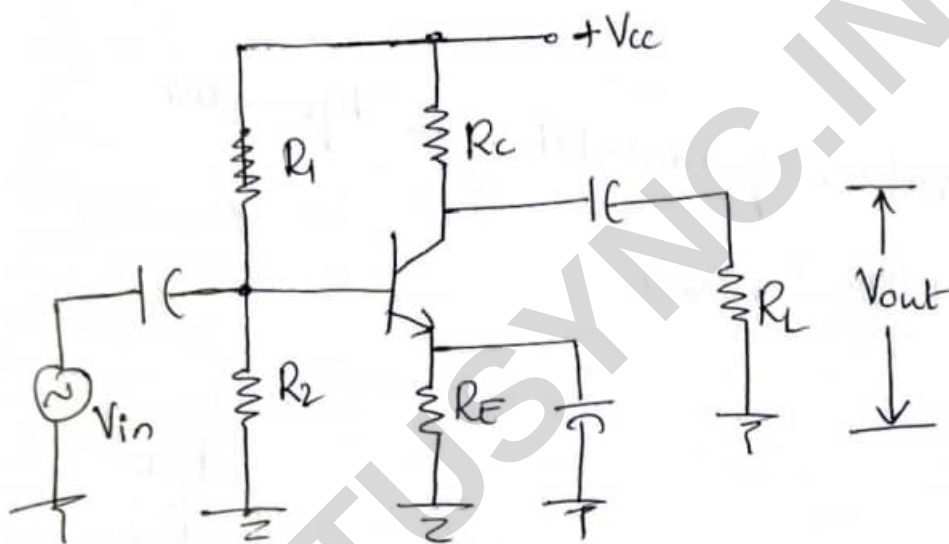
VOLTAGE AMPLIFIERS

* Voltage Gain :-

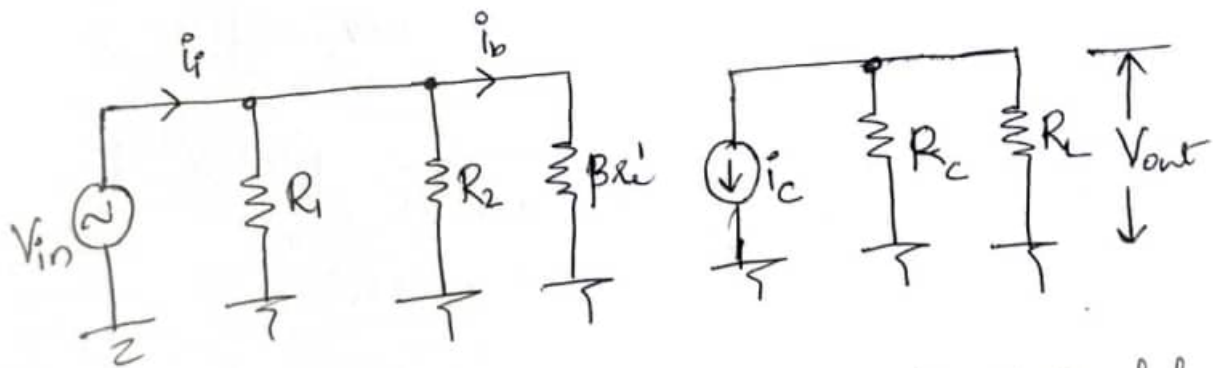
Voltage gain ~~was~~ is defined as the ac output voltage divided by ac input voltage.

* Voltage Gain for π Model

Consider VDB amplifier shown in fig(a) along with ac equivalent circuit with π -model in fig(b)



fig(a) VDB amplifier circuit



fig(b) ac equivalent ckt with π model.

→ The ac base current i_b flows through the input impedance of the base $\beta r_e'$.

Using Ohm's law, we can write

$$i_b = \frac{V_{in}}{\beta r_e'} \Rightarrow V_{in} = i_b \beta r_e'$$

→ In collector circuit, the current source pumps an ac current i_c through parallel connection of R_c & R_L .

∴ The Output Voltage V_{out} is given by

$$V_{out} = i_c (R_c \parallel R_L) \quad \text{--- (1)}$$

$$\text{wkt } i_c = \beta i_b \text{ \& } i_b = \frac{V_{in}}{\beta r_e'}$$

$$V_{out} = \beta \left(\frac{V_{in}}{\beta r_e'} \right) (R_c \parallel R_L)$$

$$\boxed{A_v = \frac{V_{out}}{V_{in}} = \frac{(R_c \parallel R_L)}{r_e'}} \quad \text{--- (2)}$$

Let $(R_c \parallel R_L) = r_c$, is the total ac load resistance seen by collector & this total resistance is called ac collector resistance (r_c)

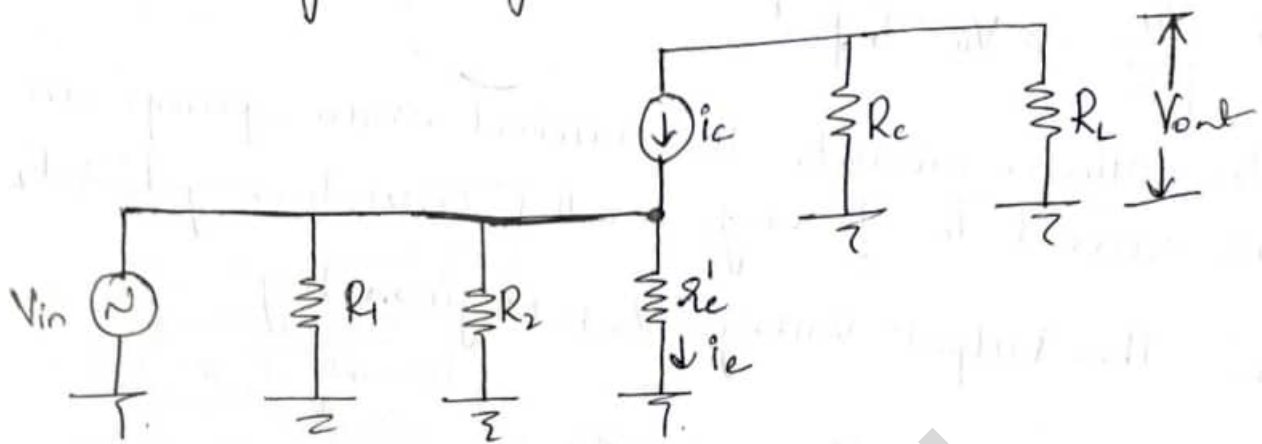
→ Now eqⁿ (2) becomes

$$\boxed{A_v = \frac{r_c}{r_e'} = \frac{\text{ac collector resistance}}{\text{ac resistance of emitter diode}}}$$

--- (3)

* Voltage Gain for T-Model

Let us draw the ac equivalent ckt for VDB amplifier shown in fig(a) using T-Model as in fig(c)



fig(c) :- ac equivalent ckt with T-Model.

→ The input voltage V_{in} appears across r_e' , with Ohm's Law we can write :-

$$V_{in} = i_e r_e'$$

→ In collector circuit, the current source pumps an ac current i_c through ac collector resistance (R_c)

∴ The ac output voltage is given by

$$V_{out} = i_c R_c$$

The Voltage gain is given by

$$A_v = \frac{V_{out}}{V_{in}} = \frac{i_c R_c}{i_e r_e'}$$

$$\text{Note } i_c \approx i_e$$

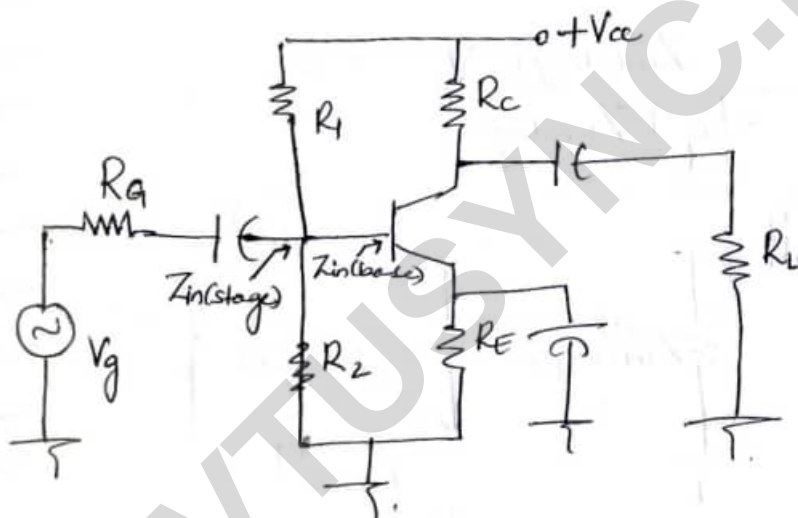
$$\therefore \boxed{A_v = \frac{R_c}{r_e'}} \quad \text{--- (4)}$$

→ Comparing eqⁿ (3) & (4) we get same equations for π & T-Models.

* The Loading Effect of Input Impedance

→ ~~In this~~ Under this topic we are considering an ac voltage source with internal resistance unlike until now we considered ideal v/g source.

Input impedance
Consider fig(a), which has an ac voltage source V_g with internal resistance of R_g .

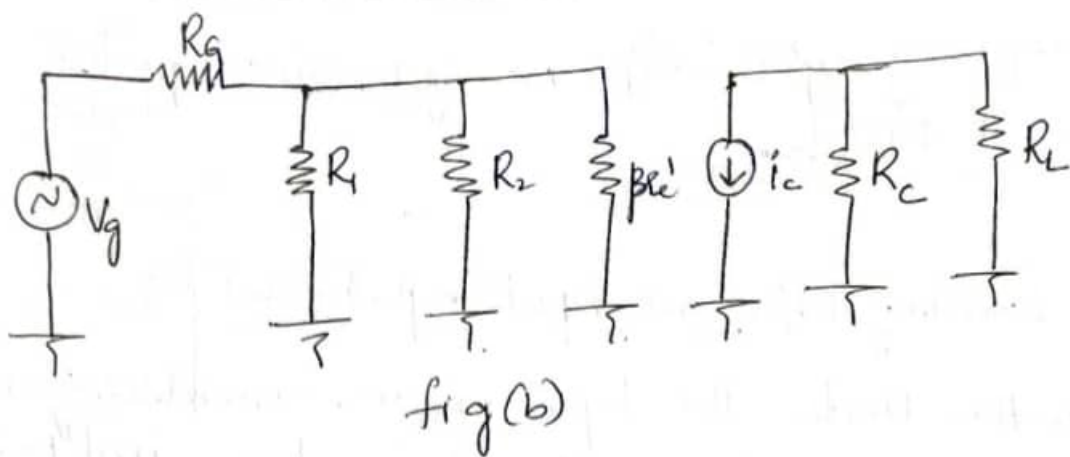


→ That is, the voltage source V_g is not stiff, in this case some of the ac source is dropped across its internal resistance. As a result, ac v/g b/w the base & gnd is less than ideal.

The input impedance $Z_{in(stage)}$ is given by (fig(b))

$$Z_{in(stage)} = R_1 \parallel R_2 \parallel \beta i_e$$

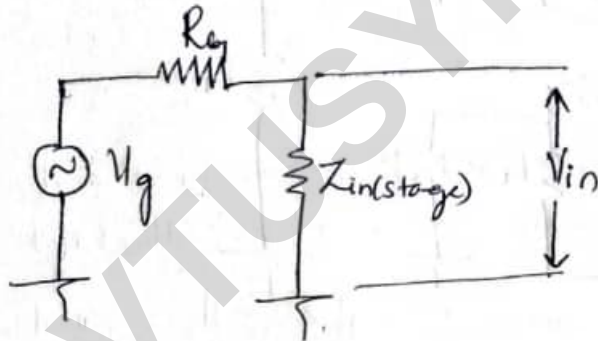
$$= R_1 \parallel R_2 \parallel Z_{in(base)}$$



Equation for Input Voltage

When the generator is not stiff, the ac input v/g V_{in} as in fig(c) is less than V_g . Using the Voltage-divider rule, we can write

$$V_{in} = \frac{Z_{in(stage)}}{R_g + Z_{in(stage)}} V_g$$

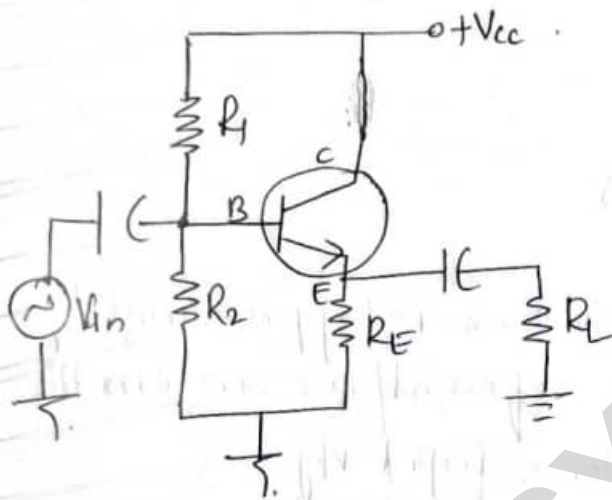


Note:- The generator (V_g) is stiff when R_g is less than $0.01 Z_{in(stage)}$.

CC AMPLIFIERS

CC Amplifiers

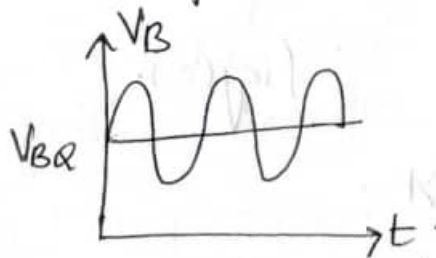
→ Common Collector (CC) Amplifier is also called Emitter follower as the input signal is coupled to the base & the output signal is taken from the emitter terminal. as in fig(a). ~~collector terminal is at ac ground~~



fig(a) :- Emitter follower ckt.

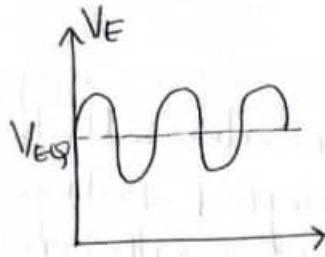
→ From fig(a), the collector is connected to dc supply $+V_{cc}$, which is ac ground while drawing ac equivalent circuit.

→ The input signal is coupled to the base & fig(b) shows the total V_B between base & gnd. It has ac component & dc component.



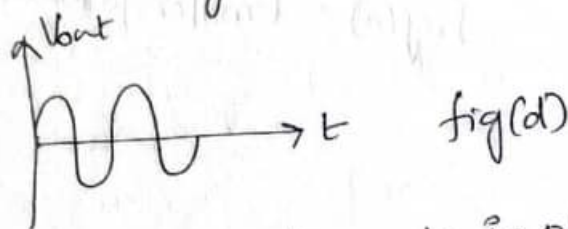
→ This input voltage at base sets up an ac emitter current & produces an ac voltage across the emitter resistor. This ac voltage is then coupled to the load resistor.

→ fig(c) shows the total voltage between the emitter & ground.



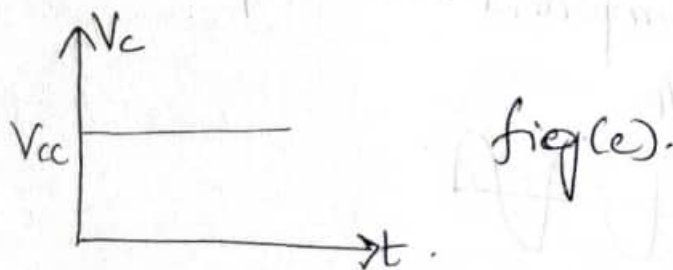
→ The ac emitter voltage is coupled to the load resistor.

→ The output voltage is shown in fig(d) which is pure ac voltage.



→ This output voltage is in phase & is approximately equal to the i/p voltage. hence referred as an emitter follower ckt → output v_o follows input v_i .

→ Since there is no resistor at collector terminal, the total v_o b/w the collector & gnd equals the supply voltage hence the o/p waveform is constant dc voltage as in fig(e)



→ There is no ac sig on the collector because it is an ac ground point.

AC Emitter Resistance

In fig(a), the ac sig coming out of the emitter sees

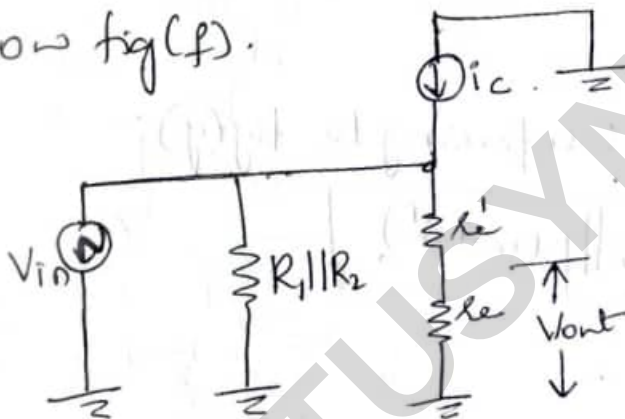
$$R_E \parallel R_L$$

$$\therefore r_e = R_E \parallel R_L$$

r_e is external ac emitter resistance, different from internal ac emitter resistance r_e' .

Voltage Gain

Let's draw ac equivalent ckt with T-model as shown below fig(f).



Using Ohm's law.

$$V_{in} = i_c(r_e + r_e')$$

$$V_{out} = i_c r_e$$

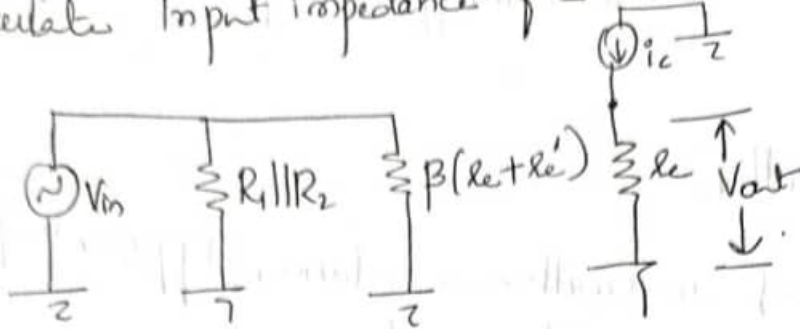
$$A_v = \frac{V_{out}}{V_{in}} = \frac{i_c r_e}{i_c(r_e + r_e')} = \frac{r_e}{r_e + r_e'}$$

$$A_v = \frac{r_e}{r_e + r_e'}$$

r_e is much greater than r_e' , so that the voltage gain is equal to 1.

Input Impedance of the Base

Let's draw an equivalent circuit using π -model to calculate input impedance of base as in fig (g).



$$Z_{in(base)} = \beta(r_e + r_e')$$

r_e is much greater than r_e'

$$\therefore Z_{in(base)} \approx \beta r_e$$

Input impedance of stage: Referring to fig (g):-

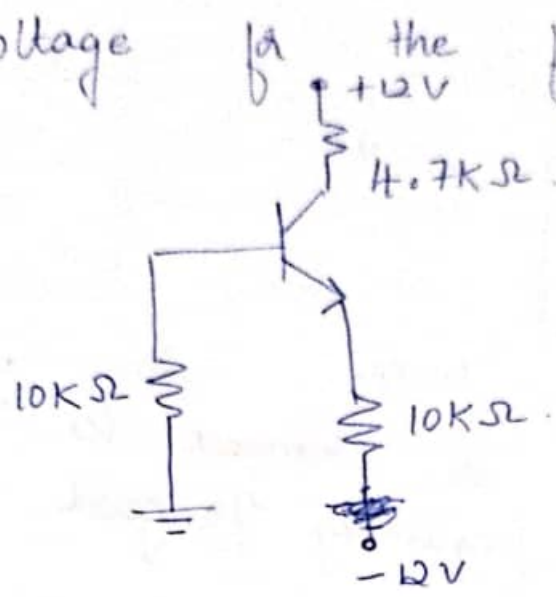
$$Z_{in(stage)} = R_1 \parallel R_2 \parallel \beta(r_e + r_e')$$

Impedance

VTUSYNC.IN

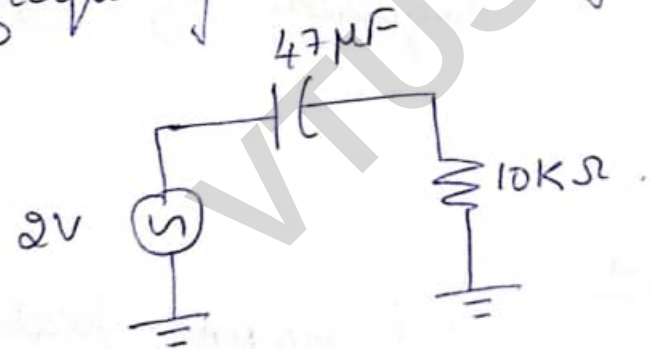
Problem on Two - supply emitter bias

1. Determine the emitter current and collector voltage for the figure below.



Problem on Base bias amplifier

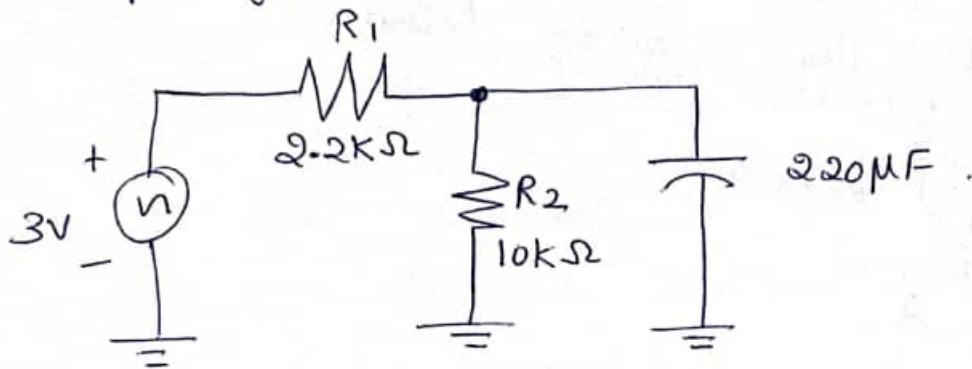
1. for the figure below, calculate the lowest frequency at which good coupling exists.



2. In the above figure, if R_L changes to $1K\Omega$, what is the lowest freq for good coupling?
3. If the capacitor is changed to $100\mu F$, what is the lowest frequency?
4. If the lowest frequency is $100Hz$, what is the C value for good coupling?

Problems on Emitter-biased amplifier

1. For the figure below, Calculate the lowest frequency at which good bypassing exists.



2. If the series resistance is changed to $10K\Omega$, what is the lowest frequency for good bypassing.
3. If the capacitor is changed to $47\mu F$, find the lowest frequency for good bypassing.
4. If the lowest input frequency is $1KHz$, find the value of C required for effective bypassing.

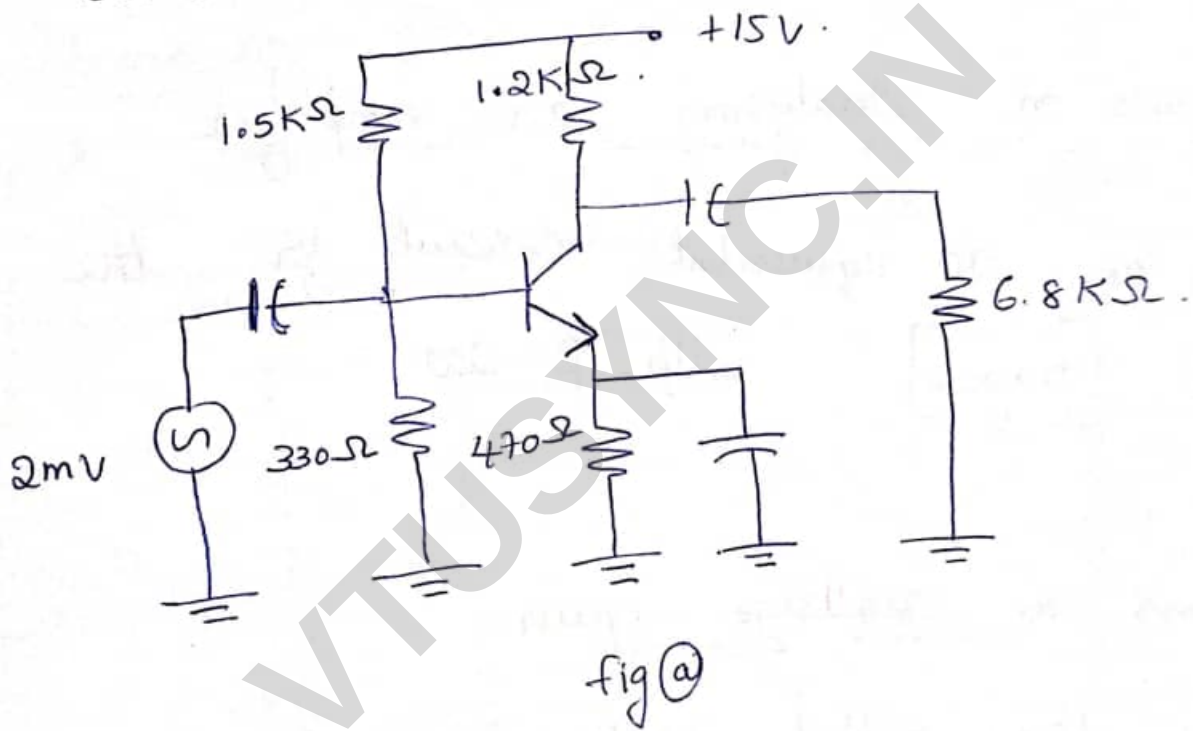
Problems on AC Beta

1. If an ac base current of $100\mu A$ produces an ac collector current of $15mA$, what is the ac beta?
2. If ac collector current is $4mA$ & ac beta is 100, what is the ac base current.

3. If the ac beta is 200 and the ac base current is $12.5 \mu A$, what is the ac collector current? (2)

Problems on Small-Signal operation

1. for the circuit to operate as small signal, what is the maximum allowable ac emitter current?



Problems on AC Resistance of the emitter diode

1. from the above figure, what is the ac resistance of the emitter diode

Problems on Two transistor models

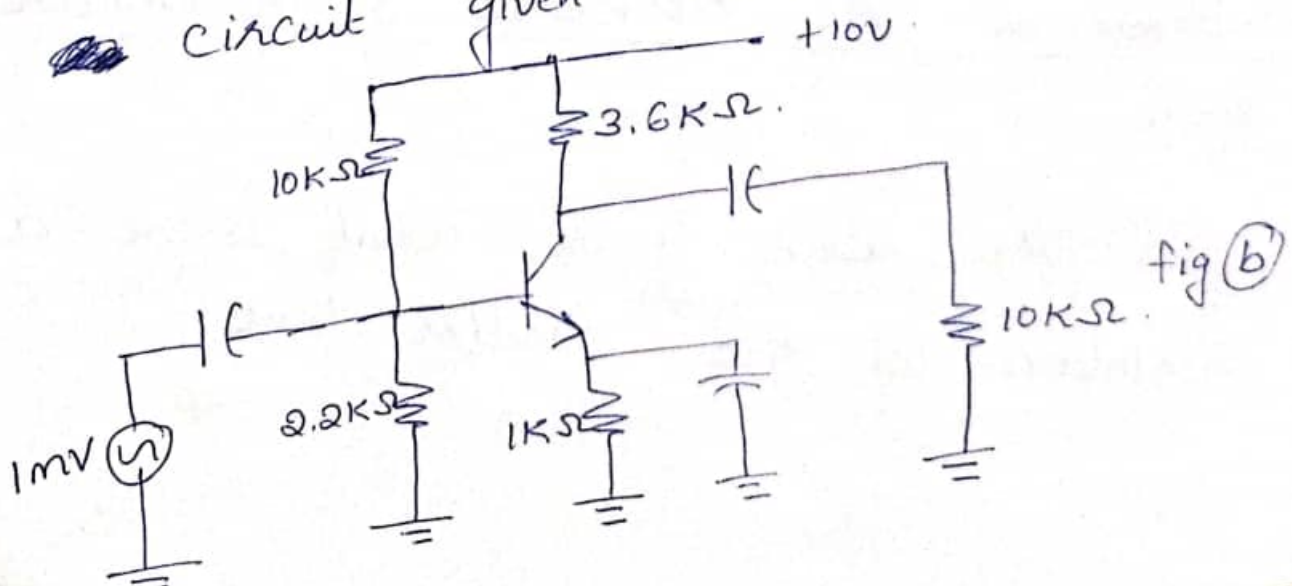
1. If $\beta = 200$ for the circuit in figure (a) [page 2]
Calculate the input impedance.
2. If $1.2K\Omega$ resistance is changed to 680Ω ,
find the input impedance of the base if
 $\beta = 200$.

Problems on Analyzing an amplifier

1. Draw an ac equivalent circuit for the
fig (a) [page 2] with $\beta = 200$.

Problems on voltage gain

1. Find the output voltage, for the ~~ac~~
circuit given below.

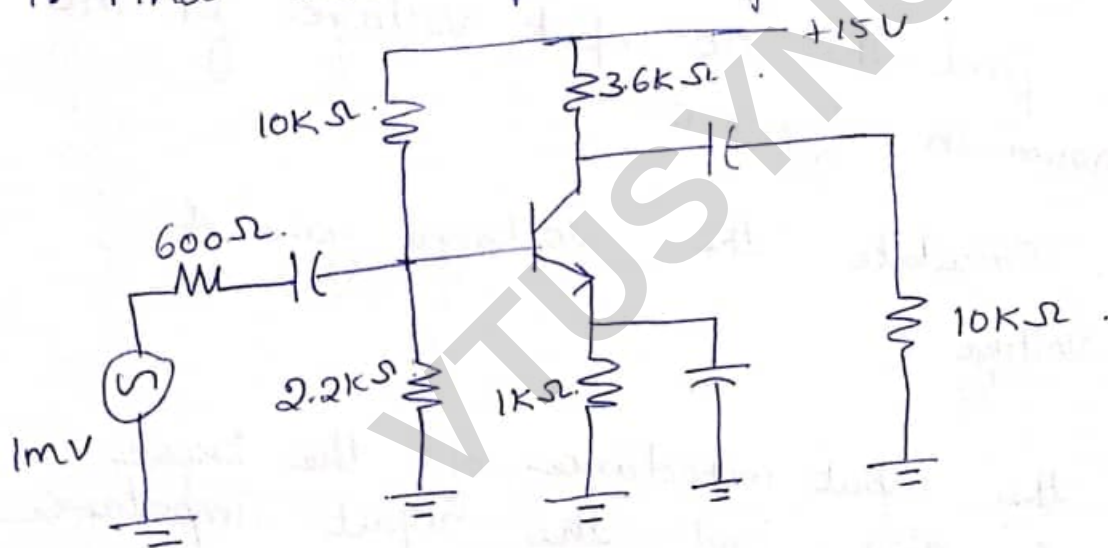


2. For the circuit in fig (b), change the load resistance to $5k\Omega$ and calculate the voltage gain. (3)

3. Find the output voltage, if the supply voltage is increased to $+15V$.

Problems on the loading effect of the input impedance

1. Find the output voltage, for the circuit below.



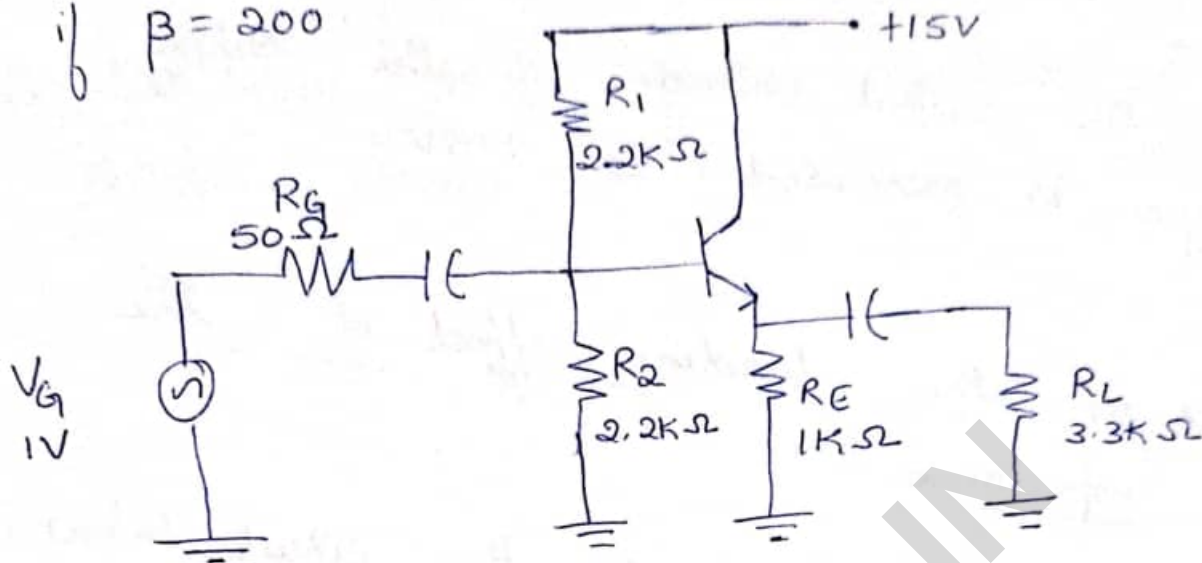
2. Find the output voltage, if the emitter resistance is $2k\Omega$.

3. If the generator resistance is 300Ω , calculate the output voltage.

Problems on CC amplifier

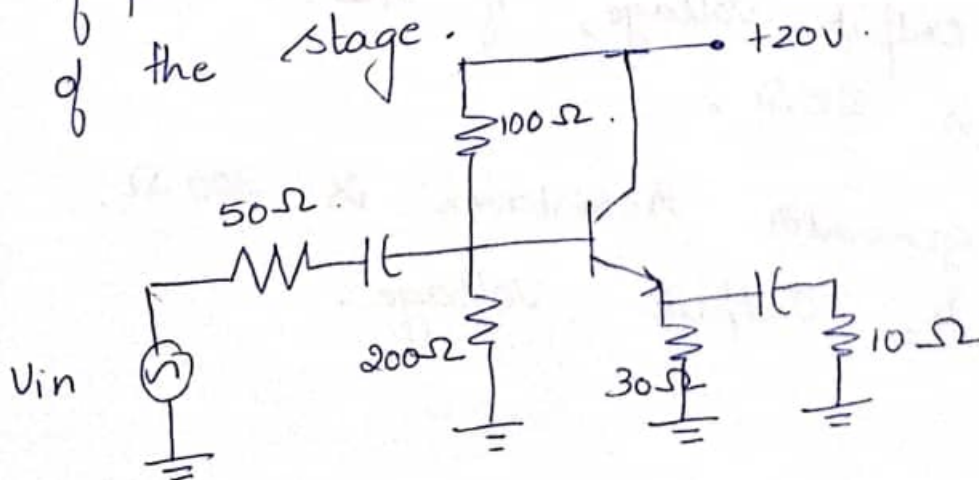
1. In the figure below, find the input impedance of the base & input impedance of the stage

if $\beta = 200$



fig(c)

2. If $\beta = 150$, find the ac input voltage for the circuit shown in figure c.
3. If $\beta = 175$, Calculate the voltage gain & ac load voltage.
4. Determine the input impedance of the base & also find the input impedance of the stage.



fig(d)

Problems on output impedance

(4)

1. Determine the output impedance for the circuit shown in fig(c) if $\beta = 200$
2. Determine the output impedance for the circuit shown in fig(d) if $\beta = 100$.

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Module - 2

Separate
→ small
→ small

$$I_c = \frac{V_{be}}{R_x} + g_m V_{be}$$

$$= \frac{V_{be}}{R_x} (1 + g_m R_x)$$

W.K.T $g_m R_x = \beta$

$$I_c = \frac{V_{be}}{R_x} (1 + \beta)$$

$$= \frac{V_{be}}{\left(\frac{R_x}{1 + \beta}\right)}$$

$$I_c = \frac{V_{be}}{R_c}$$

$$g_m = \frac{I_c}{V_T}$$

$$R_x = \frac{V_T}{I_B}$$

$$g_m R_x = \frac{I_c}{V_T} \cdot \frac{V_T}{I_B}$$

$$g_m R_x = \frac{I_c}{I_B} = \beta$$

$$\Rightarrow \beta = g_m R_x$$

A slightly different equivalent ckt model can be obtained by expressing the current of the controlled source

Biasing in MOS Amplifier ckt

- An essential step in the design of MOSFET amplifier circuit is the establishment of an appropriate dc operating point for the transistor.
- This is the step known as biasing or bias design.
- An appropriate dc operating point or bias point is characterized by a stable & predictable dc drain current I_D & by a dc drain-to-source V_{DS} that ensures operation in the saturation region for all expected i/p. s/l levels.

MOSFETs

Biasing in MOS amplifier circuits

Biasing by Fixing V_{GS}

→ Biasing a MOSFET by fixing V_{GS} is the most straightforward approach to provide desired I_D .

→ V_{GS} Value can be decided from the power supply V_{DD} through the use of an appropriate V_{GS} divider circuit.

→ Alternatively, V_{GS} can also be decided from suitable reference V_{GS} available in the system.

→ Independent of how the V_{GS} is generated, ~~this~~ fixing V_{GS} is not a good approach to bias a MOSFET. & The reason for this is recalled by the eqⁿ

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t)^2$$

where $V_t \rightarrow$ threshold V_{GS}

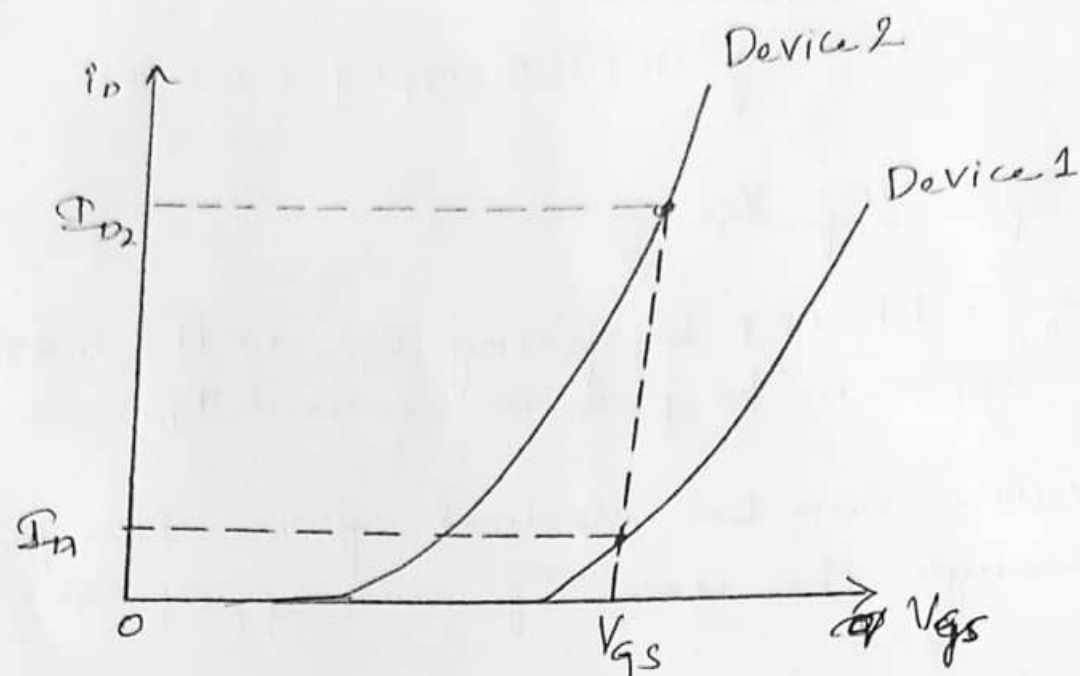
$C_{ox} \rightarrow$ capacitance oxide

$\frac{W}{L} \rightarrow$ transistor aspect ratio

Here V_t , C_{ox} & $\frac{W}{L}$ spreads in values in discrete devices & ICs, especially among devices fabricated on different wafers & b/w different batches of wafers.

→ Also V_t & μ_n depend on temperature & if we fix V_{GS} , the drain current I_D becomes very much temperature dependent.

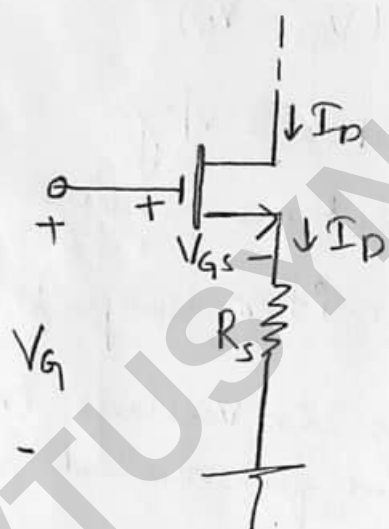
~~Also~~ The Graph I_D vs. V_{GS} characteristic curves for same batch of MOSFETs shows that for fixed value of V_{GS} , the drain current I_D spreads substantially.



Biassing by fixing V_G & connecting a Resistance in the source.

→ It is an excellent biasing technique for discrete MOSFET circuits.

→ Consider the ckt shown below



From the above ckt

$$V_G = V_{GS} + I_D R_S \quad \text{--- (1)}$$

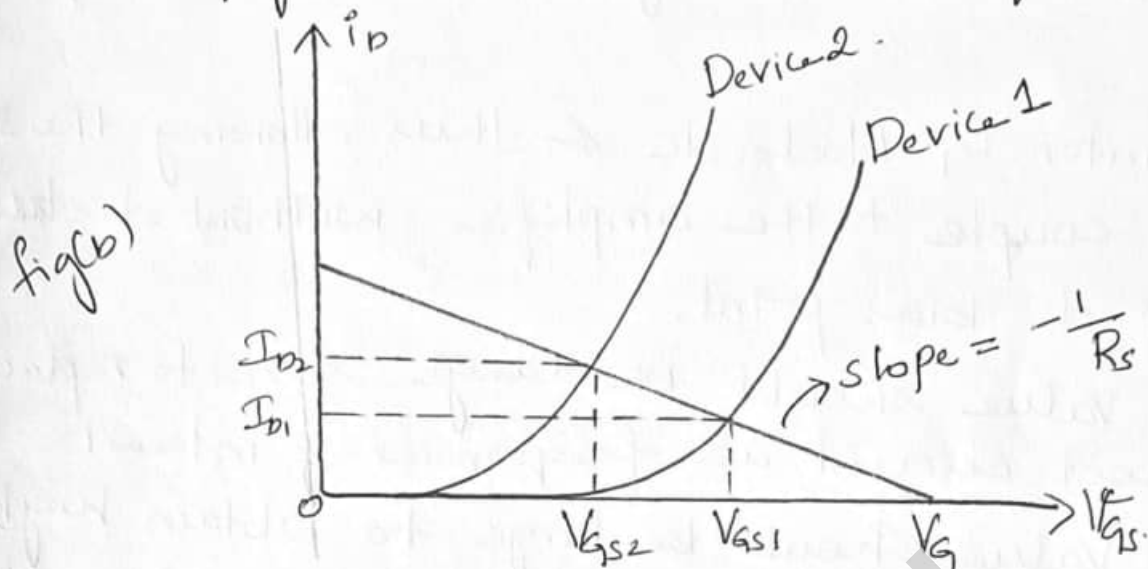
→ If V_G is greater than V_{GS} , I_D will be determined by the values of V_G & R_S .

→ Even if V_G is not much larger than V_{GS} , Resistor R_S provides negative feedback & acts to stabilize the value of the bias current I_D .

→ Equation ① indicates that since V_G is constant V_{GS} have to decrease, this intuen results in a decrease I_D . Thus the action of R_S works to keep I_D as constance.

→ This negative feedback action of R_S gives it the name degeneration Resistance.

Consider fig(b) $I_D - V_{GS}$ characteristics for 2 devices

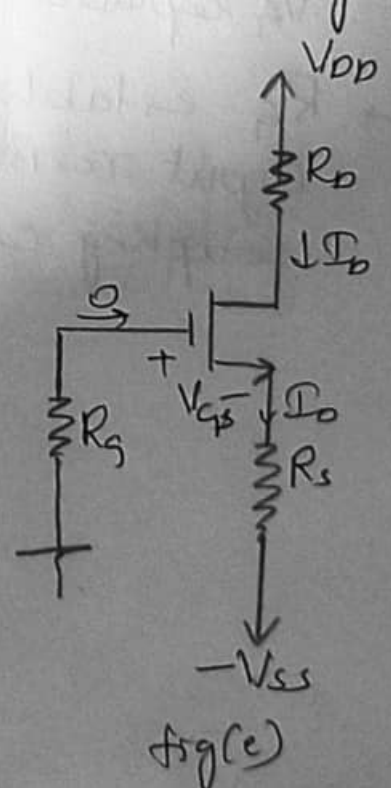
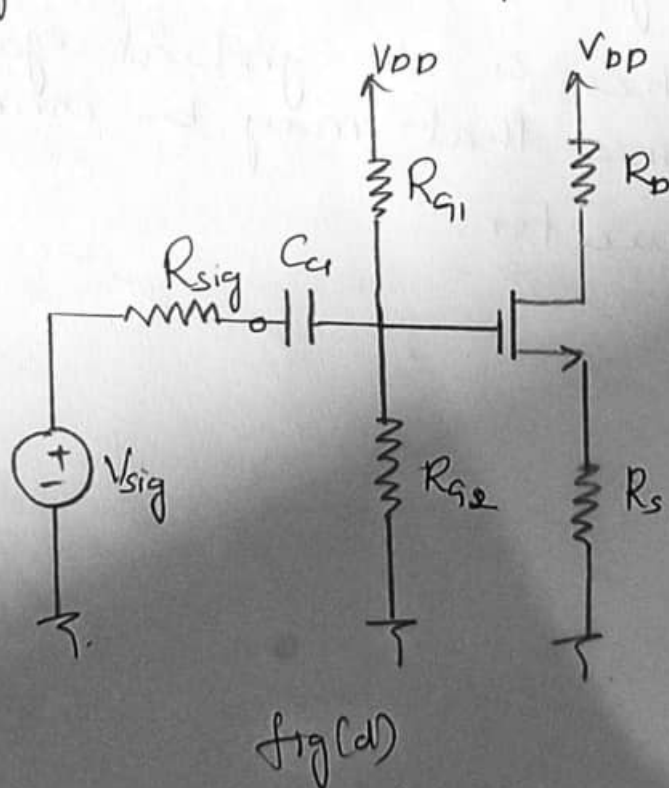
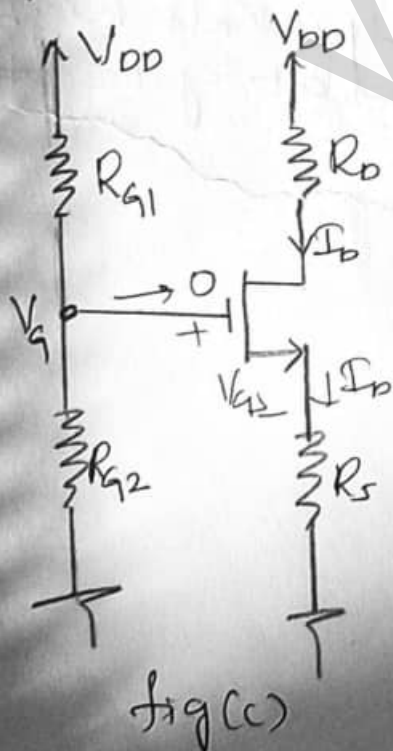


→ A straight line superimposed on device characteristic represents the constraints imposed by the bias circuit.

→ The intersection of straight line with the $I_D - V_{GS}$ ch curve provides the coordinates $(I_D \& V_{GS})$ of the bias point.

→ We can observe that the variability in I_D is much smaller than in the case of fixed V_{GS}

→ The variability decreases as V_G & R_S are made larger



Consider the 2 bias scheme shown in fig(c) & fig(e).

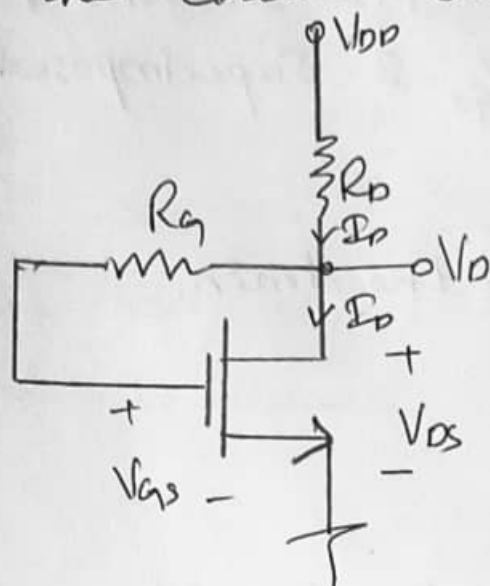
- The circuit of fig(c) uses one power-supply V_{DD} & derives V_G through a voltage divider (R_{G1}, R_{G2}).
- Since $I_G = 0$, R_{G1} & R_{G2} can be selected to be very large value allowing MOSFET with large input resistance to a signal source connected to gate terminal th^t coupling capacitor C_G .
- The capacitor C_G blocks dc & thus allowing the s^lc Sec V_{sig} to couple to the amplifier without disturbing the MOSFET dc bias point.
- Also C_G value should be large so as to replace with short circuit at frequencies of interest.
- Resistor R_D value should be large to obtain high gain but small enough to allow for the desired signal swing at the drain while keeping the MOSFET at saturation.

Now, consider the circuit in fig(d) (e) with 2 power supplies.

- This circuit is the implementation of eqⁿ (1) with V_G replaced by V_{SS} .
- R_G establishes a dc gnd at gate & presents high input resistance that may be connected at gate th^t coupling capacitor.

Biasing using a Drain-to-Gate Feedback Resistor

Consider the circuit shown



→ The large feedback resistance R_g forces the dc voltage at the gate to be equal to that at the drain ($\because I_g = 0$)

Thus we can write

$$V_{GS} = V_{DS} = V_{DD} - R_D I_D$$

$$\text{or } V_{DD} = V_{GS} + R_D I_D \quad \text{--- (1)}$$

→ In the above eqn, if I_D changes for some reason, say increases, then from eqn (1) V_{GS} must decrease.

→ The decrease in V_{GS} in turn causes a decrease in I_D , a change that is opposite in direction to the original one. Thus the negative feedback or degeneration provided by R_g works to keep the value of I_D as constant.

→ This circuit can be utilized as an AC amplifier by applying the i/p v_i to the gate via a coupling capacitor so as not to disturb the dc bias conditions established.

Small-Signal Operation & Models.

- For small-signal operation of MOS transistor, it is biased by applying a dc voltage V_{GS} & superimposed on the input sig v_{gs} to be amplified.
- The output v_{g} is taken at the drain.

* The DC Bias point

- The dc bias current I_D can be found by setting the signal v_{gs} to zero; thus

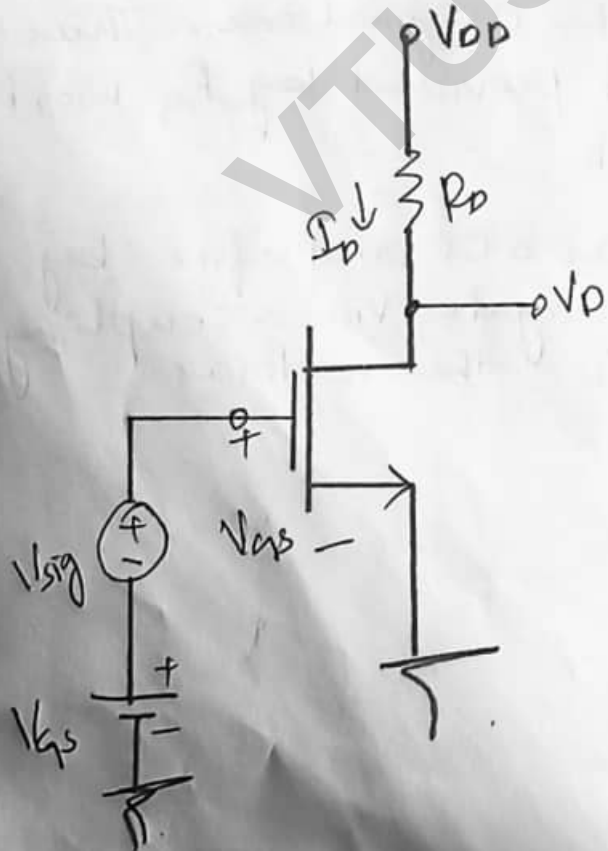
$$I_D = \frac{1}{2} k'_n \frac{W}{L} (V_{GS} - V_t)^2 \quad \text{--- (1)}$$

- The dc v_{g} at the drain V_{DS} or simply V_D (S is grounded) will be

$$V_D = V_{DD} - R_D I_D \quad \text{--- (2)}$$

To ensure saturation-region operation, we must have

$$V_D > V_{GS} - V_t$$



Since the total v_{lg} at the drain will have a signal component superimposed on V_D , V_D has to be sufficiently greater than $(V_{GS} - V_t)$ to allow for the required sll swing.

* The Signal current in the Drain Terminal

→ When the input signal v_{gs} is applied, the total instantaneous gate-to-source voltage will be

$$v_{gs} = V_{GS} + v_{gs} \sim (1)$$

resulting in the total instantaneous drain current i_D

$$i_D = \frac{1}{2} K_n' \frac{W}{L} (V_{GS} + v_{gs} - V_t)^2 \sim (2)$$

or
$$i_D = \frac{1}{2} K_n' \frac{W}{L} (V_{GS} - V_t + v_{gs})^2$$

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$\therefore i_D = \frac{1}{2} K_n' \frac{W}{L} \left\{ (V_{GS} - V_t)^2 + 2(V_{GS} - V_t)v_{gs} + v_{gs}^2 \right\}$$

$$i_D = \underbrace{\frac{1}{2} K_n' \frac{W}{L} (V_{GS} - V_t)^2}_{1^{st} \text{ term}} + \underbrace{K_n' \frac{W}{L} (V_{GS} - V_t) v_{gs}}_{2^{nd} \text{ term}} + \underbrace{\frac{1}{2} K_n' \frac{W}{L} v_{gs}^2}_{3^{rd} \text{ term}}$$

In the above eqⁿ

1st term → dc bias current I_D

2nd term → Represents a current component that is directly proportional to i/p sll.

3rd term → Current component that is proportional to the square of the input sll.

→ 3rd term is undesirable because it represents non-linear distortion.

To reduce the non linear distortion introduced by the MOSFET the i/p s/l should be kept small so that

$$\frac{1}{2} k_n' \frac{W}{L} v_{gs}^2 \ll k_n' \frac{W}{L} (V_{gs} - V_t) v_{gs}$$

$$v_{gs} \ll 2(V_{gs} - V_t)$$

or $v_{gs} \ll 2V_{ov}$

where $V_{ov} = (V_{gs} - V_t)$ is the overdrive v_g at which the transistor is operating.

If the above small-s/l condⁿ is satisfied we may neglect the 3rd term. & the i_D is expressed as

$$i_D = I_D + i_d$$

$$\therefore i_d = k_n' \frac{W}{L} (V_{gs} - V_t) v_{gs}$$

or

$$g_m \equiv \frac{i_d}{v_{gs}} = k_n' \frac{W}{L} (V_{gs} - V_t)$$

↳ transconductance.

in terms V_{ov}

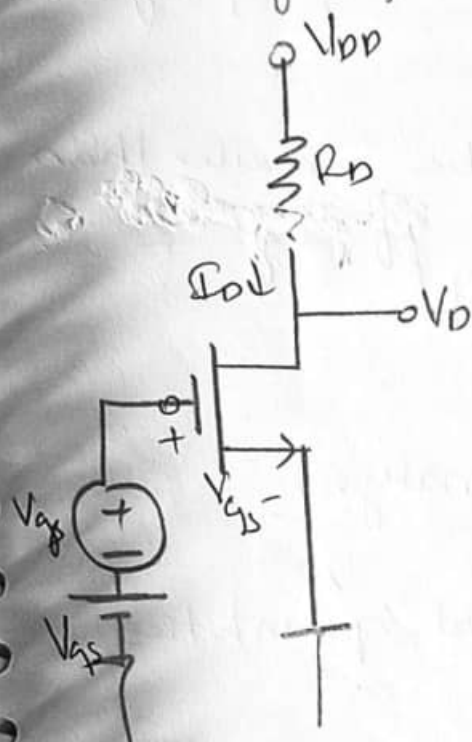
$$g_m = k_n' \frac{W}{L} V_{ov}$$

→ The graph below represents a small-s/l opⁿ of the e-MOSFET ampl^r.

→ Note that g_m is equal to slope of $i_D - V_{gs}$ ch at bias pt

$$g_m \equiv \left. \frac{i_D}{v_{gs}} \right|_{V_{gs} = V_{gs}}$$

* Voltage Gain



From the above ckt, the total instantaneous drain voltage V_D as follows:-

$$V_D = V_{DD} - I_D R_D$$

$$\text{Wkt, } \cancel{V_D = V_{DD} - I_D R_D} \quad I_D = I_{D0} + i_d$$

$$\therefore V_D = V_{DD} - (I_{D0} + i_d) R_D$$

$$= \underbrace{V_{DD} - I_{D0} R_D}_{V_{D0}} - i_d R_D$$

$$V_D = V_{D0} - i_d R_D$$

$$\therefore V_d = -i_d R_D$$

$$\text{But } i_d = g_m V_{gs}$$

$$\therefore V_d = -g_m V_{gs} R_D$$

$$\therefore \text{Voltage Gain} = A_v \equiv \frac{V_d}{V_{gs}} = -g_m R_D \quad \text{--- (1)}$$

- minus sign in eqn (1) indicates that the o/p sll v_d is 180° out of phase w/ input sll. This is illustrated in fig below.
- The i/p sll is assumed to have a triangular w/F with an amplitude smaller than $2(V_{gs} - V_t)$ to ensure linear operⁿ.
- For all time operⁿ in saturation region, the minimum value of V_D should not fall below the value of V_G by more than V_t .
- Also, the max^m value of V_D should be smaller than V_{DD} ; otherwise FET will enter cut-off regⁿ & peaks of the o/p sll will be clipped off.

* Separating the DC Analysis & the sll Analysis.

- For small-signal approximation, signal quantities are superimposed on dc quantities

For instance, the total drain current $i_D = I_D + i_d$ & total drain v/g $V_D = V_D + v_d$ & so on^m

- The Analysis & design can be greatly simplified by separating dc bias calculation from small-sll calculations
- To separate dc quantities all ac sources are set to Zero.

& Once dc operating conditions are established all ac quantities are found by setting all dc secs to Zero.

* Small-Signal Equivalent-circuit Models.

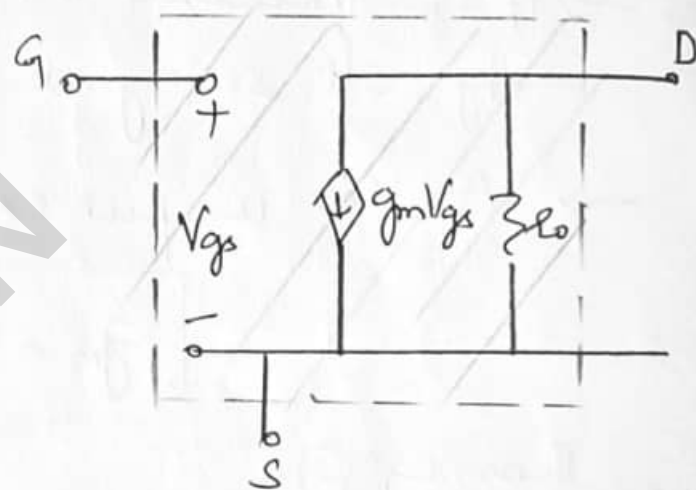
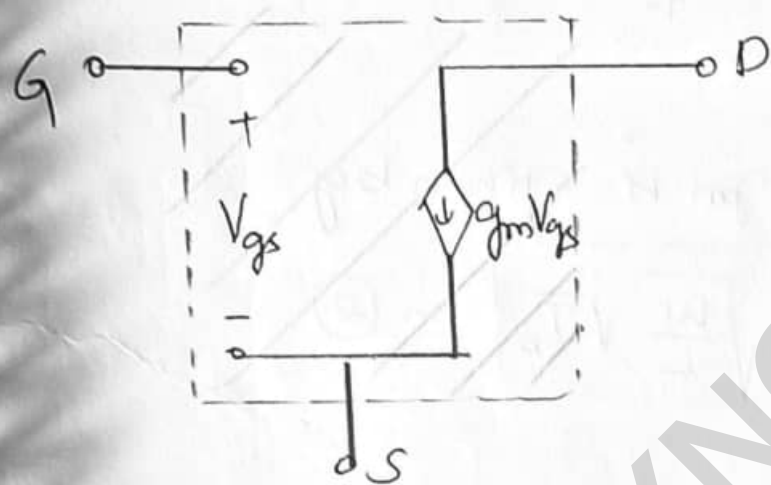
→ From a signal point of view the FET is voltage-controlled ~~device~~ current source.

→ It accepts a V_{gs} at Gate & Src terminals & provides a current $i_d = g_m V_{gs}$ at drain terminal.

→ The input resistance of FET is very high (ideally ∞).

→ Output resistance, the resistance looking into the drain is also high.

→ From above points small-sig model or equivalent ckt is shown below.



→ The r_o is given by

$$r_o = \frac{|V_A|}{I_D}$$

r_o ranges from $10\text{K}\Omega$ to $1000\text{K}\Omega$

where $V_A = \frac{1}{\lambda}$ is a MOSFET parameter proportional to MOSFET channel length.

→ I_D is given by

$$I_D = \frac{1}{2} K_n \frac{W}{L} V_{ov}^2$$

→ A_v is given by

$$A_v = \frac{V_d}{V_{gs}} = -g_m \left(\frac{R_D}{r_o} \right)$$

In above analysis K_n is for nmos, for pmos it is given by K_p

* The transconductance g_m

The transconductance is given by

$$\left. \begin{aligned} g_m &= K_n' \left(\frac{W}{L} \right) (V_{GS} - V_t) \\ \text{or } g_m &= K_n' \left(\frac{W}{L} \right) V_{ov} \end{aligned} \right\} \sim (1)$$

From above eqⁿ g_m is proportional to $K_n' = \mu_n C_{ox}$ & also to the $\frac{W}{L}$ ratio. hence to obtain relatively large g_m value the device must be short & wide.

→ g_m is also proportional to the overdrive v/g $V_{ov} = V_{GS} - V_t$

→ If g_m increases at a larger V_{GS} , it reduces the allowable v/g sll swing at the drain.

→ Another useful expⁿ for g_m is given by

$$\boxed{g_m = \sqrt{2K_n'} \sqrt{\frac{W}{L}} \sqrt{I_D}} \sim (2)$$

From eqⁿ (2)

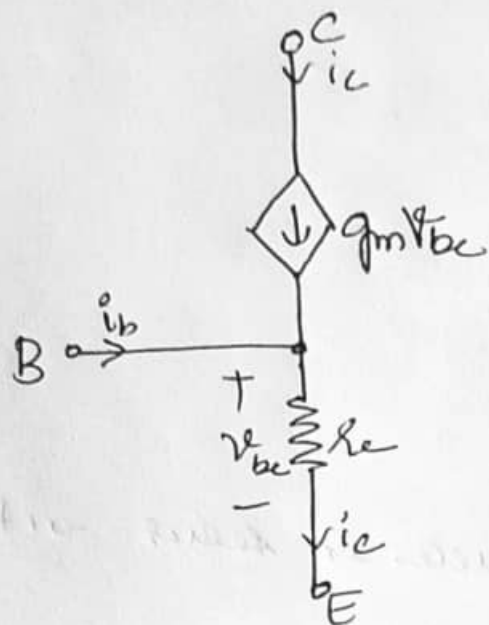
1] g_m is proportional to sqre root of the dc bias current

2] At a given bias current, $g_m \propto \sqrt{\frac{W}{L}}$

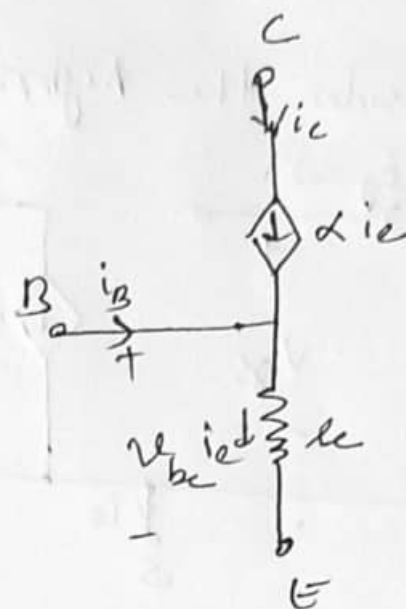
Also,

$$\boxed{g_m = \frac{2I_D}{V_{GS} - V_t} = \frac{2I_D}{V_{ov}}} \sim (3)$$

The T Model of BJT



a) Voltage controlled current source representation T-Model of BJT



b) Current controlled current source representation of T-Model of BJT

$$a) \quad g_m = \frac{I_c}{V_T}$$

$$r_e = \frac{V_T}{I_E} = \frac{\alpha}{g_m}$$

$$i_b = \frac{v_{be}}{r_e} - g_m v_{be} = \frac{v_{be}}{r_e} (1 - g_m r_e)$$

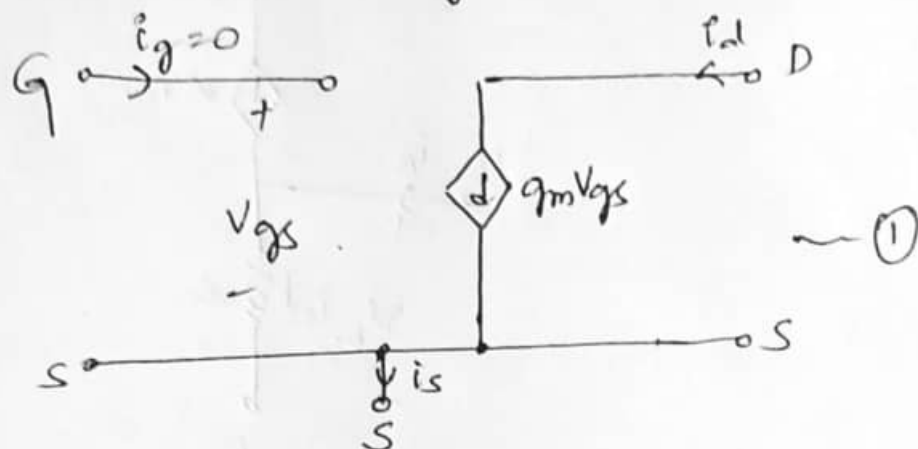
$$= \frac{v_{be}}{r_e} (1 - \alpha) = \frac{v_{be}}{r_e} \left(1 - \frac{\beta}{\beta + 1}\right)$$

$$= \frac{v_{be}}{(\beta + 1)r_e} = \frac{v_{be}}{r_{\pi}}$$

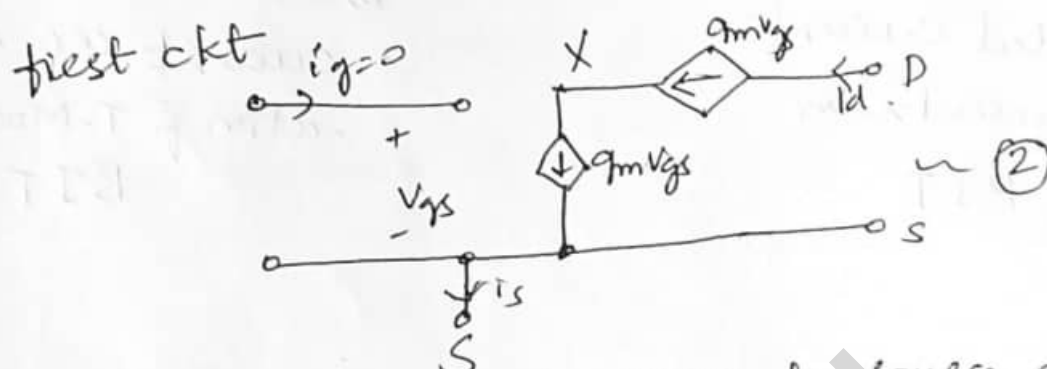
$$b) \quad g_m v_{be} = g_m (i_e r_e) \\ = (g_m r_e) i_e = \alpha i_e$$

T- Equivalent Model of MOSFET

Consider the hybrid- π Model.

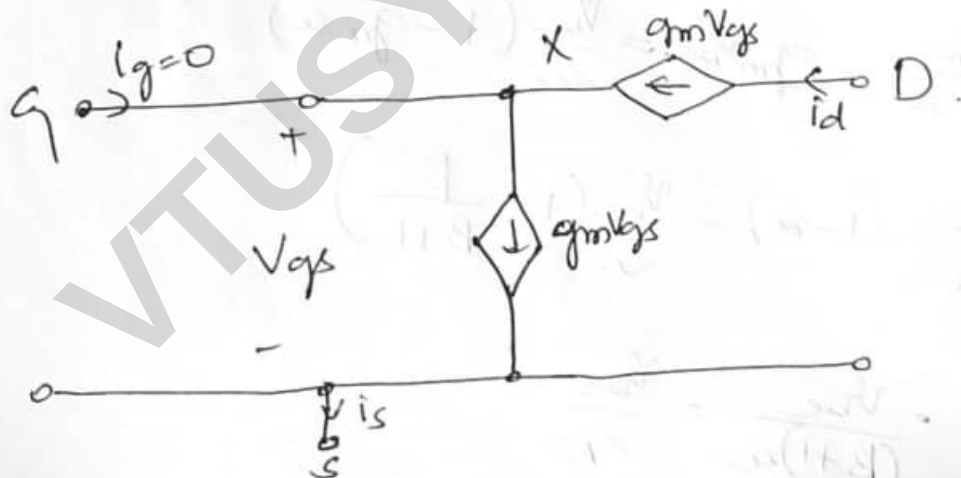


→ Now let's add second $g_m V_{gs}$ current source in series with



→ This addition of second current source does not change the terminal current & is thus allowed.

→ The newly created circuit node, labeled X, is joined to the gate terminal G in (3) ckt

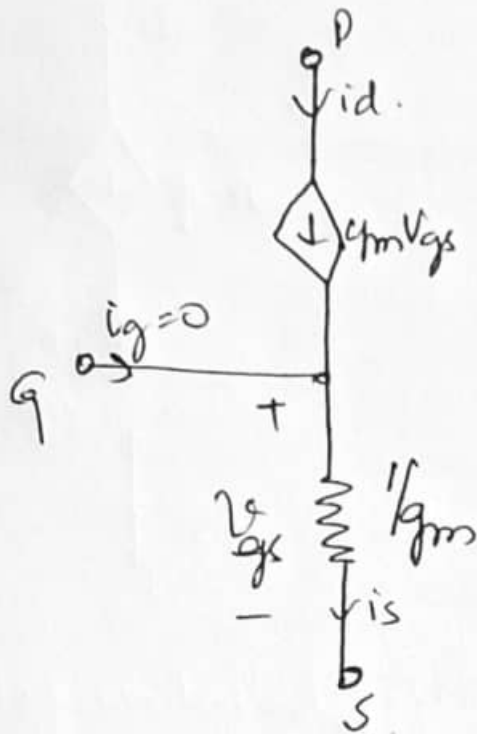


→ Observe that the gate current does not change, i.e. it remains equal to zero, & thus this connection does not alter the terminal characteristics.

→ Now we have a controlled current source $g_m V_{gs}$ connected across its control v_{gs} . We can replace this controlled

source by a resistance as long as this resistance draws an equal current as the source. The value of resistance is

$$\frac{V_{gs}}{g_m V_{gs}} = \frac{1}{g_m}$$



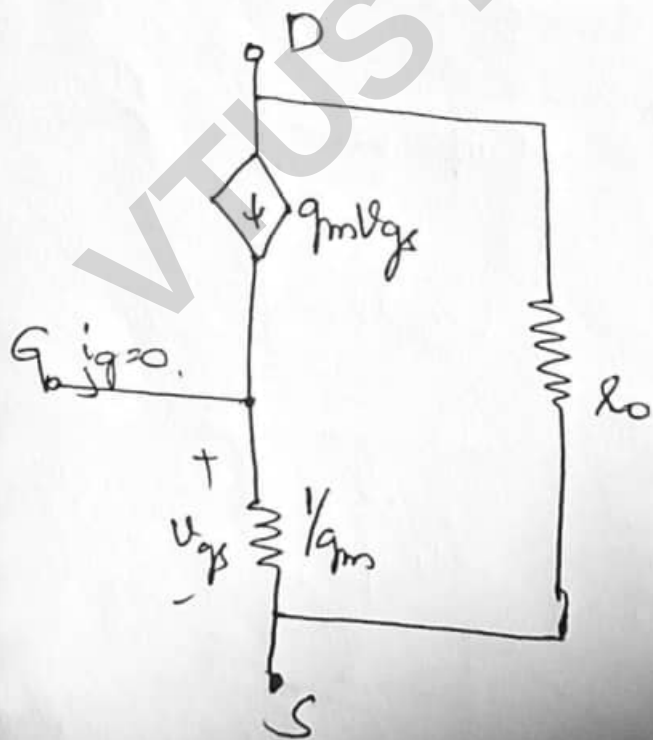
From above ckt,

$$i_g = 0 \rightarrow (1)$$

$$i_d = g_m V_{gs} \rightarrow (2)$$

$$i_s = \frac{V_{gs}}{1/g_m} = g_m V_{gs} \rightarrow (3)$$

All 3 currents remains same as Original Model.



→ MOSFET T-Model with r_o resistance.

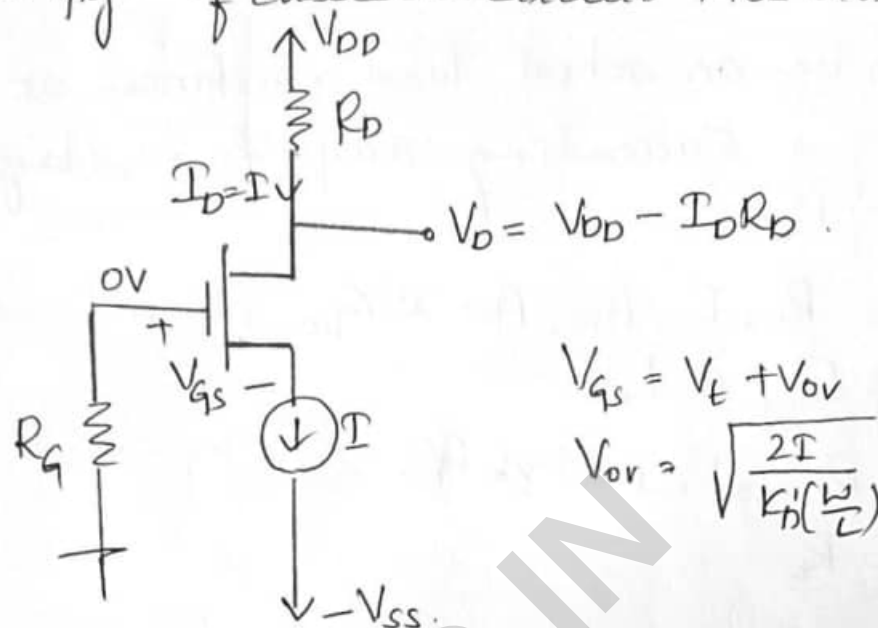
MODULE - 2

Text Book:- Microelectronic Circuits, Theory & Applications :-
Adel S Sedra, Kenneth C Smith.

MOSFET Amplifier Configuration:-

* Basic Configurations

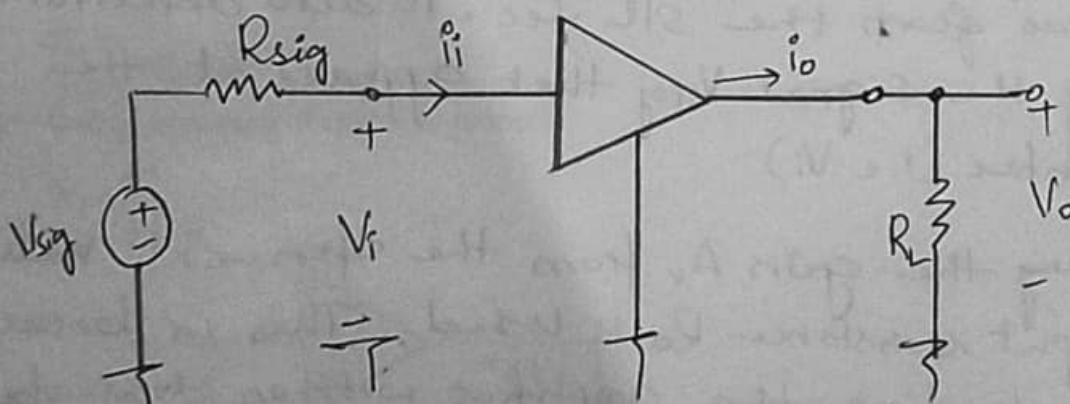
The circuit shown below is used to implement the various configs of discrete-circuit MOS amplifiers.



- This circuit is selected for its effectiveness & simplicity & employed constant current biasing.
- The circuit indicates the dc current & dc voltages at various nodes.

* Characterizing Amplifiers

Consider the amplifier shown below.



The remarks on characteristic parameters of Amplifier.

1.] Amplifier shown in fig(a) is fed with a signal source having an open-ckt vlg V_{sig} & an internal Resistance R_{sig} .

→ V_{sig} & R_{sig} can be the parameters of an actual signal source or the Thevenin equivalent of the o/p ckt of another amplifier stage preceding the one under study in a cascade amplifier.

→ Similarly, R_L can be an actual load resistance or the i/p resistance of a succeeding amplifier stage in a cascade amplr.

2.] The parameters R_i , R_o , A_{vo} , A_v & G_m do not depend on the values of R_{sig} & R_L .

In contrast, R_{in} , R_{out} , A_v , A_i , G_{vo} & G_v depend on one or both R_{sig} & R_L .

3.] For non-unilateral amplifiers, R_{in} may depend on R_L & R_{out} may depend on R_{sig} .

→ No such dependencies exist for unilateral amplifiers, for which $R_{in} = R_i$ & $R_{out} = R_o$.

4.] The loading of the amplifier on the signal source is determined by the input resistance R_{in} .

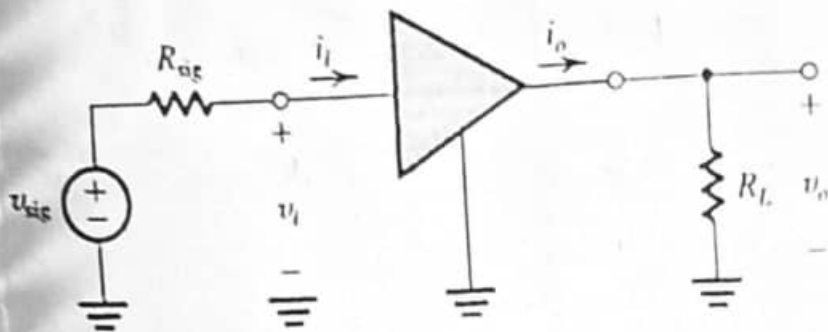
→ The value of R_{in} determines the current i_i that the amplifier draws from the sll sec. It also determines the proportion of the signal V_{sig} that appears at the i/p of the amplifier (i.e. V_i).

5.] When evaluating the gain A_v from the open-ckt value A_{vo} , the output resistance R_o is used. This is because A_v is based on feeding the amplifier with an ideal vlg V_i as shown in fig(A) of table A.

→ To evaluate the overall v/g gain G_v from its open-ckt value G_{vo} , the o/p resistance R_{out} is ~~so~~ used. This is because G_v is based on feeding the amplifier with V_{sig} , which has an internal resistance R_{sig} as shown in fig C of table A.

TABLE 4.3 Characteristic Parameters of Amplifiers

Circuit



Definitions

- Input resistance with no load:

$$R_i \equiv \frac{v_i}{i_i} \Big|_{R_L = \infty}$$

- Input resistance:

$$R_{in} \equiv \frac{v_i}{i_i}$$

- Open-circuit voltage gain:

$$A_{vo} \equiv \frac{v_o}{v_i} \Big|_{R_L = \infty}$$

- Voltage gain:

$$A_v \equiv \frac{v_o}{v_i}$$

- Short-circuit current gain:

$$A_{is} \equiv \frac{i_o}{i_i} \Big|_{R_L = 0}$$

- Current gain:

$$A_i \equiv \frac{i_o}{i_i}$$

- Short-circuit transconductance:

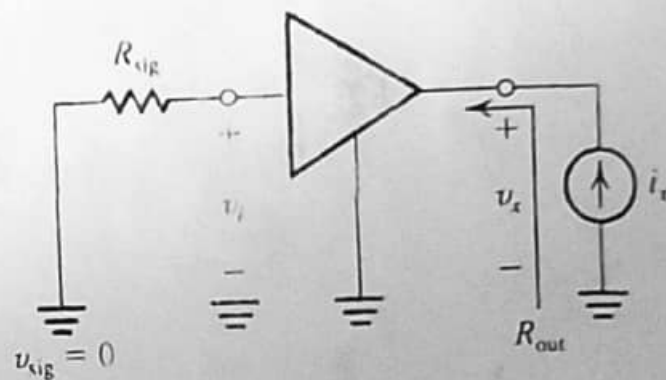
$$G_m \equiv \frac{i_o}{v_i} \Big|_{R_L = 0}$$

- Output resistance of amplifier proper:

$$R_o \equiv \frac{v_x}{i_x} \Big|_{v_i = 0}$$

- Output resistance:

$$R_{out} \equiv \frac{v_x}{i_x} \Big|_{v_{sig} = 0}$$



- Open-circuit overall voltage gain:

$$G_{vo} \equiv \frac{v_o}{v_{sig}} \Big|_{R_L = \infty}$$

- Overall voltage gain:

$$G_v \equiv \frac{v_o}{v_{sig}}$$

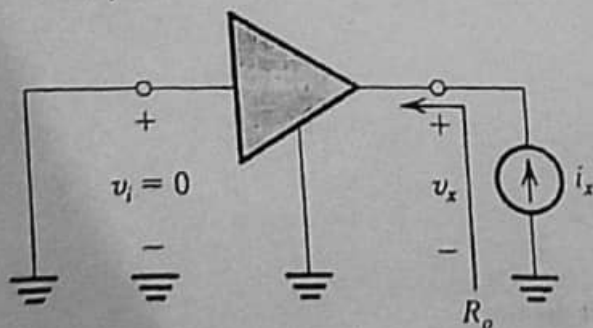
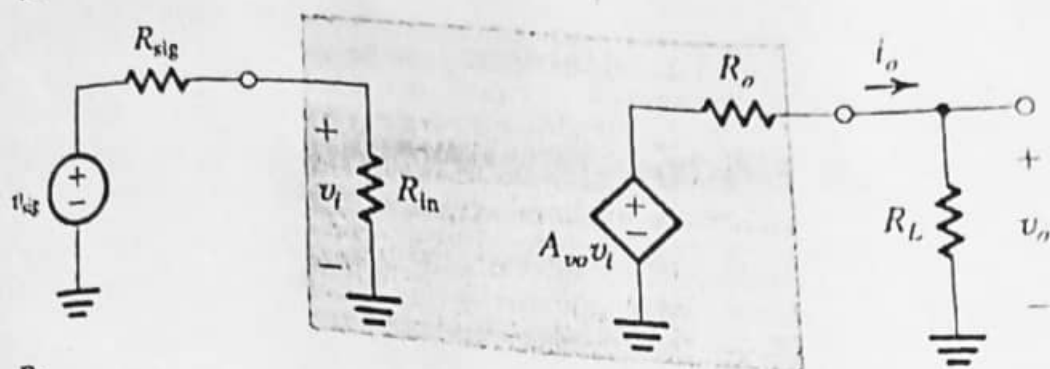


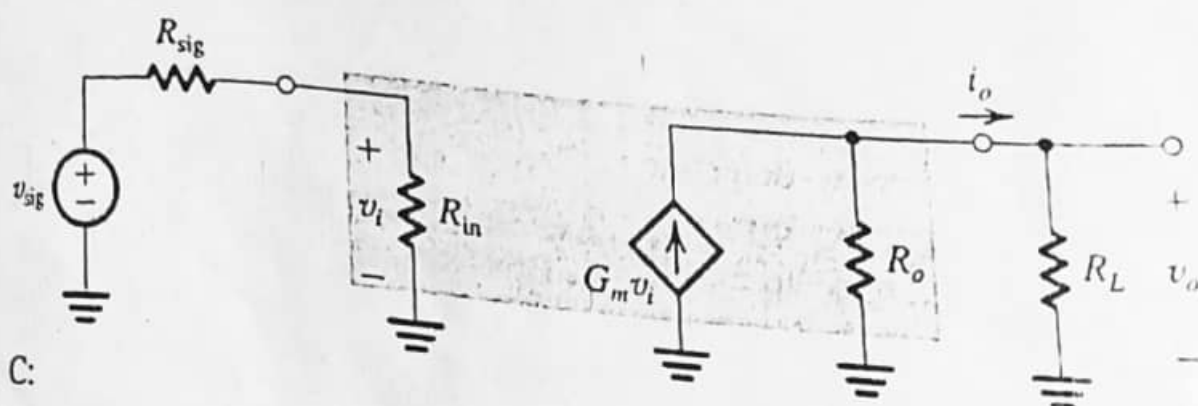
TABLE 4.3 (Continued)

Equivalent Circuits

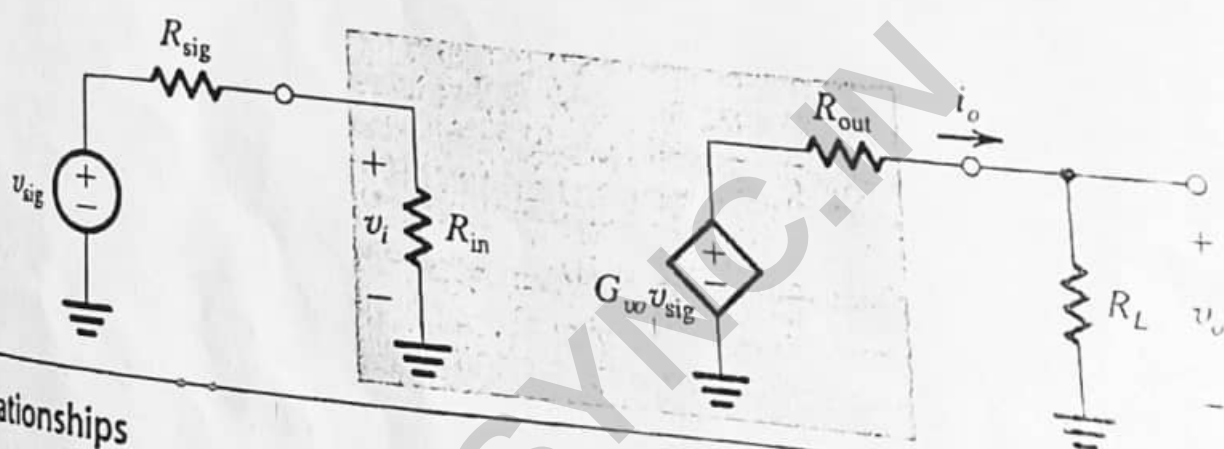
A:



B:



C:



Relationships

$$\frac{v_i}{v_{sig}} = \frac{R_{in}}{R_{in} + R_{sig}}$$

$$A_v = A_{vo} \frac{R_L}{R_L + R_o}$$

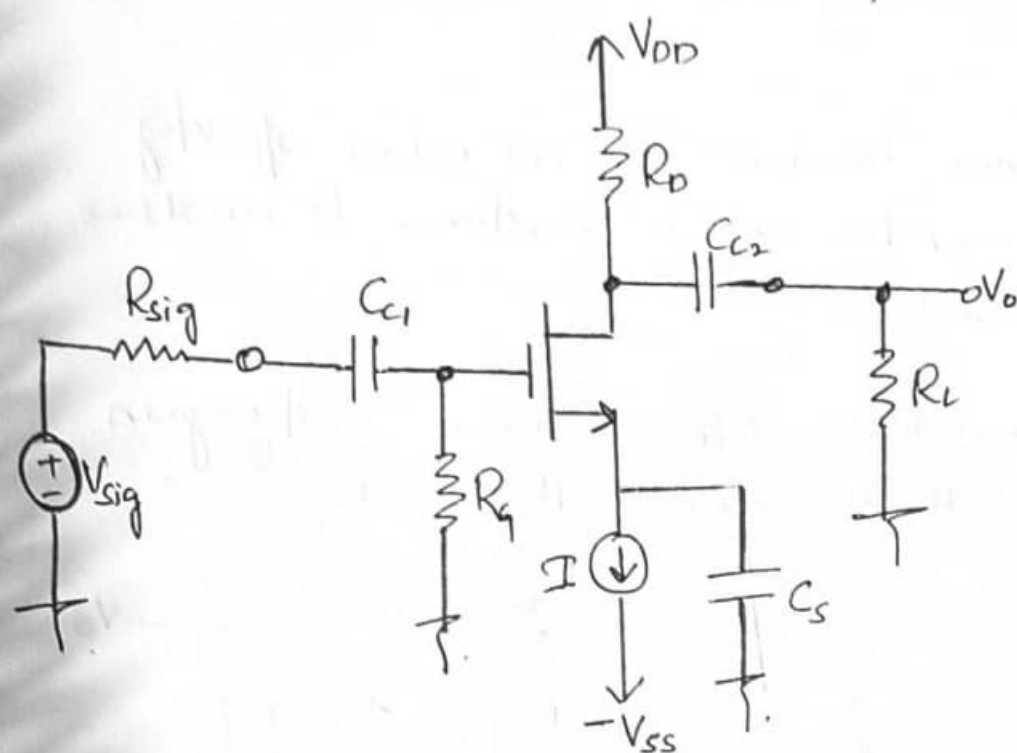
$$A_{vo} = G_m R_o$$

$$G_v = \frac{R_{in}}{R_{in} + R_{sig}} A_{vo} \frac{R_L}{R_L + R_o}$$

$$G_{vo} = \frac{R_i}{R_i + R_{sig}} A_{vo}$$

$$G_v = G_{vo} \frac{R_L}{R_L + R_{out}}$$

The Common - Source (C_s) Amplifier



A common-source amplifier realized using the above ckt.

Bypass Capacitor (C_s):- The large capacitor C_s (ranges in μF) is connected b/w source and gnd to establish signal gnd or ac gnd.

- This capacitor is required to provide very small impedance to the signal current passes th¹ C_s thus bypasses all current to ground. Hence C_s is called bypass capacitor.
- C_s is acting as a perfect short circuit & thus establishing zero sll vlg at the MOSFET source.

Coupling capacitor

C_{c1} :- In order not to disturb the dc bias current & voltages, V_{sig} is ~~connected~~ with internal Resistance R_{sig} is connected to gate through a large capacitor C_{c1} .

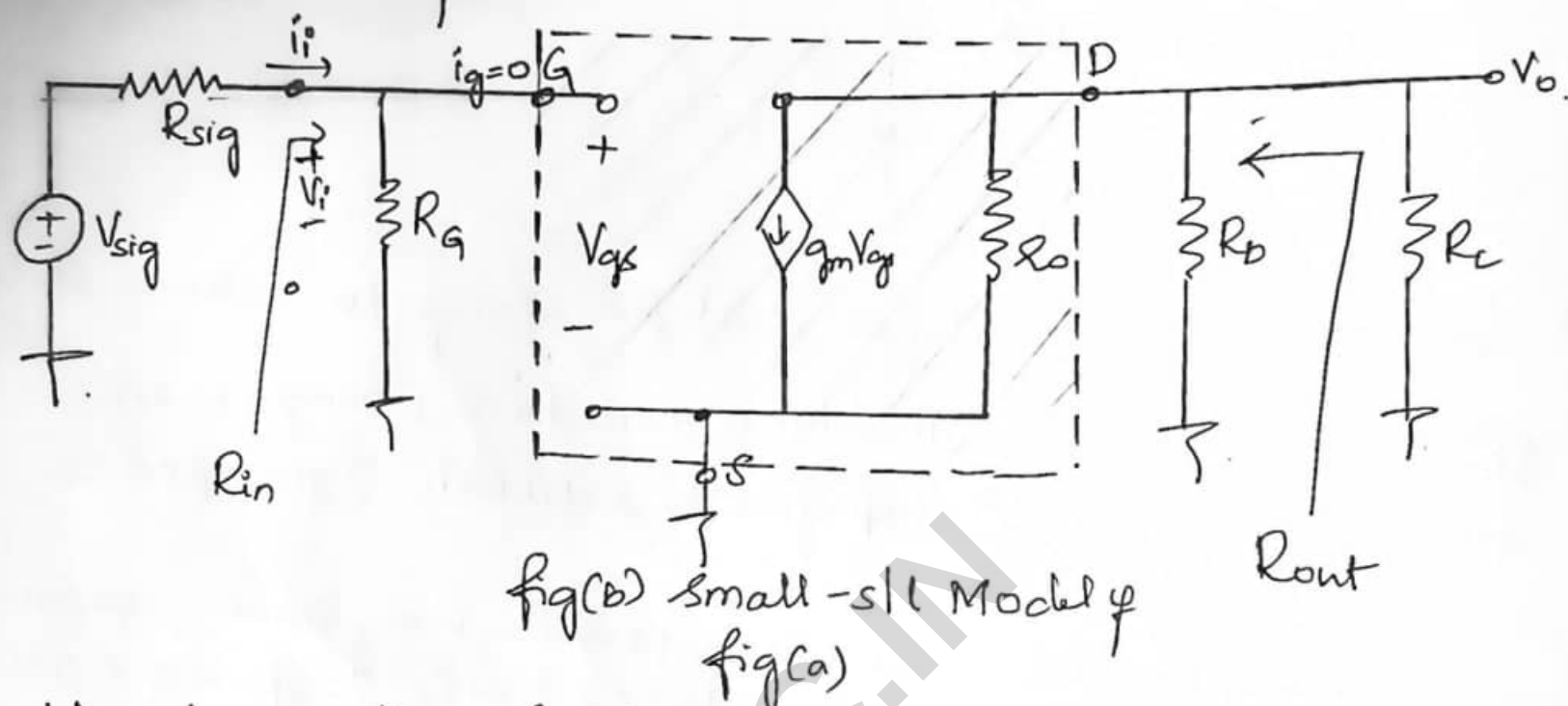
- C_{c1} is known as coupling capacitor, acts as a perfect short circuit to all ac sll while blocking dc.

C_{c2} :- The voltage signal resulting at the drain is coupled to the load resistance R_L via another coupling capacitor C_{c2} .

→ C_{cs} acts as a perfect short circuit at all signal freqs
 thus o/p v/g $V_o = V_d$

→ R_L can be either actual load resistor, at which o/p v/g is drawn or it can be input resistance to another amplifier stage in cascade case.

→ To determine i/p resistance, o/p resistance & v/g gain MOSFET is replaced with its small-sig Model.



→ We observe that fig(b) unilateral ckt

→ Therefore R_{in} does not depend on R_L & thus $R_{in} = R_i$
 Also, R_{out} does not depend on R_{sig} & thus $R_{out} = R_o$

→ At the i/p,

$$i_g = 0$$

$$R_{in} = R_G$$

$$V_i = V_{sig} \frac{R_{in}}{R_{in} + R_{sig}} = \left(\frac{R_G}{R_G + R_{sig}} \right) V_{sig}$$

if $R_G \gg R_{sig}$

$$V_i \approx V_{sig}$$

Now,

$$V_{gs} = V_i$$

&

$$V_o = -g_m V_{gs} (R_o \parallel R_D \parallel R_L)$$

Thus the v/g gain A_v is

$$A_v = \frac{V_o}{V_i} = \frac{-g_m V_{gs} (r_o \parallel R_D \parallel R_L)}{V_{gs}}$$

$$\boxed{A_v = -g_m (r_o \parallel R_D \parallel R_L)}$$

& the open-circuit v/g gain A_{vo} is

$$\boxed{A_{vo} = -g_m (r_o \parallel R_D)}$$

→ The overall v/g gain from the sll-sec to the load will be

$$G_v = \frac{R_{in}}{R_{in} + R_{sig}} A_v = -\frac{R_g}{R_g + R_{sig}} g_m (r_o \parallel R_D \parallel R_L)$$

$$\boxed{G_v = -\frac{R_g}{R_g + R_{sig}} g_m (r_o \parallel R_D \parallel R_L)}$$

→ To find R_{out} set $V_{sig} = 0$, ∴ replacing it with short circuit

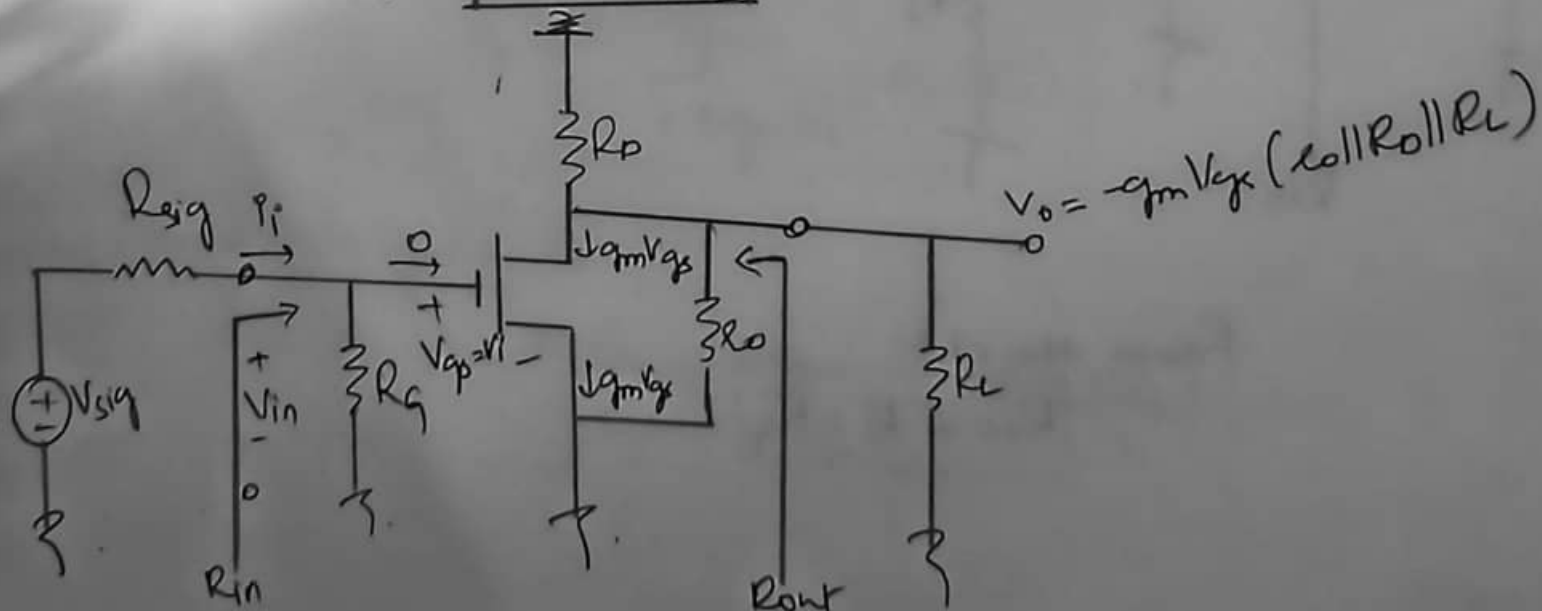
$$\boxed{R_{out} = r_o \parallel R_D}$$

if $r_o \gg R_D$

$$R_{out} = \frac{1}{\cancel{\frac{1}{r_o}}} + \frac{1}{R_D}$$

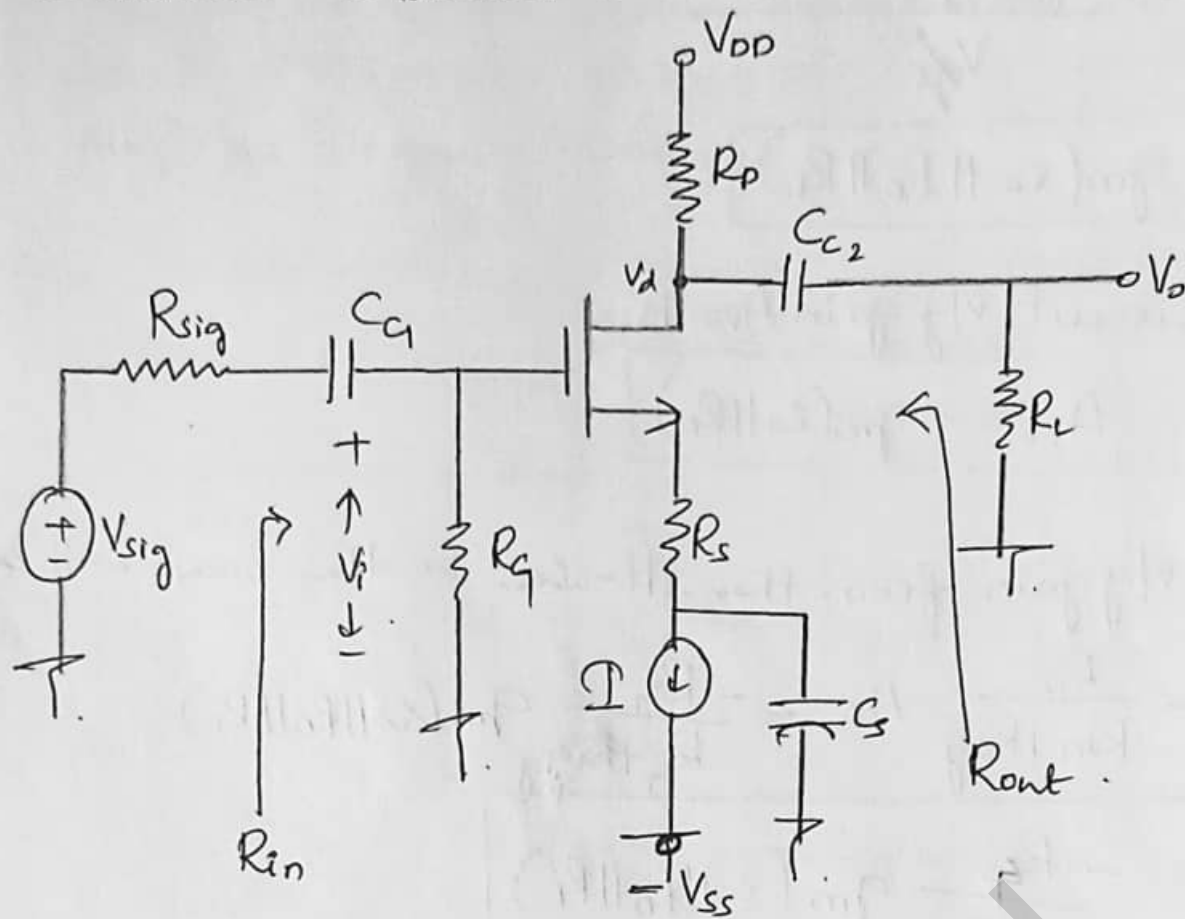
Small

$$\therefore \boxed{R_{out} \approx R_D}$$

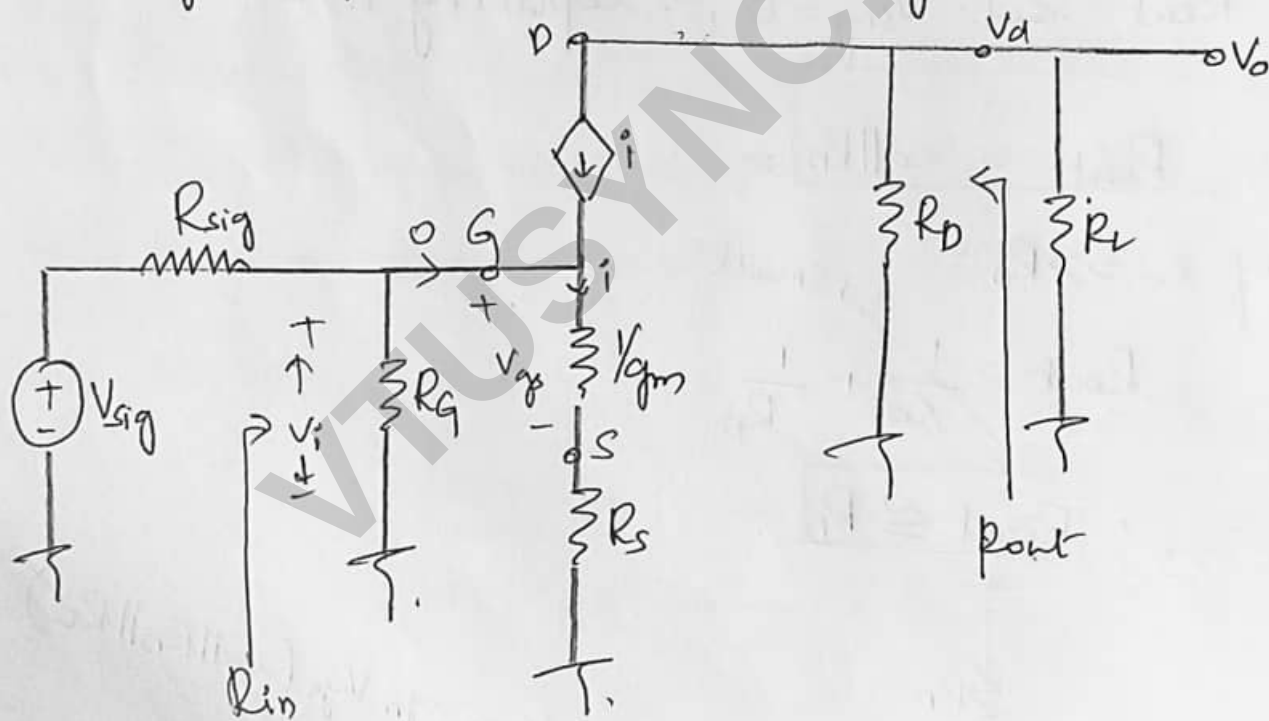


* Common-Source Amplifier with a Source Resistance

The ckt diagram for Common-Source Amplifier with a Source Resistance is shown below.



Small-signal equivalent ckt with C_o neglected is shown below.



From the ckt

$$R_{in} = R_i = R_G$$

$$\therefore V_i = V_{sig} \frac{R_g}{R_g + R_{sig}} \quad (\text{Using voltage-divider rule})$$

→ Unlike CS ckt without R_s , here V_{gs} is only a fraction of V_i . It can be determined from v/g divider composed of $\frac{1}{g_m}$ & R_s that appears across the amplifier i/p as follows.

$$V_{gs} = V_i \frac{\frac{1}{g_m}}{\frac{1}{g_m} + R_s} = V_i \frac{\frac{1}{g_m}}{\frac{1 + g_m R_s}{g_m}}$$

$$\boxed{V_{gs} = \frac{V_i}{1 + g_m R_s}}$$

→ From above eq, we observe that value of R_s controls the magnitude of V_{gs} .

→ The current i_d is equal to the current i flowing in source terminal.

$$\therefore i_d = i = \frac{V_i}{\frac{1}{g_m} + R_s} = \frac{g_m V_i}{1 + g_m R_s}$$

$$i = \frac{V}{R}$$

→ The output v/g can be found as

$$V_o = -i_d (R_D || R_L)$$

$$\boxed{V_o = -\frac{g_m V_i}{1 + g_m R_s} (R_D || R_L)}$$

Thus v/g gain is

$$\boxed{A_v = \frac{V_o}{V_i} = -\frac{g_m}{1 + g_m R_s} (R_D || R_L)}$$

↳ setting $R_L = \infty$ gives

$$A_{v0} = \frac{-g_m R_o}{1 + g_m R_s}$$

The overall v/g gain is

$$G_v = \frac{-R_G}{R_G + R_{sig}} \cdot A_v$$

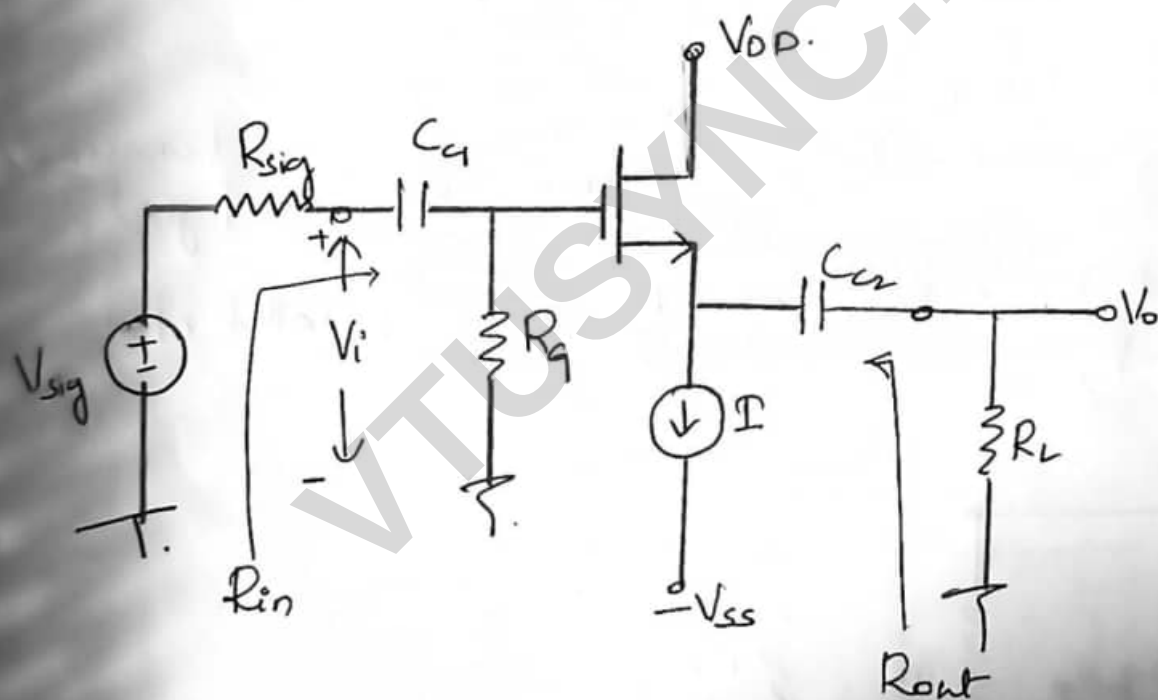
$$G_v = \frac{-R_G}{R_G + R_{sig}} \cdot \frac{g_m (R_o \parallel R_L)}{1 + g_m R_s}$$

By including resistance R_s we observe that the gain reduces by factor $(1 + g_m R_s)$, But it improves the performance of the ckt.

→ As R_s is reducing the gain, it is called source degeneration Resistance.

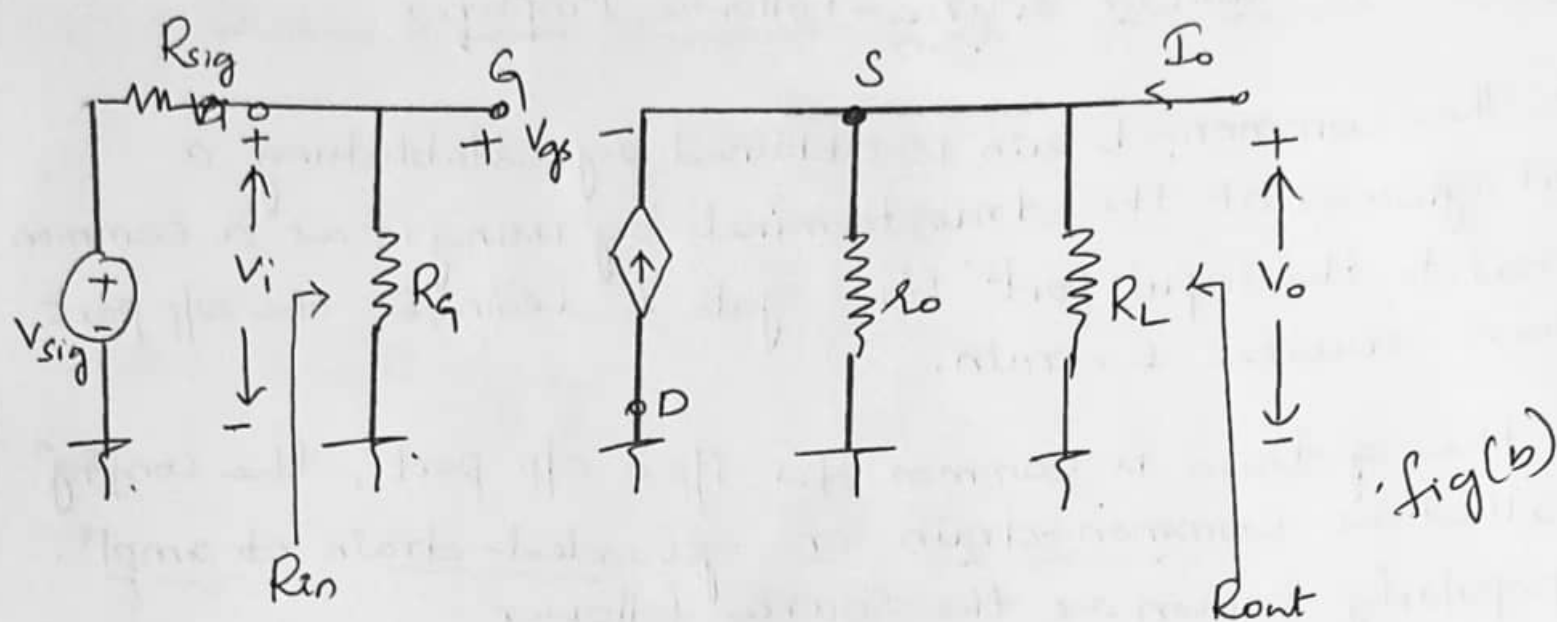
* The Common-Drain or Source-Follower Amplifier.

- ~~Here~~ The common-Drain is obtained by establishing a signal ground at the drain terminal by using it as a common terminal to the input port b/w gate & drain, & the o/p port between source & drain.
- As the drain is common b/w i/p & o/p port, the configⁿ is called as Common-drain or grounded-drain ~~amp~~ ampl^r, or popularly known as the Source follower.
- The fig(a) shows a Common-drain amplifier. The i/p signal is coupled via Capacitor C_1 to the MOSFET gate, and the o/p signal at MOSFET source is coupled via Capacitor C_2 to ~~the~~ load Resistor R_L .
- Since the drain is to funⁿ as a signal ground, there is no need for resistor R_D & it has therefore eliminated



fig(a) Source follower ckt.

- The small-sll model of fig(a) is shown in fig(b).



From the ckt,

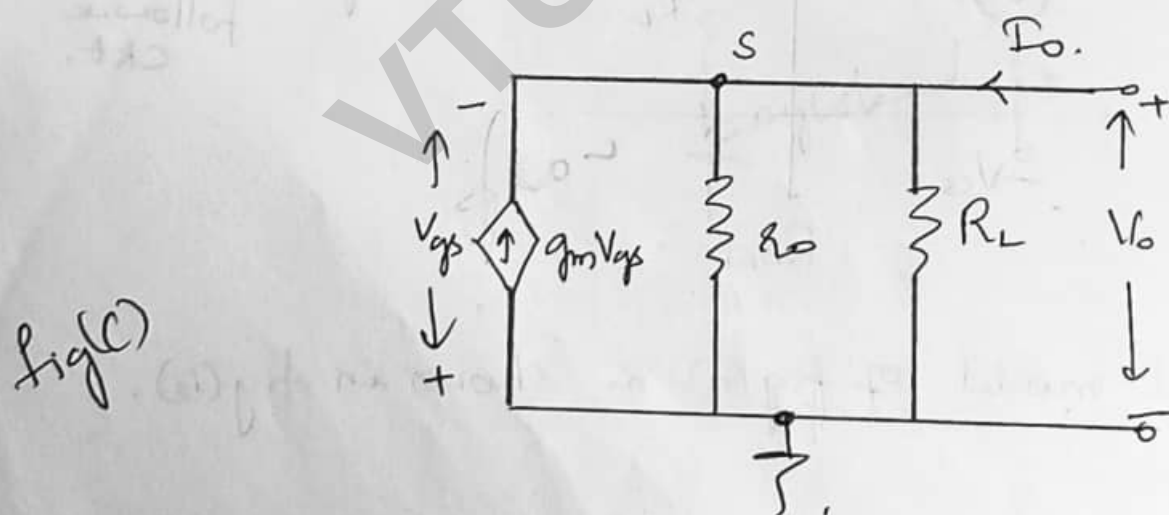
$$\boxed{R_{in} = R_g} \quad \sim (1)$$

Thus,

$$V_i = V_{sig} \frac{R_{in}}{R_{in} + R_{sig}} = V_{sig} \frac{R_g}{R_g + R_{sig}} \quad \sim (2)$$

To find R_{out} set $V_i = 0V$ results in the gate terminal being connected directly to ground shown in fig(c)

→ The fact that V_{gs} & V_o across the same parallel n/w results in $\boxed{V_o = -V_{gs}}$



Applying Kirchhoff's current law at node S, ^{fig(c)} we obtain

$$I_o + g_m V_{gs} = I_{r_o} + I_{R_L}$$

$$= \frac{V_o}{r_o} + \frac{V_o}{R_L}$$

$$\Rightarrow I_o = V_o \left[\frac{1}{r_o} + \frac{1}{R_L} \right] - g_m V_{gs}$$

$$= V_o \left[\frac{1}{r_o} + \frac{1}{R_L} \right] - g_m [-V_o]$$

$$I_o = V_o \left[\frac{1}{r_o} + \frac{1}{R_L} + g_m \right]$$

$$R_{out} = \frac{V_o}{I_o} = \frac{1}{\left[\frac{1}{r_o} + \frac{1}{R_L} + g_m \right]} = \frac{1}{\left[\frac{1}{r_o} + \frac{1}{R_L} + \frac{1}{1/g_m} \right]}$$

$$\therefore R_{out} = r_o \parallel R_L \parallel 1/g_m \quad \sim (3)$$

$r_o \gg 1/g_m$ & $r_o \gg R_L$

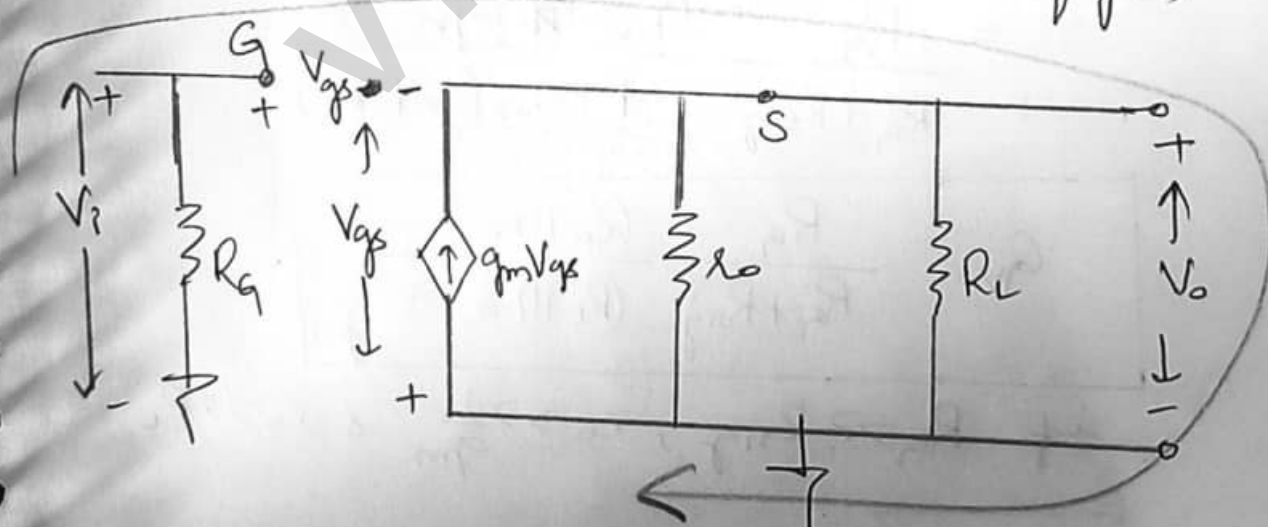
$$R_{out} \approx \frac{1}{g_m}$$

To find A_v

The output voltage V_o is determined by

$$V_o = g_m V_{gs} (r_o \parallel R_L) \quad \sim (4)$$

Consider the ckt shown below in fig(d)



Applying KVL around the perimeter of ckt shown in fig(d)

$$V_i = V_{gs} + V_o$$

$$\therefore V_{gs} = V_i - V_o \quad \sim (5)$$

→ Substituting for V_{gs} in eqⁿ (4) from eqⁿ (5)

so that, $V_o = g_m (V_i - V_o)(r_o \parallel R_L)$

$$V_o = V_i g_m (r_o \parallel R_L) - V_o g_m (r_o \parallel R_L)$$

$$V_o [1 + g_m (r_o \parallel R_L)] = V_i g_m (r_o \parallel R_L)$$

$$A_v = \frac{V_o}{V_i} = \frac{g_m (r_o \parallel R_L)}{1 + g_m (r_o \parallel R_L)} \quad \text{--- (6)}$$

$$A_v |_{r_o \gg R_L} = \frac{R_L}{R_L + 1/g_m}$$

Since A_v of eqⁿ (6) is a positive quantity, V_o & V_i are in-phase for the JFET source-follower confⁿ.

→ The open-ckt v/g gain A_{vo} (when $R_L = \infty$)

$$A_{vo} |_{R_L = \infty} = \frac{g_m r_o}{1 + g_m r_o} = \frac{r_o}{r_o + 1/g_m}$$

The overall v/g gain G_v

$$G_v = \frac{R_g}{R_g + R_{sig}} \cdot A_v$$

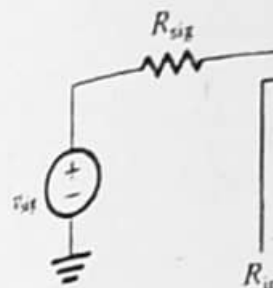
$$= \frac{R_g}{R_g + R_{sig}} \cdot \frac{(R_L \parallel r_o) g_m}{1 + g_m (r_o \parallel R_L)}$$

$$G_v = \frac{R_g}{R_g + R_{sig}} \cdot \frac{(r_o \parallel R_L)}{(R_L \parallel r_o) + 1/g_m}$$

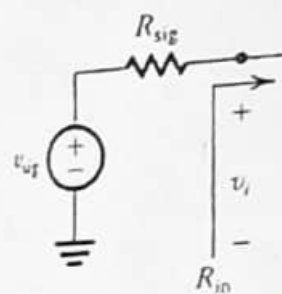
if $R_g \gg R_{sig}$, $r_o \gg 1/g_m$ & $r_o \gg R_L$

$$G_v \approx 1$$

TABLE 4.4
Common-Source



Common-Source with S



Common-Drain or Source

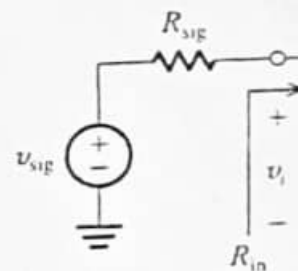
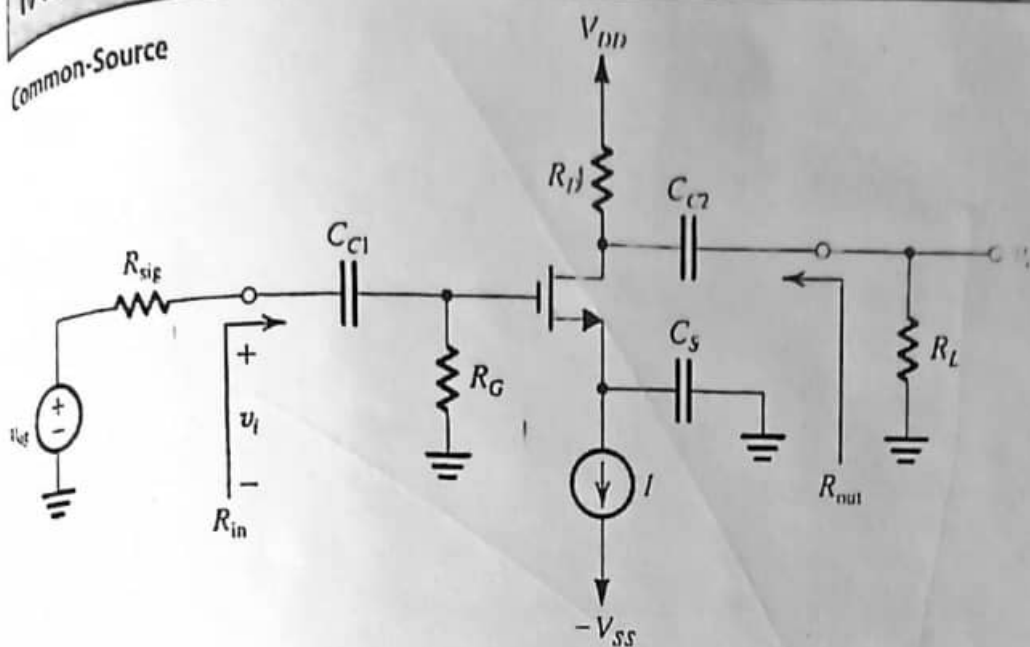


TABLE 4.4 Characteristics of Single-Stage Discrete MOS Amplifiers

Common-Source



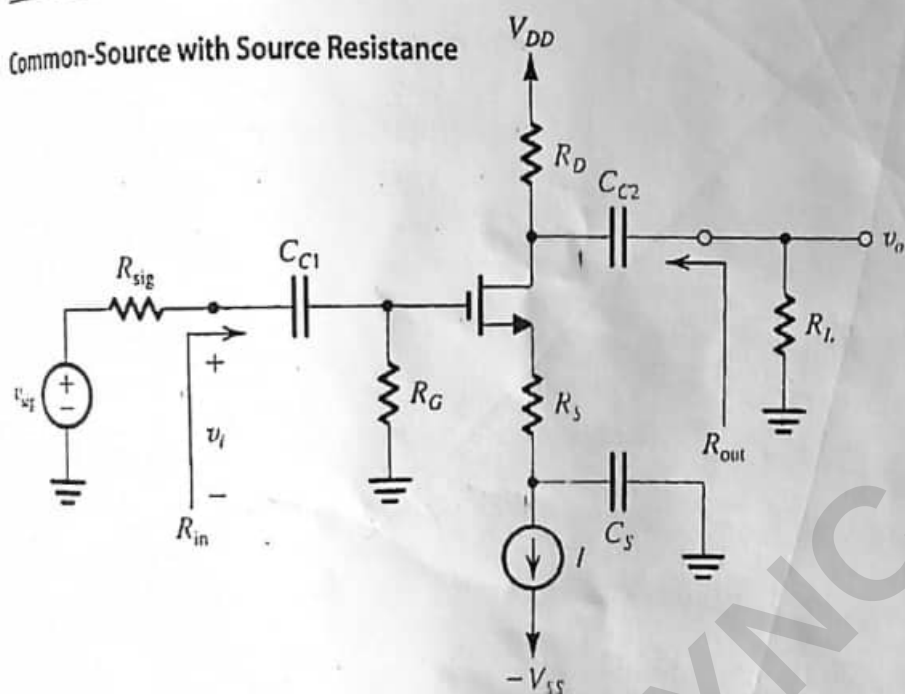
$$R_{in} = R_G$$

$$A_v = -g_m(r_o \parallel R_D \parallel R_L)$$

$$R_{out} = r_o \parallel R_D$$

$$G_v = \frac{R_G}{R_G + R_{sig}} \frac{g_m(r_o \parallel R_D \parallel R_L)}{1 + g_m R_S}$$

Common-Source with Source Resistance



Neglecting r_o :

$$R_{in} = R_G$$

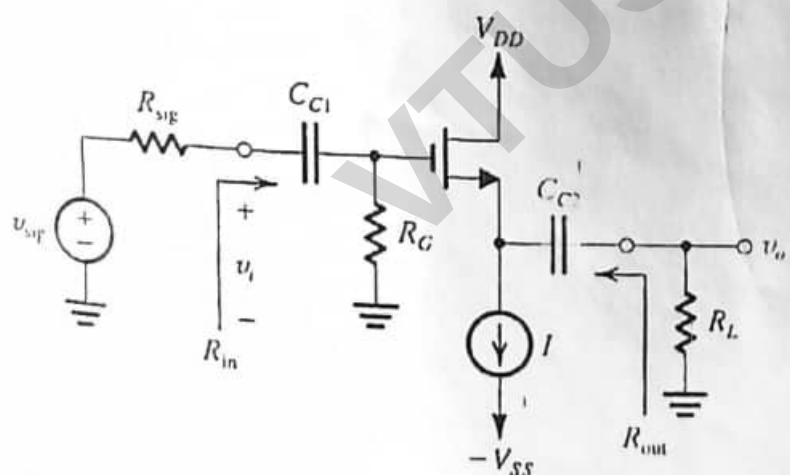
$$A_v = \frac{R_D \parallel R_L}{\frac{1}{g_m} + R_S} = \frac{g_m(R_D \parallel R_L)}{1 + g_m R_S}$$

$$R_{out} = R_D$$

$$G_v = \frac{R_G}{R_G + R_{sig}} \frac{g_m(R_D \parallel R_L)}{1 + g_m R_S}$$

$$\frac{v_{gs}}{v_i} = \frac{1}{1 + g_m R_S}$$

Common-Drain or Source Follower



$$R_{in} = R_G$$

$$A_v = \frac{r_o \parallel R_L}{(r_o \parallel R_L) + \frac{1}{g_m}}$$

$$R_{out} = \frac{1}{g_m} \parallel R_S$$

$$G_v = \frac{R_G}{R_G + R_{sig}} \frac{r_o \parallel R_L}{(r_o \parallel R_L) + \frac{1}{g_m}}$$

1) When the MOSFET is operating in the linear region, the channel will be of uniform depth. The gate-channel capacitance will be modeled by dividing it equally b/w + ends; thus

$$C_{gs} = C_{gd} = \frac{1}{2} W L C_{ox} \quad (C_{trio})$$

* The MOSFET Internal Capacitances and High-Frequency Model

→ The MOSFET has internal capacitances of 2 types.

1] The gate capacitive effect:

The gate electrode (polysilicon) forms a parallel-plate capacitor with the channel, with the oxide layer serving as the capacitor dielectric.

2] The source-body & drain-body depletion-layer capacitance:

These are the capacitances of the reverse-biased pn junction formed by the n^+ source region (also called the source diffusion) and the p-type substrate & by the n^+ drain region (the drain diffusion) and the substrate.

The above two capacitive effects can be modeled by including capacitances in the MOSFET model b/w its 4 terminals, G, D, S & B.

→ And there will be 5 capacitances in total: C_{gs} , C_{gd} , C_{gb} , C_{sb} , & C_{db} .

* The Gate Capacitive Effect

The gate capacitive effect can be modeled by the 3 capacitances C_{gs} , C_{gd} & C_{gb} . The values of these capacitances can be determined as follows.

1] When the MOSFET is operating in the triode region at small V_{DS} , the channel will be of uniform depth. The gate-channel capacitance will be WLC_{ox} and can be modeled by dividing it equally b/w the src & drain ends; thus

$$C_{gs} = C_{gd} = \frac{1}{2} WLC_{ox} \text{ (triode reg)} \quad \text{--- (1)}$$

2] When the MOSFET operates in saturation, the channel has a tapered shape & ~~it~~ is pinched off at or near the drain end.

→ In this case the gate-to-channel capacitance is approximately equal to $\frac{2}{3} W L C_{ox}$.

→ This case can be modeled by assigning this entire amount to C_{gs} and a zero amount to C_{gd}

$$\therefore \left. \begin{array}{l} C_{gs} = \frac{2}{3} W L C_{ox} \\ C_{gd} = 0 \end{array} \right\} \text{(Saturation region)} \quad \text{--- (2)}$$

3] When the MOSFET is cut off, the channel disappears, & thus $C_{gs} = C_{gd} = 0$.

→ The gate capacitive effect in this case can be modeled by assigning a capacitance $W L C_{ox}$ to gate-to-body model

$$\therefore \left. \begin{array}{l} C_{gs} = C_{gd} = 0 \\ C_{gb} = W L C_{ox} \end{array} \right\} \text{(Cutoff)} \quad \text{--- (3)}$$

4] The source & drain diffusion in previous cases extends slightly under the gate oxide & hence it should be added to C_{gs} & C_{gd} . It is referred as Overlap capacitance with overlap length is L_{ov} .
~~This overlapped capaci~~

$$\therefore C_{ov} = W L_{ov} C_{ox}$$

Typically, $L_{ov} = 0.05$ to 0.1μ .

The Junction Capacitances.

The depletion-layer capacitances of the 2 reverse-biased pn-junctions are formed b/w each of the source & the drain diffusions & the body.

∴ For the source diffusion, the source-body capacitance is given by

$$C_{sb} = \frac{C_{sbo}}{\sqrt{1 + \frac{V_{sb}}{V_0}}}$$

where $C_{sbo} \rightarrow$ Value of C_{sb} at Zero body-source bias,
 $V_{sb} \rightarrow$ Magnitude of the reverse bias V_{gs} .
 $V_0 \rightarrow$ Junction built-in V_{bi} (0.6V to 0.8V)

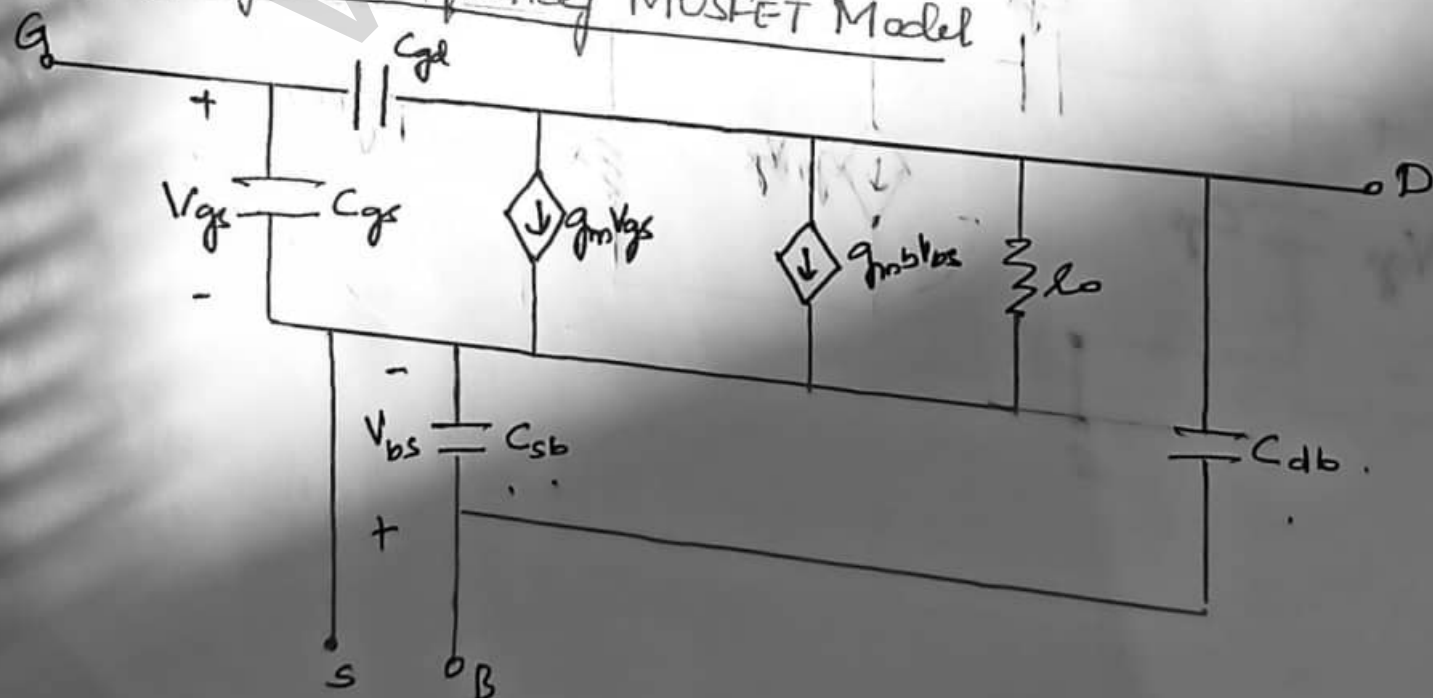
Similarly, for the drain diffusion, the drain-body capacitance C_{db}

$$C_{db} = \frac{C_{dbo}}{\sqrt{1 + \frac{V_{db}}{V_0}}}$$

where, $C_{dbo} \rightarrow$ Capacitance value to Zero reverse-bias V_{gs}
 $V_{db} \rightarrow$ Magnitude of the reverse-bias V_{ds} .

→ It is assumed that for both junctions, the grading co-eff $m = \frac{1}{2}$.

* The High-Frequency MOSFET Model



Module - 3

Module - 33.1:- The Summing Amplifiers

→ Whenever we need to combine two or more analog signals into a single output, the summing amplifier is a choice.

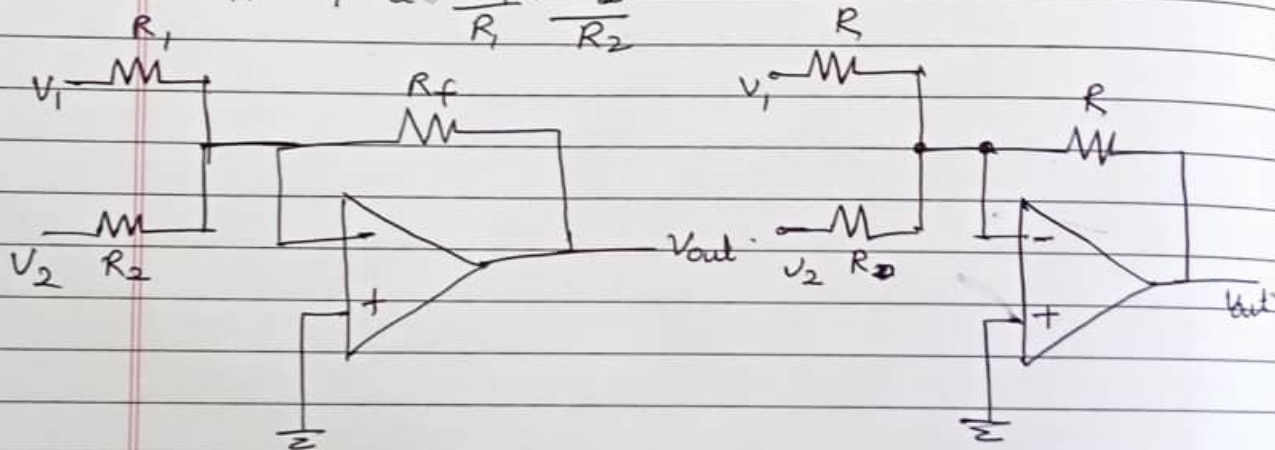
→ The summing circuit combines all the amplified input signals into a single output.

$$V_{out} = A_{V1(CL)} V_1 + A_{V2(CL)} V_2$$

$$A_{V1(CL)} = \frac{-R_f}{R_1}$$

$$A_{V2(CL)} = \frac{-R_f}{R_2}$$

$$i_{in} = i_1 + i_2 = \frac{V_1}{R_1} + \frac{V_2}{R_2}$$



Because of the virtual ground, all the current flows through the feedback resistor.

$$V_{out} = (i_1 + i_2) R_f = \left(\frac{R_f V_1}{R_1} + \frac{R_f V_2}{R_2} \right)$$

In some applications, all resistors are equal, each channel has a closed-loop voltage gain of unity.

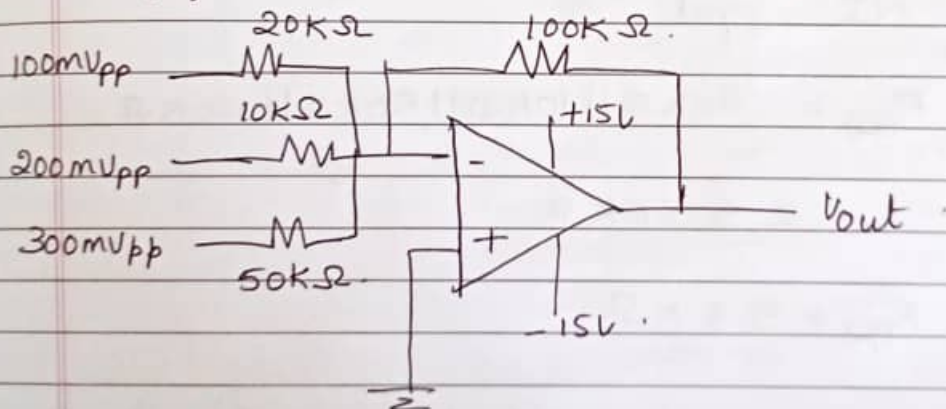
$$V_{out} = -(V_1 + V_2 + \dots + V_n)$$

→ If the summing circuit needs to be compensated by adding an equal resistance to the non-inverting input, the resistance to use is ~~the~~ given by the parallel equivalent of all resistances connected to the virtual ground.

$$R_{B2} = R_1 \parallel R_2 \parallel R_f \parallel \dots \parallel R_n$$

Problem

1) Three audio signals drive the summing amplifier, the output ~~voltage~~ voltage



$$A_{V1(CL)} = - \frac{R_f}{R_1} = - \frac{100K\Omega}{20K\Omega} = -5$$

$$A_{V2(CL)} = - \frac{R_f}{R_2} = - \frac{100K\Omega}{10K\Omega} = -10$$

$$A_{V3(CL)} = - \frac{R_f}{R_3} = - \frac{100K\Omega}{50K\Omega} = -2$$

$$\begin{aligned} V_{out} &= A_{V1(CL)} V_1 + A_{V2(CL)} V_2 + A_{V3(CL)} V_3 \\ &= (-5)(100m) + (-10)(200m) + (-2)(300m) \end{aligned}$$

$$V_{out} = -3.1 V_{pp}$$

to compensate of i/p bias R_B added to NI input is

$$R_{B2} = 20K\Omega || 10K\Omega || 50K\Omega || 100K\Omega$$

$$= 5.56K\Omega$$

$$\underline{R_{B2} = 5.6 K\Omega}$$

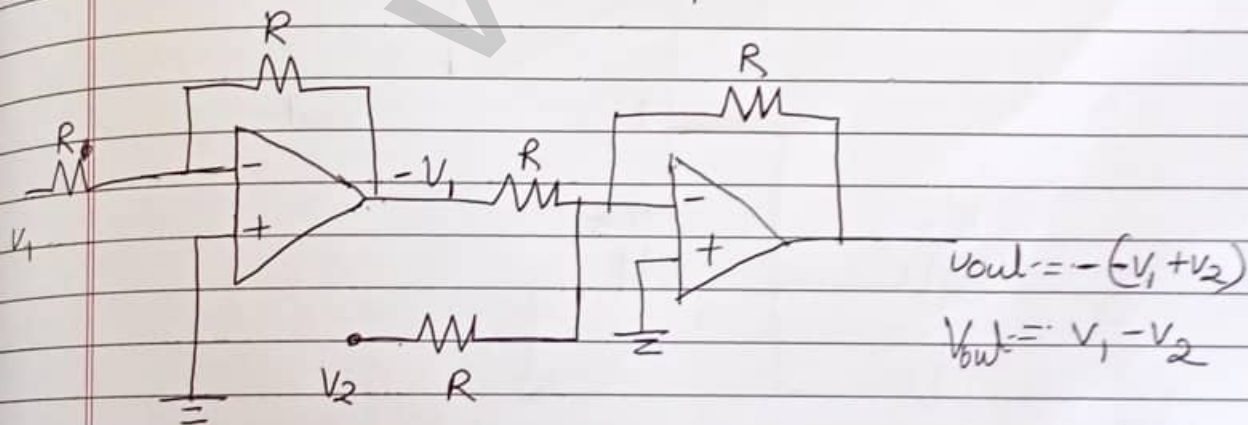
32 The Subtractor:

→ Figure below shows a circuit that subtracts two ~~the~~ input voltages to produce an output voltage equal to the difference of V_1 and V_2 .

→ Input V_1 drives an inverter with a voltage gain of unity.

→ Output of the first stage is $-V_1$.

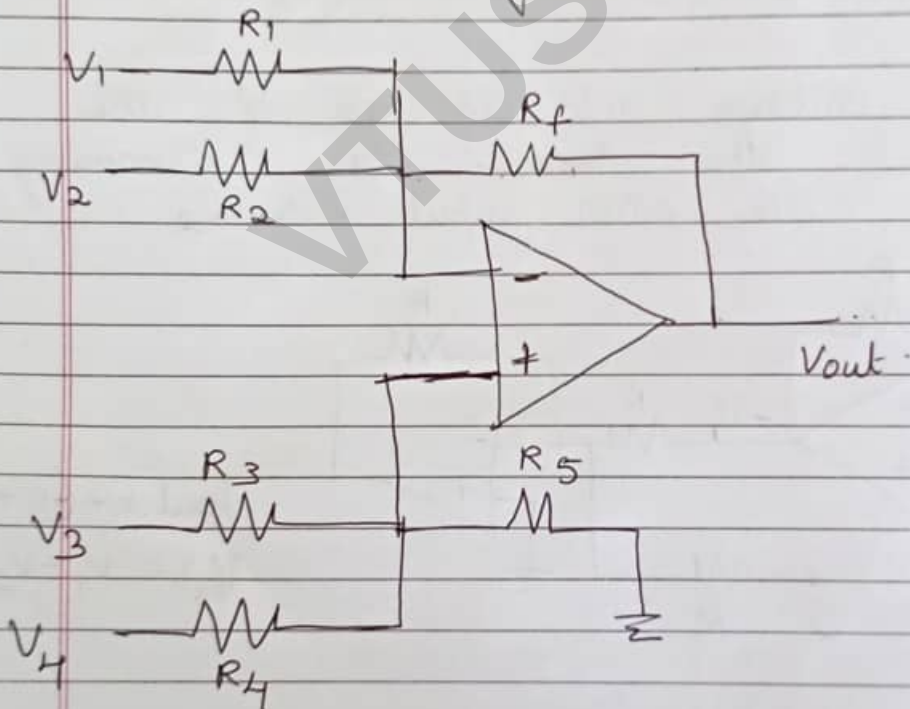
→ This voltage acts as one of the input to the second-stage summing circuit. The other input is V_2 .



The gain of each circuit is unity, the final output voltage is $V_1 - V_2$.

Summing on both inputs

- Summer circuit can be with both inverting and noninverting inputs.
- The inverting has two input channels & the non-inverting has two input channels.
- The total gain is the superposition of the channel gains.



Gain of each inverting channel is the ratio of the feedback resistor R_f to the input channel resistance R_1/R_2 .

$$A_{V1} = \frac{-R_f}{R_1}$$

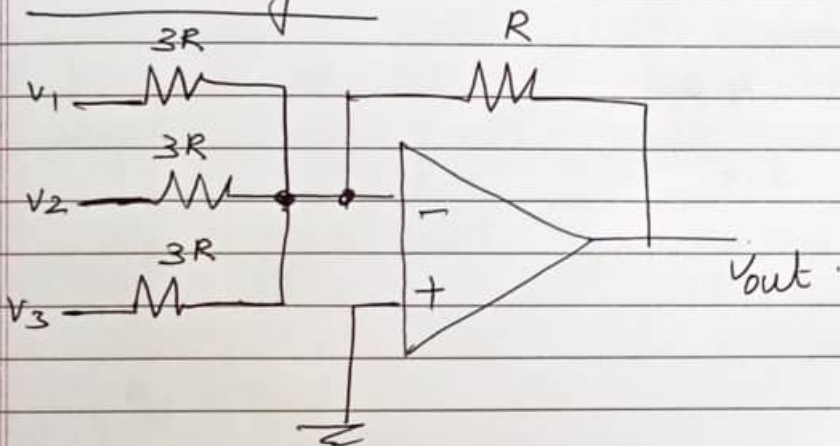
$$A_{V2} = \frac{-R_f}{R_2}$$

Gain of noninverting channel is

$$A_{V3} = \left(\frac{R_f + 1}{(R_1 \parallel R_2)} \right) \cdot \left(\frac{R_4 \parallel R_5}{R_3 + R_4 \parallel R_5} \right)$$

$$A_{V4} = \left(\frac{R_f + 1}{(R_1 \parallel R_2)} \right) \cdot \left(\frac{R_3 \parallel R_5}{R_4 + R_3 \parallel R_5} \right)$$

The Averager



$$V_{out} = - \frac{V_1 + V_2 + V_3}{3}$$

An averager, is a circuit whose output equals the average of the input voltages.

$$A_v = \frac{R}{3R} = \frac{1}{3}$$

when all the amplified outputs are added, we get the average of all input voltages.

Problem 2.

for the summing amplifier, ckt, $R_1 = 1K\Omega$, $R_2 = 2K\Omega$, $R_3 = 3K\Omega$, $R_4 = 4K\Omega$, $R_5 = 5K\Omega$ & $R_f = 6K\Omega$.
find V_{tg} gain of each channel?

$$A_{v1} = \frac{-R_f}{R_1} = \frac{-6K}{1K} = -6$$

$$A_{v2} = \frac{-R_f}{R_2} = \frac{-6K}{2} = -3$$

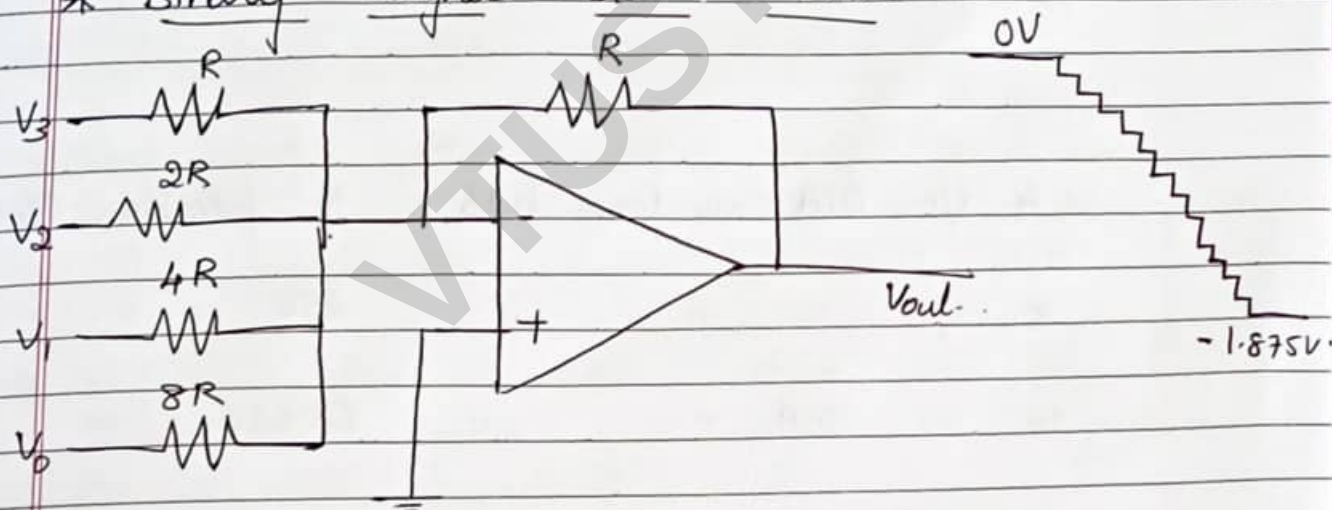
$$A_{v3} = 4.26$$

$$A_{v4} = 3.19$$

D/A Converter

- Digital - to - analog (D/A) Converter takes a binary represented value and converts it into a Voltage & Current.
- This voltage / current will be proportional to the input binary value.
- Two methods of D/A conversion are
 - * binary - weighted D/A
 - * $R/2R$ ladder D/A

* Binary - weighted D/A Converter



The binary weighted D/A converter circuit produces an output voltage equal to the weighted sum of the inputs.

$$A_{V_0} = -\frac{R}{8R} = -0.125 \quad A_{V_1} = -\frac{R}{4R} = -0.25 \quad A_{V_2} = -\frac{R}{2R} = -0.5$$

$$A_{V_3} = -1$$

The input with 4 inputs, with 16 possible combinations is

i/p	Val
0 0 0 0	0
0 0 0 1	- 0.125
0 0 1 0	- 0.25
0 0 1 1	- (0.125 + 0.25) = 0.375
0 1 0 0	- 0.5
0 1 0 1	- 0.625
0 1 1 0	- 1
0 1 1 1	- 1.125
1 0 0 0	- 1.25
1 0 0 1	- 1.375
1 0 1 0	- 1.5
1 0 1 1	- 1.625
1 1 0 0	- 1.75
1 1 0 1	- 1.875
1 1 1 0	- 2
1 1 1 1	- 2.125

4 i/p D/A Converter has 16 possible o/p

8 i/p " " " 256 " "

16 " D/A " " 65,536 " "

A binary weighted D/A Converter can be used in application where the number of inputs are limited and where high precision is not required.

- When a higher number of inputs ~~is~~ ~~used~~ is used, a higher no of different resistor values is required.
- The accuracy & stability of D/A converter depends on the absolute accuracy of the resistors & their ability to track each other with temperature variations.
- Because the input resistors have different values, identical tracking characteristics is difficult to obtain.
- R/2R Ladder D/A Converter:
 - overcomes the limitations of the binary-weighted D/A Converter & is the method most often used in integrated-circuit D/A converters.
 - As only two resistor values are required, this method most often used in integrated circuit D/A converters.
 - Because only two resistor values are required this method lends itself to ICs with 8 bit & higher binary inputs & provides a higher degree of accuracy.

For a 4-bit D/A converter, the switches $D_0 - D_3$ would be some type of active switch.

The four switches connect the four inputs to either ground (logic 0) or V_{ref} (logic 1).

The ladder network converts the possible binary input values from 0000 through 1111 to one of 16 unique output levels.

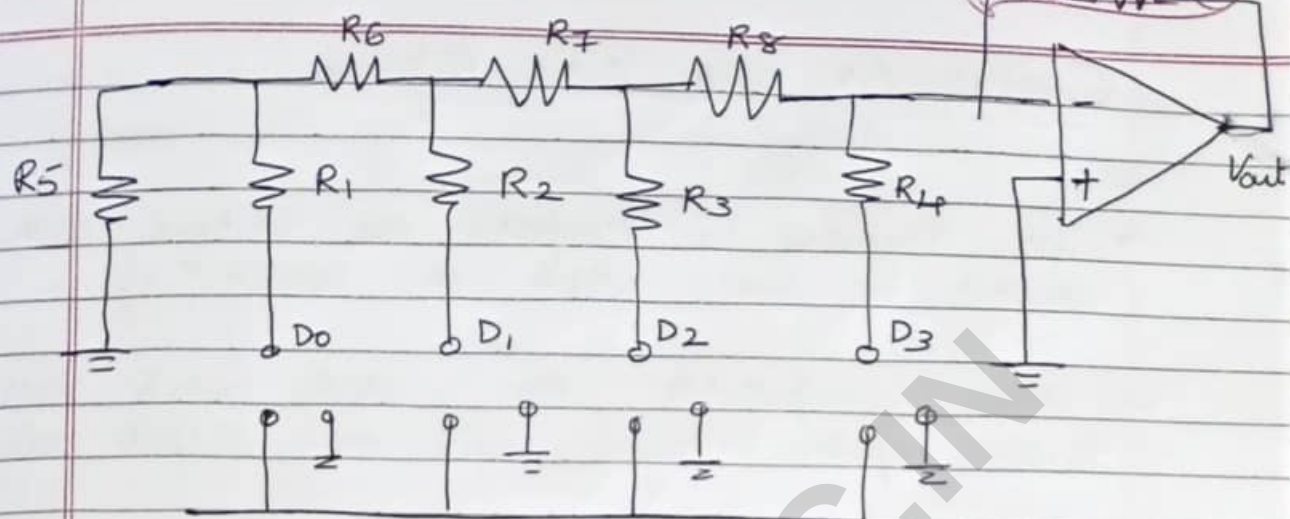
D_0 - LSB & D_3 - MSB.

To find the D/A converter's output voltage first change the binary input value to its decimal equivalent value, BIN.

$$BIN = (D_0 \times 2^0) + (D_1 \times 2^1) + (D_2 \times 2^2) + (D_3 \times 2^3)$$

$$\therefore V_{out} = - \left(\frac{BIN}{2^N} \times 2V_{ref} \right)$$

N - no of inputs.



$V_{ref} = -5V$

Problem

$D_0 = 1, D_1 = 0, D_2 = 0$ & $D_3 = 1$ using $V_{ref} = +5V$.
determine the decimal equivalent of the binary input (BIN) & the output voltage of the converter

$$\begin{aligned} \text{BIN} &= (D_0 \times 2^0) + (D_1 \times 2^1) + (D_2 \times 2^2) + (D_3 \times 2^3) \\ &= (1 \times 2^0) + (0 \times 2^1) + (0 \times 2^2) + (1 \times 2^3) \\ &= (1 + 8) = 9 \end{aligned}$$

$$V_{out} = - \frac{\text{BIN}}{2^N} \times 2 \times V_{ref} = - \frac{9}{2^4} \times 2 \times 5$$

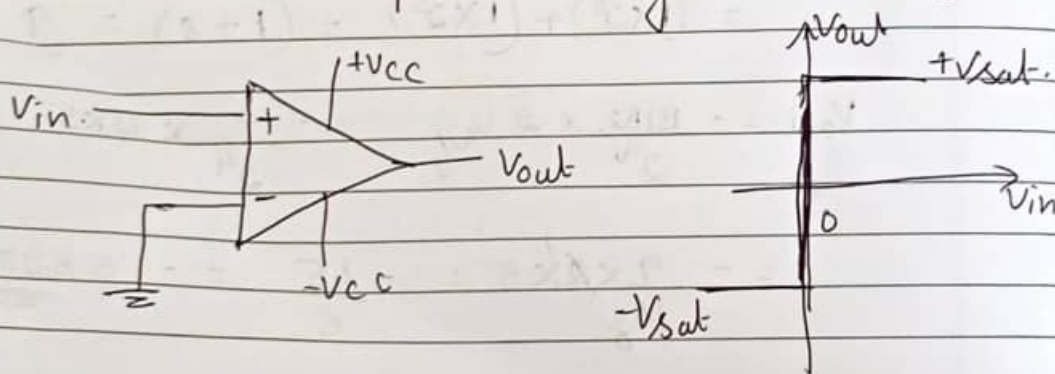
$$= - \frac{9 \times 2 \times 5}{16} = - \frac{45}{8} = - 5.625V$$

Comparators with Zero Reference

- We require to compare one voltage with another to see which is larger.
- Hence comparators are used, which has two input voltages and one output voltage.

* Basic Idea:

- The simplest way to build a comparator is to connect an op-amp without feedback resistors.
- Because of high open-loop voltage gain a positive input voltage produces positive saturation and a negative input voltage produces negative saturation.
- The comparator is called zero-crossing detector because the output voltage ideally switches from low to high or vice versa whenever the input voltage crosses zero.



The minimum input voltage that produces saturation is $V_{in(min)} = \frac{\pm V_{sat}}{A_{vol}}$

if $V_{sat} = \pm 14V$, $A_{vol} = 100,000$.

$$V_{in(min)} = \frac{\pm 14V}{100,000} = \pm 0.14mV$$

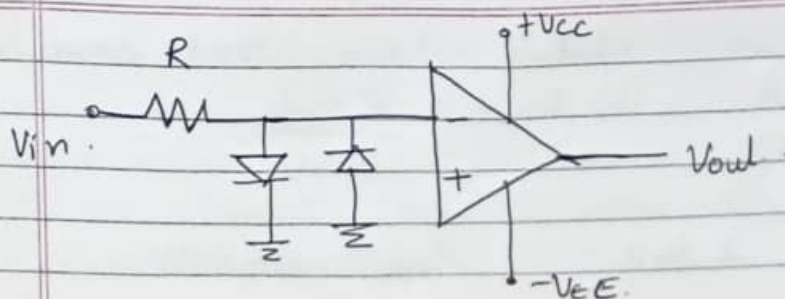
i.e. if an input voltage is more positive than $+0.014mV$ drives the comparator into positive saturation and an input voltage more negative than $-0.014mV$ drives it into negative saturation.

→ A Lissajous pattern appears on the oscilloscope when harmonically related signals are applied to the horizontal and vertical inputs.

* Inverting Comparator

→ The inverting input forms the input of the comparator. The noninverting input is grounded.

→ Here, a slightly positive input voltage produces a maximum negative output & a slightly negative input voltage produces a maximum positive output.

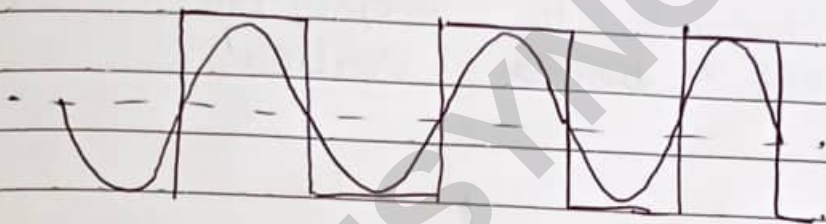


- two diode clamps protecting the comparators against excessively large input voltages.
- Some comparators, have maximum input voltage rating as little as $\pm 5V$, whereas with others, it may be more than $\pm 30V$.
- In any case, the comparators can be protected against destructively large input voltages by using the diode clamps.
- These diodes have no effect on the operation of the circuit as long as the magnitude of the input voltage is less than $0.7V$.
- when the magnitude of the input voltage is greater than $0.7V$, one of the diodes will turn on and clamp the magnitude of the inverting input voltage to approximately $0.7V$.

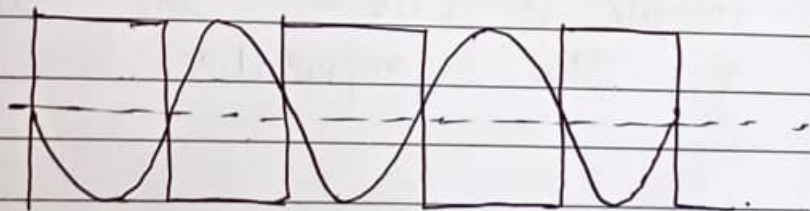
Converting Sine waves to Square waves

→ The trip point / threshold / reference of a comparator is the input voltage that causes the output voltage to switch states

→ For a noninverting comparator, with sine wave input and threshold of $0V$, the output will be square wave.



→ For an inverting comparator, with sine wave input and threshold of $0V$, the output will be 180° out of phase.



Linear Region

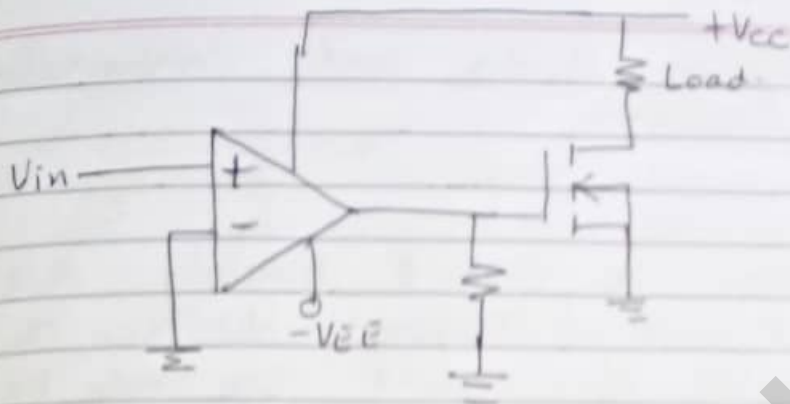
- If the comparator had an infinite open-loop gain, the transition between negative and positive saturation would be vertical.
- It appears to be vertical, as the sensitivity of the 2 channel is 10mV/DIV.
- When the sensitivity of the channel is changed to 200mV/DIV, the transition is not ~~linear~~ vertical.



- The narrow input region between -100 to $+100\text{mV}$ is called the linear region of the comparator.

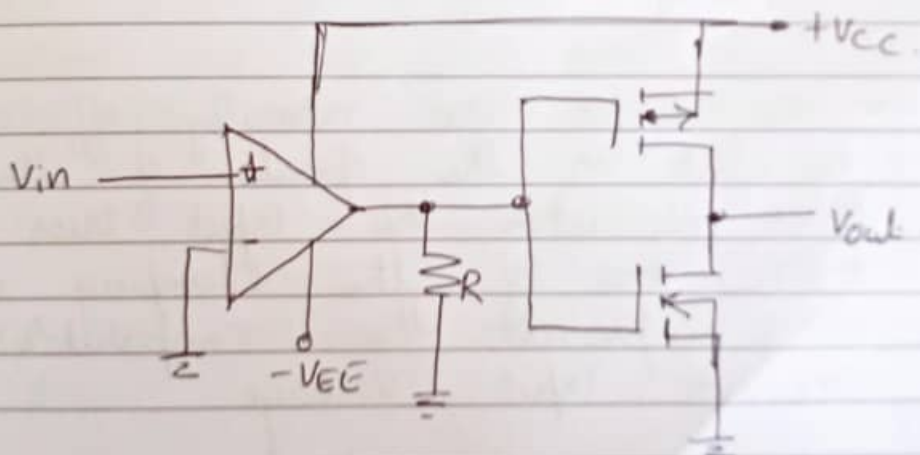
Interfacing Analog and Digital Circuits

- Comparators interface their outputs with digital circuits such as CMOS, EMOs / TTL



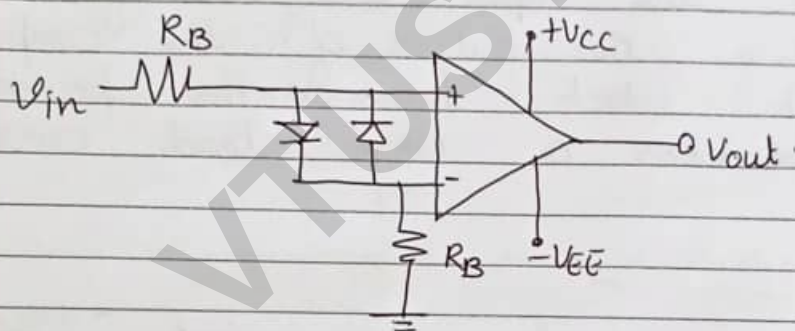
→ The above figure shows a zero-crossing detector interface with an CMOS circuit.

→ Whenever the input voltage is greater than zero, the output of the comparator is high, which turns on the power FET and produces a large load current.



Clamping diodes and compensating resistors

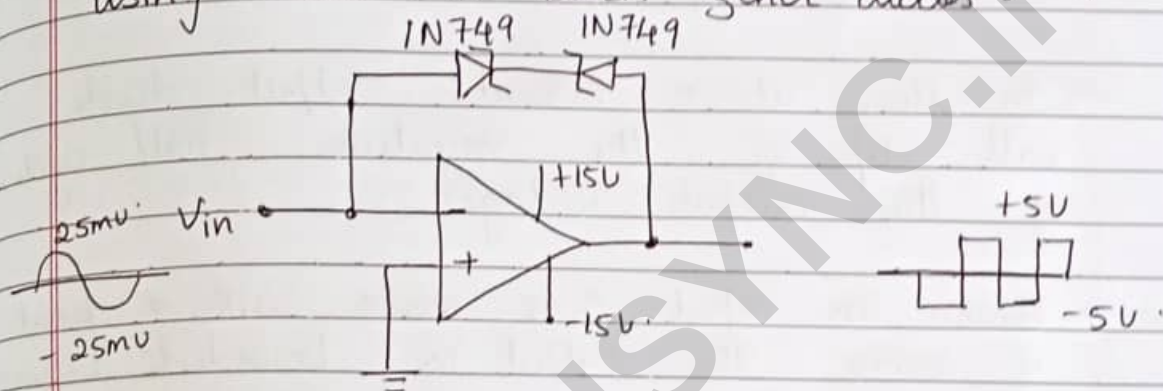
- When a current-limiting resistor is used with clamping diodes, a compensating resistor of equal size may be used on the other input of the comparator.
- The circuit has a compensating resistor to eliminate the effect of input bias current.



- The diodes are normally off and have no effect on the operation of the circuit. It is only when the input tries to exceed $\pm 0.7V$, one of the clamping diodes turns on & protects the comparator against excessive input voltage.

Bounded output

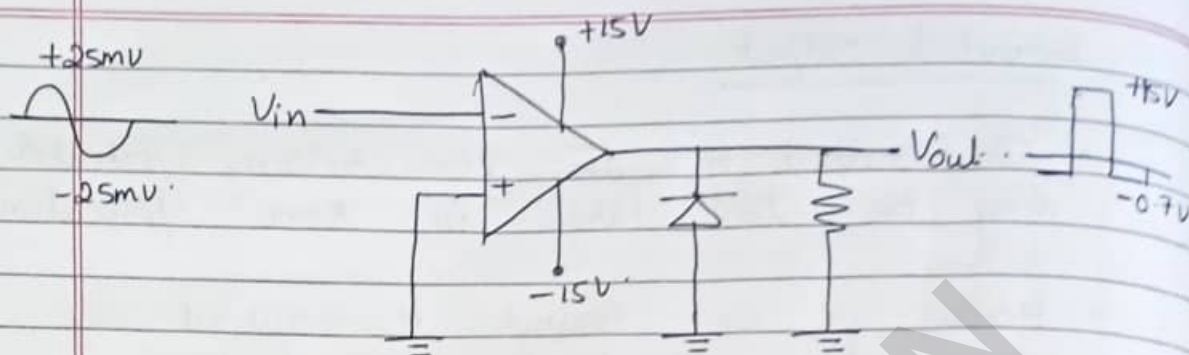
- The output of the zero crossing detector may be too large in some applications.
- Hence we can bound the output by using back-to-back zener diodes.



- In this circuit, the inverting comparator has a bounded output because one of the diodes will be conducting in the forward direction and the other will be operating in the breakdown region.

Eg: $1\text{N}749$ has a zener voltage 4.3V .

hence the voltage across the diodes will be $\pm 5\text{V}$.



- In the above circuit, output diode will clip off the negative half cycles of the output voltage.
- Given an input sine wave with a peak of 25mV, the output is bounded between -0.7V to +15V.

3.4:- Comparators with Nonzero references

- In some applications, threshold voltage different from zero may be preferred.
- The voltage divider produces the following reference voltage at the inverting input.

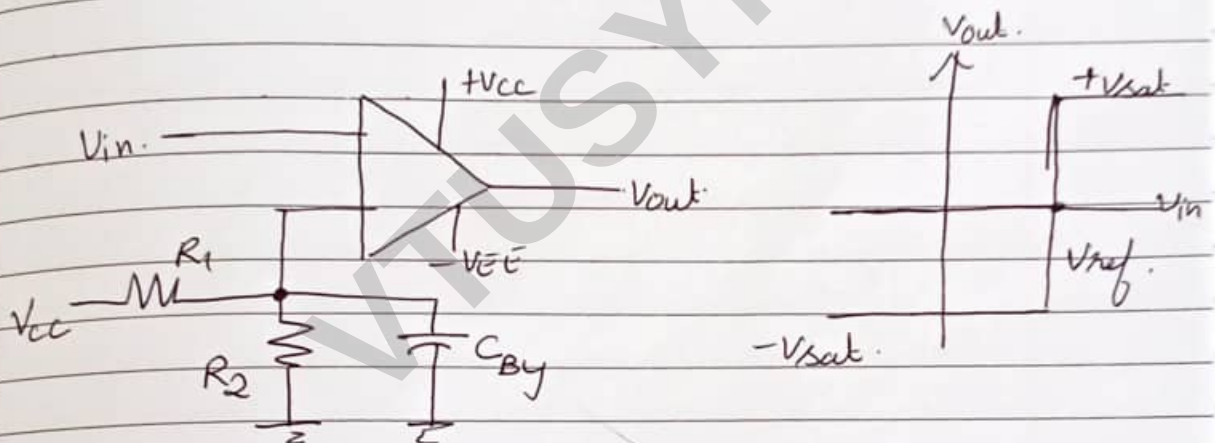
$$V_{ref} = \frac{R_2}{R_1 + R_2} \cdot V_{cc}$$

When V_{in} is greater than V_{ref} , the i/p voltage is positive.

4 hence the output voltage is high.

→ when V_{in} is less than V_{ref} , the i/p is negative and hence the output voltage is low.

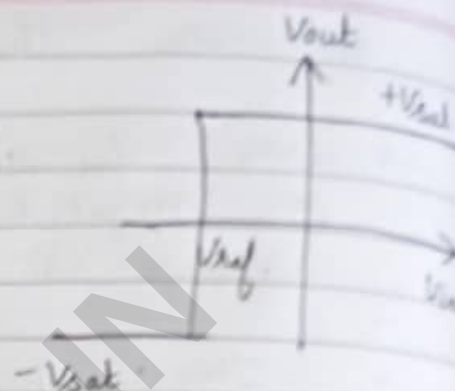
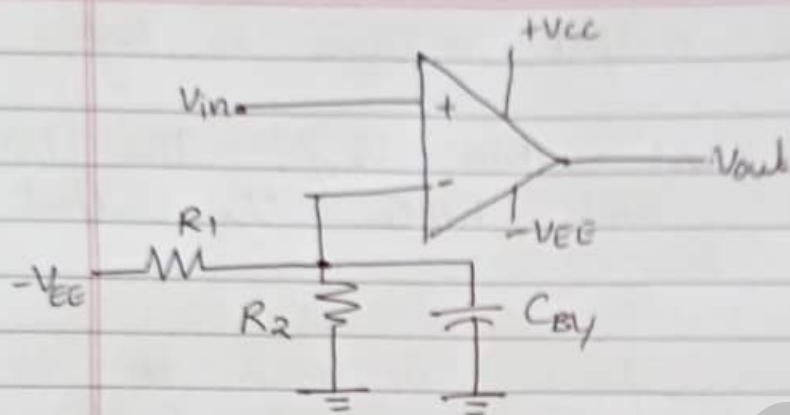
→ The bypass capacitor is used to reduce the amount of power-supply ripple and other noise appearing at the inverting input.



Transfer char

$$f_c = \frac{1}{2\pi(R_1 \parallel R_2)C_{By}}$$

From the transfer characteristics, when V_{in} is greater than V_{ref} , the output of the comparator goes into positive saturation. also when V_{in} is less than V_{ref} , the output goes into negative saturation.



Module - 4

Module - 4

①

Negative feedback

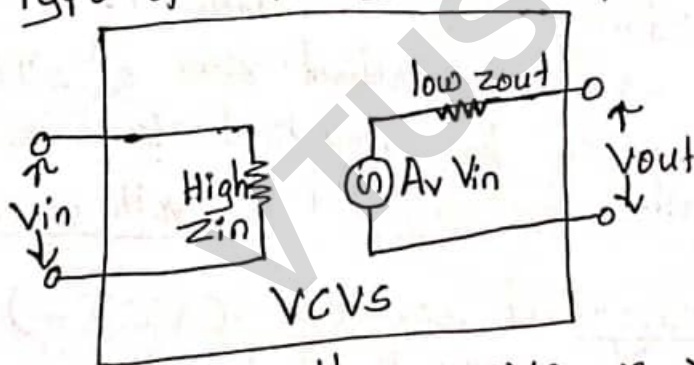
- Stabilizes the voltage gain.
- Increases input impedance.
- Decreases output " " .

Four Types of Negative feedback

1. Voltage - controlled voltage source (VCVS)
2. Current - " " " (ICVS)
3. Voltage - " " " (VCIS)
4. Current - " " " (CICIS)

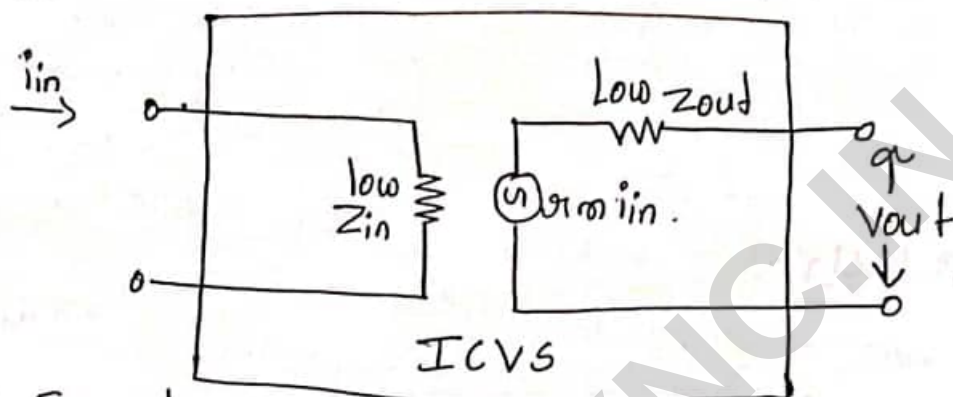
1. Voltage controlled voltage source (VCVS)

- The circuit with negative feedback that accepts voltage input & produces voltage output, is called voltage controlled voltage source.
- Type of voltage amplifier.



- Fig shows the VCVS, a voltage amplifier.
- It is an ideal voltage amplifier because it has ideal characteristics like stabilized voltage gain, infinite input impedance & zero output impedance.
- Practical circuit has high- Z_{in} , low- Z_{out} .
- A_v is the voltage gain.
- As Z_{out} approaches zero, output side of a VCVS is a stiff voltage source to any practical load resistance.

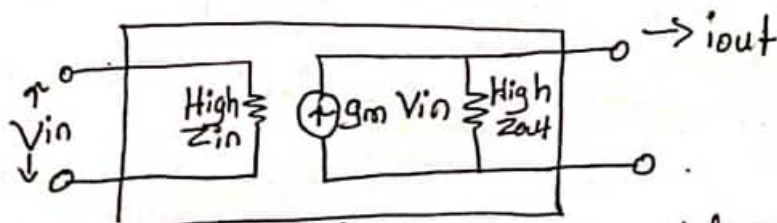
2. Current controlled voltage source (ICVS)
- Type of the -ve feedback circuit that accepts current as a input & gives voltage as a output is called current controlled voltage source.
 - Input current controls an output voltage.
 - Also called as transresistance amplifier.
 - Because V_{out}/I_{in} has unit ohms & trans means taking ratio of an output quantity to an input quantity.
 - Type of transresistance amplifier.



- Fig shows ICVS.
- It has $Z_{in} = \text{low}$ $Z_{out} = \text{low}$.
- g_m is the transresistance, unit is ohm(Ω).
- As Z_{out} approaches zero, the output side of a ICVS is a stiff voltage source for practical load resistances.
- This circuit is also called as current to voltage converter.

3. Voltage controlled current source (VCCS)

- Type of the -ve feedback circuit that accepts voltage as a input & gives current as an output is called voltage controlled current source.
- It is also called as voltage to current converter.
- " " " " " transconductance amplifier.
- For trans → ratio, conductance → (I_{out}/V_{in}) mhos or Siemens.

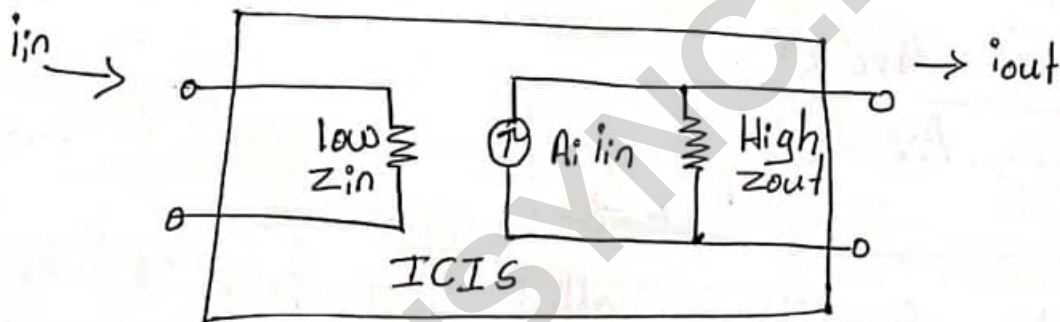


- Type of transconductance amplifier.

- Fig shows a VCIS.
- It has $Z_{in} = \text{high}$, $Z_{out} = \text{high}$
- g_m represents transconductance.
- As Z_{out} approaches infinity, the output side of a VCIS is a stiff current source for load circuit.

4. Current-controlled current source (CCIS)

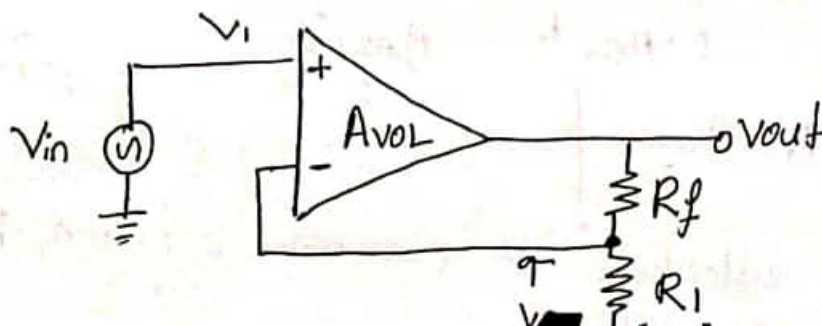
- The type of a negative feedback that accepts current as a input & gives current as a output is called current-controlled current source.
- It is a type of current amplifier.
- It is a ideal current amplifier with stabilized current gain, zero input impedance & infinite output impedance.



- Fig shows an ICIS.
- It has $Z_{in} = \text{low}$, $Z_{out} = \text{high}$.
- A_i is the current gain.
- Z_{out} approaches infinity, the output side of a VCIS is a stiff current source to any practical load resistance.

VCVS voltage gain

- non inverting amplifier circuit act as VCVS, therefore consider this to derive voltage gain.



Exact closed loop voltage gain

- A_{VOL} is the open loop voltage gain of the amplifier.
- V_{out} is feed back to the non-inverting input using voltage divider network called feedback network. It has a feedback gain or feedback fraction B .

$$\therefore B = \frac{\text{output}}{\text{input}} = \frac{V_2}{V_{out}}$$

- B is also called as feedback attenuation factor.
- Closed loop voltage gain is given by,

$$A_{V(CCL)} = \frac{A_{VOL}}{1 + A_{VOL} B}$$

$$A_V = A_{V(CCL)}$$

$$\therefore A_V = \frac{A_{VOL}}{1 + A_{VOL} B}$$

Loop gain - $A_{VOL} B$ is called loop gain. It is the voltage gain of the forward and feedback paths.

- Larger loop gain stabilizes the loop gain and produces enhanced (better) gain stability, distortion, offsets, input impedance & output impedance.

Ideal closed loop voltage gain

- For the better performance of the MCVS it is good to keep loop gain $\gg 1$ i.e. $A_{VOL} B \gg 1$

$$\therefore A_V = \frac{A_{VOL}}{1 + A_{VOL} B} \approx \frac{A_{VOL}}{A_{VOL} B}$$

$$A_V \approx \frac{1}{B}$$

- Formula to calculate % error between ideal & exact voltage gain.

$$\% \text{ error} = \frac{100\%}{1 + A_{VOL} \cdot B}$$

Using Ideal Equation - we can derive B & A_v in terms of resistance.

→ according to voltage divider rule.

$$V_2 = \frac{R_f}{R_f + R_1} \cdot V_{out}$$

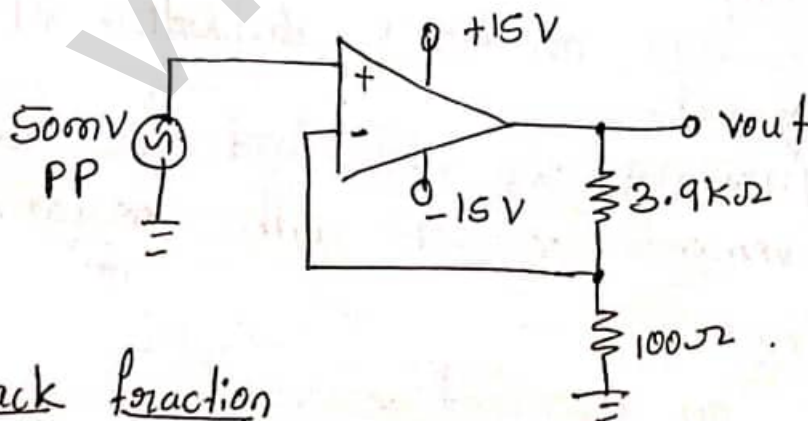
$$\frac{V_2}{V_{out}} = B = \frac{R_1}{R_f + R_1}$$

$$A_v \equiv \frac{1}{B} \quad \therefore A_v = \frac{R_f + R_1}{R_1}$$

$$A_v = \frac{R_f}{R_1} + 1$$

which is same as voltage gain of non-inverting amplifier.

Example - Calculate the feedback fraction, the ideal closed loop voltage gain, the percent error and the exact closed-loop voltage gain. Use a typical A_{VOL} of 100,000 for the 741C.



Feedback fraction

$$B = \frac{R_1}{R_1 + R_f} = \frac{100\Omega}{100\Omega + 3.9k\Omega} = 0.025$$

Ideal closed loop gain

$$A_v = \frac{1}{B} = \frac{1}{0.025} = 40$$

The % error

$$\% \text{ error} = \frac{100\%}{1 + A_{VOL} B} = \frac{100\%}{1 + (100,000)(0.025)} = 0.04\%$$

exact closed loop gain (A_V)

$$A_V = A_V - (\% \text{ error}) \cdot A_V$$

$$= 40 - (0.04\%) \cdot 40$$

$$A_V = 39.984$$

OR

$$A_V = \frac{A_{VOL}}{1 + A_{VOL} B} = \frac{100,000}{1 + (100,000)(0.025)} = 39.984$$

Practice - change the feedback resistor from $2.9k\Omega$ to $4.9k\Omega$. calculate the feedback fraction, the ideal-closed loop voltage gain, the % error & the exact closed loop gain.

Other VCVS Equations

- Negative feedback in a VCVS amplifier increases the input impedance, decreases the output impedance and reduce any nonlinear distortion of the amplified signal.
- We with following we will just find out how much improvements occurs with negative feedback.

Gain stability

- It depends on percent error
- The smaller the percent error, the better the stability.
- The worst-case error of closed loop voltage gain occurs when the open-loop voltage gain is minimum.

$$\% \text{ Maximum error} = \frac{100\%}{500}$$

$$1 + A_{VOL}(\min) B$$

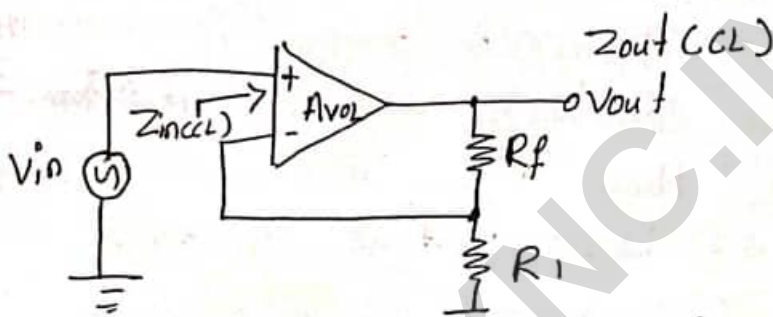
According to data sheet of 741 IC

$$A_{VOL}(\min) = 20,000$$

$$\therefore \% \text{ maximum error} = \frac{100\%}{500} = 0.2\%$$

→ In mass production the closed-loop voltage gain of any VCVS amplifier is within 0.2% of ideal value.

Closed-loop input impedance



→ Exact equation for the closed-loop input impedance of the VCVS amplifier.

$$Z_{in}(CCL) = (1 + A_{VOL} B) R_{in} \parallel R_{cm}$$

R_{in} = the open loop input resistance of the op-amp.

R_{cm} = the common-mode input resistance of the op-amp.

From the data sheet $R_{in} = 2 \text{ M}\Omega$

" " " " $R_{cm} = R_E = 100 \text{ M}\Omega$

→ often R_{cm} is ignored because it is large.

$$\therefore Z_{in}(CCL) \approx (1 + A_{VOL} B) R_{in}$$

→ In practical VCVS amplifier $(1 + A_{VOL} B) \gg 1$.
 $Z_{in}(CCL)$ will be extremely high.

→ ultimate limit on $Z_{in}(CCL)$ is

$$Z_{in}(CCL) = R_{cm}$$

→ $Z_{in}(CCL)$ is larger than R_{in} but less than ultimate limit on R_{cm} .

Closed-loop output impedance

→ Overall output impedance looking back into the VCVS amplifier is called closed loop output impedance. ($Z_{out}(CL)$)

→ Equation for $Z_{out}(CL) = \frac{R_{out}}{1 + A_{VOL} B}$

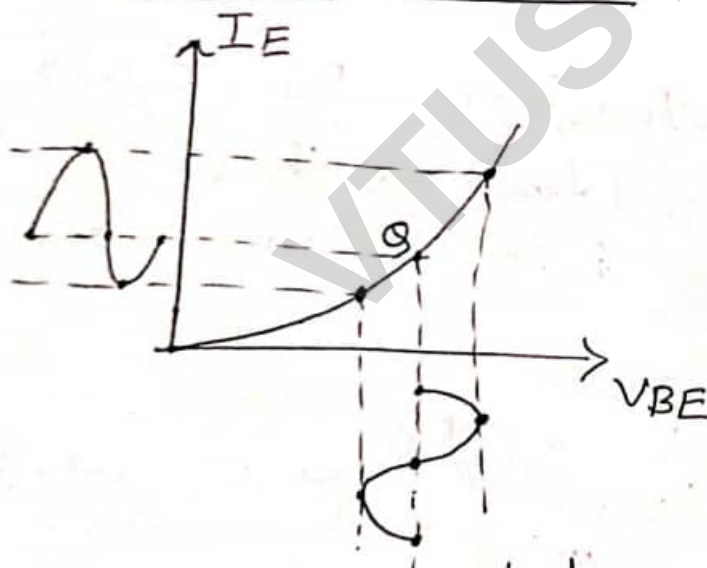
→ R_{out} → Open-loop output resistance of the opamp
(as per data sheet of 741C $R_{out} = 75 \Omega$)

→ As $(1 + A_{VOL} B) \gg 1$ in practical VCVS amplifiers $Z_{out}(CL)$ is less than 1Ω .

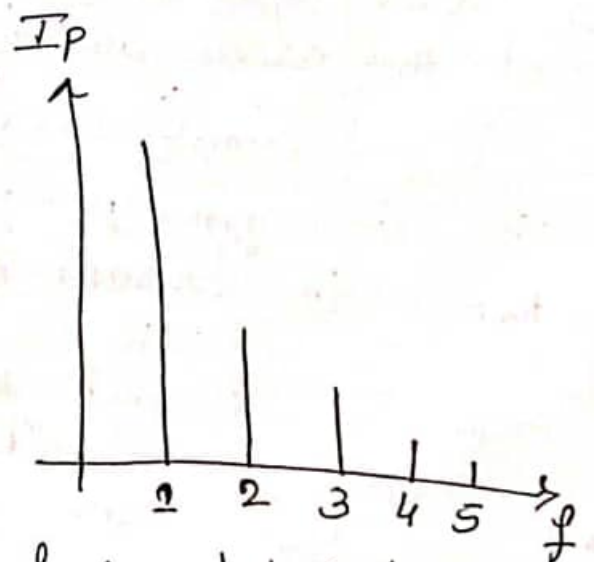
→ $Z_{out}(CL)$ may approach zero in a voltage follower.

→ VCVS negative feedback reduces it to values much smaller than 1Ω . Therefore $Z_{out}(CL)$ approaches an ideal voltage source.

Nonlinear Distortion



non-linear distortion



fundamental & harmonics

→ In fig nonlinear graph of the base-emitter diode distorts a large signal by elongating the positive half cycle & compressing the -ve half cycle.

→ If 1 kHz is the fundamental frequency then 2, 3, 4, 5 kHz are the harmonics.

→ RMS values of all the harmonics measured together

⑨
→ $\therefore$ Nonlinear distortion is called as harmonic distortion.

→ Harmonic distortion can be measured by the instrument called a distortion analyzer.

→ Total Harmonic Distortion (THD) is given by the formula
$$THD = \frac{\text{Total harmonic voltage}}{\text{Fundamental voltage}} \times 100\%$$

Ex - If $THV = 0.1V_{rms}$ & $FV = 1V$ then $THD = 10\%$

→ Negative feedback reduces harmonic distortion.
equation for closed loop harmonic distortion is

$$THD_{CL} = \frac{THD_{OL}}{1 + A_{VOL} B}$$

where $THD_{OL} \rightarrow$ open loop harmonic distortion

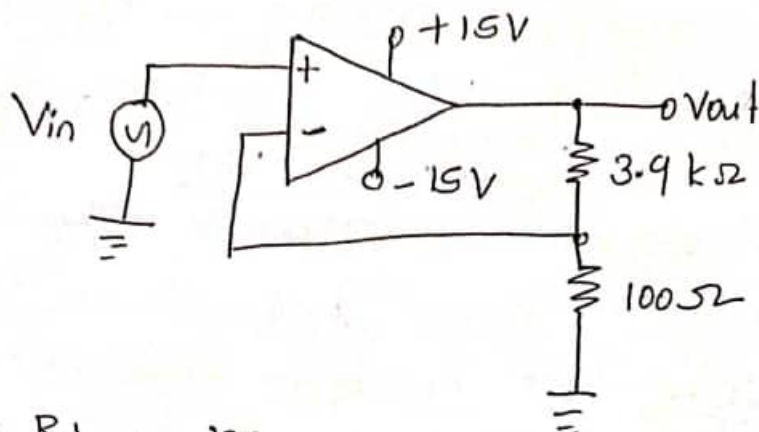
$THD_{CL} \rightarrow$ closed-loop " "

→ As $1 + A_{VOL} B \gg 1$, so it reduces the harmonic distortion to negligible level.

→ In stereo amplifiers, we hear high-fidelity (HF) music instead of distorted sounds.

Discrete Negative feedback amplifier

Example - In fig the 741C has an R_{in} of $2M\Omega$ & R_{cm} of $200M\Omega$. what is the closed-loop input impedance? Use $A_{VOL} = 1,00,000$ for 741C.



$$B = \frac{R_1}{R_1 + R_f} = \frac{100}{100 + 3.9k} = 0.025$$

$$\therefore 1 + A_{VOL} B = 1 + 1,00,000 \times 0.025 = 2500$$

$$Z_{in(CCL)} \approx (1 + A_{VOL} B) \cdot R_{in} = 2500 \times 2 \times 10^6$$

$$Z_{in(CCL)} \approx 5000 M\Omega$$

When $Z_{in(CCL)} > 100 M\Omega$ use the formula

$$Z_{in(CCL)} = \left(\frac{1}{1 + A_{VOL} B} \right) R_{in} \parallel R_{cm}$$

$$= (5000 \parallel 200 M\Omega)$$

$$Z_{in(CCL)} = 192 M\Omega$$

high $Z_{in(CCL)}$ represents that VCVS approaches an ideal voltage amplifier.

* In fig change the $3.9 k\Omega$ resistor to $4.9 k\Omega$ & solve for $Z_{in(CCL)}$

Example - with values $A_{VOL} = 1,00,000$, $R_f = 3.9k$, $R_1 = 100\Omega$, $R_{out} = 75\Omega$ calculate the closed loop output impedance

$$B = \frac{R_1}{R_1 + R_f} = \frac{100}{100 + 3.9k} = 0.025$$

$$1 + A_{VOL} B = 1 + 1,00,000 \cdot 0.025 = 2500$$

$$Z_{out(CCL)} = \frac{R_{out}}{(1 + A_{VOL} B)} = \frac{75}{2500} = 0.03\Omega$$

Low output impedance means that a VCVS approaches an ideal voltage amplifier.

Exple - Suppose the amplifier has an open-loop total harmonic distortion of 7.5 percent. What is the closed loop total harmonic distortion?

$$THD_{CL} = \frac{THD_{OL}}{1 + A_{VOL} B} = \frac{7.5\%}{2500} = 0.003\%$$

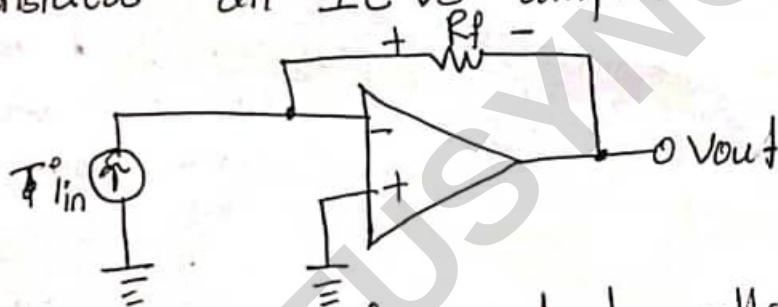
* Repeat the same problem with 3.9k Ω resistor changed to 4.9k Ω .

* The ICVS amplifier

→ It is an almost perfect current-to-voltage converter because it has zero input impedance & zero output impedance.

Output voltage

→ Consider an ICVS amplifier.



→ exact equation for output voltage is

$$v_{out} = -\left(i_{in} R_f \cdot \frac{A_{VOL}}{1 + A_{VOL}}\right)$$

as $A_{VOL} \gg 1$, $v_{out} = -i_{in} R_f \frac{A_{VOL}}{A_{VOL}}$

$$\boxed{v_{out} = -i_{in} R_f}$$

Non-inverting Input & output Impedances

→ The exact equations for closed-loop input & output impedance are

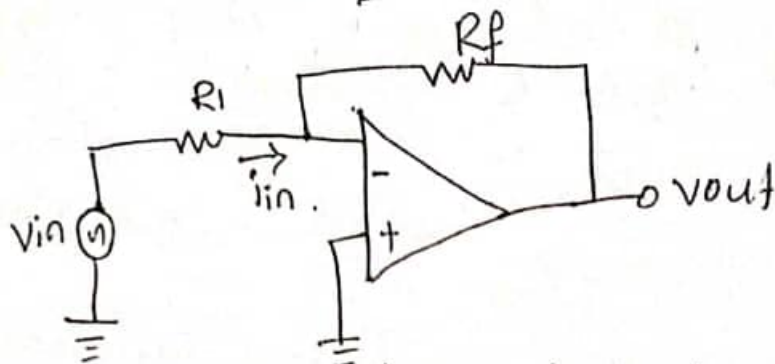
$$\boxed{Z_{in(CL)} = \frac{R_f}{1 + A_{VOL}}}$$

$$\boxed{Z_{out(CL)} = \frac{R_{out}}{1 + A_{VOL}}}$$

RK

→ Large denominator will reduce the impedance to a very low value.

The inverting amplifier

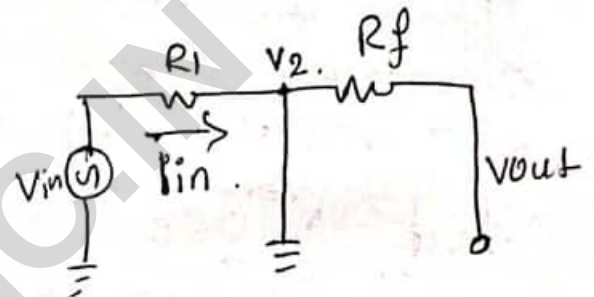


→ inverting amplifier has closed loop voltage gain

$$A_v = -\frac{R_f}{R_1}$$

→ It uses ICVS negative feedback. Because of the virtual ground on the inverting input, the input current equals

$$i_{in} = \frac{-V_{in}}{R_1}$$

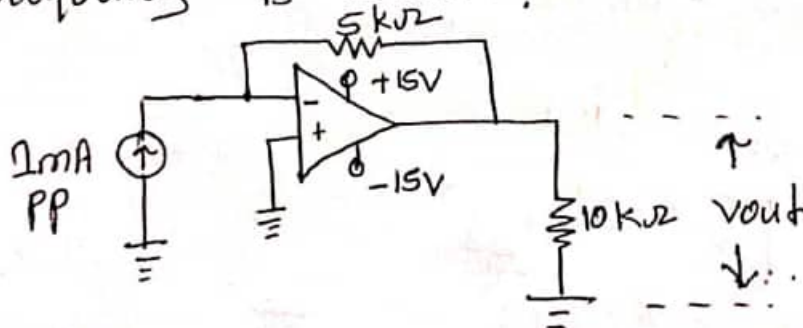


$$\frac{V_2 - V_{in}}{R_1} = i_{in} \quad \text{and} \quad V_2 = 0$$

$$\frac{-V_{in}}{R_1} = i_{in}$$

$$i_{in} = \frac{-V_{in}}{R_1} //$$

Example - What is the output voltage if the input frequency is 1 kHz?



$$\begin{aligned} V_{out} &= -i_{in} R_f \\ &= -1\text{mA} (5\text{k}\Omega) \\ &= -5\text{V}_{PP} \end{aligned}$$

* practice the same with $R_f = 2\text{k}\Omega$.

Example - For the above circuit what are the closed-loop input and output impedance?
Use 741C parameters.

$$R_f = 5k\Omega \quad R_{out} = 75\Omega$$

$$A_{VOL} = 1,00,000$$

$$Z_{in(CCL)} = \frac{R_f}{1 + A_{VOL}} = \frac{5k\Omega}{1 + 1,00,000} \approx \frac{5k\Omega}{1,00,000} = 0.05\Omega$$

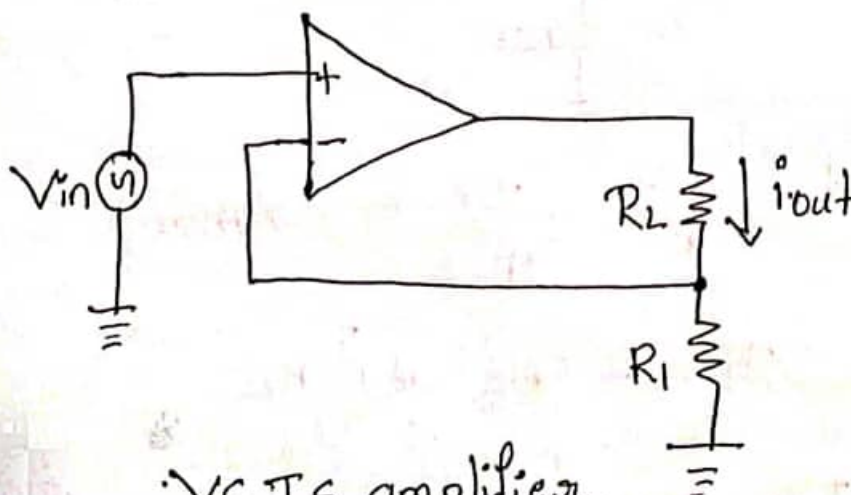
$$Z_{out(CCL)} = \frac{R_{out}}{1 + A_{VOL}} = \frac{75}{1 + 1,00,000} \approx \frac{75}{1,00,000} = 0.00075\Omega$$

* Repeat the example with $A_{VOL} = 200,000$.

The VCIS amplifier

- It converts an input voltage to a precise value of output current.
- Fig shows VCIS amplifier. It is same as VCVS except R_L & feedback resistor R_1 .
- Output is the current through R_L .
- Output current is stabilized, exact equation for i_{out} is

$$i_{out} = \frac{V_{in}}{R_1 + (R_1 + R_L)/A_{VOL}}$$



VCIS amplifier.

→ In a practical circuit, $(R_1 + R_2) / A_{VOL} \ll R_1$

$$\therefore \boxed{i_{out} = \frac{V_{in}}{R_1}}$$

also written as $\boxed{i_{out} = g_m V_{in}}$ where $g_m = \frac{1}{R_1}$

→ As the output side of a VCIS is same as input side of VCVS amplifier the closed loop input impedance is given by

$$\boxed{Z_{in(CCL)} = (1 + A_{VOL} B) R_{in}}$$

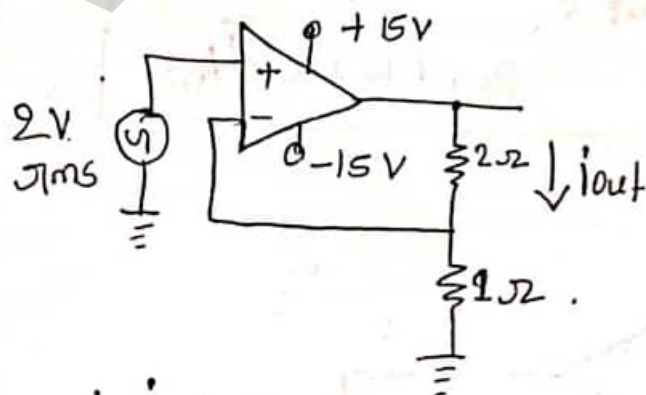
→ output impedance is given by

$$\boxed{Z_{out(CCL)} = (1 + A_{VOL}) R_1}$$

→ The circuit is an almost perfect voltage to current converter because it has very high $Z_{in(CCL)}$ & $Z_{out(CCL)}$.

Example - what is the load current in fig? The load power? what happens if the load resistance changes to 4Ω ?

Load



$$\text{load current } i_{out} = \frac{V_{in}}{R_1} = \frac{2V_{7ms}}{2\Omega} = 2A_{7ms}$$

$$\begin{aligned} \text{Load Power } P_L &= \frac{1}{2} R_L (i_{out})^2 \cdot R_L \\ &= (2A_{7ms})^2 (2\Omega) \end{aligned}$$

→ If load resistance changes to 4Ω .

(15)

$$P_L = (i_{out})^2 R_L$$

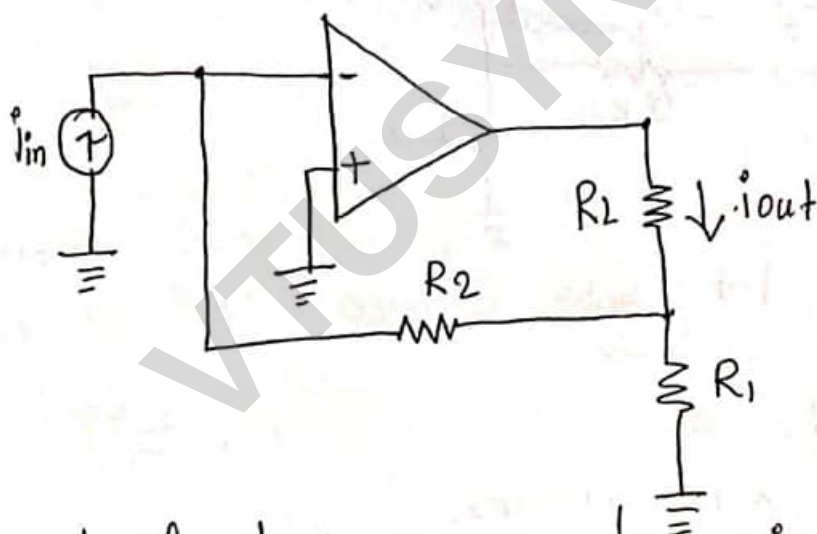
$$= (2)^2 4$$

$$P_L = 16W$$

* Practice the problem by changing the input voltage to $3V_{rms}$ & solve for i_{out} & P_L .

The ICIS amplifier.

- It acts as a perfect current amplifier.
- It has heavy negative feedback.
- It has low input impedance & a very high output impedance $Z_{out}(CL)$.
- Fig shows ICIS amplifier.



→ The closed-loop current gain is stabilized & given by

$$A_i = \frac{A_{VOL} (R_1 + R_2)}{R_L + A_{VOL} R_1}$$

→ usually $A_{VOL} R_1 \gg R_L$

$$\therefore A_i = \frac{A_{VOL} (R_1 + R_2)}{A_{VOL} R_1} = \frac{R_1 + R_2}{R_1} = 1 + \frac{R_2}{R_1}$$

$$A_i = 1 + \frac{R_2}{R_1}$$

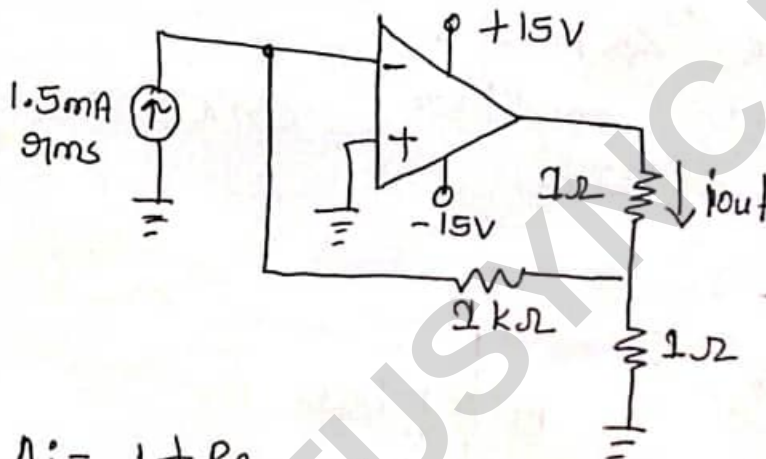
$$Z_{in(CCL)} = \frac{R_2}{1 + A_{VOL} B}$$

feedback fraction B $B = \frac{R_1}{R_1 + R_2}$

$$Z_{out(CCL)} = (1 + A_{VOL}) R_1$$

→ Large A_{VOL} produces small Z_{in} & large Z_{out} so it acts as a perfect current amplifier.

Example - What is the load current in fig? the load power? If the load resistance is changed to 2Ω . What are the load current & power?



$$A_i = 1 + \frac{R_2}{R_1} = 1 + \frac{2k\Omega}{2\Omega} = 1000$$

$$i_{out} = A_i \cdot i_{in}$$

$$= 1000 \times 1.5 \text{ mA rms}$$

$$i_{out} = 1.5 \text{ A rms}$$

Load Power $P_L = (i_{out})^2 R_L$

$$P_L = (1.5 \text{ A})^2 (1\Omega)$$

$$P_L = 2.25 \text{ W}$$

For $R_L = 2\Omega$.

$$i_{out} = 1.5 \text{ A}$$

$$P_L = (i_{out})^2 R_L$$

$$= (1.5 \text{ A})^2 2\Omega$$

$$P_L = 4.5 \text{ W}$$

Practice - Using above fig
change i_{in} to 2mA
calculate i_{out} & P_L .

Active Filters

(17)

→ A filter is an electronic circuit that passes one band of frequencies while rejecting others.

→ 2 types - Active filter, Passive filter.

→ Passive filter.

→ Built with R, L & C

→ Used above 1 MHz

→ Do not have power gain.

→ Difficult to tune

Active filters

→ Built with R, C & op-amp.

→ Useful below 1 MHz

→ Have power gain.

→ Easy to tune

Ideal Responses

There are five types of filters.

1. Low-pass filter

2. High-pass "

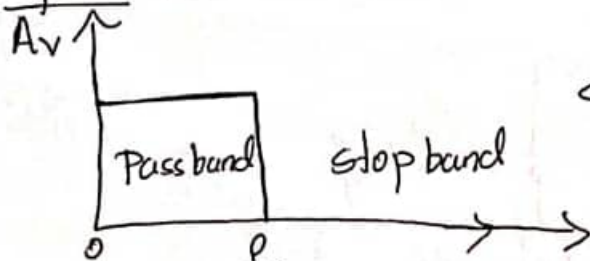
3. Band- " "

4. Bandstop " "

5. All pass " "

Note - Frequency response of a filter is the graph of its voltage gain versus frequency.

1. Low Pass Filter.



← ideal lowpass filter.

→ The filter that passes lower frequencies & stops higher frequency is called low pass filter.

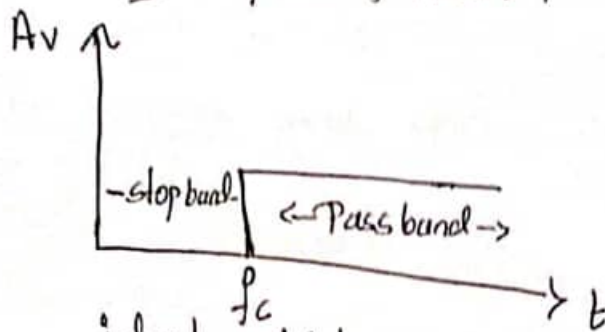
→ It passes all the frequencies from zero to the cut off frequency ' f_c ' called pass band.

→ Blocks all frequencies above f_c called stop band.

→ An ideal low-pass filter has zero attenuation (signal loss) in pass band. Infinite attenuation in the stopband & vertical attenuation center.

2. High-pass filter

→ A filter that passes the frequency above ' f_c ' and stops the signal frequency below ' f_c ' is called high-pass filter.

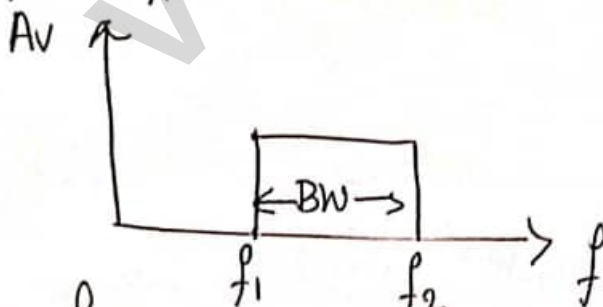


ideal high-pass response

- blocks all frequencies from zero to f_c called stop band
- Passes " " above f_c - called pass band
- ideal HPF has infinite attenuation in stop band & zero attenuation in pass band & vertical attenuation.

3. Band pass filter

A filter that passes all the frequencies between the lower & upper cutoff frequencies is called Band pass filter.



← ideal response of bandpass.

- Passes frequencies between f_1 & f_2 called pass band.
- Blocks frequencies below f_1 (lower cut off) & above f_2 (higher cut off) are called stop band.
- Band width = $f_2 - f_1$
- An ideal bandpass filter has zero attenuation in pass band, infinite attenuation in the stopband & two vertical transition.

2. $\rightarrow$ Center frequency $f_0 = \sqrt{f_1 f_2}$

(19)

$$Q = \frac{f_0}{BW}$$

If $Q > 10$ $f_0 \approx \frac{f_1 + f_2}{2}$

$\rightarrow$ If $Q < 1$ band pass filter is called wide band filter

$\rightarrow$ If $Q > 1$: " " " " " Narrow

Band stop Filter

$\rightarrow$ A filter that passes all frequencies from zero up to lower cutoff frequency & above upper cutoff frequency is called band stop filter.



Ideal response of BSF

$\rightarrow$ 0 to f_1 & above f_2 is the pass band

$\rightarrow$ f_1 to f_2 is the stop band

$\rightarrow$ Ideal band stop filter has infinite attenuation in the stop band, no attenuation in the pass band & two, vertical transition.

$$BW = f_2 - f_1$$

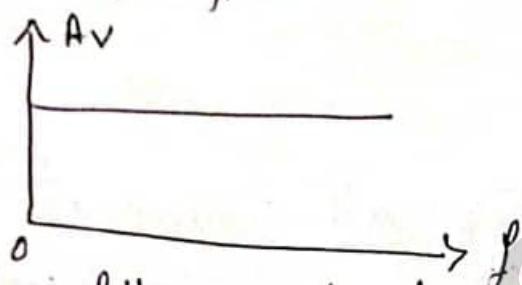
$$f_0 = \sqrt{f_1 f_2}$$

$$Q = \frac{f_0}{BW}$$

→ It is sometimes called as notch filter because it notches out (removes) all frequencies in the stopband.

All-Pass Filter

- A filter that passes all the frequency is called All pass filter.
- It has passband and no stopband.
- It passes all frequencies between zero & ∞ frequency.
- Since it has zero attenuation for all frequencies it is called as filter.



→ In all pass filter, each frequency can be shifted by a certain amount as it passes through the filter.

First Order Stages

- It is also called as one-pole ^{active} filter.
- Stages have only one capacitor it is called as 1st order.

Low-pass stages

→ Below are 3 ways to implement low pass stages.

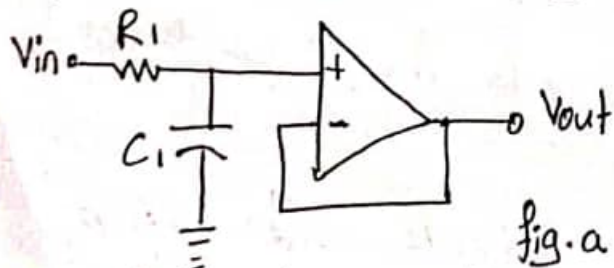


fig. a

Non inverting unity gain
1st order low pass filter

$$A_v = 1$$

$$f_c = \frac{1}{2\pi R_1 C_1} \quad (3\text{-dB cutoff frequency})$$

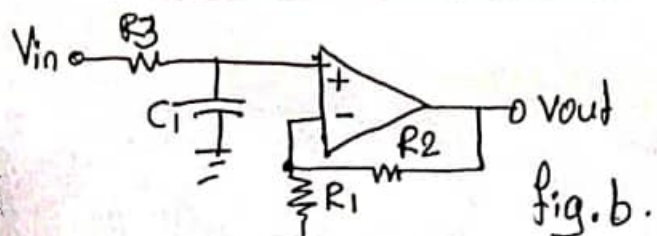
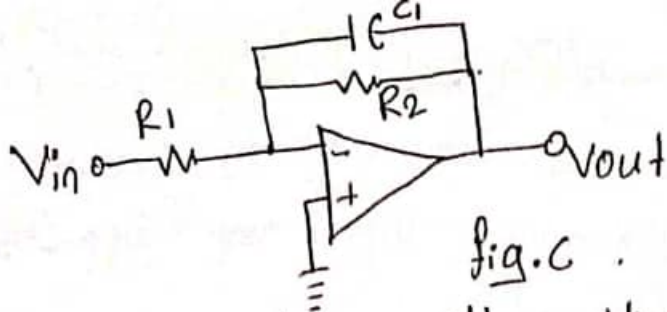


fig. b.

$$A_v = \frac{R_2}{R_1} + 1$$

$$f_c = \frac{1}{2\pi R_3 C_1}$$

Non-inverting with voltage gain 1st order LPF



$$A_v = \frac{-R_2}{R_1}$$

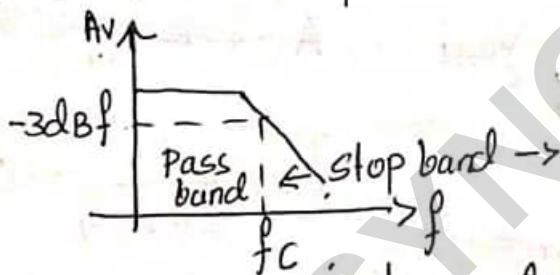
$$f_c = \frac{1}{2\pi R_2 C_1}$$

Non Inverting with voltage gain
1st order LPF.

→ In fig. a as the frequency increases above cut off frequency f_c , $X_C \downarrow$ decreases & reduces the non inverting input voltage.

→ so output is roll off after ' f_c '.

→ As frequency reaches ' ∞ ' capacitor becomes short & there is zero input voltage & zero output voltage.

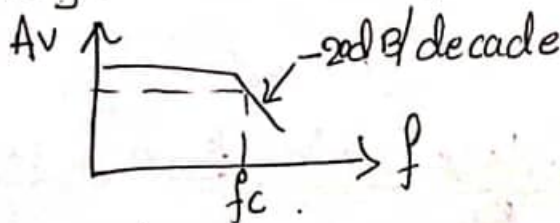


→ Fig. b shows non inverting first order lowpass filter.

→ It produces an voltage gain

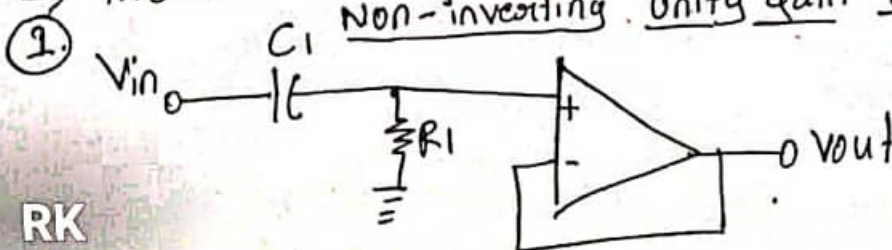
$$A_v = \frac{R_2}{R_1} + 1$$

→ $R_3 C_1$ lag circuit is outside the feedback loop, the output voltage rolls off at a rate of 20 dB per decade



High pass stage

There are 3 ways to implement high pass stages
① Non-inverting unity gain highpass stage $A_v = 1$ unity gain.

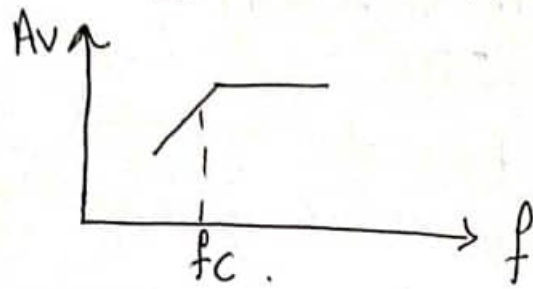


3dB cut off frequency

$$f_c = \frac{1}{2\pi R_1 C_1}$$

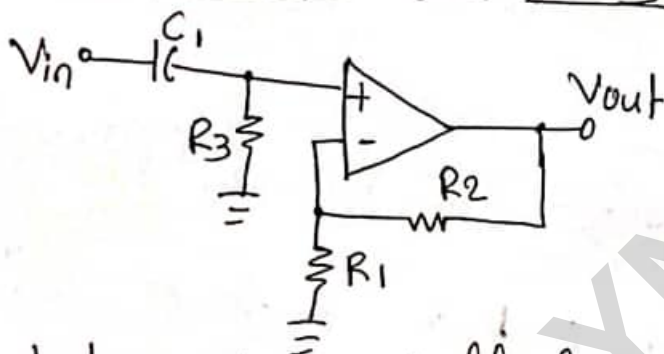
Noninverting unity gain

→ when frequency decreases below the cut off frequency. X_C - capacitive reactance increases. and reduces the non inverting input voltage. So output voltage rolls off. As the input →



→ As the frequency approaches zero, capacitor becomes open & there is zero input voltage.

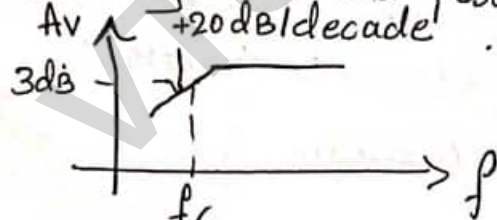
②. Non-inverting with voltage gain · high-pass stage



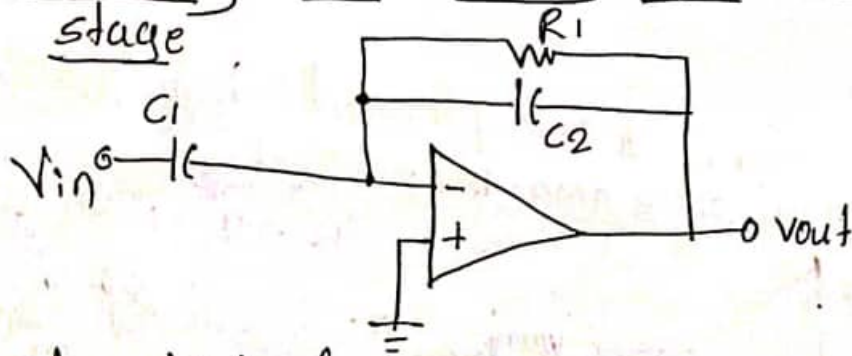
$$A_V = \frac{R_2}{R_1} + 1 \text{ Gain}$$

$$f_c = \frac{1}{2\pi R_3 C_1} \text{ 3dB cut off frequency.}$$

→ below the cut off frequency R_C circuit reduces the non inverting input voltage. So output voltage starts roll off increasing 20 dB per decade.



③. Inverting stage with voltage gain first order high pass



$$A_V = -\frac{C_1}{C_2} \text{ gain}$$

$$f_c = \frac{1}{2\pi R_1 C_2} \text{ 3dB cut off frequency}$$

→ At high frequency the circuit act as an inverting amplifier with a voltage gain of

RK

$$A_V = \frac{-X_{C2}}{X_{C1}} = -\frac{C_1}{C_2}$$

$$X_{C1} = \frac{1}{\omega C_1} = C_1 \frac{1}{\omega C_2} = C_2$$

→ As the frequency decreases. X_C $\uparrow$ & reduces the input signal & the feed back.

→ voltage gain also reduces.

→ As the frequency approaches zero, the capacitor becomes open and there is no input signal.

VCVS Unity-Gain Second-Order Low-Pass Filters (LPF)

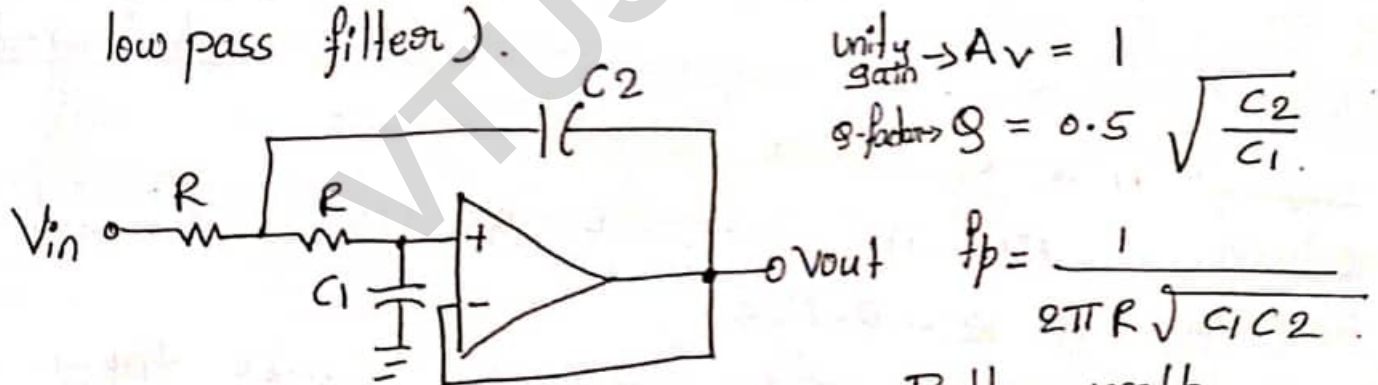
→ It is a second order or 2-pole filter.

→ called as Sallen-Key LPF (named after inventors).

→ called as VCVS filters because the op-amp is used as a voltage controlled voltage source.

→ VCVS LPF circuit can implement 3 basic approximation: Butterworth, Chebyshev, and Bessel.

Second-order VCVS stage for Butterworth and Bessel (Sallen-Key Second-order low pass filter).



Butterworth
 $Q = 0.707$
 $K_C = 1$

Bessel
 $Q = 0.577$
 $K_C = 0.786$

→ In the circuit, two resistors have same value. But the two capacitors have different value.

→ Lag circuit is connected to non-inverting terminal. feedback (output) is connected through capacitor C_2 .

- When frequency is low both capacitors appear as open circuit.
- Circuit is connected as a voltage follower $\therefore A_v = 1$
- As when frequency increases, impedance of C_1 decreases & the noninverting input voltage decreases.
- As C_2 is connected in the feedback provides the signal to the noninverting input which is in phase. So feedback signal adds the source signal, feedback is positive.
- $C_2 > C_1$ to get positive feedback. It increases 'Q'.

Pole frequency - A special frequency used in the design of active filters. is called pole frequency (f_p)

Pole frequency is given by $f_p = \frac{1}{2\pi \sqrt{R_1 R_2 C_1 C_2}}$
as $R_1 = R_2 = R$.

$$f_p = \frac{1}{2\pi R \sqrt{C_1 C_2}}$$

If $Q = 0.707$, the response we get is a Butterworth response and k_c value is 1.

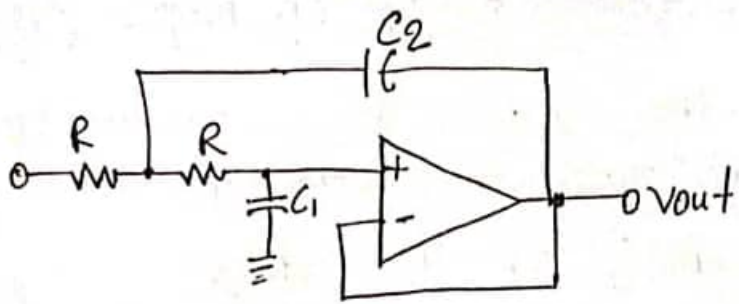
If $Q = 0.577$, the response we get is a Bessel response & $k_c = 0.786$.

→ cut off frequency - A frequency at which the attenuation is 3dB is given by

$$f_c = k_c f_p$$

Second order VCVS stage for $Q > 0.707$
(Peaked response).

→ Fig analyze the circuit when $Q > 0.707$.



$$A_v = 1$$

$$Q = 0.5 \sqrt{\frac{C_2}{C_1}}$$

$$f_p = \frac{1}{2\pi R \sqrt{C_1 C_2}}$$

$$Q > 0.707$$

$$f_0 = k_0 f_p$$

$$f_c = k_c f_p$$

$$f_{3dB} = k_3 f_p$$

→ Resonant frequency where peaking appears

$$f_0 = k_0 f_p$$

→ Edge frequency

$$f_c = k_c f_p$$

→ 3dB frequency

$$f_{3dB} = k_3 f_p$$

→ For $Q > 10$

$$k_0 = 1$$

$$k_c = 1.414$$

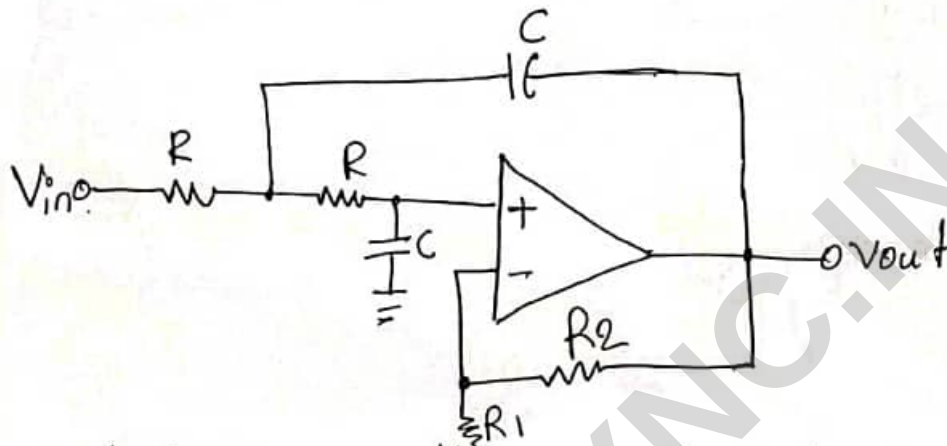
$$k_3 = 1.55$$

$$A_p = 20 \log Q$$

➤ Gain-Bandwidth product of op Amps (GBW)

- Not to affect the filter performance op-amps must have enough gain-bandwidth product (GBW).
- Limited GBW (inverted) of the stage
- " " change performance of a filter.
- Using predistortion method limited GBW can be corrected.

VCVS Equal-Component Low pass filters



$$A_v = \frac{R_2}{R_1} + 1$$

$$Q = \frac{1}{3 - A_v}$$

$$f_p = \frac{1}{2\pi RC}$$

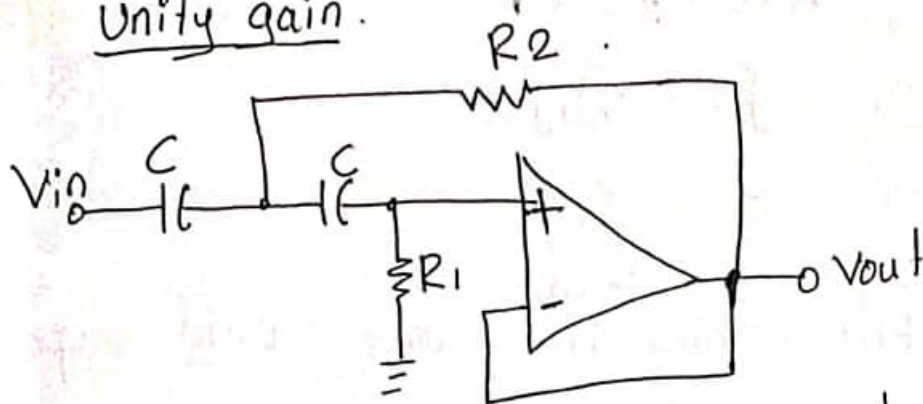
- It is another Sallen-Key second order low-pass filter.
- It uses same values of 'R' & same values of 'C'.
- This is why the circuit called a Sallen-Key equal component filter.
- Voltage gain of the circuit is $A_v = \frac{R_2}{R_1} + 1$.
- Q of the circuit is $Q = \frac{1}{3 - A_v}$.
- Condition is A_v can be no smaller than unity (1 to 3) & minimum Q is 0.5.
- When $A_v > 3$, positive feedback becomes too large and it breaks in to oscillation. When A_v approaches 3 component tolerance & drift causes A_v to exceed 3.

VCVS High-Pass filters

(27)

There are two configuration.

① Second order VCVS high-pass filter stages - Unity gain.



$$A_v = 1$$

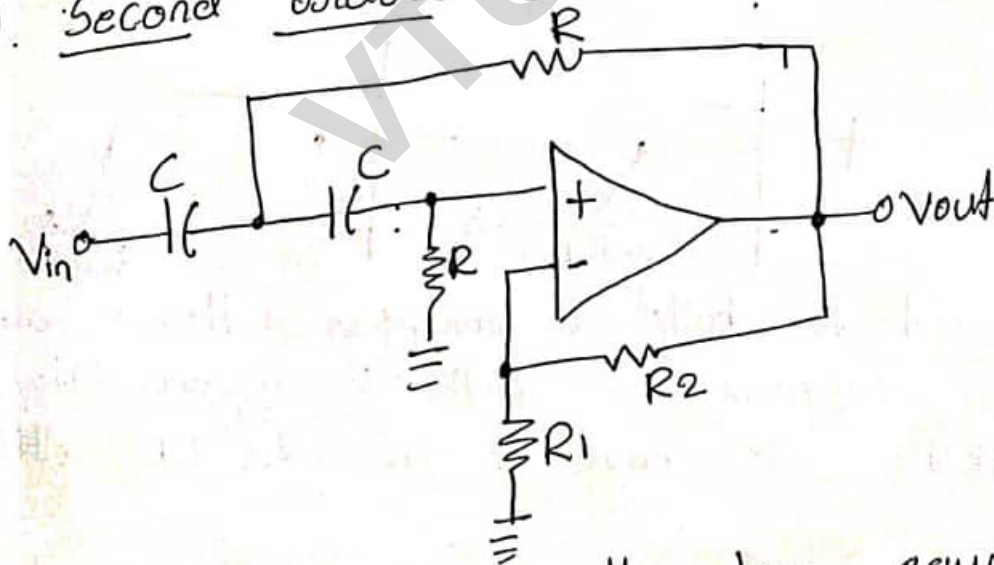
$$Q = 0.5 \sqrt{\frac{R_1}{R_2}}$$

$$f_p = \frac{1}{2\pi C \sqrt{R_1 R_2}}$$

- Fig shows the sallen-key unity gain high pass filter and its equation.
- Here position of R & C are reversed.
- Q depends on value of R.
- cut off frequency of a high pass filter

$$f_c = \frac{f_p}{K_c}$$

② Second order VCVS high pass stages - voltage gain



$$A_v = \frac{R_2}{R_1} + 1$$

$$Q = \frac{1}{3 - A_v}$$

$$f_p = \frac{1}{2\pi R C}$$

- Fig shows the sallen-key equal component high pass filter and its equation.

Band pass Filters

→ A band pass filter has a center frequency & a bandwidth.

$$\text{Band width, } BW = f_2 - f_1$$

$$\text{Center frequency, } f_0 = \sqrt{f_1 f_2}$$

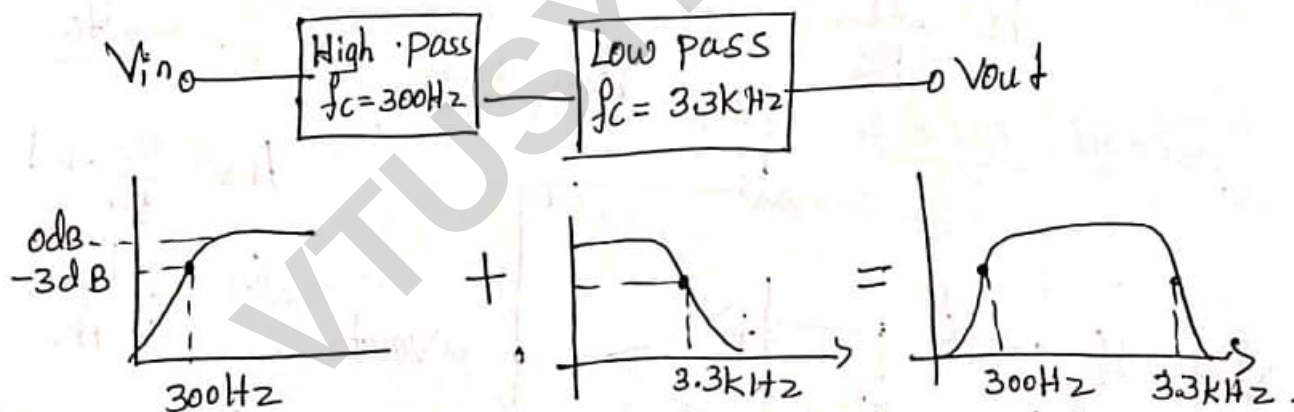
$$Q = \frac{f_0}{BW}$$

When $Q > 1$ Filter has a narrow band response
" $Q < 1$ " " Wide band response

Wide Band Filters

→ When $Q < 1$ Filter has a wide band response

→ we can construct wide band filter by cascading low-pass & high pass stages.



→ suppose we want to build a bandpass filter with a lower cutoff frequency of 300 Hz & upper cutoff frequency 3.3 kHz. we have to cascade LPF & HPF as above.

$$f_0 = \sqrt{f_1 f_2} = \sqrt{300\text{Hz} (3.3\text{kHz})} = 995\text{Hz}$$

$$BW = f_2 - f_1 = 3.3\text{kHz} - 300\text{Hz} = 3\text{kHz}$$

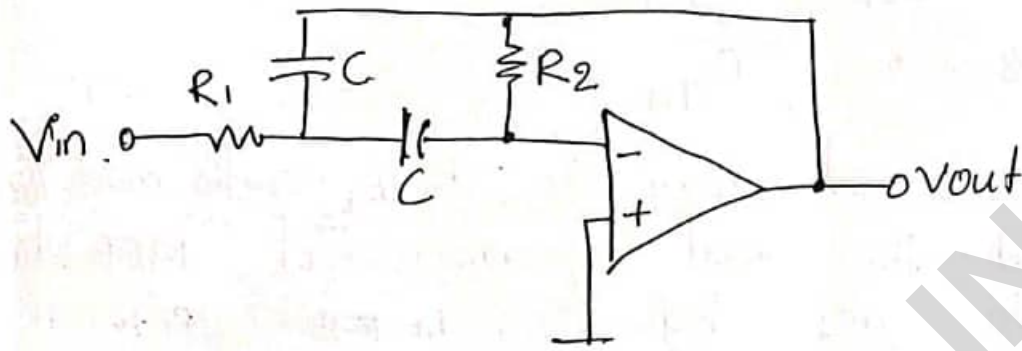
$$Q = \frac{f_0}{BW} = \frac{995}{3\text{kHz}} = 0.332$$

Q is less than 1.

- HPF has a cut $f_c = 300\text{Hz}$.
- LPF " " $f_c = 3.3\text{kHz}$.
- when we add the responses we get bandpass filter with cutoff frequencies of 300Hz & 3.3kHz .

Narrow band Filters (Multiple feedback (MFB))

- when $Q > 1$. multiple-feedback filter is used.



$$A_v = \frac{-R_2}{2R_1}$$

$$Q = 0.5 \sqrt{\frac{R_2}{R_1}}$$

$$f_0 = \frac{1}{2\pi C \sqrt{R_1 R_2}}$$

- V_{in} is given to the inverting terminal.
- Circuit has two feedback path one through 'C' & another through 'R2'.
- At lower frequencies (when f is low) capacitor act as open switch. Therefore input signal cannot reach the opamp and the output is zero.
- At high frequency (when f is high) capacitor act as a closed switch. so voltage gain is zero as capacitor has zero impedance.
- Between extremes low and high frequency, there is a band of frequencies where the circuit acts like an inverting amplifier. voltage gain at the center frequency is given by $A_v = \frac{-R_2}{2R_1}$.

Q of the circuit is given by

$$Q = 0.5 \sqrt{\frac{R_2}{R_1}}$$

$$Q = 0.707 \cdot \sqrt{-A_v}$$

If $A_v = -100$.

$Q = 0.707 \sqrt{100} = 7.07$.

→ Greater the voltage gain higher the Q .

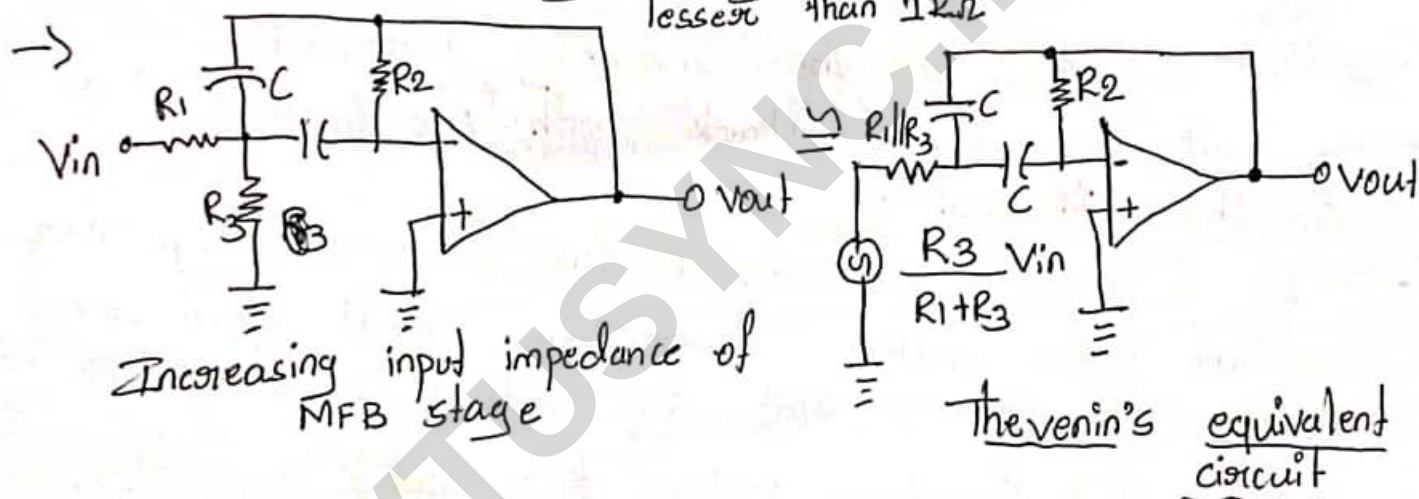
$f_0 = \frac{1}{2\pi \sqrt{R_1 R_2 C_1 C_2}}$ Since $C_1 = C_2$

$f_0 = \frac{1}{2\pi C \sqrt{R_1 R_2}}$

Increasing the input Impedance.

We have $Q = 0.5 \sqrt{R_2/R_1}$.

→ $Q \propto R_2/R_1$, To get high Q s; R_2/R_1 ratio must be kept high. But the input impedance of MFB stage is too low to get high Q s. $R_2 \approx 100k\Omega$, R_1 has to be lesser than $1k\Omega$.



→ Fig shows an MFB bandpass filter that increases the input impedance. Extra resistor R_3 is used to divide the voltage.

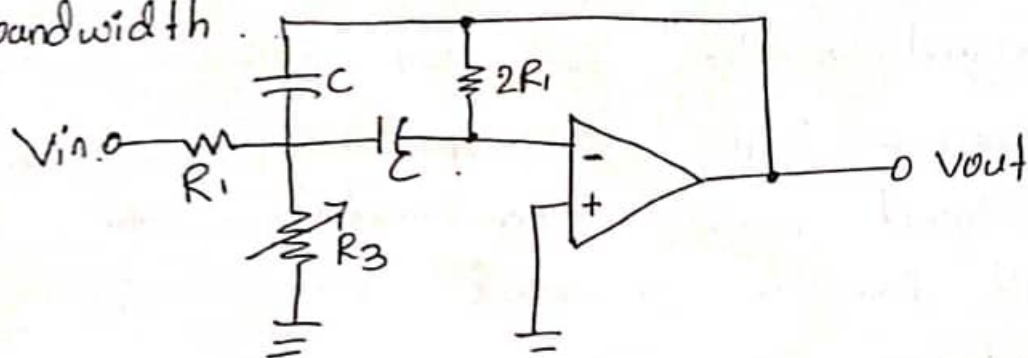
Voltage gain $A_v = \frac{-R_2}{2R_1}$.

Q factor $Q = 0.5 \sqrt{\frac{R_2}{R_1 || R_3}}$

$f_0 = \frac{1}{2\pi C \sqrt{(R_1 || R_3) R_2}}$

Tunable Center Frequency with constant Bandwidth. (31)

→ Fig shows the modified circuit that has $A_v = 1$ & it has a variable center frequency and constant bandwidth.



→ In the above circuit, $R_2 = 2R_1$ & R_3 is variable. The circuit has following equations.

$$A_v = -1$$

$$Q = 0.707 \sqrt{\frac{R_1 + R_3}{R_3}}$$

$$f_0 = \frac{1}{2\pi C \sqrt{2R_1(R_1 + R_3)}}$$

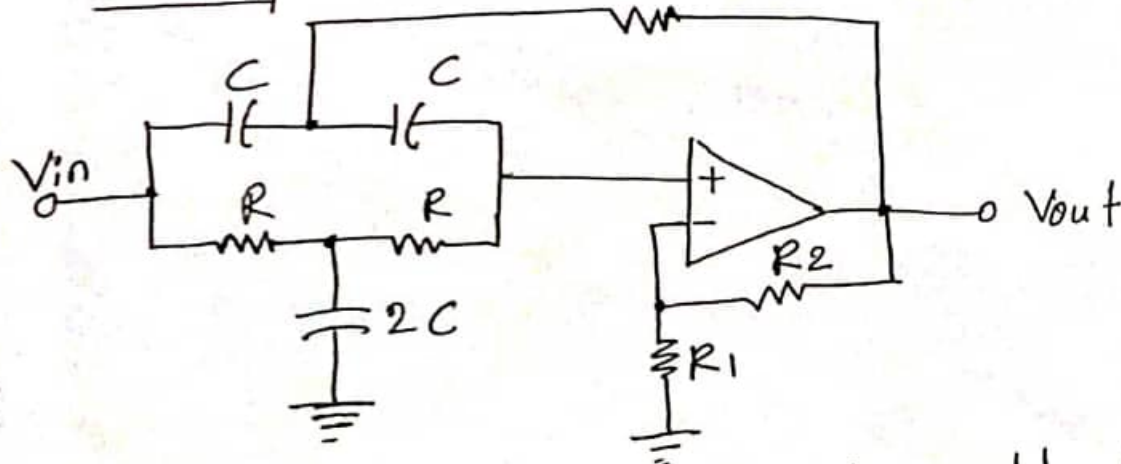
f_0 depends on R_3
 R_3 changes center frequency
 f_0 changes.

Since $BW = \frac{f_0}{Q}$

$$BW = \frac{1}{2\pi R_1 C}$$

BW does not depend on R_3
 $\therefore$ even though R_3 changes BW remains constant.

Bandstop Filters: R/2.



Sallen-key second-order notch filter

→ Fig shows a Sallen-Key second order notch filter & its equations.

→ Bandstop filter block only a single frequency (notch filter)

→ When frequency is low all capacitors are open, all input signal reaches the non-inverting input.

→ When frequency is high, all capacitors are shorted, all input signal reaches non-inverting input.

→ The circuit has a passband voltage gain

$$A_v = \frac{R_2}{R_1} + 1$$

→ Center frequency (f_0) is between extreme low & high frequency.

$$f_0 = \frac{1}{2\pi RC}$$

→ At f_0 feedback signal returns with the correct amplitude and phase to attenuate the signal on the non-inverting input.

→ The Q of the circuit

$$Q = \frac{0.5}{2 - A_v}$$

→ $A_v < 2$ to avoid oscillations. & Q must be less than 10

First Order stages.

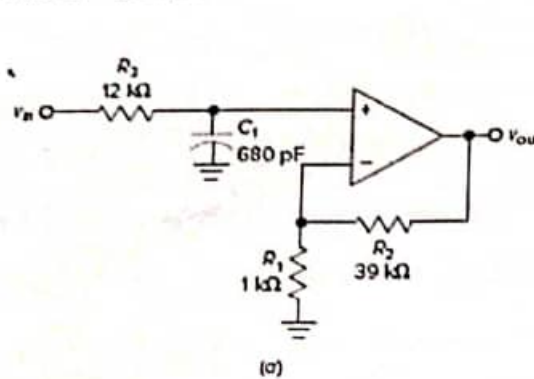
Example 19-1

1100 001 001 100 000

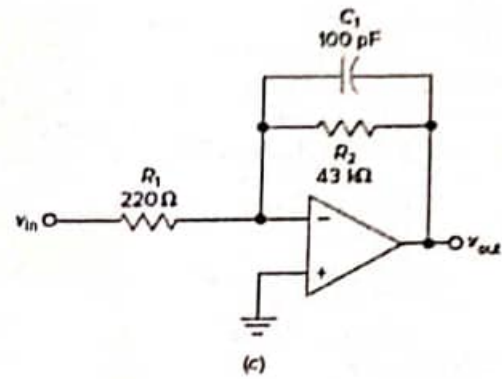
What is the voltage gain in Fig. 19-23a? What is the cutoff frequency? What is the frequency response?

Figure 19-23 Example.

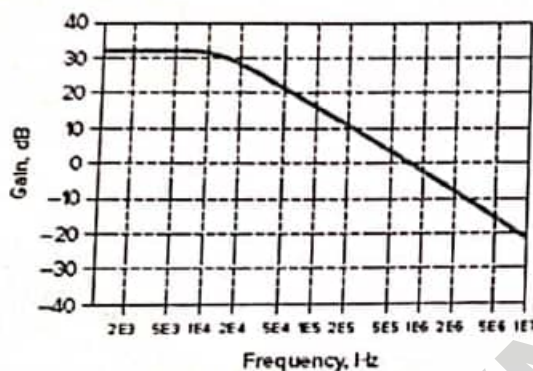
9



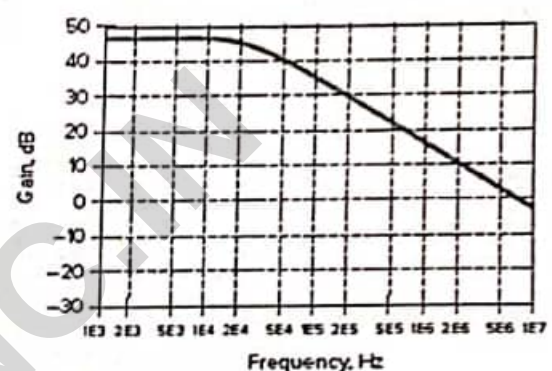
(a)



(c)



(b)



(d)

SOLUTION This is a noninverting first-order low-pass filter. With Eqs. (19-10) and (19-11), the voltage gain and cutoff frequencies are:

$$A_v = \frac{39 \text{ k}\Omega}{1 \text{ k}\Omega} + 1 = 40$$

$$f_c = \frac{1}{2\pi(12 \text{ k}\Omega)(680 \text{ pF})} = 19.5 \text{ kHz}$$

Figure 19-23b shows the frequency response. The voltage gain is 32 dB in the passband. The response breaks at 19.5 kHz and then rolls off at a rate of 20 dB per decade.

PRACTICE PROBLEM 19-1 Using Fig. 19-23a, change the 12-kΩ resistor to 6.8 kΩ. Find the new cutoff frequency.

Example 19-2

What is the voltage gain in Fig. 19-23c? What is the cutoff frequency? What is the frequency response?

SOLUTION This is an inverting first-order low-pass filter. With Eqs. (19-12) and (19-13), the voltage gain and cutoff frequencies are:

$$A_v = \frac{-43 \text{ k}\Omega}{220 \Omega} = -195$$

$$f_c = \frac{1}{2\pi(43 \text{ k}\Omega)(100 \text{ pF})} = 37 \text{ kHz}$$

Figure 19-23d shows the frequency response. The voltage gain is 45.8 dB in the passband. The response breaks at 37 kHz and then rolls off at a rate of 20 dB per decade.

PRACTICE PROBLEM 19-2 In Fig. 19-23c, change the 100-pF capacitor to 220 pF. What is the new cutoff frequency?

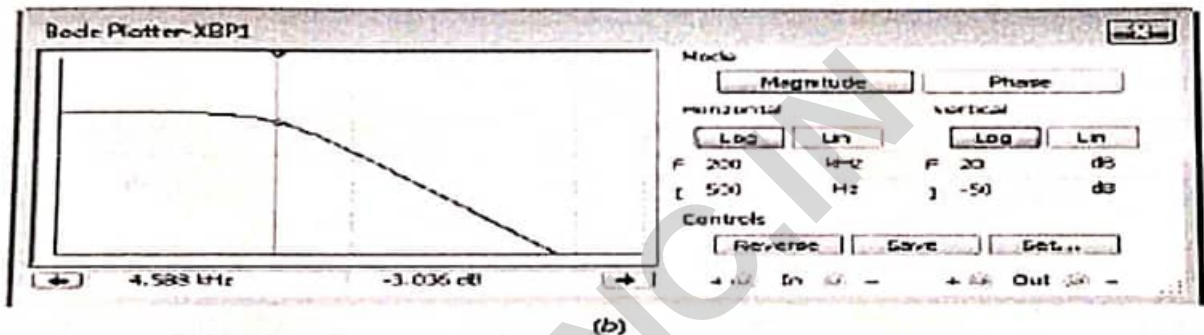
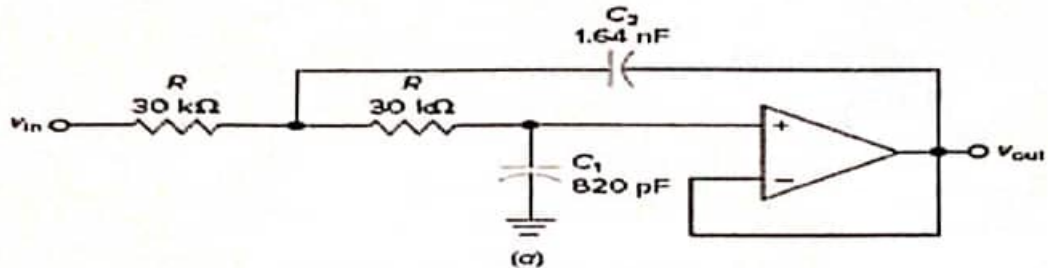
Second order stages

Application Example 19-3

Multisim

What are the pole frequency and Q of the filter shown in Fig. 19-27? What is the cutoff frequency? Show the frequency response using a Multisim Bode plotter.

Figure 19-27 (a) Butterworth unity-gain example; (b) Multisim frequency response.



Courtesy of National Instruments.

SOLUTION The Q and pole frequency are:

$$Q = 0.5 \sqrt{\frac{C_2}{C_1}} = 0.5 \sqrt{\frac{1.64 \text{ nF}}{820 \text{ pF}}} = 0.707$$

$$f_p = \frac{1}{2\pi R \sqrt{C_1 C_2}} = \frac{1}{2\pi (30 \text{ k}\Omega) \sqrt{(820 \text{ pF})(1.64 \text{ nF})}} = 4.58 \text{ kHz}$$

The Q value of 0.707 tells us that this is a Butterworth response, so the cutoff frequency is the same as the pole frequency:

$$f_c = f_p = 4.58 \text{ kHz}$$

$$f_c = k_c f_p$$

The response of the filter breaks at 4.58 kHz and rolls off at a rate of 40 dB per decade because $n = 2$. Figure 19-27b shows the Multisim frequency response plot.

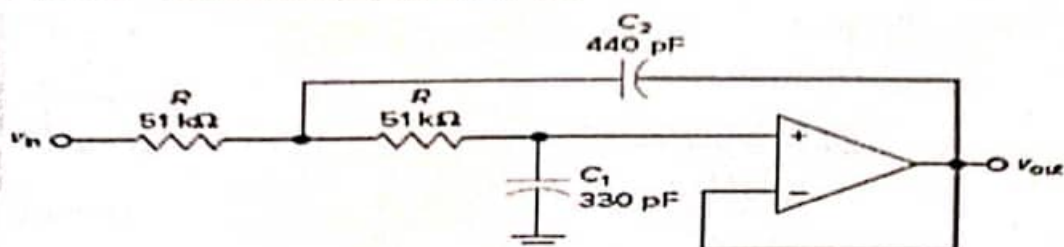
PRACTICE PROBLEM 19-3 Repeat Application Example 19-3 with the resistor values changed to 10 kΩ.

second order LPF .

Example 19-4

In Fig. 19-28, what are the pole frequency and Q ? What is the cutoff frequency?

Figure 19-28 Bessel unity-gain example.



SOLUTION The Q and pole frequency are:

$$Q = 0.5 \sqrt{\frac{C_2}{C_1}} = 0.5 \sqrt{\frac{440 \text{ pF}}{330 \text{ pF}}} = 0.577$$

$$f_p = \frac{1}{2\pi R \sqrt{C_1 C_2}} = \frac{1}{2\pi (51 \text{ k}\Omega) \sqrt{(330 \text{ pF})(440 \text{ pF})}} = 8.19 \text{ kHz}$$

The Q value of 0.577 tells us that this is a Bessel response. With Eq. (19-21), the cutoff frequency is given by:

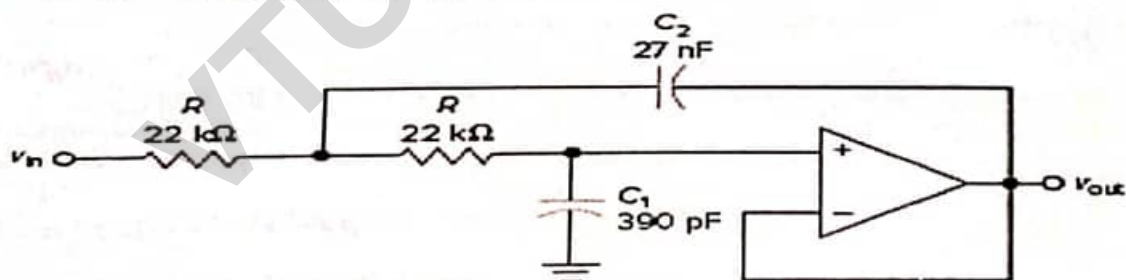
$$f_c = K_c f_p = 0.786(8.19 \text{ kHz}) = 6.44 \text{ kHz}$$

PRACTICE PROBLEM 19-4 In Example 19-4, if the value of C_1 changed to 680 pF, what value should C_2 be to maintain a Q of 0.577?

Example 19-5

What are the pole frequency and Q in Fig. 19-29? What are the cutoff and 3-dB frequencies?

Figure 19-29 Unity-gain example with $Q > 0.707$.



SOLUTION The Q and pole frequency are:

$$Q = 0.5 \sqrt{\frac{C_2}{C_1}} = 0.5 \sqrt{\frac{27 \text{ nF}}{390 \text{ pF}}} = 4.16$$

$$f_p = \frac{1}{2\pi R \sqrt{C_1 C_2}} = \frac{1}{2\pi (22 \text{ k}\Omega) \sqrt{(390 \text{ pF})(27 \text{ nF})}} = 2.23 \text{ kHz}$$

Referring to Fig. 19-26, we can read the following approximate K and A_p values:

$$\left. \begin{aligned} K_0 &= 0.99 // \\ K_c &= 1.38 \\ K_3 &= 1.54 \end{aligned} \right\} \\ A_p = 12.5 \text{ dB}$$

The cutoff or edge frequency is:

$$f_c = K_c f_p = 1.38(2.23 \text{ kHz}) = 3.08 \text{ kHz}$$

and the 3-dB frequency is:

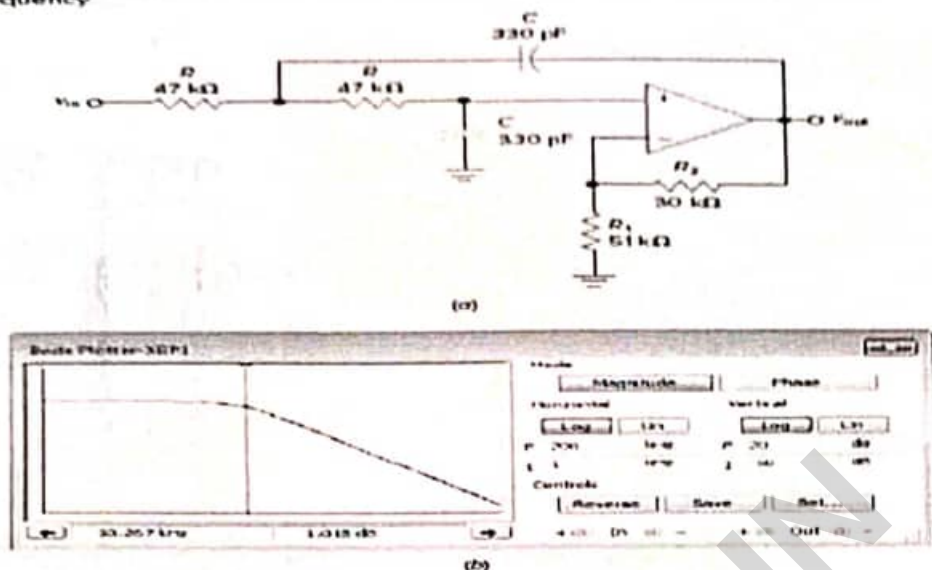
$$f_{3\text{dB}} = K_3 f_p = 1.54(2.23 \text{ kHz}) = 3.43 \text{ kHz}$$

PRACTICE PROBLEM 19-5 In Fig. 19-29, change the 27-nF capacitor to 14 nF and repeat Example 19.5.

Application Example 19-6

What are the pole frequency and Q of the filter shown in Fig. 19-32? What is the cutoff frequency? Show the frequency response using a Multisim Bode plotter.

Figure 19-32 (a) Butterworth equal-component example, (b) Multisim pole frequency



Courtesy of National Instruments

SOLUTION The A_v , Q , and f_p are:

$$A_v = \frac{30 \text{ k}\Omega}{51 \text{ k}\Omega} + 1 = 1.59$$

$$Q = \frac{1}{3 - A_v} = \frac{1}{3 - 1.59} = 0.709$$

$$f_p = \frac{1}{2\pi RC} = \frac{1}{2\pi(47 \text{ k}\Omega)(330 \text{ pF})} = 10.3 \text{ kHz}$$

It takes a Q of 0.77 to produce a ripple of 0.1 dB. Therefore, a Q of 0.709 produces a ripple of less than 0.003 dB. For all practical purposes, the calculated Q of 0.709 means that we have a Butterworth response to a very close approximation.

The cutoff frequency of a Butterworth filter is equal to the pole frequency of 10.3 kHz. Notice in Fig. 19-32b that the pole frequency is at approximately 1 dB. This value is 3 dB down from the passband gain of 4 dB.

PRACTICE PROBLEM 19-6 In Application Example 19-6, change the 47-k Ω resistors to 22 k Ω and solve for A_v , Q , and f_p .

Example 19-7

Example 19-7 In Fig. 19-33, what are the pole frequency and Q ? What is the cutoff frequency?

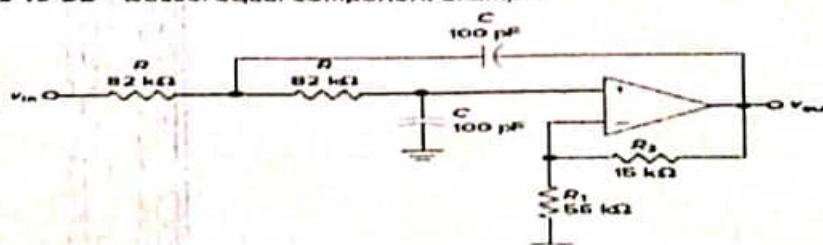
SOLUTION The A_v , Q , and f_p are:

$$A_v = \frac{15 \text{ k}\Omega}{16 \text{ k}\Omega} + 1 = 1.27$$

$$Q = \frac{1}{3 - A_V} = \frac{1}{3 - 1.27} = 0.578$$

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(2 \text{ k}\Omega)(100 \text{ pF})} = 19.4 \text{ kHz}$$

Figure 19-33 Bessel equal-component example



This is the Q of a Bessel second-order response. Therefore, $K_e = 0.786$ and the cutoff frequency is:

$$f_c = 0.786 f_p = 0.786(19.4 \text{ kHz}) = 15.2 \text{ kHz}$$

PRACTICE PROBLEM 19-7 Repeat Example 19-7 with the capacitance equal to 330 pF and the R value set to 100 k Ω .

Example 19-8

What are the pole frequency and Q in Fig. 19-34? What are the resonant, cutoff, and 3-dB frequencies? What is the ripple depth in decibels?

SOLUTION The A_v , Q , and f_p are:

$$A_v = \frac{39 \text{ k}\Omega}{20 \text{ k}\Omega} + 1 = 2.95$$

$$Q = \frac{1}{3 - A_v} = \frac{1}{3 - 2.95} = 20$$

$$f_p = \frac{1}{2\pi RC} = \frac{1}{2\pi(56 \text{ k}\Omega)(220 \text{ pF})} = 12.9 \text{ kHz}$$

Figure 19-26 has Q s only between 1 and 10. In this case, we need to use Eqs. (19-25) through (19-28) to get the K and Q values:

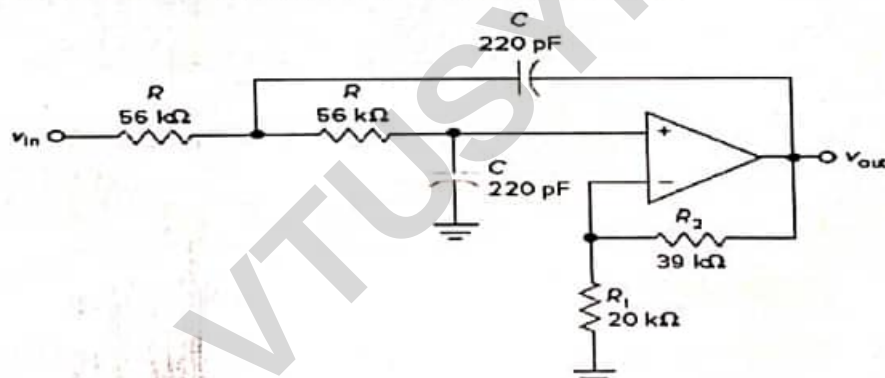
$$K_0 = 1$$

$$K_e = 1.414$$

$$K_3 = 1.55$$

$$A_p = 20 \log Q = 20 \log 20 = 26 \text{ dB}$$

Figure 19-34 Equal-component example with $Q > 0.707$.



The resonant frequency is:

$$f_0 = K_0 f_p = 12.9 \text{ kHz}$$

The cutoff or edge frequency is:

$$f_c = K_e f_p = 1.414 (12.9 \text{ kHz}) = 18.2 \text{ kHz}$$

and the 3-dB frequency is:

$$f_{3dB} = K_3 f_p = 1.55 (12.9 \text{ kHz}) = 20 \text{ kHz}$$

The circuit produces a 26-dB peak in the response at 12.9 kHz, rolls off to 0 dB at the cutoff frequency, and is down 3 dB at 20 kHz.

A Sallen-Key circuit like this is impractical because the Q is too high. Since the voltage gain is 2.95, any error in the values of R_1 and R_2 can cause large increases in Q . For instance, if the tolerance of the resistors is ± 1 percent, the voltage gain can be as high as:

$$A_v = \frac{1.01(39 \text{ k}\Omega)}{0.99(20 \text{ k}\Omega)} + 1 = 2.989$$

This voltage gain produces a Q of:

$$Q = \frac{1}{3 - A_v} = \frac{1}{3 - 2.989} = 90.9$$

The Q has changed from a design value of 20 to an approximate value of 90.9, which means that the frequency response is radically different from the intended response.

Even though the Sallen-Key equal-component filter is simple compared to other filters, it has the disadvantage of component sensitivity when high Q s are used. This is why more complicated circuits are typically used for high- Q stages. The added complexity reduces the component sensitivity.

Application Example 19-9

III Edition

What are the pole frequency and Q of the filter shown in Fig. 19-36a? What is the cutoff frequency? Show the frequency response using a Multisim Bode plotter.

SOLUTION The Q and pole frequency are:

$$Q = 0.5 \sqrt{\frac{R_1}{R_2}} = 0.5 \sqrt{\frac{24 \text{ k}\Omega}{12 \text{ k}\Omega}} = 0.707$$

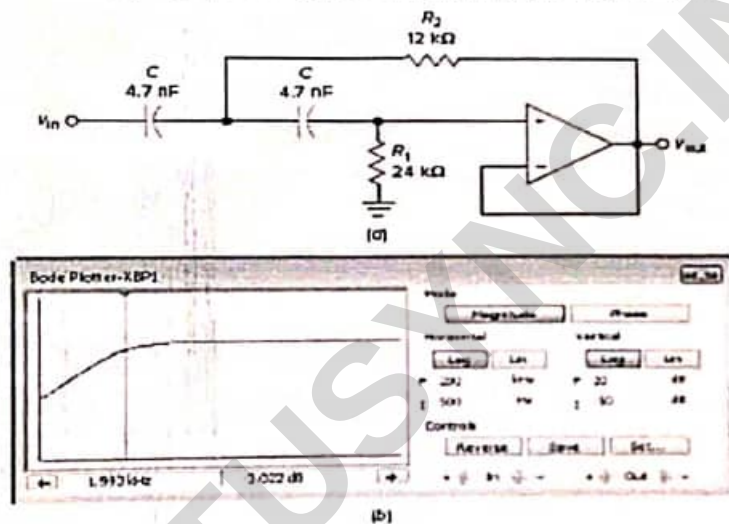
$$f_p = \frac{1}{2\pi C \sqrt{R_1 R_2}} = \frac{1}{2\pi (4.7 \text{ nF}) \sqrt{(24 \text{ k}\Omega)(12 \text{ k}\Omega)}} = 2 \text{ kHz}$$

Since $Q = 0.707$, the filter has a Butterworth second-order response and:

$$f_c = f_p = 2 \text{ kHz}$$

The filter has a high-pass response with a break at 2 kHz, and it rolls off at 40 dB per decade below 2 kHz. Figure 19-36(b) shows the Multisim frequency response plot.

Figure 19-36 (a) High-pass Butterworth example; (b) Multisim cutoff frequency.



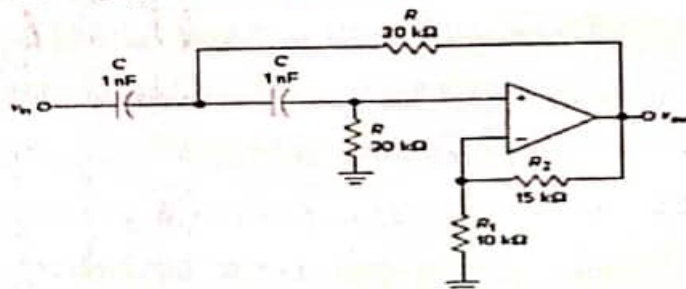
Courtesy of National Instruments

PRACTICE PROBLEM 19-9 In Fig. 19-36, double the two resistor values. Find the circuit's Q , f_p , and f_c values.

Example 19-10

What are the pole frequency and Q in Fig. 19-37? What are the resonant, cutoff, and 3-dB frequencies? What is the ripple depth or peaking in decibels?

Figure 19-37 High-pass example with $Q > 1$.



SOLUTION The A_v , Q , and f_p are:

$$A_v = \frac{15 \text{ k}\Omega}{10 \text{ k}\Omega} + 1 = 2.5$$

$$Q = \frac{1}{3 - A_v} = \frac{1}{3 - 2.5} = 2$$

$$f_p = \frac{1}{2\pi RC} = \frac{1}{2\pi (30 \text{ k}\Omega)(1 \text{ nF})} = 5.31 \text{ kHz}$$

In Fig. 19-26, a Q of 2 gives the following approximate values:

$$K_0 = 0.94$$

$$K_c = 1.32$$

$$K_3 = 1.48$$

$$A_p = 20 \log Q = 20 \log 2 = 6.3 \text{ dB}$$

The resonant frequency is:

$$f_0 = \frac{f_p}{K_0} = \frac{5.31 \text{ kHz}}{0.94} = 5.65 \text{ kHz}$$

The cutoff frequency is:

$$f_c = \frac{f_p}{K_c} = \frac{5.31 \text{ kHz}}{1.32} = 4.02 \text{ kHz}$$

The 3-dB frequency is:

$$f_{3\text{dB}} = \frac{f_p}{K_3} = \frac{5.31 \text{ kHz}}{1.48} = 3.59 \text{ kHz}$$

The circuit produces a 6.3-dB peak in the response at 5.65 kHz, rolls off to 0 dB at the cutoff frequency of 4.02 kHz, and is down 3 dB at 3.59 kHz.

PRACTICE PROBLEM 19-10 Repeat Example 19-10 with the 15-k Ω resistor changed to 17.5 k Ω .

Example 19-11

The gate voltage of Fig. 19-42 can vary the JFET resistance from 15 to 80 Ω . What is the bandwidth? What are the minimum and maximum center frequencies?

SOLUTION Equation (19-42) gives the bandwidth:

$$\text{BW} = \frac{1}{2\pi R_1 C} = \frac{1}{2\pi (18 \text{ k}\Omega)(8.2 \text{ nF})} = 1.08 \text{ kHz}$$

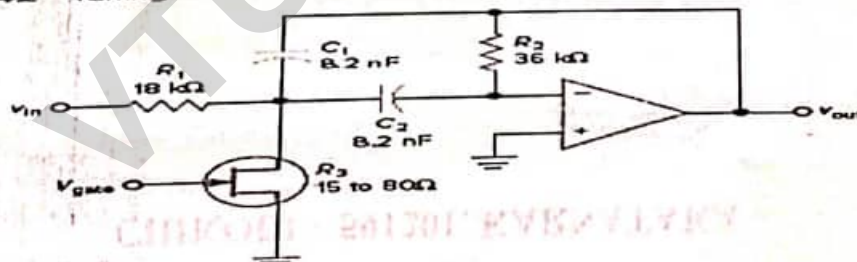
With Eq. (19-41), the minimum center frequency is:

$$\begin{aligned} f_0 &= \frac{1}{2\pi C \sqrt{2R_1(R_1 \parallel R_3)}} \\ &= \frac{1}{2\pi (8.2 \text{ nF}) \sqrt{2(18 \text{ k}\Omega)(18 \text{ k}\Omega \parallel 80 \Omega)}} \\ &= 11.4 \text{ kHz} \end{aligned}$$

The maximum frequency is:

$$f_0 = \frac{1}{2\pi (8.2 \text{ nF}) \sqrt{2(18 \text{ k}\Omega)(18 \text{ k}\Omega \parallel 15 \Omega)}} = 26.4 \text{ kHz}$$

Figure 19-42 Tuning an MFB filter with a voltage-controlled resistance.



PRACTICE PROBLEM 19-11 Using Fig. 19-42, change R_1 to 10 k Ω , change R_2 to 20 k Ω , and repeat Example 19-11.

Example 19-12

What are the voltage gain, center frequency, and Q for the bandstop filter shown in Fig. 19-43 if $R = 22 \text{ k}\Omega$, $C = 120 \text{ nF}$, $R_1 = 13 \text{ k}\Omega$, and $R_2 = 10 \text{ k}\Omega$?

SOLUTION With Eqs. (19-43) through (19-45):

$$A_v = \frac{10 \text{ k}\Omega}{13 \text{ k}\Omega} + 1 = 1.77$$

$$f_0 = \frac{1}{2\pi(22 \text{ k}\Omega)(120 \text{ nF})} = 60.3 \text{ Hz}$$

$$Q = \frac{0.5}{2 - A_v} = \frac{0.5}{2 - 1.77} = 2.17$$

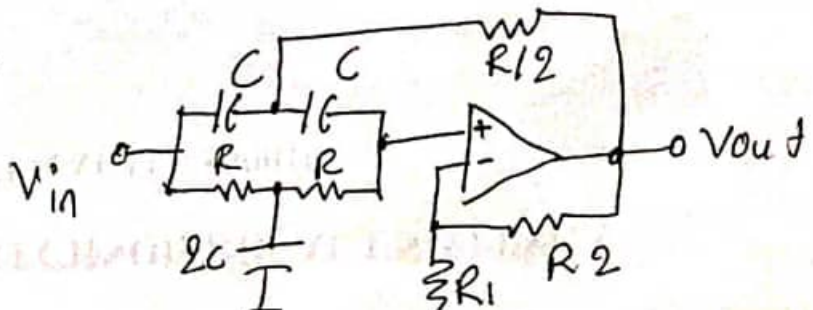


Figure 19-44 (a) Second-order notch filter at 60 Hz; (b) notch filter with $n = 20$.

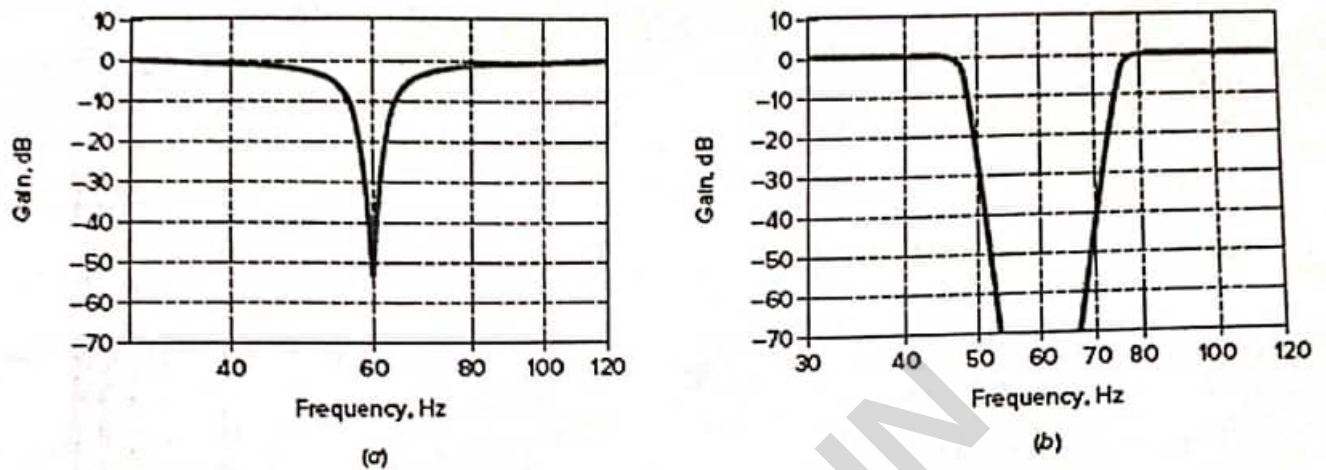


Figure 19-44a shows the response. Notice how sharp the notch is for a second-order filter.

By increasing the order of the filter, we can broaden the notch. For instance, Fig. 19-44b shows the frequency response for a notch filter with $n = 20$. The broader notch reduces component sensitivity and guarantees that the 60-Hz hum will be heavily attenuated.

PRACTICE PROBLEM 19-12 In Fig. 19-43, change R_2 to obtain a Q value of 3. Also, change the C value for a center frequency of 120 Hz.

Module - 5

Chapter-1 Power Amplifiers

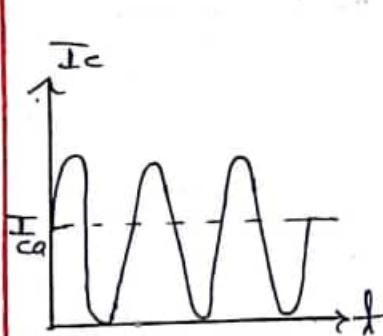
5.1

Amplifier TermsClasses of Operation

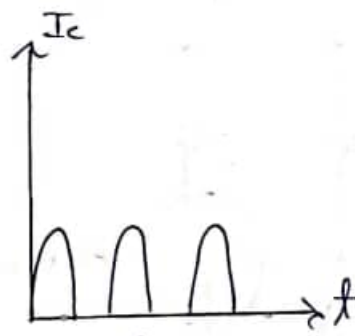
There are 4 Types:

- 1) Class A Operation
- 2) Class B Operation
- 3) Class C Operation
- 4) Class D Operation

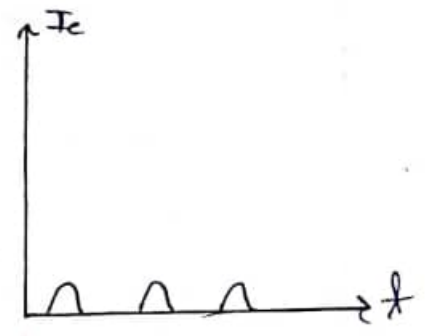
<u>Parameter</u>	<u>Class A</u>	<u>Class B</u>	<u>Class C</u>
<u>Operation</u>	Full 360°	Half the cycle 180°	flows for less than 180°
<u>Efficiency</u>	25% to 50%	78.56%	90°



a) Class A



b) Class B



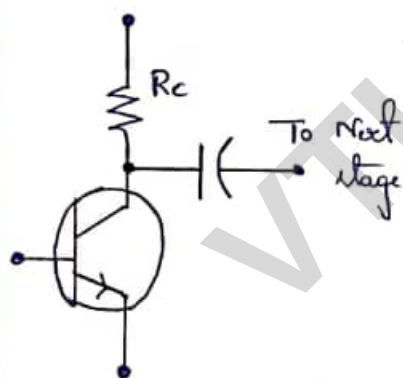
c) Class C

* Class D Operation

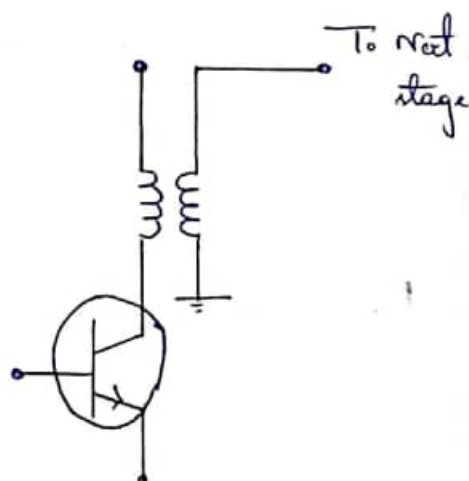
A Class D amplifier is one whose output is switched on and off that is, it essentially spends zero time during each input cycle in linear region of operation. A Class D amplifier is often used as a pulse-width modulator, which is a circuit whose output pulses have widths that are proportional to the amplitude level of the amplifiers input signal.

Types of Coupling:- Capacitive, transformer, direct.

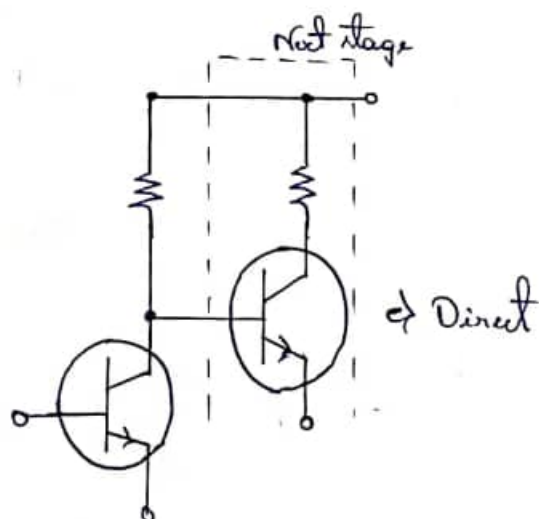
* Direct Coupling



a) Capacitive



b) Transformer



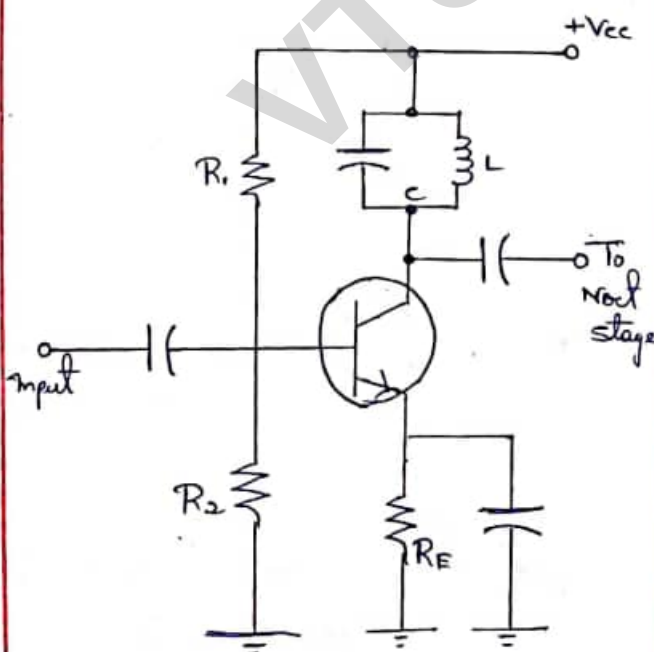
c) Direct

Range of Frequency

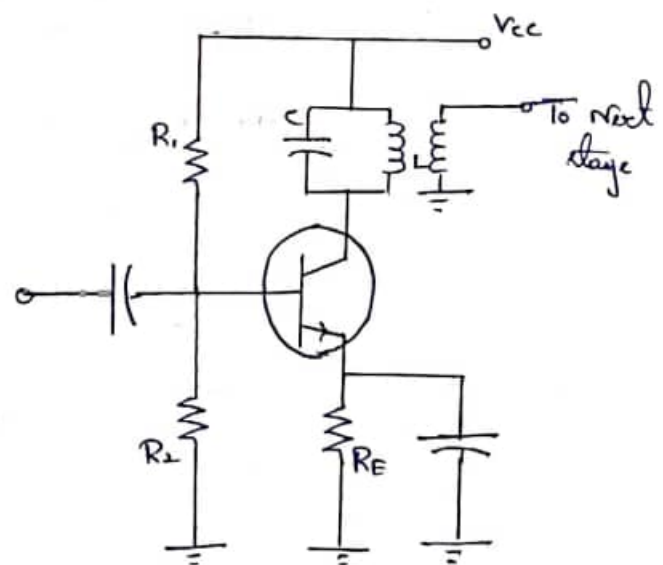
- Amplifiers are classified as Narrowband or wideband.
- A Narrow-band amplifier works over a small frequency range like 450 to 460 kHz.
- A Wideband amplifier operates over a large frequency range like 0 to 1 MHz.

<u>Amplifiers</u>	<u>frequency range</u>
Audio Amplifier	20 Hz to 20 kHz
Radio frequency Amplifier	above 20 kHz
FM Radios	88 and 108 MHz

AM Radio
= 535 to 1605 kHz



a) Capacitive coupling



b) Transformer coupling

5.2

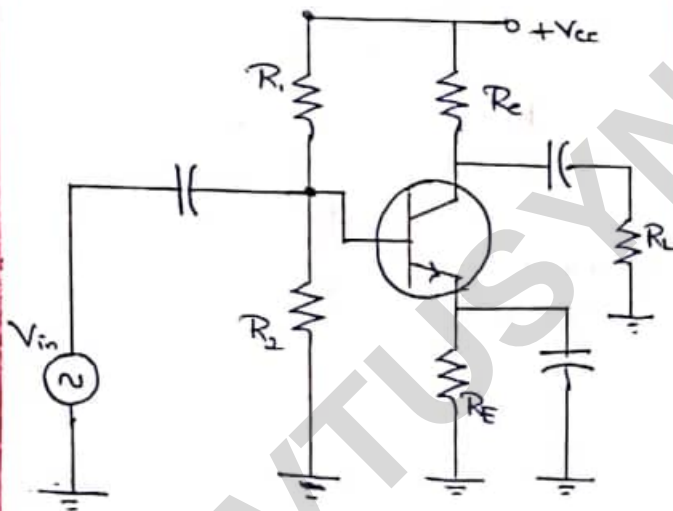
Two load lines

• DC load line:-

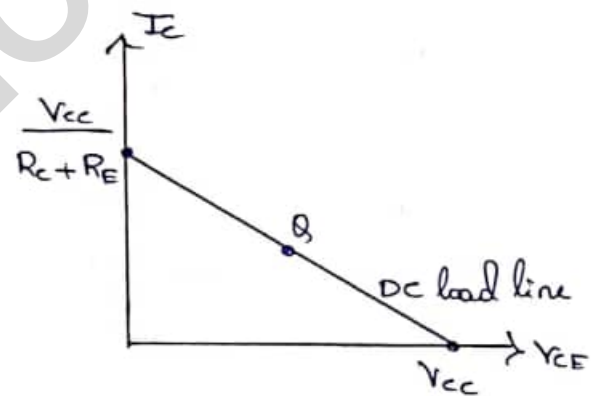
$$- I_{C[sat]} = \frac{V_{CC}}{R_C + R_E}$$

$$- V_{CE[cutoff]} = V_{CC}$$

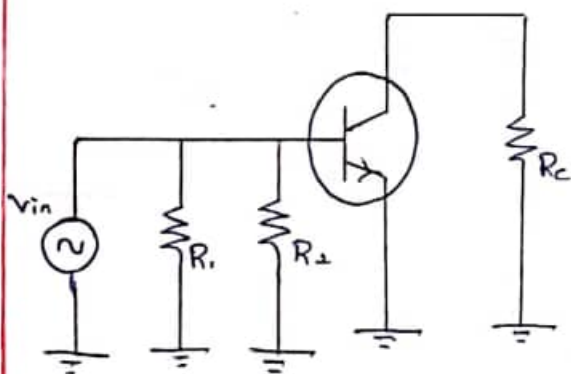
• AC load line:-



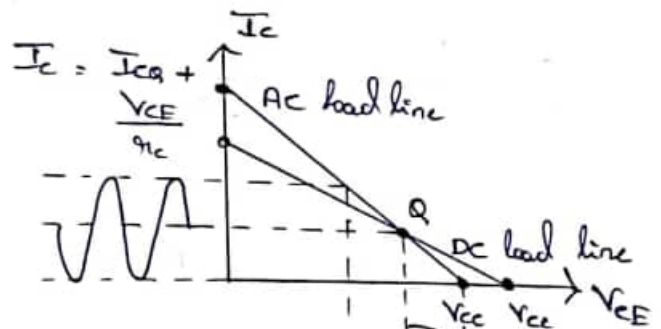
of VDB Amplifier



b) DC load line



c) AC equivalent circuit



d) AC load line

$$V_{CE} + i_c r_c = 0$$

[OR]

$$i_c = -\frac{V_{CE}}{r_c} \rightarrow \textcircled{\star}$$

The ac collector current is given by:

$$i_c = \Delta I_c = I_c - I_{CQ}$$

and the ac collector voltage is:

$$V_{ce} = \Delta V_{CE} = V_{CE} - V_{CEQ}$$

When substituting these expressions into eqⁿ $\textcircled{\star}$ and rearranging, we arrive at:

$$I_c = I_{CQ} + \frac{V_{CEQ}}{r_c} - \frac{V_{CE}}{r_c} \rightarrow \textcircled{1}$$

This is the equation of ac load line. When the transistor goes into saturation, V_{CE} is zero, and eqⁿ $\textcircled{1}$ gives us

$$i_c[\text{sat}] = I_{CQ} + \frac{V_{CEQ}}{r_c}$$

where, $i_c[\text{sat}] =$ ac saturation current

$I_{CQ} =$ dc collector current

$V_{CEQ} =$ dc collector-emitter voltage

$r_c =$ ac resistance seen by the collector

when the transistor goes into cutoff, I_c equals zero, since

$$V_{ce}[\text{cutoff}] = V_{CEQ} + \Delta V_{CE}$$

and

$$\Delta V_{CE} = [\Delta I_c][r_c]$$

we can substitute to get:

$$\Delta V_{CE} = [I_{CQ} - 0A][r_c]$$

resulting in:

$$V_{ce} [\text{cutoff}] = V_{CEQ} + I_{CQ} R_c$$

Because the ac load line has a higher slope than the dc load line, the Maximum peak to peak [MPP] output is always less than the supply voltage. As a formula.

$$MPP < V_{CC}$$

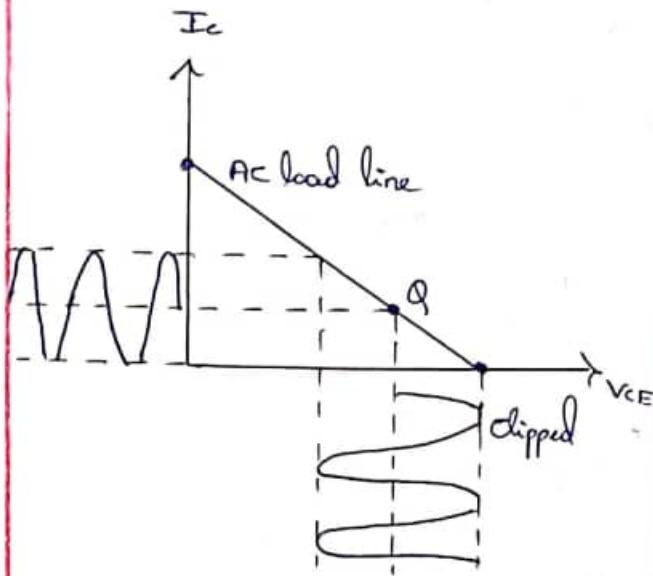
- Maximum Output

$$MP = I_{CQ} R_c \text{ or } V_{CEQ}$$

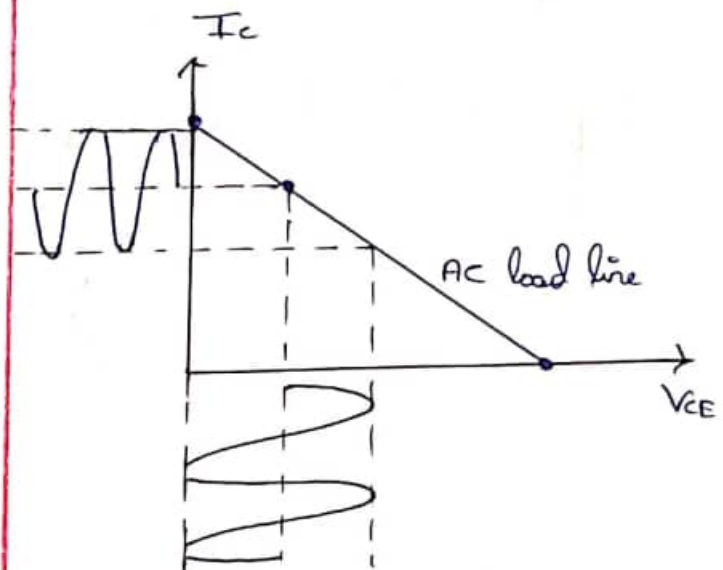
$$MPP = 2MP$$

- Clipping of large Signals

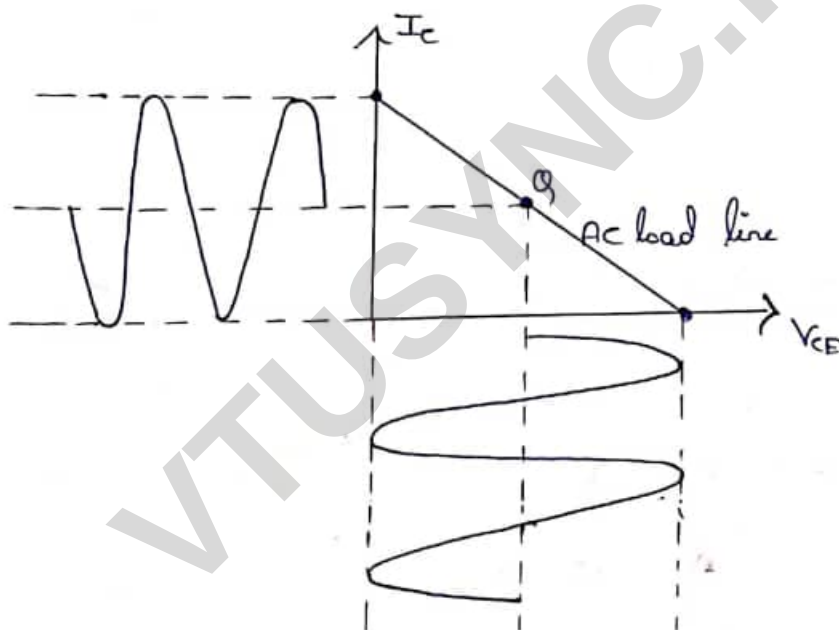
- When the Q point is at centre of dc load line, the ac signal cannot use all the ac load line without clipping.
- If the Q Point is moved higher, a large signal will drive the transistor into saturation. In this case, we get saturation clipping. Both cutoff and saturation clipping are undesirable because they distort the signal.
- If the Q-Point is at middle of ac load line, in this case, we get a maximum peak-to-peak unclipped output.



a) Cutoff clipping



b) Saturation clipping



c) Optimum Q-Point

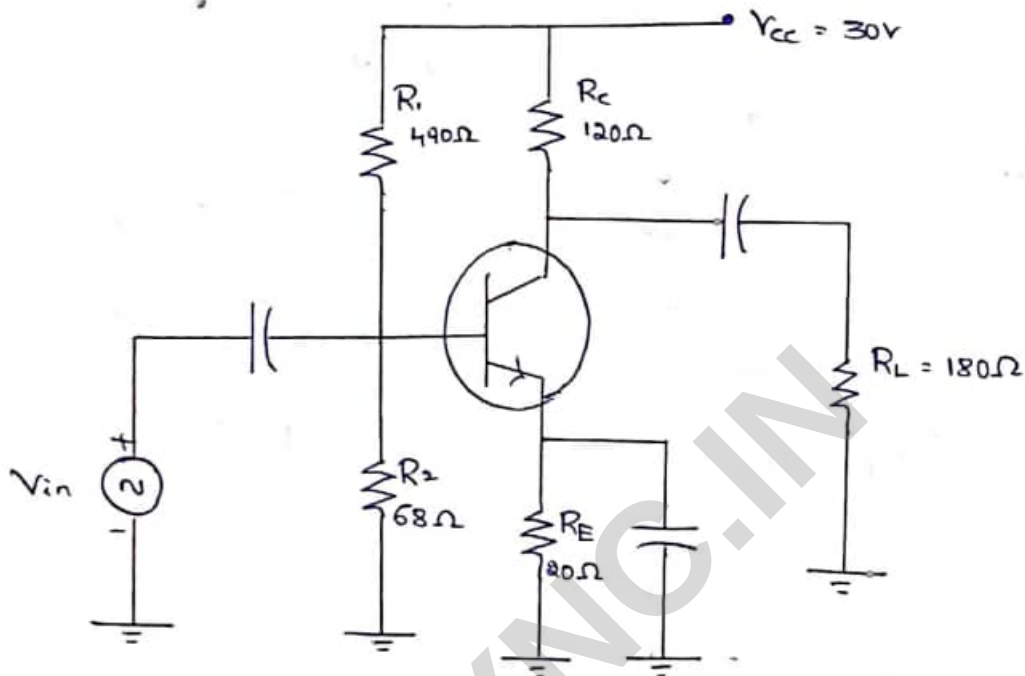
when the Q-Point is at centre of ac load line

$$I_{CQ} r_c = V_{CEQ}$$

$$R_E = \frac{R_c + r_c}{V_{CC}/V_E - 1}$$

Problems

1. What are the values of I_{CQ} , V_{CEQ} and r_c



Soln:-

$$V_B = \frac{68}{68 + 490} [30] = \underline{3.7V}$$

$$\begin{aligned} V_E &= V_B - 0.7V \\ &= 3.7V - 0.7V = \underline{3V} \end{aligned}$$

$$I_E = \frac{V_E}{R_E} = \frac{3V}{20\Omega} = \underline{150mA}$$

$$I_{CQ} \cong I_E = \underline{150mA}$$

$$\begin{aligned} V_{CEQ} &= V_C - V_E \\ &= 12V - 3V = \underline{9V} \end{aligned}$$

$$\begin{aligned} r_c &= R_C \parallel R_L \\ &= 120\Omega \parallel 180\Omega \\ &= \underline{72\Omega} \end{aligned}$$

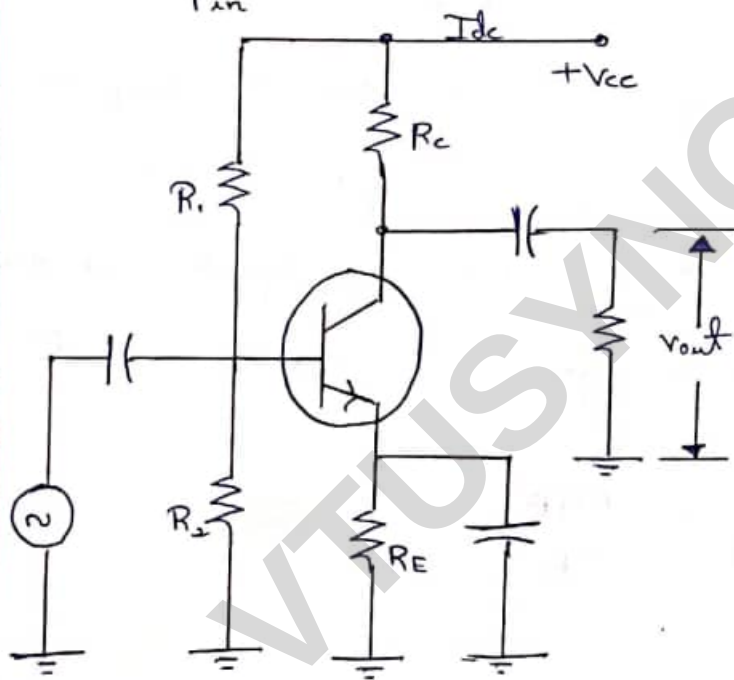
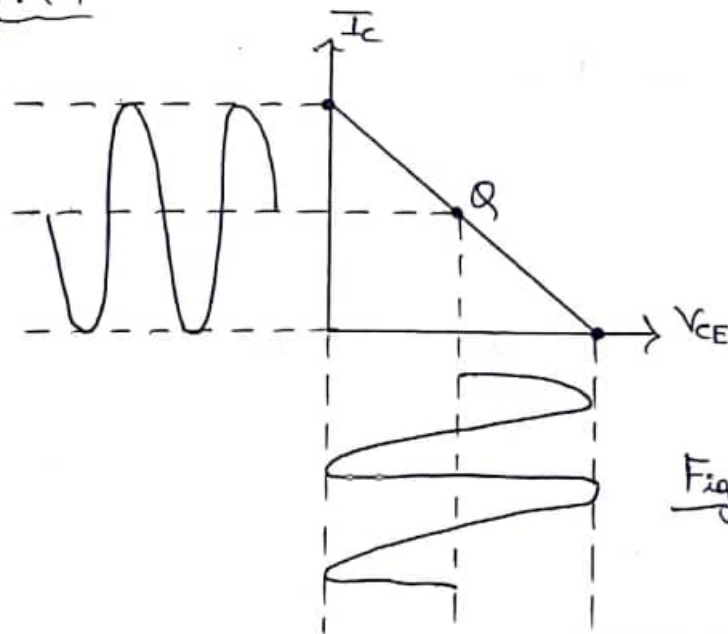
5.3

Class A Operation

The VDB amplifier of fig (a) is a class A amplifier as long as the output signal is not clipped.

Power gain

$$A_p = \frac{P_{out}}{P_{in}}$$

Fig. (a)Fig. (b)

Output Power

- The Output power is given by

$$P_{out} = \frac{V_{rms}^2}{R_L}$$

$$P_{out} = \frac{V_{out}^2}{8R_L}$$

- The factor of 8 in the denominator occurs because $V_{pp} = 2\sqrt{2}V_{rms}$, where you square $2\sqrt{2}$ is equal to 8.
- The maximum output power occurs when the amplifier is producing the maximum peak-to-peak output voltage, as shown in fig. (b). In this case, V_{pp} equals the maximum peak to peak output voltage and maximum output power is:

$$P_{out(max)} = \frac{MPP^2}{8R_L}$$

Transistor Power Dissipation

$$\Rightarrow P_{DQ} = V_{CEQ} I_{CQ}$$

Efficiency:-

The dc power supplied to an amplifier by the dc source is:

$$P_{dc} = V_{cc} I_{dc}$$

$$\eta = \frac{P_{out}}{P_{dc}} \times 100\%$$

The maximum efficiency of a class A amplifier with a dc collector resistance and a separate load resistance is 25 percent.

Conclusion:-

- Efficiency can also be defined as amplifier's ability to convert its dc input power to useful ac output power.

Problems:-

- 2) If the peak to peak output voltage is 18V and the input impedance of base is 100Ω . What is the power gain in Fig (a).

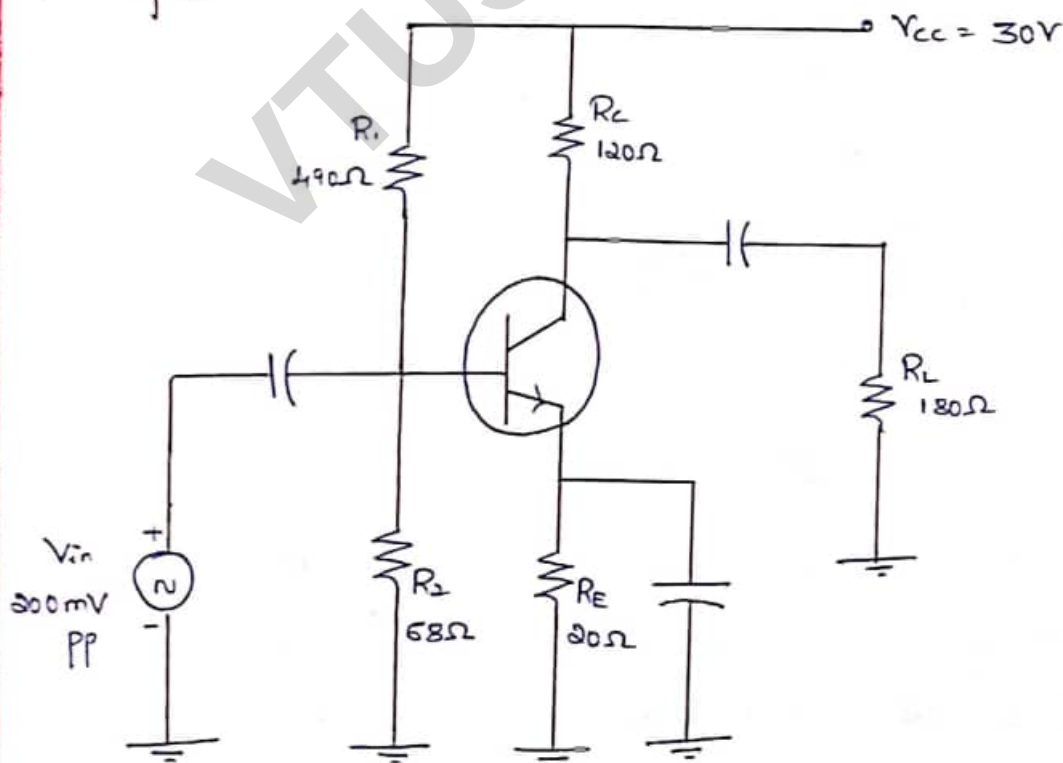
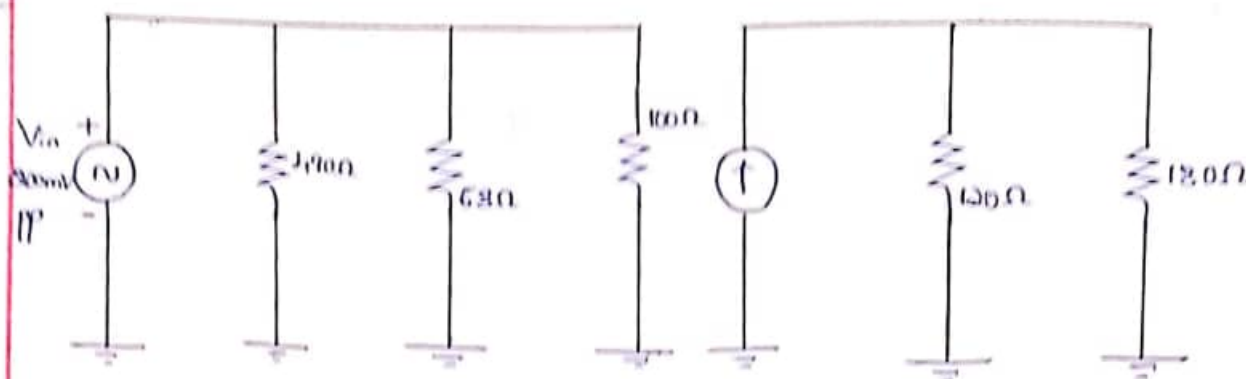


Fig (a)

Solution:



As shown in above fig

$$Z_{in}[\text{stage}] = 1k\Omega \parallel 5k\Omega \parallel 100\Omega = \underline{\underline{37.1\Omega}}$$

The ac input power is:

$$P_{in} = \frac{[200mV]^2}{8[37.1]} = \underline{\underline{133.7\mu W}}$$

The ac output power is:

$$P_{out} = \frac{[18V]^2}{8[120\Omega]} = \underline{\underline{225mW}}$$

The Power gain is:

$$A_p = \frac{225mW}{133.7\mu W} = \underline{\underline{1683}}$$

Emitter - Follower Power Amplifier

When the emitter follower is used as class A power amplifier at the end of a system, a designer will usually locate the Q point at the centre of ac load line to get maximum peak-to-peak [MPP] Output.

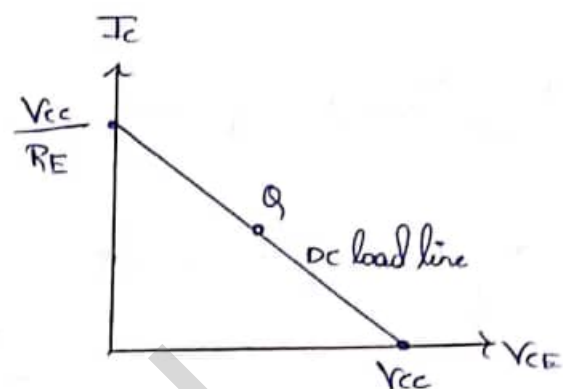
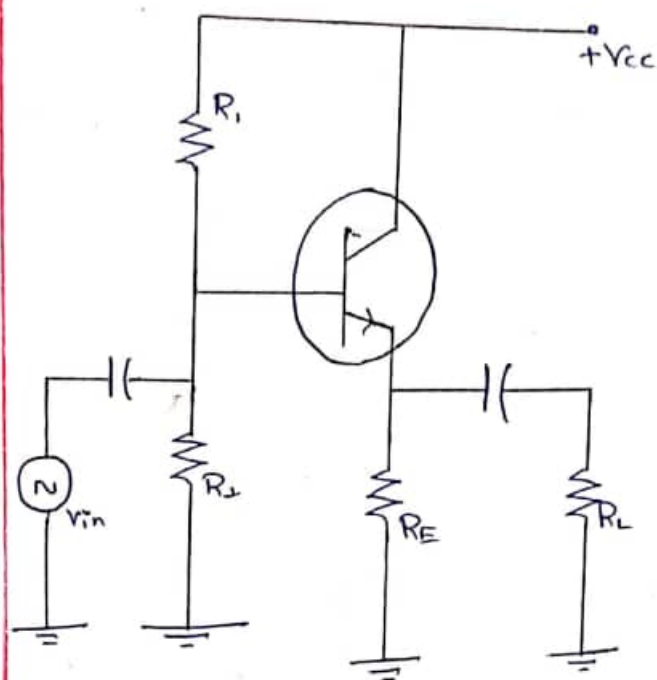


Fig (b)

Fig (a)

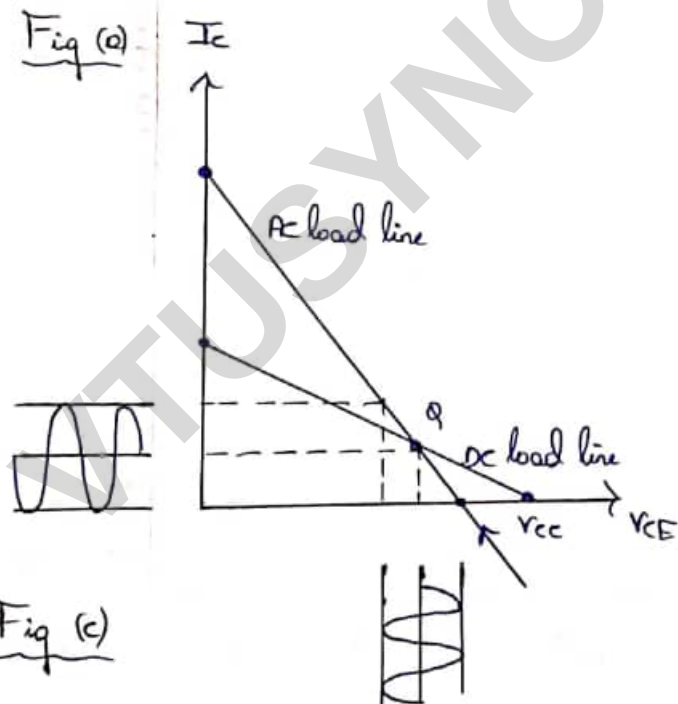


Fig (c)

$$I_C [sat] = \frac{V_{CC}}{R_E}$$

$$V_{CE} [cut-off] = V_{CC}$$

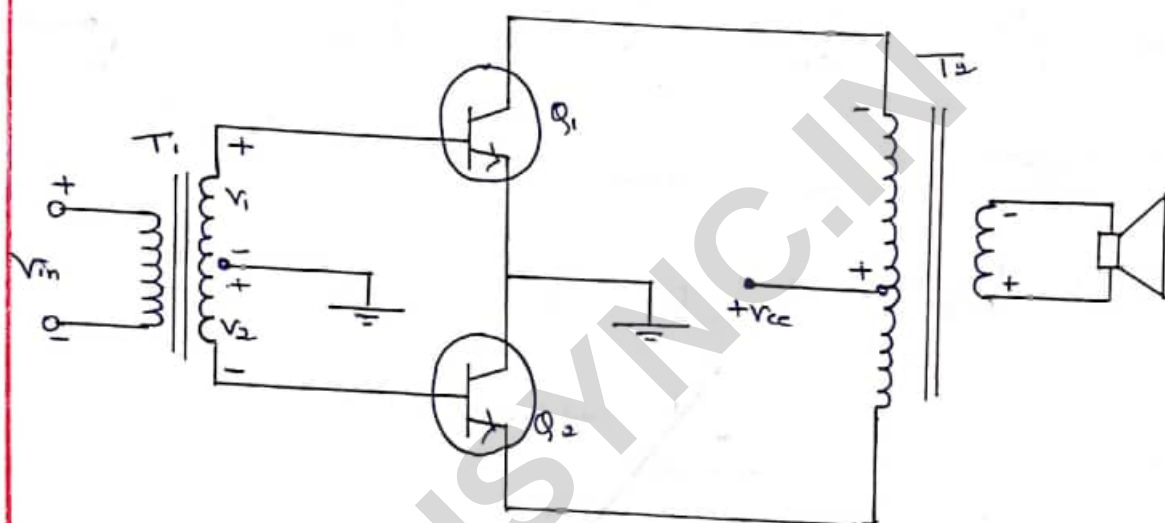
$$i_c [sat] = I_{CQ} + \frac{V_{CE}}{r_{ce}}$$

and

$$V_{CE} [\text{cutoff}] = V_{CE} + I_{CQ} r_{ce}$$

5.4 Class B Operation

- Push-pull Circuit



- Positive Half cycle of input voltage :-

The secondary winding T_2 has voltage v_1 and v_2 as shown, Therefore, the upper transistor conducts and the lower one cuts off. The collector current through Q_1 flows through the upper half of output primary winding. This produces an amplified and inverted voltage, which is transformed-coupled to the loudspeaker.

• Next half cycle of input voltage :-

The Polarities reverse, Now the lower transistor turns on and the upper transistor turns off. The lower transistor amplifies the signal, and the alternate half cycle appears across the loudspeaker.

Since each transistor amplifies one-half of input cycle, the loudspeaker receives a complete cycle of amplified signal.

Advantages of Push-Pull Amplifier

- * Since there is no bias, each transistor is at cut-off when there is no input signal, an advantage because there is no current drain when the signal is zero.
- * Another advantage is improved efficiency where there is an input signal
- * The Maximum efficiency of Class B push-pull amplifier is 78.5 percent.

Disadvantages of Push-Pull Amplifier

- * The Main disadvantage of amplifier is the use of transformers. Audio transformers are bulky and Expensive

5.5

Class B Push-Pull Emitter follower

Class B operation means that the collector current flows for only 180° of ac cycle.

The advantage of class B amplifiers is lower current drain and higher stage efficiency.

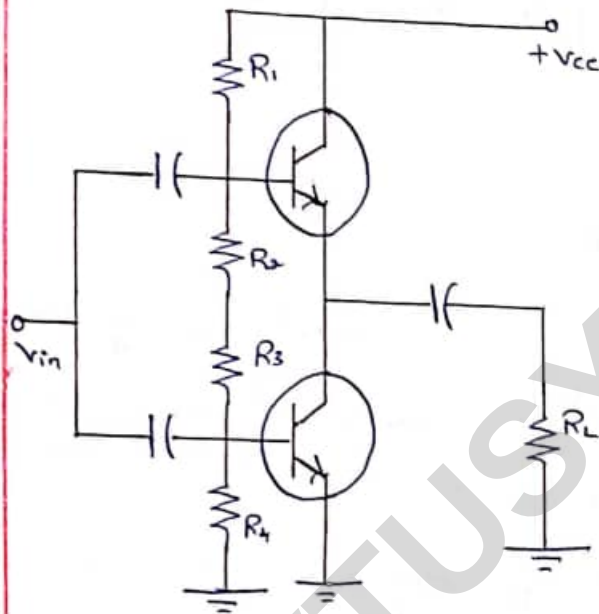
Push-Pull Circuit

Fig (a). Complete circuit

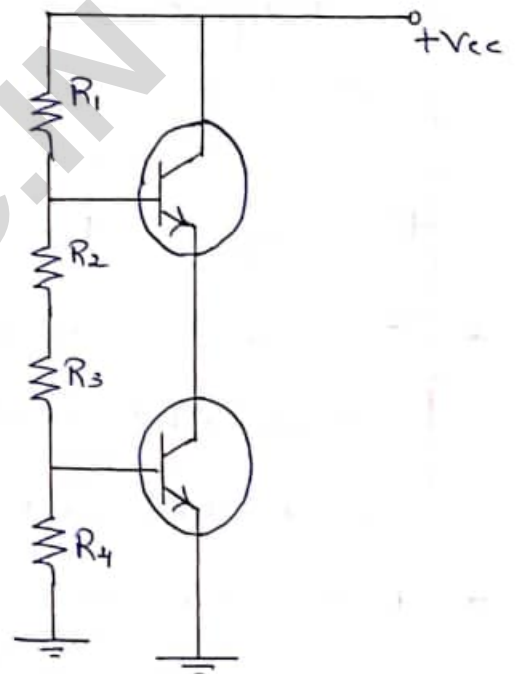


Fig (b) dc Equivalent

$$I_{CQ} = 0$$

$$V_{CEQ} = \frac{V_{CC}}{2}$$

AC load line

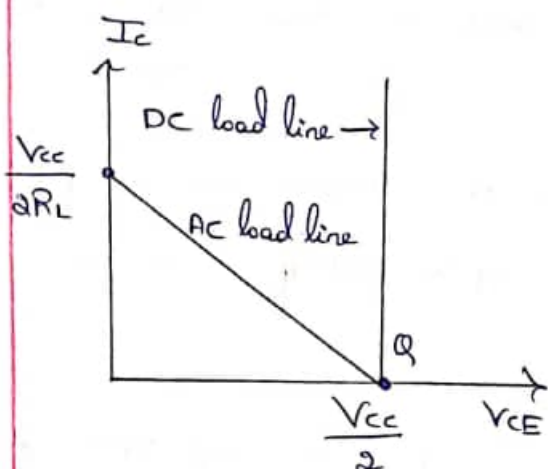


Fig (a) DC and AC load lines

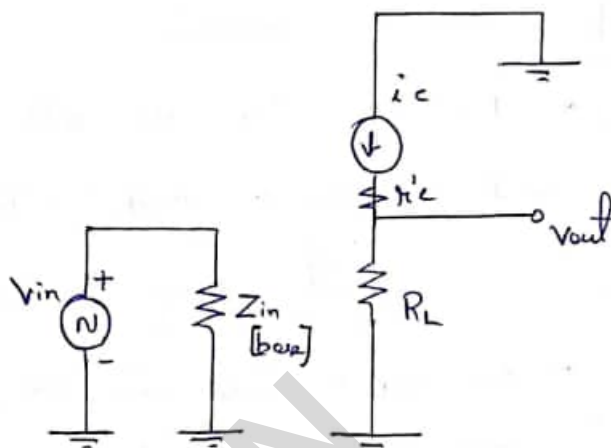


Fig (b) ac equivalent circuit

Maximum Peak-to-Peak output is:
 $MPP = V_{CC}$

AC Analysis

The voltage gain is $A_v \approx 1$ and input impedance of base is
 $Z_{in} [base] \approx \beta R_L$

Overall Action

- Positive Half Cycle:- * The upper transistor conducts and the lower-one cuts off.
* The upper transistor acts like an ordinary emitter follower so that the output voltage approximately equals the input voltage.

• Negative Half Cycle of Input Voltage:-

- * The upper transistor cuts off and the lower transistor conducts.
- * The lower transistor acts like an ordinary emitter follower and produces a load voltage approximately equal to input voltage.
- * The upper transistor handles the positive half cycle of input voltage, and the lower transistor takes care of the negative half cycle.
- * During either half cycle, the source sees a high input impedance looking into either base.

Conclusion:-

Some power amplifiers are biased to operate as class AB amplifiers to improve the linearity of output signal.

A Class AB amplifier has a conduction angle of roughly 210° . The improved linearity of output signal does not come without price. However, a reduction in the circuit's efficiency.

Crossover Distortion

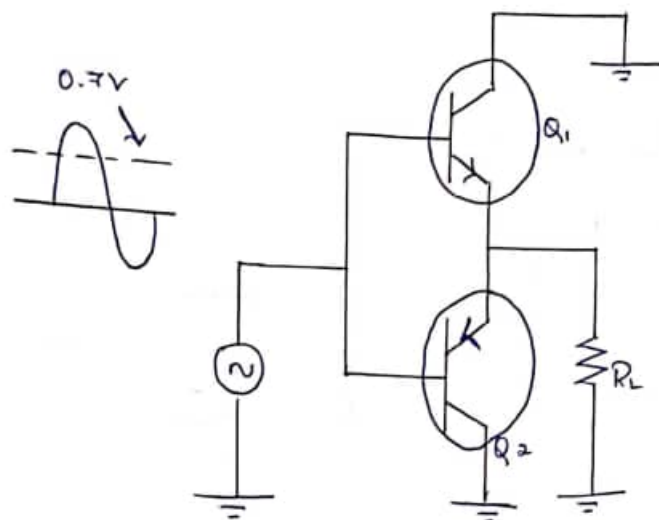


Fig (a) AC Equivalent circuit

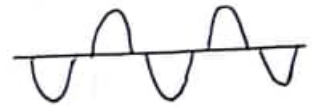


Fig (b) Crossover distortion

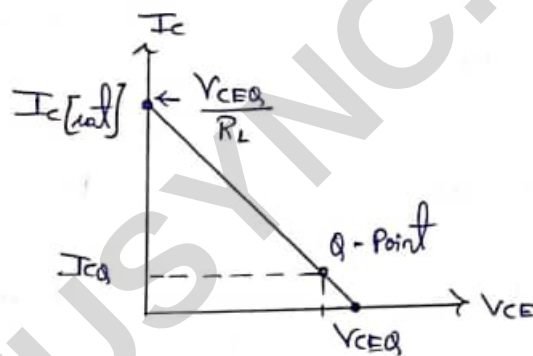


Fig (c) Q-Point is slightly

Class AB

Defined as a conduction angle between 180° and 360°

Transistor Power Dissipation

The maximum transistor power dissipation of each transistor

$$P_D [\max] = \frac{MPP^2}{40R_L}$$

Power Formulas

Amplifier Power Formulas	
Equation	Value
$A_p = \frac{P_{out}}{P_{in}}$	Power gain
$P_{out} = \frac{V_{out}^2}{8R_L}$	AC Output power
$P_{out(max)} = \frac{MPP^2}{8R_L}$	Maximum ac output power
$P_{dc} = V_{cc} I_{dc}$	DC input power
$\eta = \frac{P_{out}}{P_{in}} \times 100\%$	Efficiency

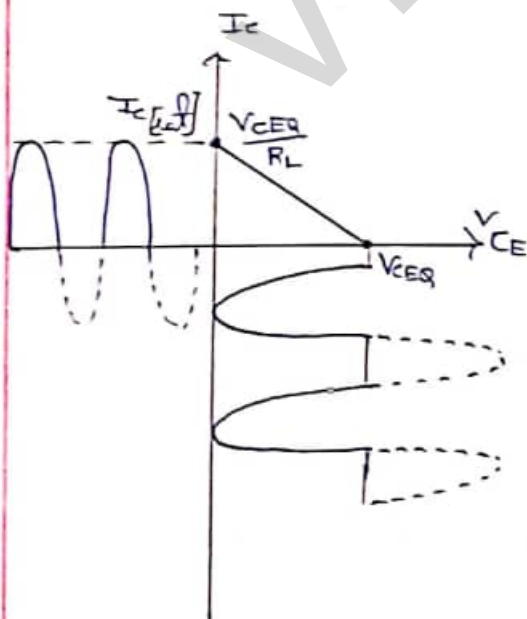


Fig (a) Class B load line

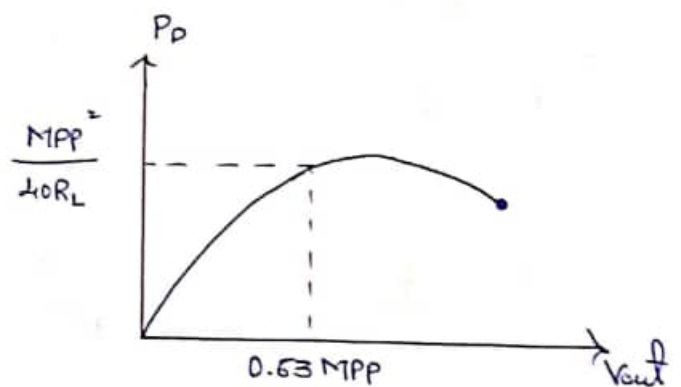
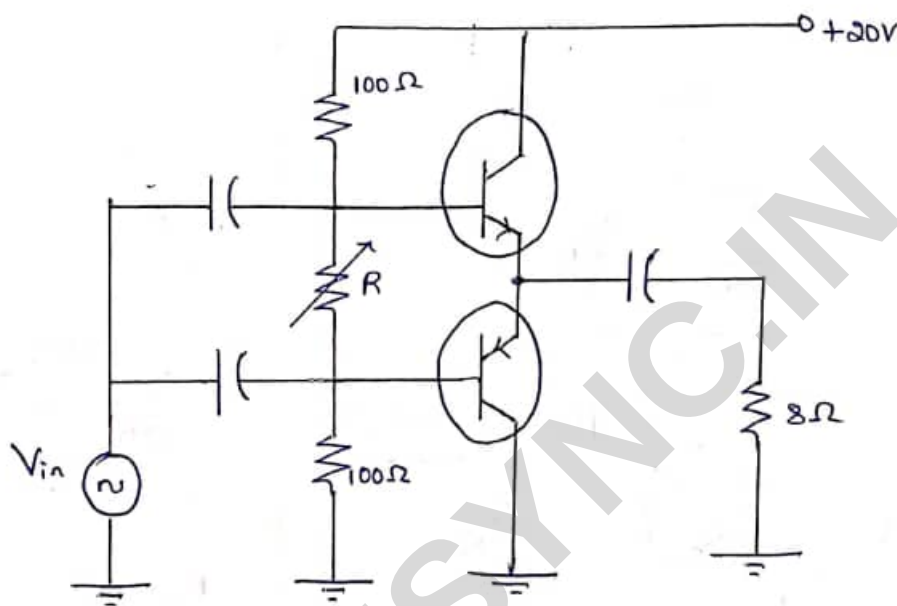


Fig (b) transistor Power dissipation

Problem:-

- 3) The Adjustable resistor sets both emitter diodes on verge of conduction. What is the maximum transistor power dissipation? The maximum output power?



Soln:-

The maximum Peak-to-Peak output is:-

$$MPP = V_{CC} = 20V$$

By Transistor Power Dissipation,

$$P_o [max] = \frac{MPP^2}{40R_L} = \frac{[20V]^2}{40(8\Omega)} = \underline{\underline{1.25W}}$$

The maximum output power is

$$P_{out} [max] = \frac{MPP^2}{8R_L} = \frac{[20V]^2}{8[8\Omega]} = \underline{\underline{6.25W}}$$

5.6 Class C Operation

- Class C amplifiers is called as tuned Amplifiers

Resonant frequency

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

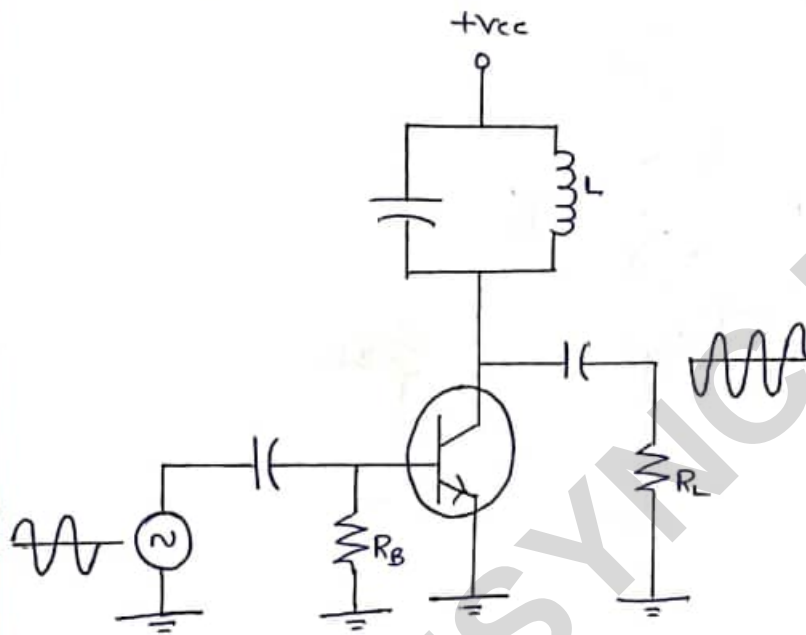


Fig (a) Tuned class C Amplifier

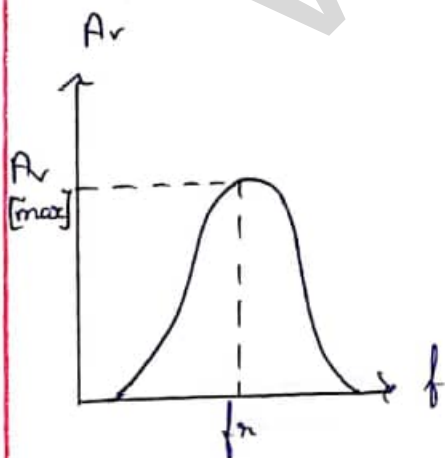


Fig (b) Voltage gain versus frequency

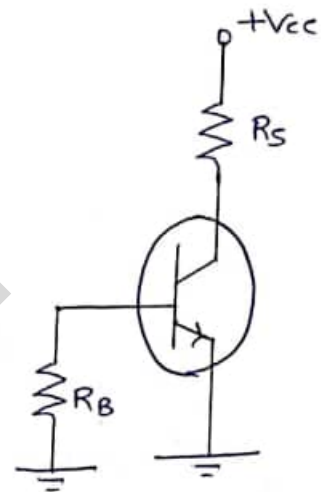


Fig (c) dc equivalent circuit unbiased

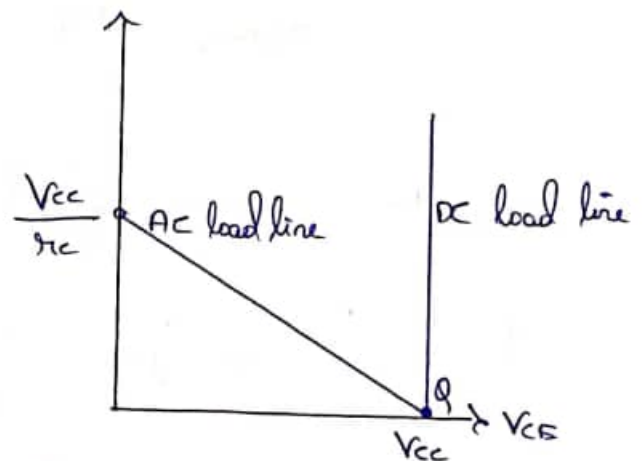


Fig (d) Two load lines

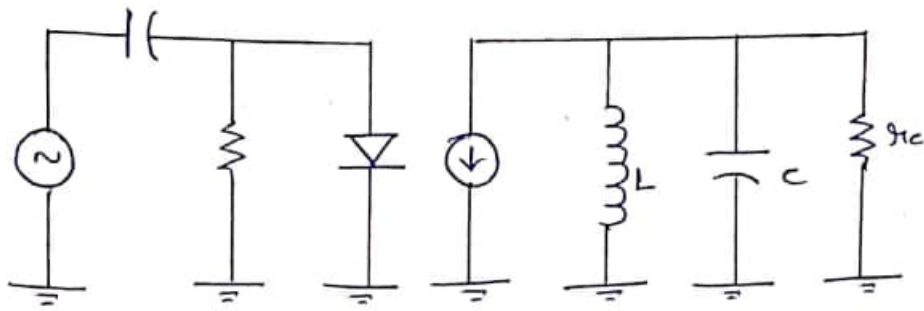


Fig (e) ac equivalent circuit

Filtering the Harmonics

f = Fundamental frequency

Harmonics = $2f, 3f, \dots, nf$

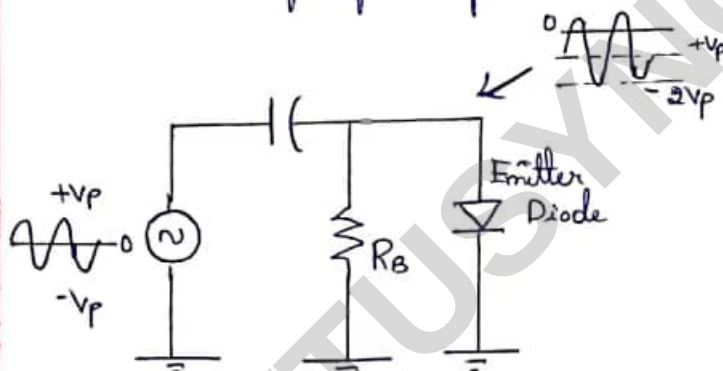


Fig (e) Input signal is negatively clamped at base

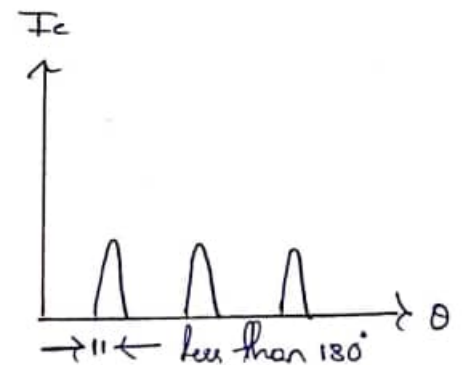


Fig (b) Collector current flows in pulses

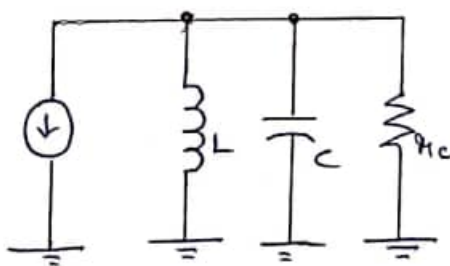


Fig (c) ac collector circuit

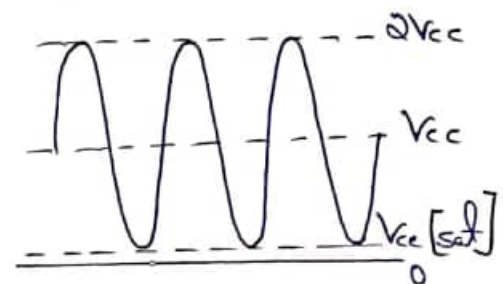


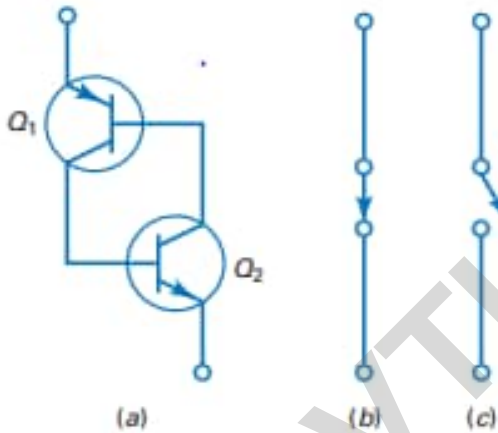
Fig (d) Collector voltage waveform

VTUSYNC.IN

Thyristor

The Four-Layer Diode

Figure 13-1 Transistor latch.



- Thyristor operation :
- The upper transistor Q1 is a pnp device, and the lower transistor Q2 is an npn device.
- The collector of Q1 drives the base of Q2. Similarly, the collector of Q2 drives the base of Q1.

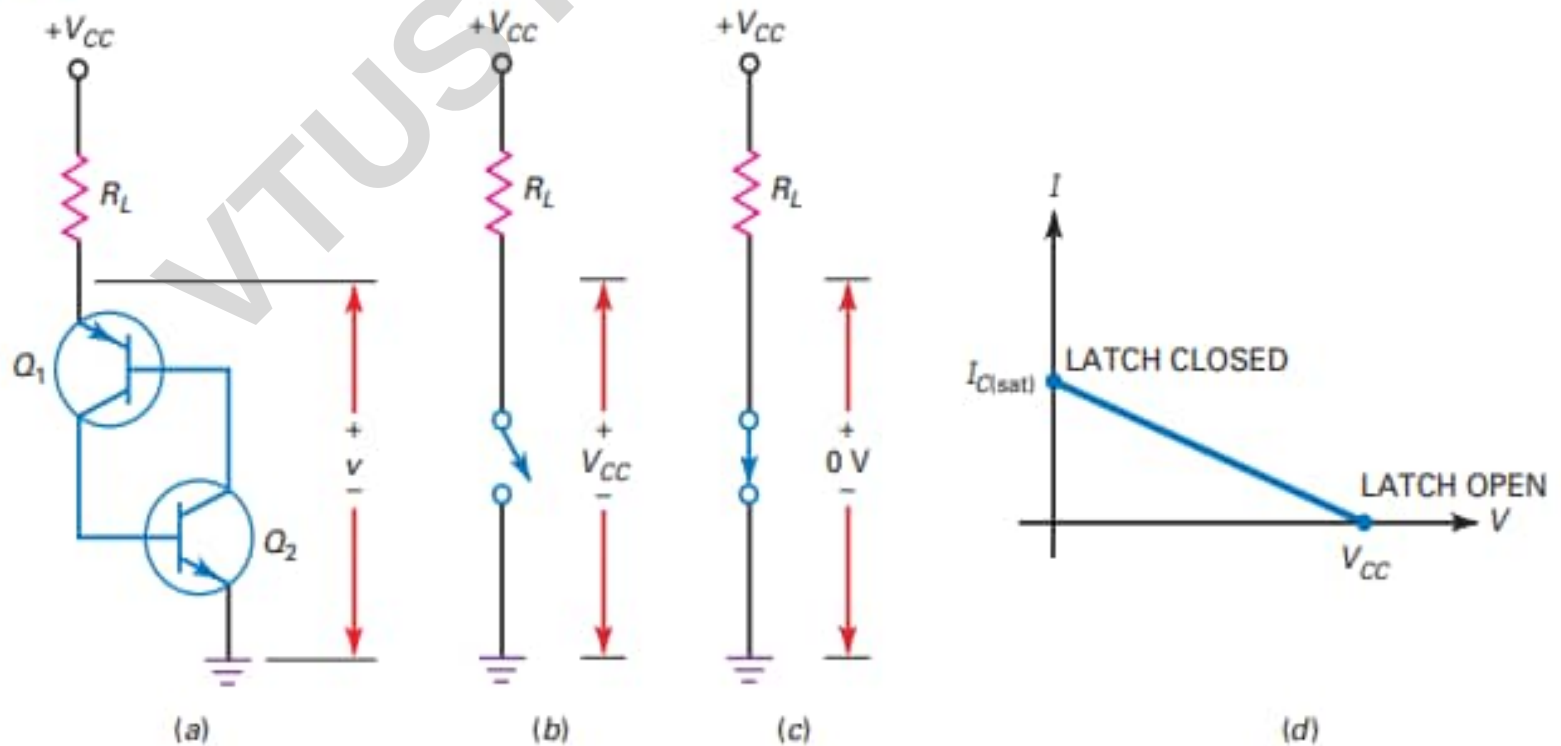
- Positive Feedback: This uses a positive feedback.
- If the base current of Q2 increases, the collector current of Q2 increases. This increases the base current of Q1 and the collector current of Q1. This amplify-and-feedback action continues until both transistors are driven into saturation and circuit acts like a closed switch (Fig. 13-1b).

The Four-Layer Diode

- On the other hand, if something causes the base current of Q2 to decrease, the collector current of Q2 decreases, the base current of Q1 decreases, the collector current of Q1 decreases, and the base current of Q2 decreases further. This action continues until both transistors are driven into cutoff. Then, the circuit acts like an open switch (Fig. 13-1c).
- Therefore the circuit of Fig. 13-1a is stable in either of two states: open or closed. It will remain in either state indefinitely until acted on by an outside force.

The Four-Layer Diode

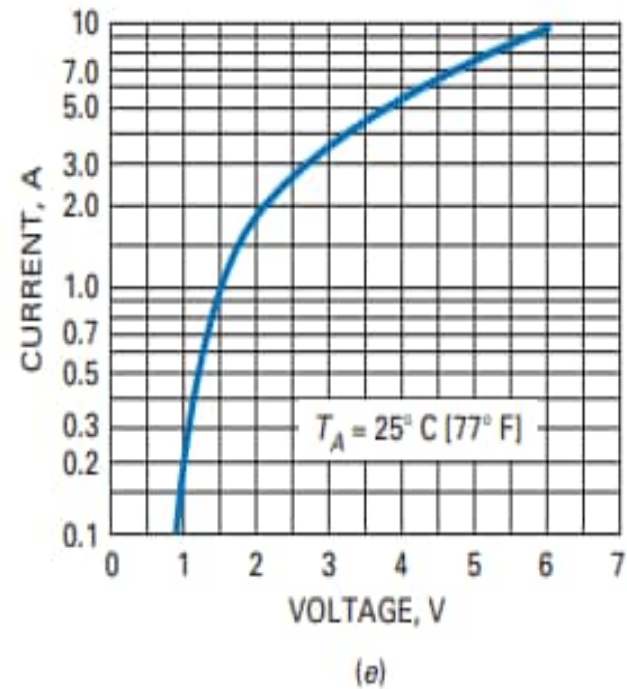
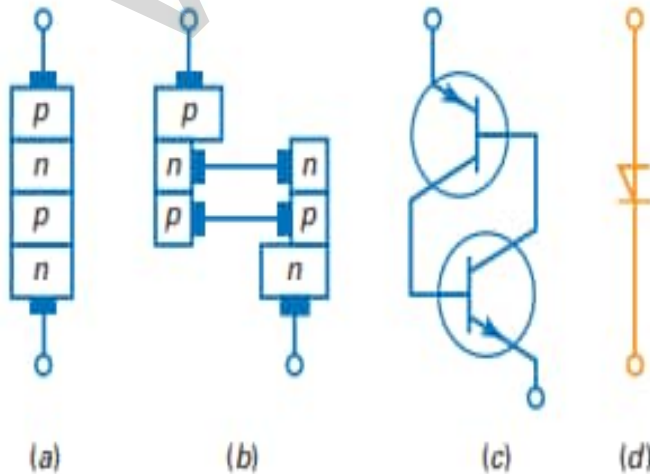
Figure 13-2 Latching circuit.



- Figure 13-2a shows a latch connected to a load resistor with a supply voltage of V_{CC} .
- Let the latch be open, as shown in Fig. 13-2b. As there is no current through the load resistor, the voltage across the latch equals the supply voltage.
- So, the operating point is at the lower end of the dc load line (Fig. 13-2d).
- The only way to close the latch of Fig. 13-2b is by breakover. This means using a large enough supply voltage V_{CC} to break down the Q1 collector diode.
- Since the collector current of Q1 increases the base current of Q2, the positive feedback will start. This drives both transistors into saturation.
- When saturated, both transistors ideally look like short-circuits, and the latch is closed (Fig. 13-2c).
- Now the latch has zero voltage across it when it is closed and the operating point is at the upper end of the load line (Fig. 13-2d).
- Again to open the latch reduce the V_{CC} supply to zero. This forces the transistors to switch from saturation to cutoff.

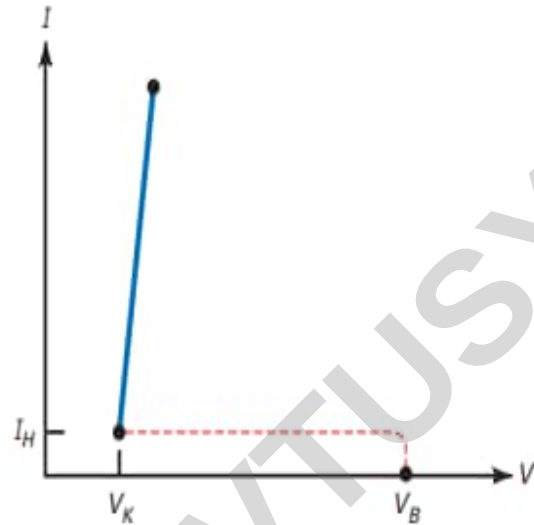
The Shockley Diode

Figure 13-3 Four-layer diode.



- Schockley diode, also known as four-layer diode, pnpn diode, and silicon unilateral switch (SUS).
- This device lets current flow in only one direction.
- The only way to close a four-layer diode is by breakover. The only way to open it is by low-current drop-out, which means reducing the current to less than the holding current (given on data sheets).
- The holding current is the low value of current where the transistors switch from saturation to cutoff

Figure 13-4 Breakover characteristic.



- Figure 13-4 shows the graph of current versus voltage of a four-layer diode.
 - The device has two operating regions: cutoff and saturation.
 - The dashed line is the transition path between cutoff and saturation.
 - It is dashed to indicate that the device switches rapidly between the off and on states
-
- When the device is at cutoff, it has zero current.
 - If the voltage across diode tries to exceed V_B , the device breaks over and moves rapidly along the dashed line to the saturation region.
 - When the diode is in saturation, it is operating on the upper line.
 - As long as the current through it is greater than the holding current I_H , the diode remains latched in the on state.
 - If the current becomes less than I_H , the device switches into cutoff.
 - The value of knee voltage V_K , approximately 0.7 V

The Silicon Controlled Rectifier

- The SCR is the most widely used thyristor. It can switch very large currents on and off. Because of this, it is used to control motors, ovens, air conditioners, and induction heaters.

Figure 13-7 Transistor latch with trigger input.

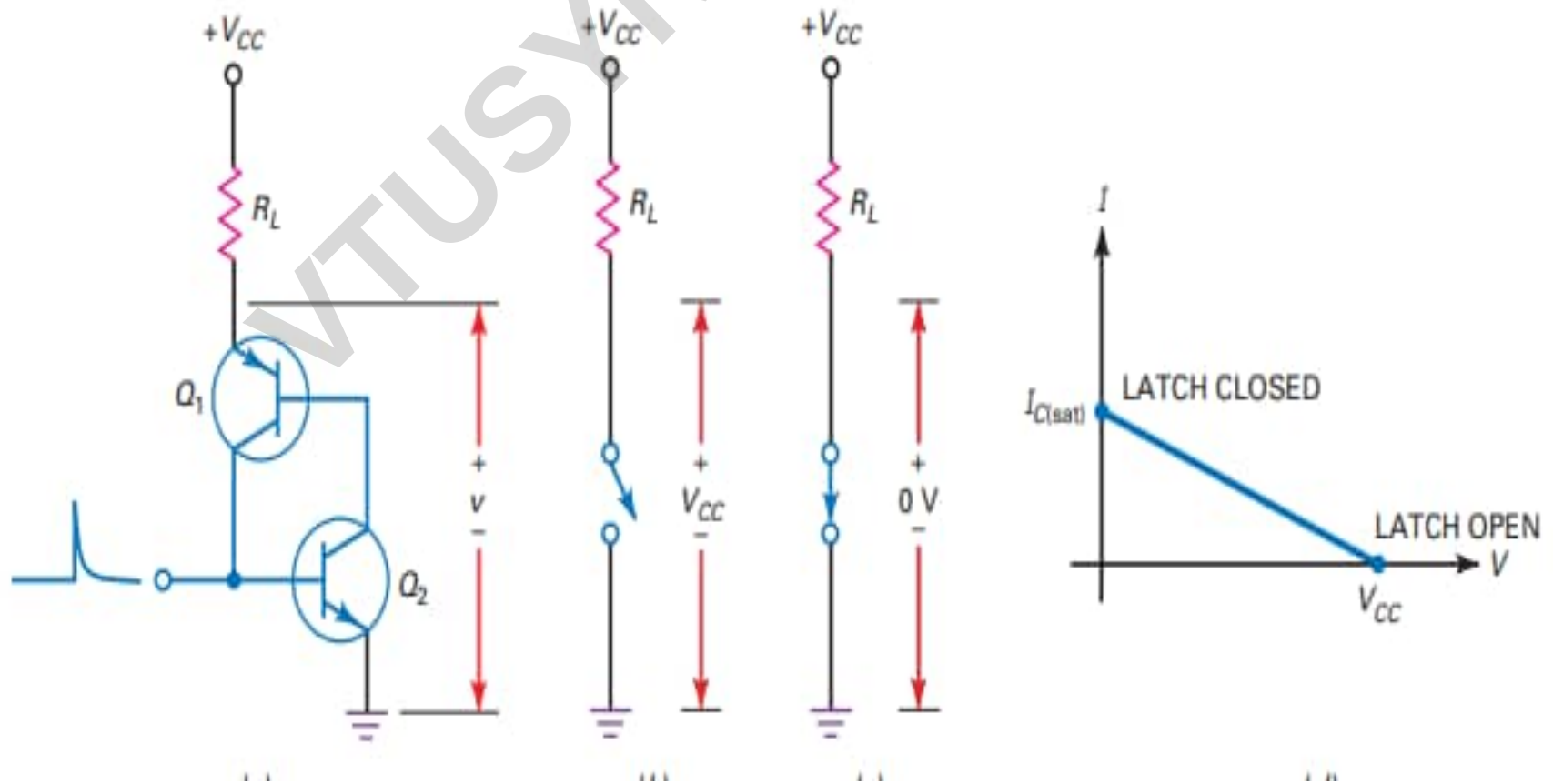
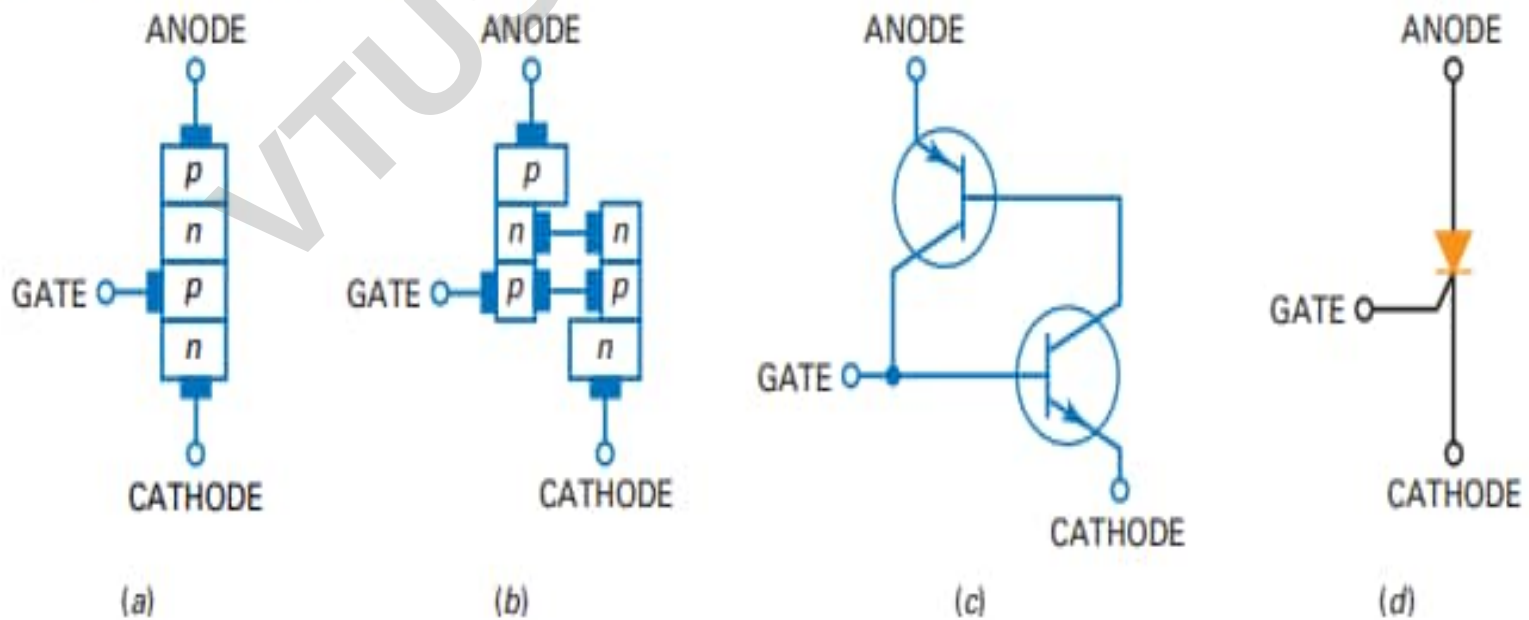
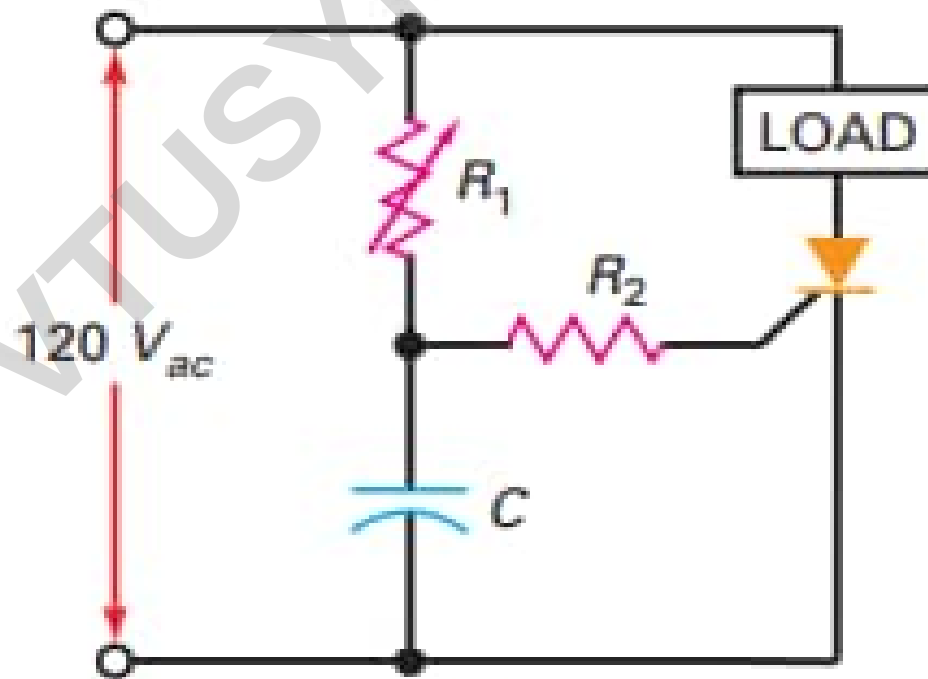


Figure 13-8 Silicon controlled rectifier (SCR).



SCR phase control.



(a)

RC Circuit Controls Phase Angle

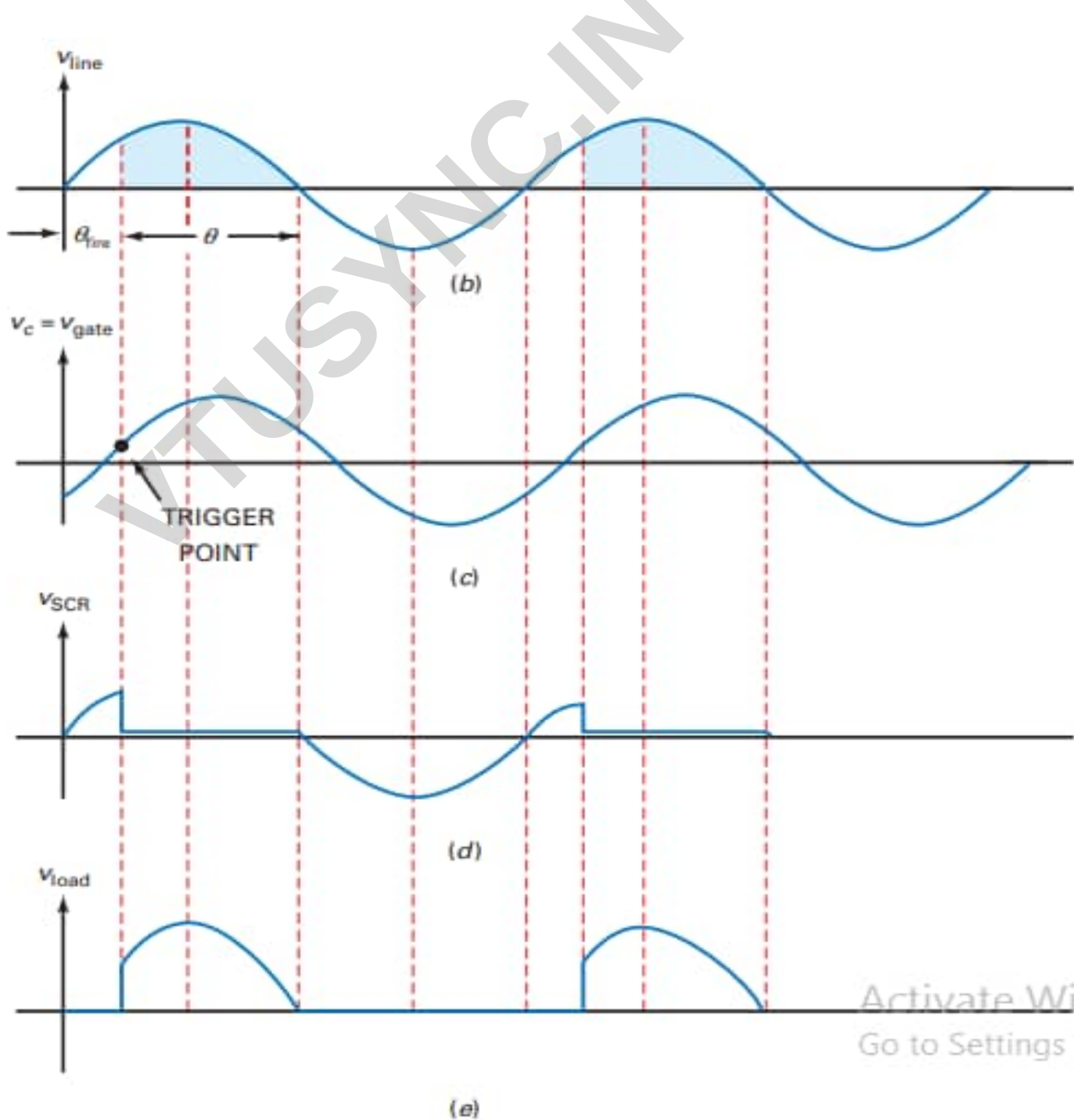
- Figure 13-22a shows ac line voltage being applied to an SCR circuit that controls the current through a heavy load. In this circuit, variable resistor R1 and capacitor C shift the phase angle of the gate signal.
- When R1 is zero, the gate voltage is in phase with the line voltage, and the SCR acts like a half-wave rectifier. R2 limits the gate current to a safe level.
- When R1 increases, however, the ac gate voltage lags the line voltage by an angle between 0° and 90° , as shown in Figs. 13-22b and c.

RC Circuit Controls Phase Angle

- Before the trigger point shown in Fig. 13-22c, the SCR is off and the load current is zero.
- At the trigger point, the capacitor voltage is large enough to trigger the SCR.
- When this happens, almost all of the line voltage appears across the load and the load current becomes high.
- Ideally, the SCR remains latched until the line voltage reverses polarity. This is shown in Figs. 13-22c and d

RC Circuit Controls Phase Angle

- The angle at which the SCR fires is called the firing angle, shown as α in Fig. 13-22a. The angle between the start and end of conduction is called the conduction angle, shown as θ . The RC phase controller of Fig. 13-22a can change the firing angle between 0° and 90° , which means that the conduction angle changes from 180° to 90° .



Bidirectional Thyristors

Figure 13-24 Diac.

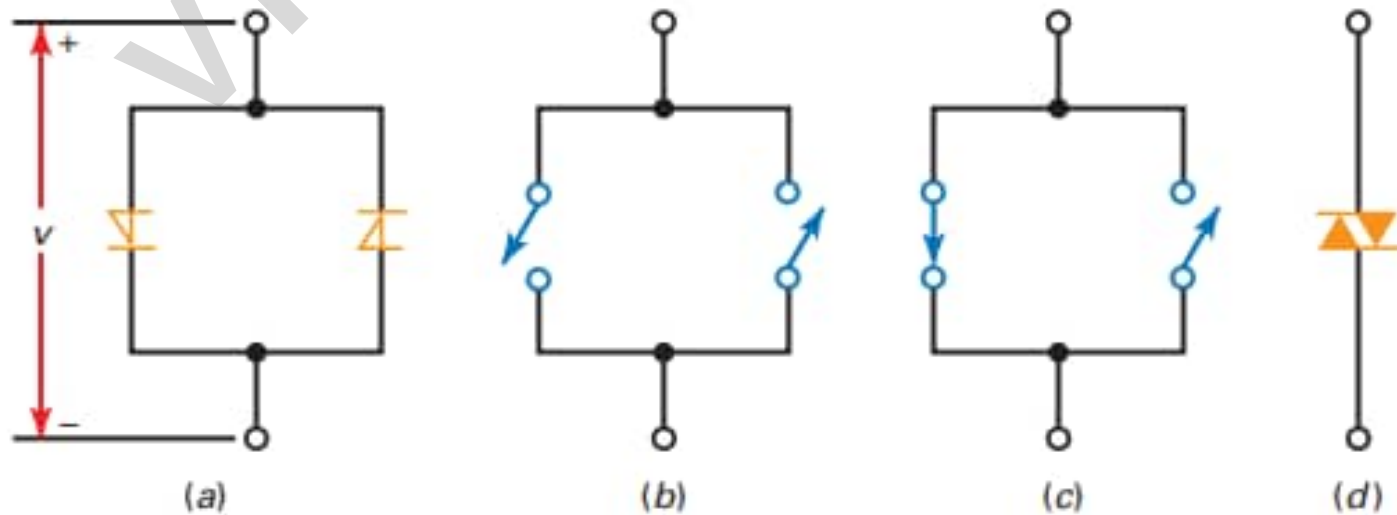
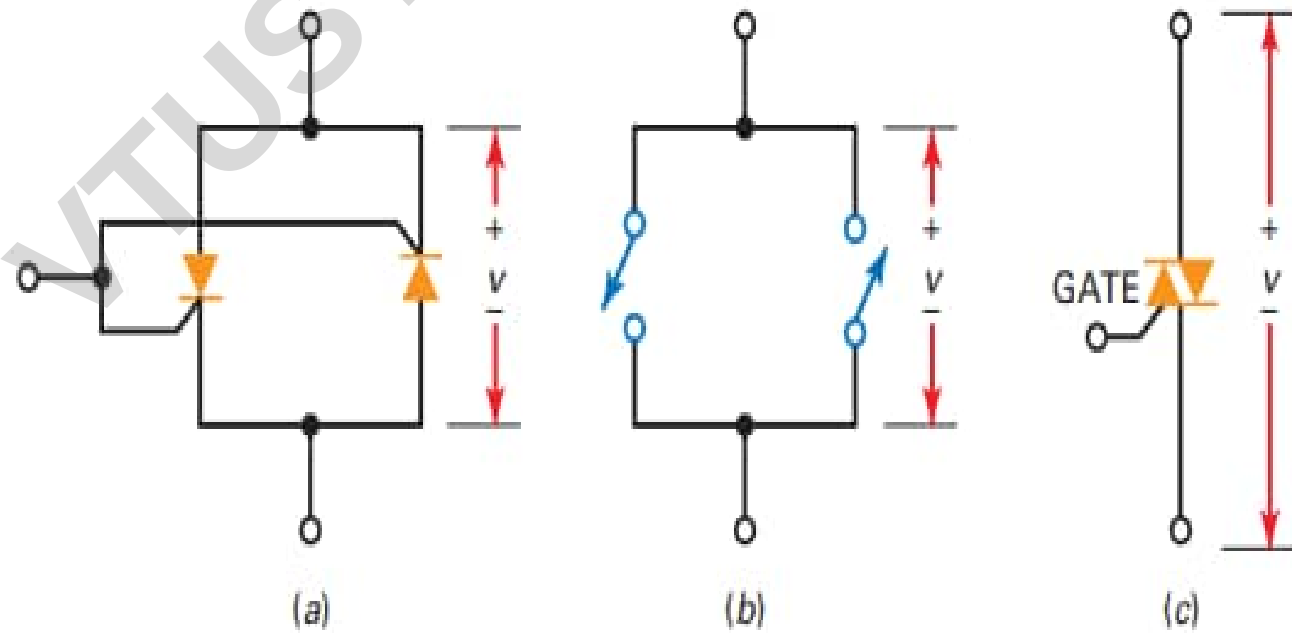
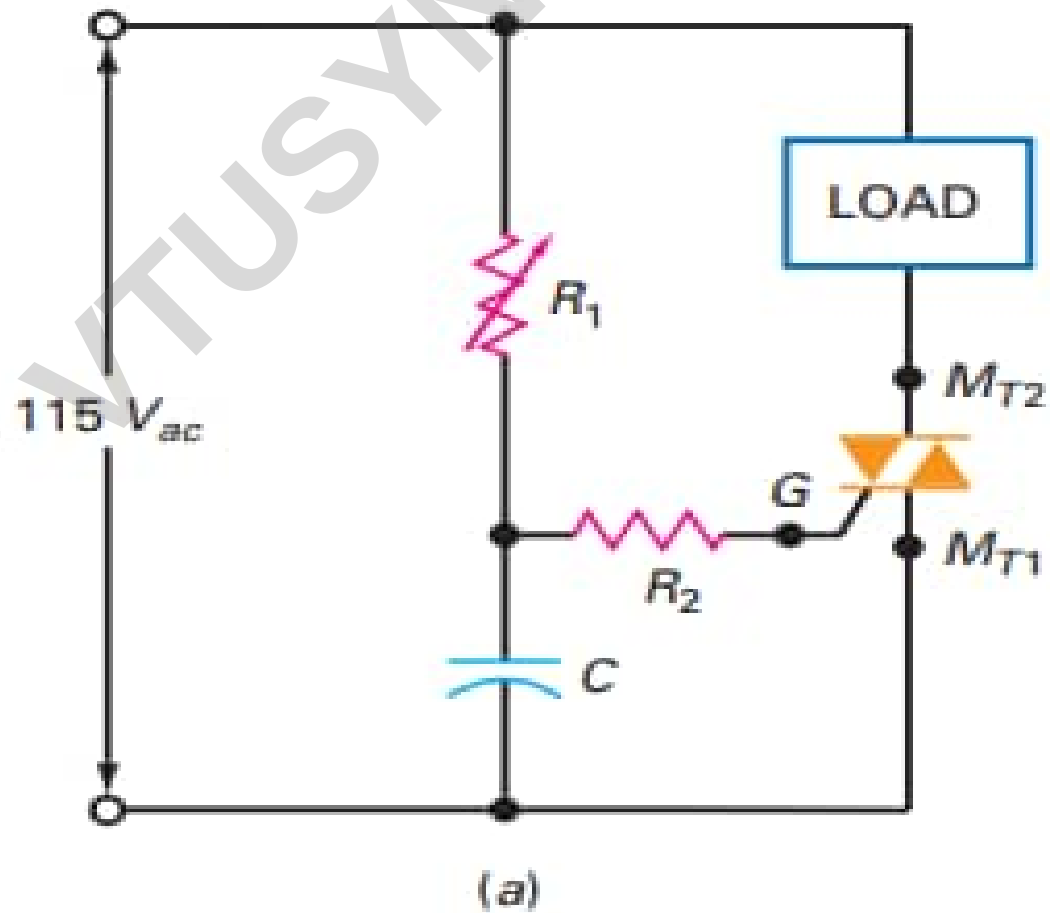
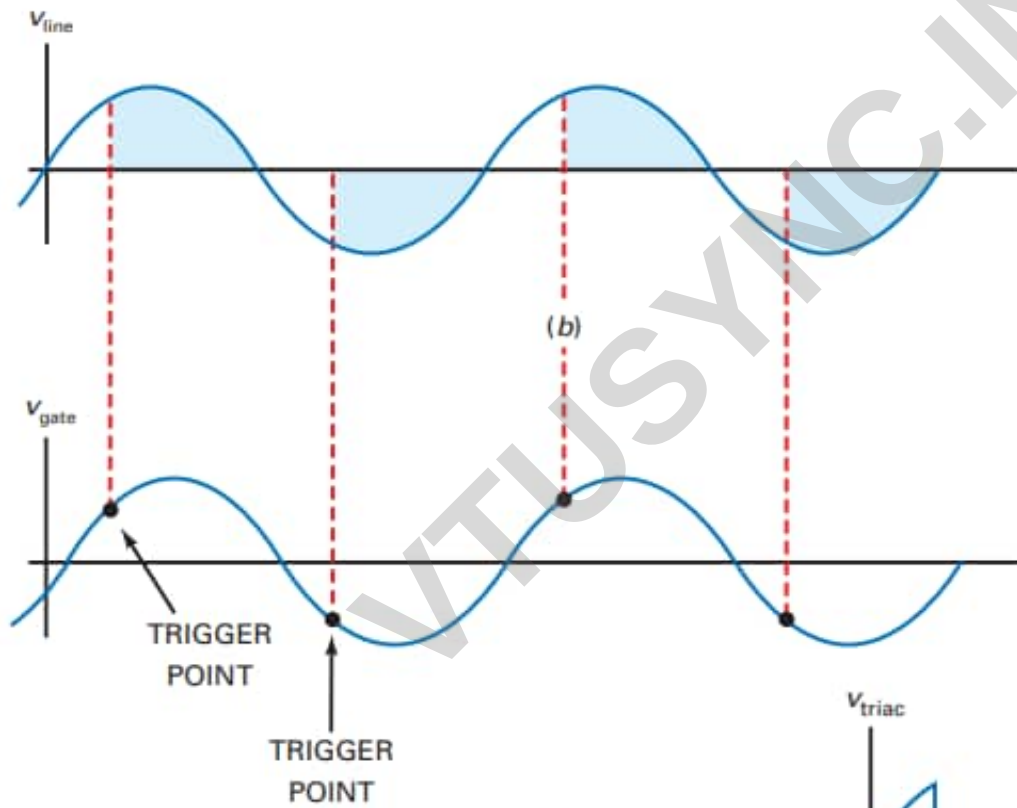


Figure 13-25 Triac.

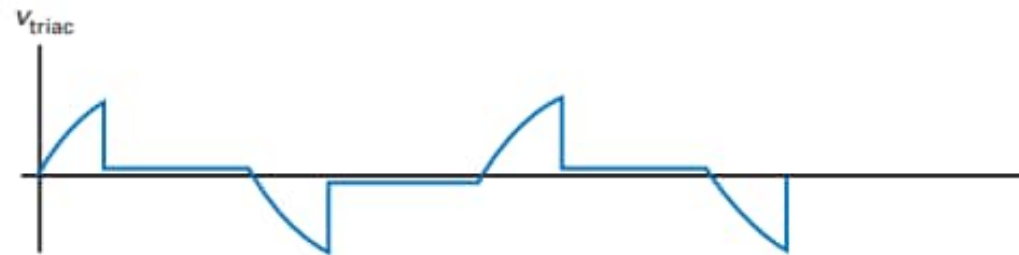


Triac phase control.

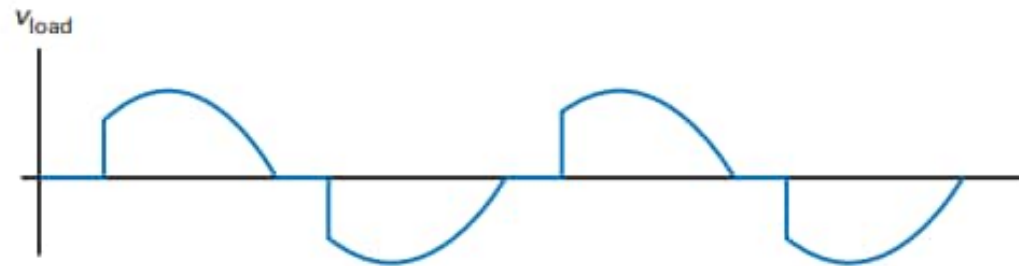




(c)

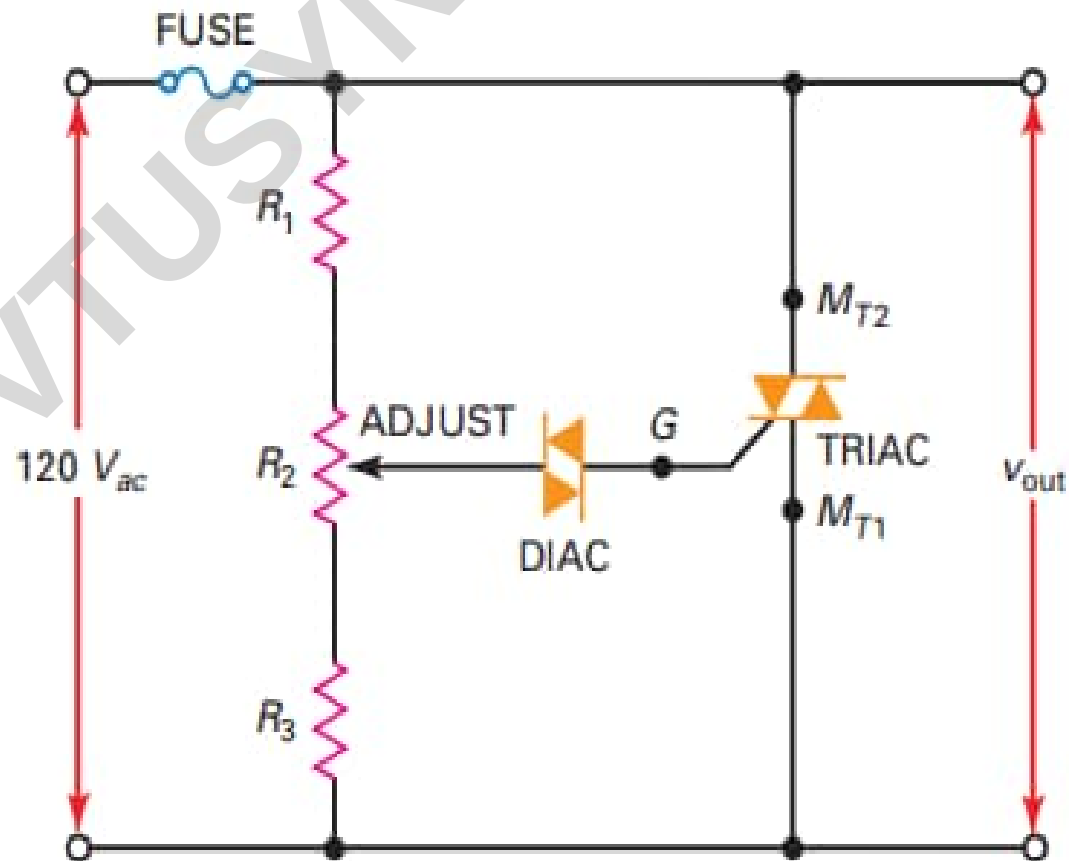


(d)



(e)

Figure 13-28 Triac crowbar.

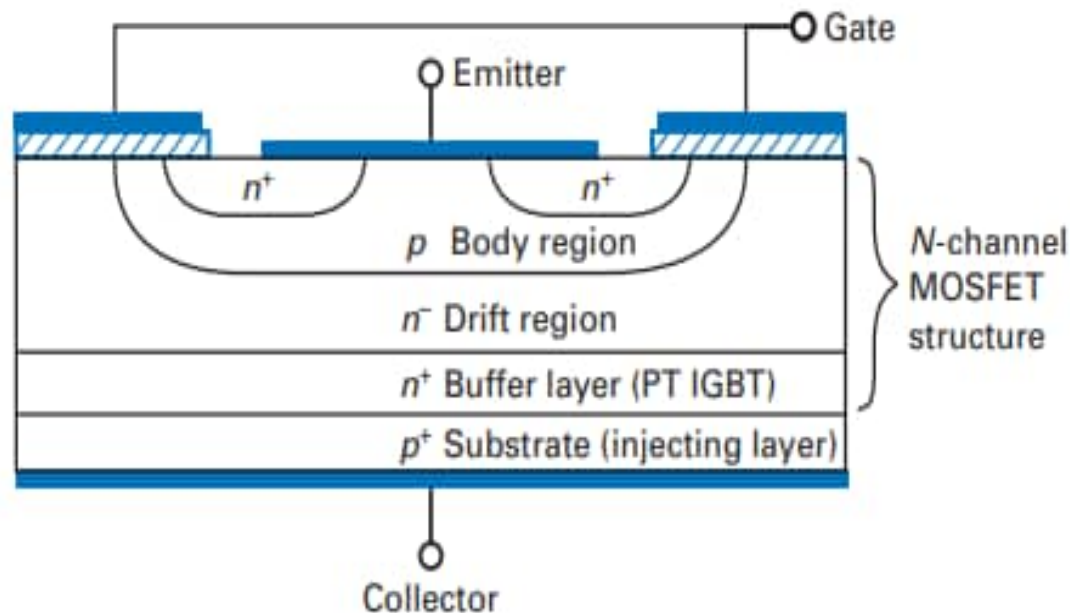


IGBTs-Insulated-Gate Bipolar Transistors

- Power MOSFETs and BJTs can both be used in high-power switching applications. The MOSFET has the advantage of greater switching speed, and the BJT has lower conduction losses. By combining the low conduction loss of a BJT with the switching speed of a power MOSFET, we can begin to approach an ideal switch.

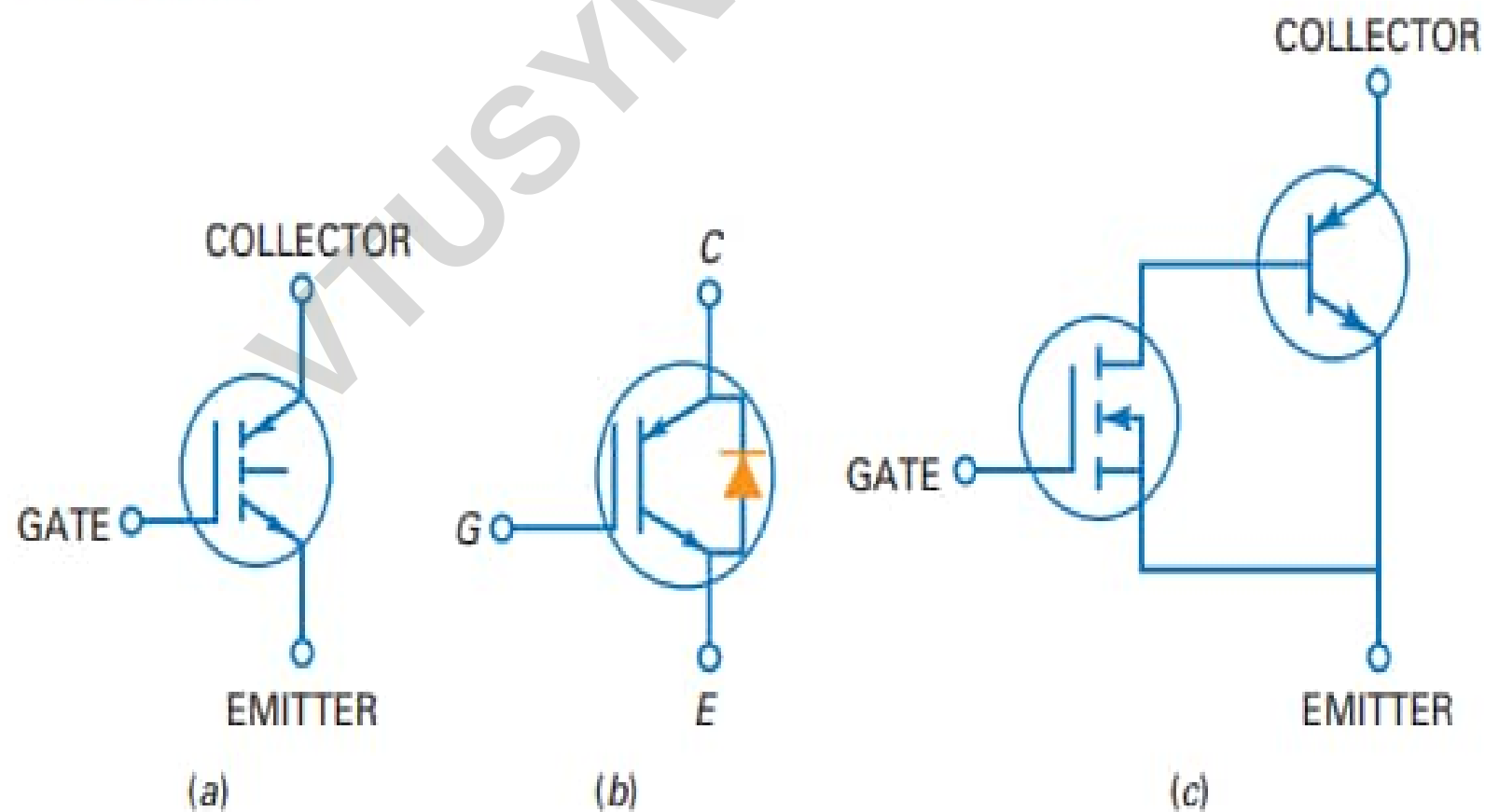
- This hybrid device exists and is called an insulated-gate bipolar transistor (IGBT).
- Its structure and operation closely resembles a power MOSFET.
- The basic structure of an n-channel IGBT is shown. Its structure resembles an n-channel power MOSFET constructed on a p-type substrate. As shown, it has gate, emitter, and collector leads.

- Two versions of IGBTs:
 - Punch-through (PT) and
 - Nonpunch-through (NPT) IGBTs.
- Figure shows the structure of a PT IGBT.
- The PT IGBT has an n^+ buffer layer between its p^+ and n^- regions, and the NPT device has no n^+ buffer layer.



- NPT versions have higher conduction $V_{CE(on)}$ values than PT versions and a positive temperature coefficient.
- The positive temperature coefficient makes the NPT suited for paralleling.
- PT versions, with the extra n^+ layer, have the advantage of higher switching speeds. They also have a negative temperature coefficient.

Figure 13-31 IGBTs: (a) and (b) Schematic symbols; (c) simplified equivalent circuit.



- Figures 13-31a and b show two common schematic symbols for an n-channel IGBT.
- In Fig.b there is intrinsic body-diode.
- This built-in diode is similar to the body-diodes found in power FETs.
- Also, Fig. 13-31c shows a simplified equivalent circuit for this device.
- The IGBT is essentially a power MOSFET on the input side and a BJT on the output side.
- The input control is a voltage between the gate and emitter leads.
- Just as in a power FET, the gate drive circuits for an IGBT must be able to quickly charge up and discharge the IGBT's input capacitance for fast switching speeds.
- The output is a current between the collector and emitter leads. Because the output of the IGBT relies on BJT construction, this results in a slower device turn-off speed than a power FET.

- The IGBT is a normally off high-input impedance device.
- When the input voltage V_{GE} is large enough, collector current will begin to flow.
- This minimum voltage value is the gate threshold voltage $V_{GE(th)}$.
- The typical $V_{GE(th)}$ is 5.0 V when I_C 5 60 mA.
- The maximum continuous collector current is shown to be 60 A.
- Another important characteristic is its collector to emitter saturation voltage $V_{CE(sat)}$.
- The typical $V_{CE(sat)}$ value, shown on the data sheet, is 1.5 V at a collector current of 10 A and 2.5 V at a collector current of 60 A

IGBT Advantages

1. Conduction losses of IGBTs are related to the forward voltage drop of the device, and the MOSFET conduction loss is based on its $R_{DS(on)}$ values. For low-voltage applications, power MOSFETs can have extremely low $R_{D(on)}$ resistances.
2. IGBTs also have a much higher collector-emitter breakdown voltage as compared to the V_{DSS} maximum value of MOSFETs.

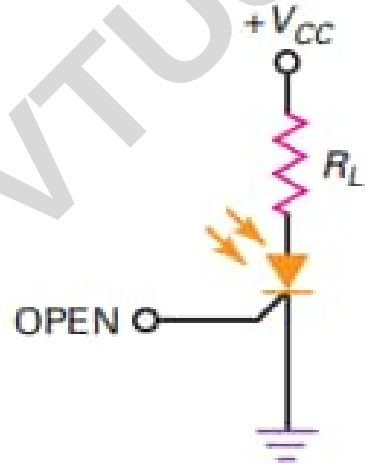
This is important in applications using higher-voltage inductive loads, such as inductive heating (IH) applications. This makes the IGBT ideal for high-voltage full H-bridge and half-bridge circuits

IGBT Advantages

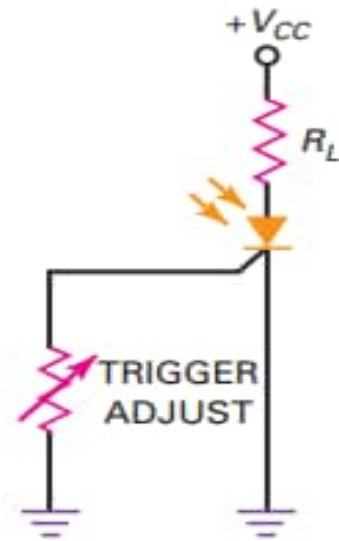
4. As compared to BJTs, IGBTs have a much higher input impedance and have much simpler gate drive requirements. Although the IGBT cannot match the switching speed of the MOSFET, new IGBT families, such as FS IGBTs, are being developed for high-frequency applications. IGBTs are, therefore, effective solutions for high-voltage and current applications at moderate frequencies.

Other Thyristors

- **Photo-SCR(light-activated SCR):**
- The arrows represent incoming light that passes through a window and hits the depletion layers.
- When the light is strong enough, valence electrons are dislodged from their orbits and become free electrons.
- The flow of free electrons starts the positive feedback, and the photo-SCR closes.
- After a light trigger has closed the photo-SCR, it remains closed, even though the light disappears. For maximum sensitivity to light, the gate is left open, as shown in Fig. 13-34a.
- To get an adjustable trip point, we can include the trigger adjust shown in Fig. 13-34b. The resistance between the gate and ground diverts some of the light-produced electrons and reduces the sensitivity of the circuit to the incoming light.



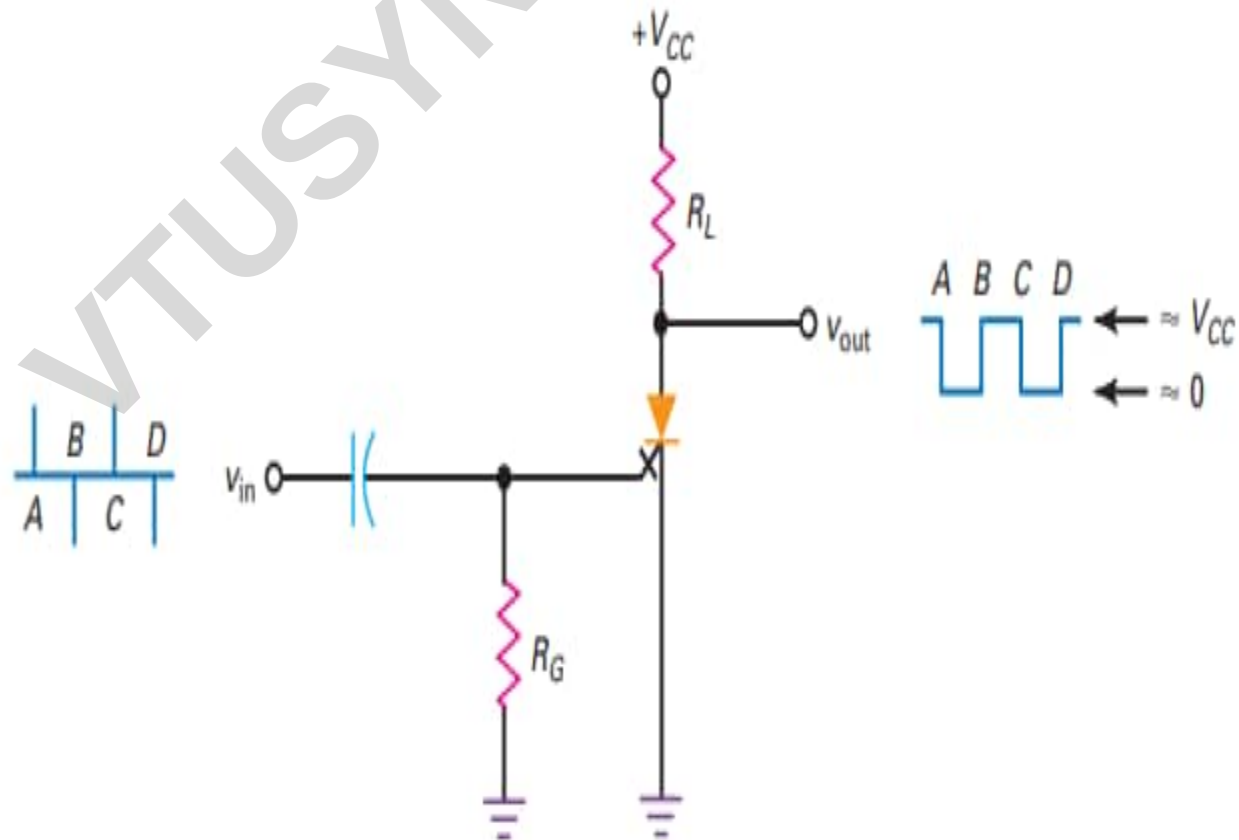
(a)



(b)

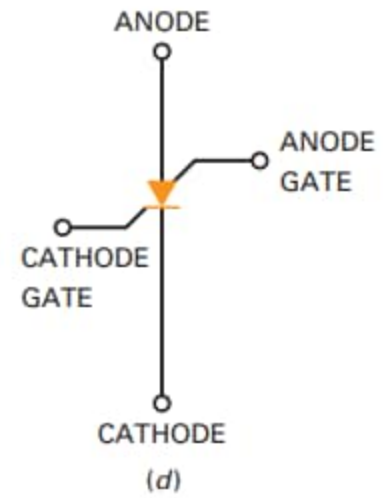
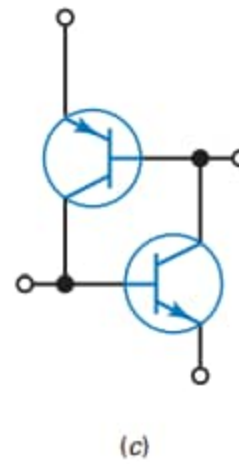
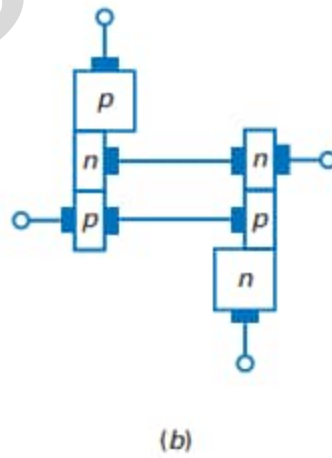
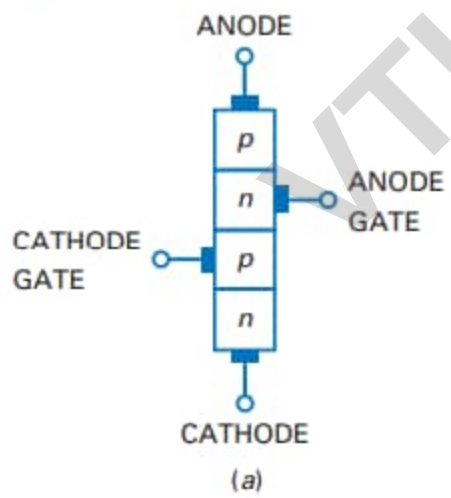
- Gate-Controlled Switch:
- low-current drop-out is the normal way to open an SCR. But the gate-controlled switch is designed for easy opening with a reversebiased trigger.
- A gate-controlled switch is closed by a positive trigger and opened by a negative trigger. Figure 13-35 shows a gate-controlled circuit.
- Each positive trigger closes the gate-controlled switch, and each negative trigger opens it. Because of this, we get the square-wave output shown. The gate-controlled switch has been used in counters, digital circuits, and other applications in which a negative trigger is available

Figure 13-35 Gate-controlled switch.



Silicon Controlled Switch

- Figure 13-36a shows the doped regions of a silicon controlled switch. Now an external lead is connected to each doped region.
- Visualize the device separated into two halves (Fig. 13-36b). Therefore, it's equivalent to a latch with access to both bases (Fig. 13-36c).
- A forward-bias trigger on either base will close the silicon controlled switch.
- Likewise, a reverse-bias trigger on either base will open the device.
- Figure 13-36d shows the schematic symbol for a silicon controlled switch
- . The lower gate is called the cathode gate, and the upper gate is the anode gate.
- The silicon controlled switch is a low-power device compared to the SCR. It handles currents in milliamperes rather than amperes.



Unijunction Transistor and PUT

Figure 13-37 Unijunction transistor.

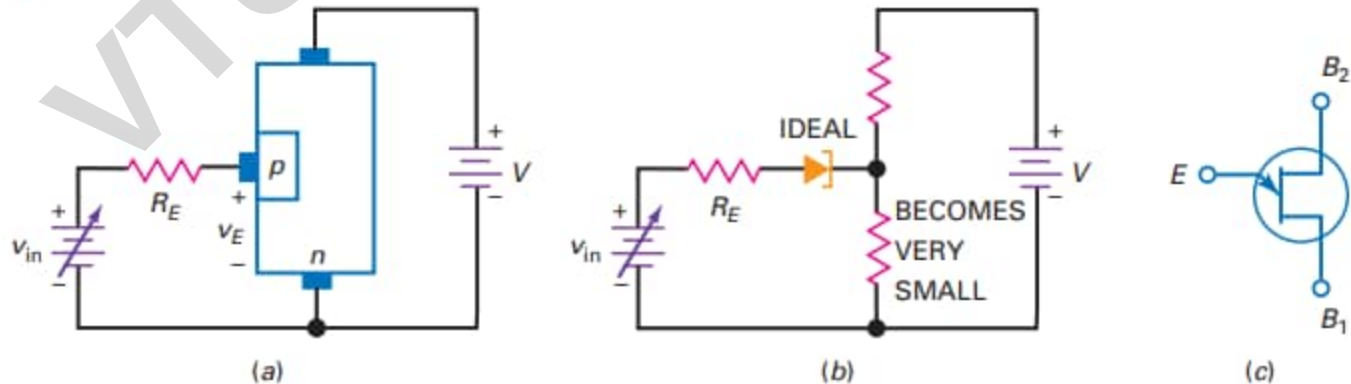
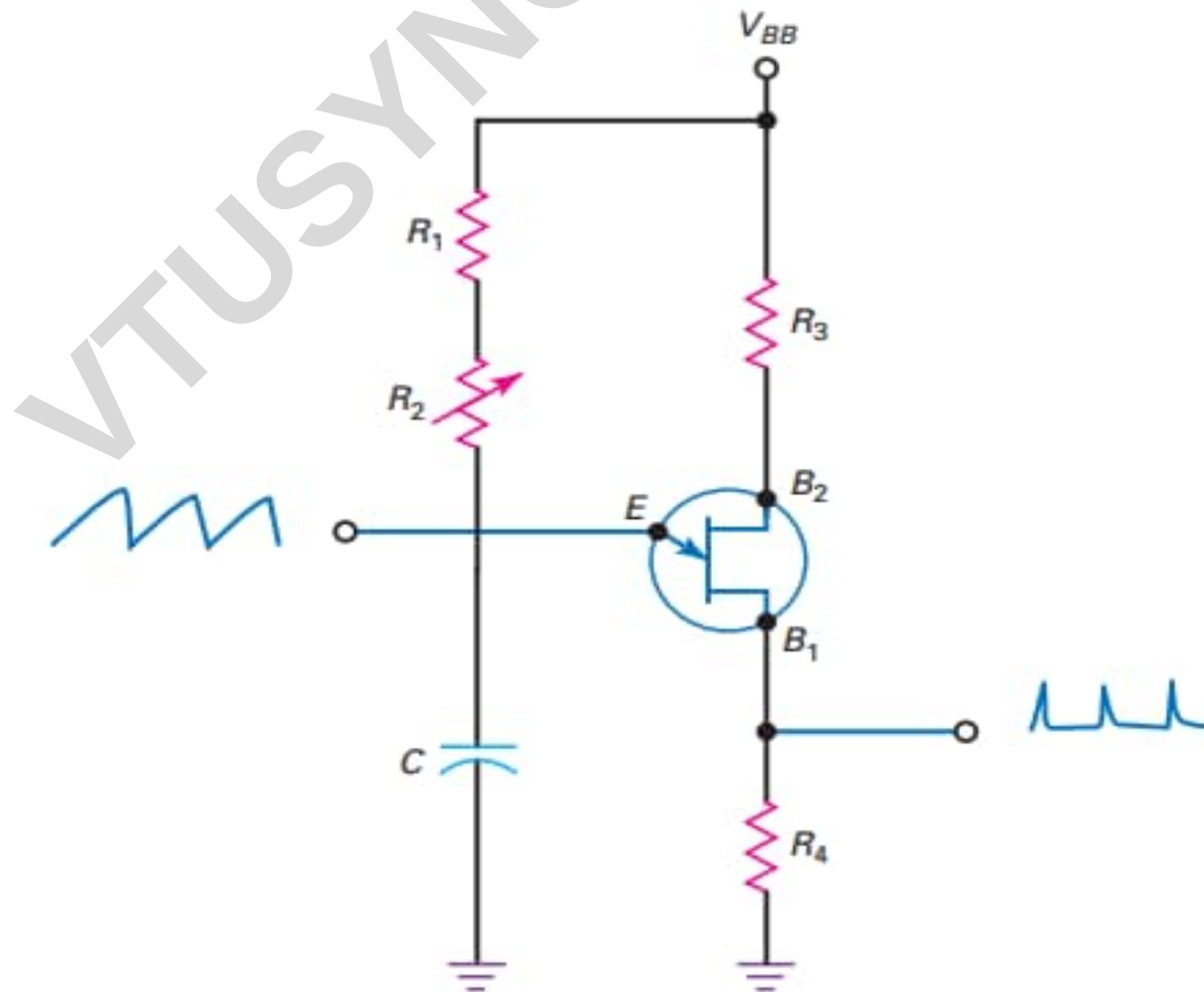


Figure 13-38 UJT relaxation oscillator.



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"To become a recognized world class Women Educational Institution, by imparting professional education to the students, creating technical opportunities through academic excellence and technical achievements, with ethical values"

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- To empower women with the additional skill for professional future career
- To enrich students with research blends in order to fulfill the International challenges
- To create multidisciplinary center of excellence
- To achieve Accreditation standards towards International education recognition.
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- To increase Doctorates numbers towards the Research quality of academics.